Machine Learning — Capstone Projects

How to use this book

A **capstone** is a project big enough to pull together several chapters at once: you take a real dataset, frame a prediction problem, build and evaluate a model honestly, and (from the intermediate level on) put it behind an API, a pipeline or a monitor. Reading about cross-validation is one thing; finding out that your Titanic score drops five points once you stop leaking the test set into your imputer is what makes the idea stick. The 40 briefs below run from a first regression model you can finish in a weekend to production-style systems (two-tower recommenders, streaming fraud scoring, feature stores with drift monitoring) that would take a small team weeks.

How to pick one. Start from the Project map. Choose a project whose prerequisite chapters you have read (or are reading now), and whose time box you can actually give it. If you are new to ML, do at least three beginner projects in order (1, 2 and 9 are a good trio: regression, classification with proper validation, and your first served model). If you are preparing for ML system design interviews, the advanced projects map one-to-one onto the case studies in Chapters 17–18. Build at least one of them end to end, because interviewers can hear the difference between someone who has drawn a feature store and someone who has debugged one.

How to use a brief. Each brief describes *what* to build and *why*. It does not give you the code — you write it. Work through it like this:

1. Read *What it is*, *The problem it solves* and *Requirements*, then sketch your own architecture before reading ours.
2. Build the **milestones in order**. Each one ends with a "done when" check. Do not move on until the check passes, because later milestones assume it.
3. Read *Deep dive: the hard parts* before you hit them. These are the places where people usually lose a week.
4. Write the tests described in *Testing strategy*. ML code fails silently (a model trains fine on leaked data and looks wonderful), so tests and sanity checks matter more here than in ordinary code.
5. Only then try the **stretch goals**.
6. Finish by answering *In the interview* out loud, without looking.

Colour coding. In the PDF the recurring sections have their own colour so you can skim: the framing sections (*What it is*, *The problem it solves*, *Real-world examples*), the design sections (*Requirements*, *The architecture*), the *Deep dive*, the *Trade-offs*, *In the interview*, *Self-check* and *TL;DR* each stand out. Use the colours to jump straight to the part you need when you revisit a project.

Levels.

Level	What it means	Typical time
Beginner	One dataset, one notebook or script, scikit-learn or a small PyTorch model, a clean evaluation. Ch 1–9 are enough.	1–2 weekends
Intermediate	A stronger model (boosting, transfer learning, embeddings), real evaluation pitfalls, and usually an API, a pipeline or a dashboard.	2–3 weeks
Advanced	A multi-stage system: offline training plus online serving, feature freshness, monitoring, experiments. Mirrors Ch 15–19.	3–8 weeks

The stack. Unless a brief says otherwise, use Python 3.14, pandas or polars for data, scikit-learn for classic models, PyTorch for deep learning, MLflow for experiment tracking and the model registry, FastAPI for serving and PostgreSQL for anything persistent. Python 3.14 is still fairly new for the ML ecosystem, so a few libraries with compiled extensions (for example some versions of SHAP, numba-based tools, TensorFlow-only packages or niche RL and federated-learning libraries) may not ship 3.14 wheels yet. When that happens, keep a separate 3.12 or 3.13 virtual environment for that one library rather than fighting the build.

Datasets are real. Every brief names a public dataset (UCI, Kaggle, MovieLens, academic benchmarks) and points to engineering blogs or papers that describe how real companies solved the same problem. Respect each dataset's licence and terms: several Kaggle datasets require you to accept competition rules before

downloading, and some popular sets are non-commercial only (MovieLens is for research use, and the Criteo uplift and RetailRocket datasets are CC BY-NC-SA 4.0), which is fine for study but matters if you reuse them at work.

> The briefs describe the problem, the design and the traps. The code, the experiments and the mistakes are yours. That is the point.

Project map

#	Project	Level	Time	Prerequisites (chapters)
1	House-price regression	Beginner	1 weekend	Ch 1, 3, 4, 5, 6
2	Titanic survival with honest cross-validation	Beginner	1 weekend	Ch 4, 5, 6, 7
3	SMS spam filter with Naive Bayes	Beginner	1 weekend	Ch 2, 6, 8
4	Customer segmentation with k-means	Beginner	1 weekend	Ch 4, 9
5	Handwritten digit recognizer	Beginner	1–2 weekends	Ch 10, 11
6	Credit-default scorecard	Beginner	1–2 weekends	Ch 5, 6
7	Tumour diagnosis with k-NN and SVM	Beginner	1 weekend	Ch 6, 8
8	Bike-share demand with tree ensembles	Beginner	1–2 weekends	Ch 4, 7
9	Your first model API (wine quality)	Beginner	1–2 weekends	Ch 3, 6, 16
10	Book "more like this" finder	Beginner	1 weekend	Ch 8, 13
11	Churn model with gradient boosting and SHAP	Intermediate	2 weeks	Ch 6, 7, 19
12	Pet-breed classifier with transfer learning	Intermediate	2 weeks	Ch 10, 11
13	Movie-review sentiment, TF-IDF to transformer	Intermediate	2–3 weeks	Ch 8, 12
14	Movie recommender with matrix factorisation	Intermediate	2–3 weeks	Ch 6, 13
15	Retail demand forecasting	Intermediate	2–3 weeks	Ch 6, 7, 12
16	Metric anomaly detection	Intermediate	2 weeks	Ch 9, 12
17	A/B test analysis toolkit	Intermediate	2 weeks	Ch 2, 16
18	Semantic document search	Intermediate	2 weeks	Ch 12, 13
19	Pedestrian detector by fine-tuning	Intermediate	2–3 weeks	Ch 11
20	Headline-testing bandit	Intermediate	2 weeks	Ch 14, 16
21	Predictive maintenance (remaining useful life)	Intermediate	2–3 weeks	Ch 7, 12
22	Retraining pipeline with registry and batch scoring	Intermediate	2–3 weeks	Ch 16
23	Uplift model for a marketing campaign	Intermediate	2 weeks	Ch 6, 7
24	Spoken keyword spotter	Intermediate	2–3 weeks	Ch 11, 12
25	Toxic-comment classifier with bias evaluation	Intermediate	2–3 weeks	Ch 12, 19
26	Two-tower retrieval and ranking recommender	Advanced	4–6 weeks	Ch 13, 15, 17
27	Social feed ranking with a multi-task model	Advanced	4–6 weeks	Ch 15, 17
28	Real-time fraud detection with streaming features	Advanced	4–6 weeks	Ch 16, 18
29	Search ranking with learning to rank	Advanced	4–5 weeks	Ch 6, 7, 17
30	Ad click-through-rate prediction	Advanced	4–5 weeks	Ch 6, 13, 18
31	Feature store, model serving and drift monitoring	Advanced	5–8 weeks	Ch 16
32	Reinforcement-learning agent: DQN to PPO	Advanced	3–5 weeks	Ch 14
33	Fairness audit of an income model	Advanced	3–4 weeks	Ch 6, 19
34	Visual search ("shop the look")	Advanced	4–6 weeks	Ch 11, 13, 18
35	People You May Know (link prediction)	Advanced	4–6 weeks	Ch 13, 18

#	Project	Level	Time	Prerequisites (chapters)
36	Trip ETA prediction	Advanced	4–6 weeks	Ch 12, 16
37	Session-based next-item recommender	Advanced	4–5 weeks	Ch 12, 13
38	Retrieval-augmented question answering with evaluation	Advanced	4–6 weeks	Ch 12, 13, 19
39	Federated next-word prediction	Advanced	4–6 weeks	Ch 12, 19
40	Similar listings and real-time personalisation	Advanced	4–6 weeks	Ch 13, 17, 18

1. House-price regression

Level: Beginner · **Time:** 1 weekend · **Prerequisites:** Ch 1, 3, 4, 5, 6

What it is

A model that predicts the sale price of a house from its features: size, number of rooms, neighbourhood, year built, quality ratings. This is **regression** — predicting a number rather than a category. You will start with a simple **baseline** (a model so dumb it sets the floor, such as "always predict the median price"), then linear regression, then regularised linear models, and you will learn to tell real improvement from noise.

The problem it solves

Buyers, sellers, lenders and tax assessors all need a fair estimate of what a property is worth before a sale happens. A human appraiser is slow and inconsistent. A model trained on past sales gives an instant, repeatable estimate and — just as important — tells you which features drive the price.

Real-world examples

- The **Ames Housing** dataset was assembled by Dean De Cock specifically as a teaching replacement for the old Boston data; his paper describes the 2,930 sales and 80 explanatory variables: Ames, Iowa: Alternative to the Boston Housing Data.
- Kaggle's long-running House Prices — Advanced Regression Techniques competition uses a subset of Ames and scores on the error of the *logarithm* of the price, the same trick you will use here.
- scikit-learn ships the smaller California housing dataset (block-level census data), handy as a second dataset to check your pipeline generalises.

Requirements

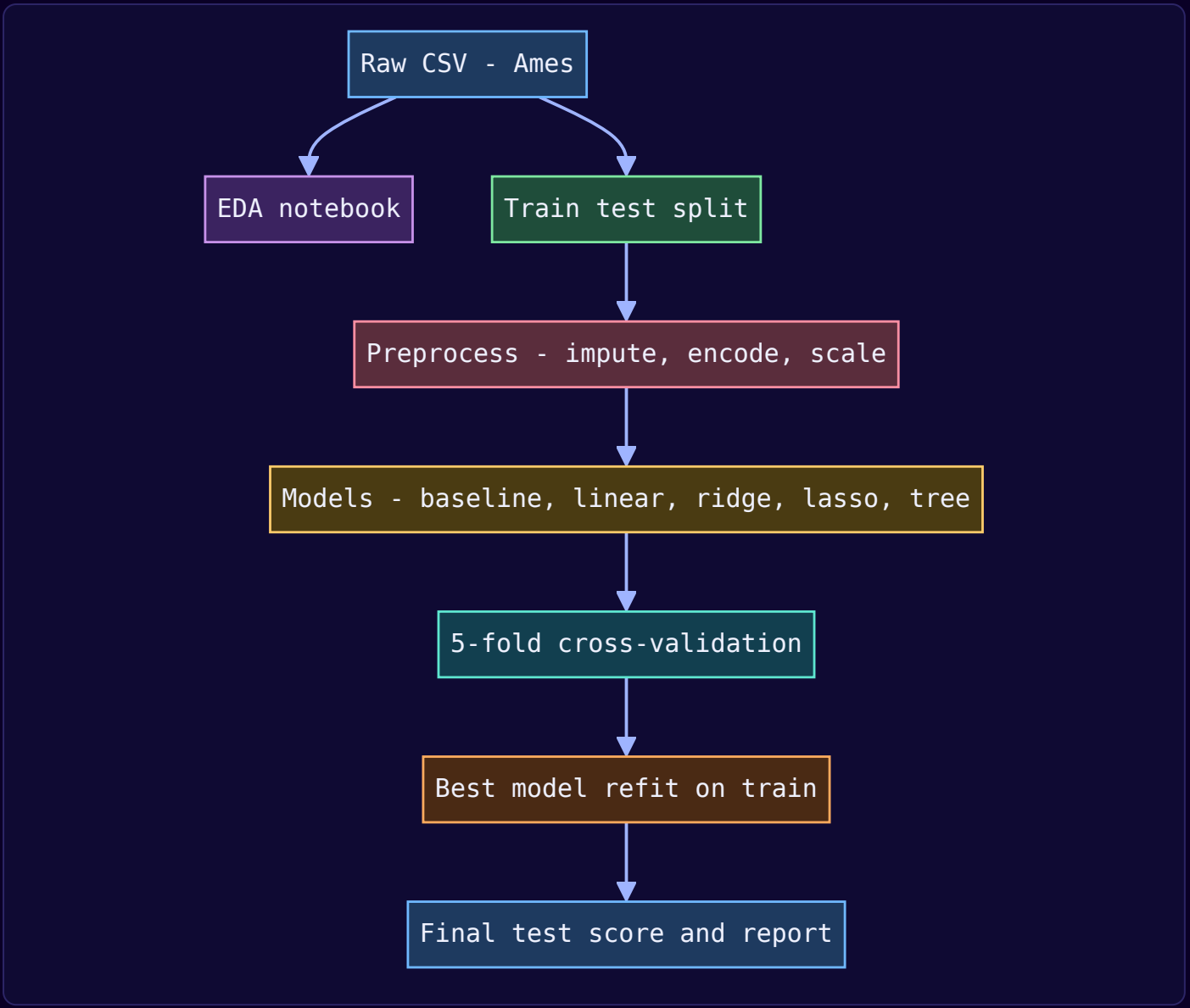
Functional

- Load the Ames data, explore it, and document every column you drop and why.
- Train and compare: median baseline, linear regression, Ridge, Lasso, and one tree model.
- Report RMSE (root mean squared error, the typical size of a miss in dollars) on the log price, plus MAE (mean absolute error) in dollars.
- Produce a short report of the top drivers of price.

Non-functional

- Whole pipeline runs from raw CSV to metrics in under 1 minute on a laptop.
- Results reproducible: fixed random seeds, same numbers on every run.
- One `Pipeline` object holds every preprocessing step so nothing leaks from the test set.

The architecture



- **EDA notebook**: exploratory data analysis — histograms, scatter plots, missing-value counts. It guides choices, it never touches the test set's labels.
- **Train/test split**: hold back 20% of rows that you look at exactly once, at the end.
- **Preprocess**: impute (fill) missing values, one-hot encode categories (turn "Neighborhood = NAmes" into a 0/1 column), scale numbers. Wrapped in a `ColumnTransformer`.
- **Models + cross-validation**: compare candidates on the training part only, using 5-fold CV (train on four fifths, score on the fifth, rotate).
- **Final test score**: the honest estimate you report.

Data model

Item	Type	Notes
`SalePrice`	float, target	Right-skewed, so model `log1p(SalePrice)`
Numeric features	float	`GrLivArea`, `TotalBsmtSF`, `YearBuilt`, `GarageCars` ...
Ordinal features	string to int	Quality scales such as `ExterQual` (Po to Ex) map to 1–5
Nominal features	string	`Neighborhood`, `MSZoning` — one-hot encoded
"NA means none" columns	string	`PoolQC = NA` means "no pool", not "unknown"

Tech stack

Component	Choice	Why
Data	pandas or polars	Fast tabular wrangling
Models	scikit-learn	`Pipeline`, `ColumnTransformer`, `RidgeCV`, `LassoCV`
Plots	matplotlib	Residual plots, coefficient bars
Tracking	MLflow (optional)	Log each model's CV score

Milestones

1. **EDA.** Plot the target, the top correlated features and missingness. *Done when* you can name the five strongest predictors and explain why you log-transform the target.
2. **Baseline.** Median predictor scored with CV. *Done when* you have a single RMSE number that every later model must beat.
3. **Linear pipeline.** `ColumnTransformer` + `LinearRegression`. *Done when* CV RMSE beats the baseline and the pipeline runs with no warnings.
4. **Regularisation.** Ridge and Lasso with alpha chosen by inner CV. *Done when* you can show how many coefficients Lasso set to zero.
5. **Tree model and final test.** A `HistGradientBoostingRegressor`, then pick the winner and score once on the test set. *Done when* the test RMSE is within a reasonable distance of the CV estimate.

Deep dive: the hard parts

- **"NA" is not missing.** In Ames, many `NA` values mean "this house has no basement/garage/pool". Imputing them with the column mode invents basements. Replace them with an explicit `"None"` category and set the matching area columns to 0.
- **Skew and outliers.** Prices are right-skewed: a few mansions dominate squared error. Fitting on `log1p(price)` makes errors roughly proportional (a 10% miss counts the same on a cheap and an expensive house). De Cock's paper flags a few very large houses sold cheaply (partial sales); decide on a written rule for them *before* looking at test results.
- **Leakage through preprocessing.** If you compute the mean for imputation or the scaler on the full dataset before splitting, the test set has quietly influenced training. Putting every step inside the `Pipeline` and passing the whole pipeline to `cross_val_score` fixes this.
- **Reading coefficients.** Coefficients are only comparable when features are on the same scale, and correlated features (garage cars vs garage area) share credit unpredictably. Say "associated with", never "causes".

Testing strategy

- Unit test the custom "NA means none" transformer on a three-row frame.
- Assert the pipeline's output has no NaNs and the expected number of columns.
- Sanity test: shuffle the target and retrain; CV RMSE should collapse to roughly baseline level. If it does not, something is leaking.
- Check that predicted prices are all positive after inverting the log.

Trade-offs, scaling and failure modes

- Linear models are transparent and fast but miss interactions (a big house in a bad area); boosting captures them at the cost of interpretability.
- The model learns one city in one period (2006–2010 sales). Applied to another city or another decade it will be confidently wrong — this is **distribution shift**.
- Small data (about 1,500–3,000 rows) means CV scores have real variance; report the mean and standard deviation across folds, not a single number.

Stretch goals

- Add prediction intervals with quantile regression (gradient boosting with `loss="quantile"`).
- Engineer features (total square footage, house age at sale) and measure each one's CV gain.

- Re-run the same pipeline on California housing with no code changes except the column lists.
- Wrap the best model in a FastAPI endpoint (see Project 9).

Skills practised

- Ch 1 — What Machine Learning Is: regression framing, the end-to-end workflow.
- Ch 4 — Data, EDA and Feature Engineering: imputation, encoding, `ColumnTransformer`, leakage.
- Ch 5 — Linear and Logistic Regression: least squares, Ridge, Lasso.
- Ch 6 — Evaluation and Tuning: baselines, RMSE vs MAE, cross-validation.

In the interview

Q: Why model the log of the price? **A:** Prices are right-skewed and errors scale with price. On the log scale, residuals are more symmetric, linear models fit better and the metric penalises relative error, so a mansion does not dominate the loss.

Q: Ridge or Lasso? **A:** Ridge shrinks all coefficients and handles correlated features gracefully. Lasso can zero out features, which helps when you suspect many are useless. ElasticNet mixes the two. Choose by cross-validation.

Q: Your CV RMSE is great but the test RMSE is much worse. What happened? **A:** Usually leakage (preprocessing fitted on all data, or a feature computed using the target), or the test set differs from training. Check that every step lives inside the pipeline and compare feature distributions.

TL;DR

Predict a skewed number honestly: log the target, wrap every preprocessing step in one pipeline, beat a dumb baseline under cross-validation, and touch the test set exactly once.

2. Titanic survival with honest cross-validation

Level: Beginner · **Time:** 1 weekend · **Prerequisites:** Ch 4, 5, 6, 7

What it is

A **binary classifier** (two possible answers) that predicts whether a Titanic passenger survived, from their class, sex, age, fare, family aboard and port of embarkation. The dataset is tiny and famous, which makes it perfect for one lesson: how to measure a model properly so that the score you report is the score you would actually get on new data.

The problem it solves

Most beginner projects overfit their own evaluation. People tune on the test set, engineer features using all rows, and celebrate a number that will not repeat. This project forces you to build a **validation protocol** — nested cross-validation, stratified folds, a leakage-free pipeline — that you will reuse in every later project.

Real-world examples

- Kaggle's Titanic — Machine Learning from Disaster is the classic starter competition: 891 labelled training rows and 418 unlabelled test rows.
- The scikit-learn cross-validation user guide documents stratified k-fold, nested CV and the "fit preprocessing inside the loop" rule this project drills.
- Medical and credit models use the same protocol at larger scale: stratified splits for imbalance, nested CV when data is small.

Requirements

Functional

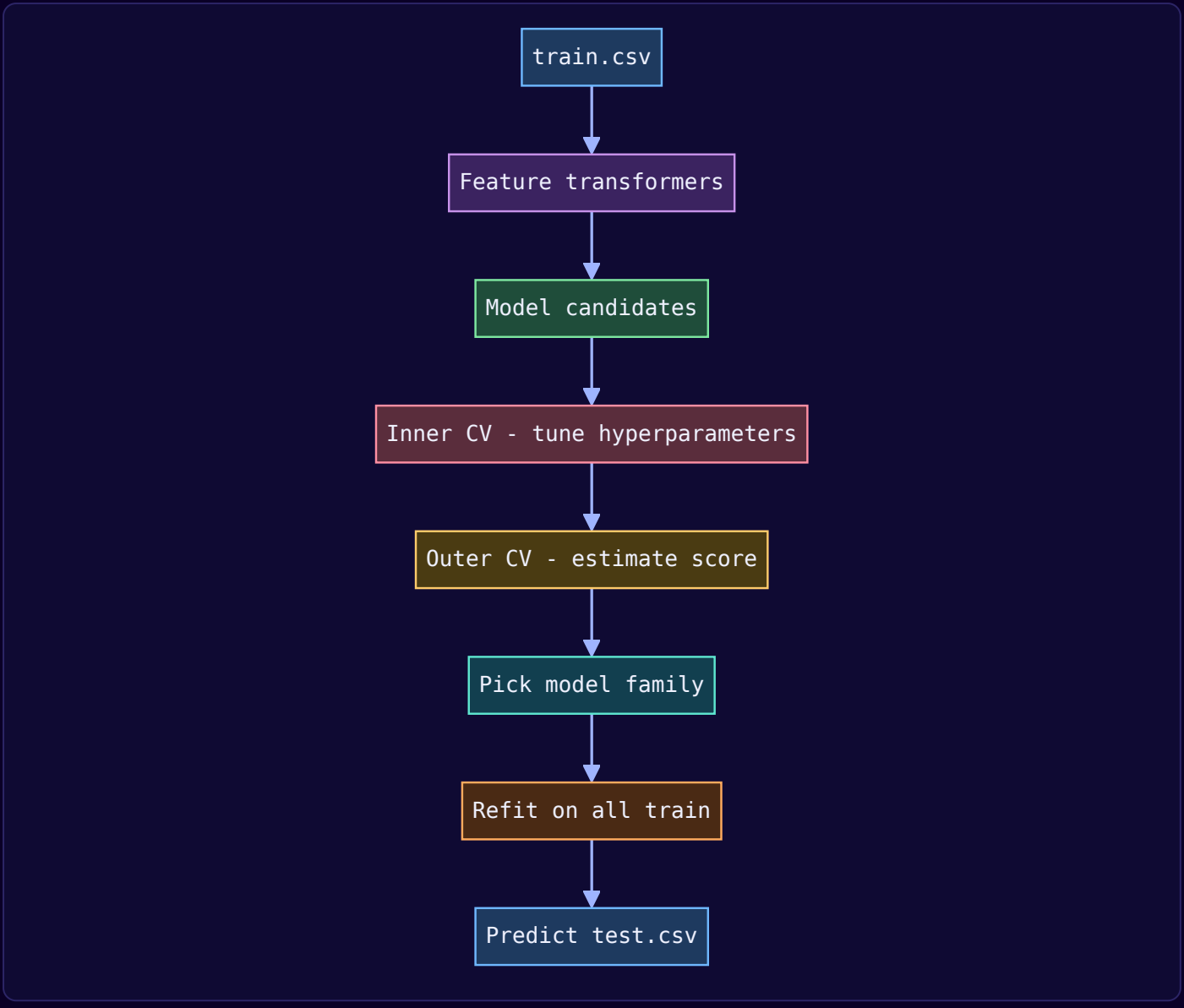
- Build features: title extracted from the name (Mr, Mrs, Miss, Master), family size, cabin known or not, fare per person.

- Compare logistic regression, a random forest and gradient boosting.
- Use **stratified** 5-fold CV (each fold keeps the same survival rate) and **nested** CV for hyperparameter search.
- Report accuracy, ROC AUC and a confusion matrix.

Non-functional

- All feature engineering implemented as scikit-learn transformers, so it is re-fitted inside every fold.
- Run time under 2 minutes for the full nested search.
- Report mean ± standard deviation over folds.

The architecture



- **Feature transformers**: custom classes with `fit` and `transform`, so title mapping and age imputation are learned only from the training fold.
- **Inner CV**: searches hyperparameters (tree depth, regularisation strength) within each outer training fold.
- **Outer CV**: scores the *whole procedure*, including tuning, on folds the search never saw. This is the honest number.
- **Refit**: once you have chosen the approach, refit it on all labelled data before predicting the unlabelled test file.

Data model

Column	Type	Handling
`Survived`	0/1 target	About 38% survived in the training file
`Pclass`, `Sex`, `Embarked`	category	One-hot encode, impute the two missing `Embarked`
`Age`	float, about 20% missing	Impute by title median inside the pipeline, add a "was missing" flag
`SibSp`, `Parch`	int	Combine into `family_size`
`Name`, `Ticket`, `Cabin`	string	Extract title, group size, deck letter

Tech stack

Component	Choice	Why
Data	pandas	Small CSV
Models	scikit-learn	`LogisticRegression`, `RandomForestClassifier`, `HistGradientBoostingClassifier`
Search	`GridSearchCV` inside `cross_validate`	Gives nested CV in a few lines
Plots	matplotlib	ROC curves, learning curves

Milestones

1. **Baselines.** "Everyone dies" and "women survive, men die". *Done when* you have both CV accuracies written down (the second is surprisingly strong).
2. **Leak-free pipeline.** Custom transformers + `ColumnTransformer` + logistic regression. *Done when* `cross_val_score` runs on the raw dataframe without any manual preprocessing beforehand.
3. **Nested CV.** Wrap a grid search in an outer CV loop for three model families. *Done when* you have outer-fold mean ± std for each.
4. **Diagnose.** Learning curves and a confusion matrix for the winner. *Done when* you can say whether more data or a simpler model would help.
5. **Submit.** Refit, predict the test file, compare the leaderboard score with your outer-CV estimate. *Done when* you can explain the gap.

Deep dive: the hard parts

- **Why nested CV.** If you pick hyperparameters by CV score and then report that same CV score, you have selected the luckiest configuration and reported its luck. Nested CV separates "choose" (inner loop) from "measure" (outer loop). With 891 rows, the optimism of the non-nested number is visible.
- **Group leakage.** Families and groups travelling on the same ticket tend to survive or die together. If a family is split across train and validation folds, the model can "recognise" it through shared ticket or surname features. Try `GroupKFold` with ticket as the group and see how much the score drops.
- **Variance on tiny data.** One fold contains about 178 passengers, so one passenger is about 0.6 percentage points of accuracy. Differences under 1–2 points between models are noise; use repeated stratified CV to see the spread.
- **Accuracy vs AUC.** Accuracy depends on the 0.5 threshold. ROC AUC (the probability that a random survivor is ranked above a random non-survivor) measures ranking quality independent of threshold.

Testing strategy

- Unit test the title extractor on odd names ("Rothes, the Countess. of").
- Assert transformers are stateless between fits: fitting on fold A then fold B gives fold-B statistics only.
- Shuffle-label test: accuracy should fall to about the majority-class rate.

- Check the submission file has 418 rows and the right columns.

Trade-offs, scaling and failure modes

- More features is not always better on 891 rows: high-cardinality features (raw ticket, raw cabin) let trees memorise.
- Leaderboard chasing is overfitting to the public test split. Trust the outer CV.
- The protocol scales: nested CV becomes expensive on big data, where a single fixed validation split is usually enough because variance shrinks.

Stretch goals

- Calibrate probabilities with `CalibratedClassifierCV` and draw a reliability diagram.
- Build a stacking ensemble and check, with nested CV, whether it really beats the best single model.
- Use permutation importance to rank features on held-out folds.

Skills practised

- Ch 4 — Data, EDA and Feature Engineering: custom transformers, missing values, leakage.
- Ch 5 — Linear and Logistic Regression: logistic regression and its coefficients.
- Ch 6 — Evaluation and Tuning: stratified and nested CV, ROC AUC, learning curves.
- Ch 7 — Decision Trees and Ensembles: random forests and boosting on small tabular data.

In the interview

Q: What is nested cross-validation and when do you need it? **A:** An inner CV chooses hyperparameters and an outer CV scores the whole choose-and-train procedure. You need it when data is small and you want an unbiased estimate after tuning.

Q: Why stratify? **A:** So every fold has the same class balance as the full data. Without it, a fold can by chance have very few positives and its score swings wildly.

Q: How would leakage show up here? **A:** Imputing age with the global median before splitting, or letting family members appear on both sides of a split. The symptom is a CV score that the hidden test set does not reproduce.

TL;DR

On small data the protocol matters more than the model: stratified, nested, leak-free cross-validation gives a number you can trust, and a strong simple baseline keeps you honest.

3. SMS spam filter with Naive Bayes

Level: Beginner · **Time:** 1 weekend · **Prerequisites:** Ch 2, 6, 8

What it is

A text classifier that labels short messages as **spam** or **ham** (legitimate). You turn each message into word counts, train a **multinomial Naive Bayes** model (which applies Bayes' rule assuming each word contributes independently), and serve it as a small command-line tool or API. It is the "hello world" of text classification and still a strong baseline today.

The problem it solves

Unwanted messages cost time and carry phishing links. Hand-written rules ("block messages containing FREE") are easy to evade and block legitimate messages. A learned filter adapts to the words spammers actually use and gives a probability you can threshold according to how much a false block hurts.

Real-world examples

- The SMS Spam Collection on UCI contains 5,574 English SMS messages labelled spam or ham (also mirrored on Kaggle).

- Paul Graham's essay A Plan for Spam (2002) popularised Bayesian spam filtering in email clients and explains why per-word probabilities work so well against spam.

Requirements

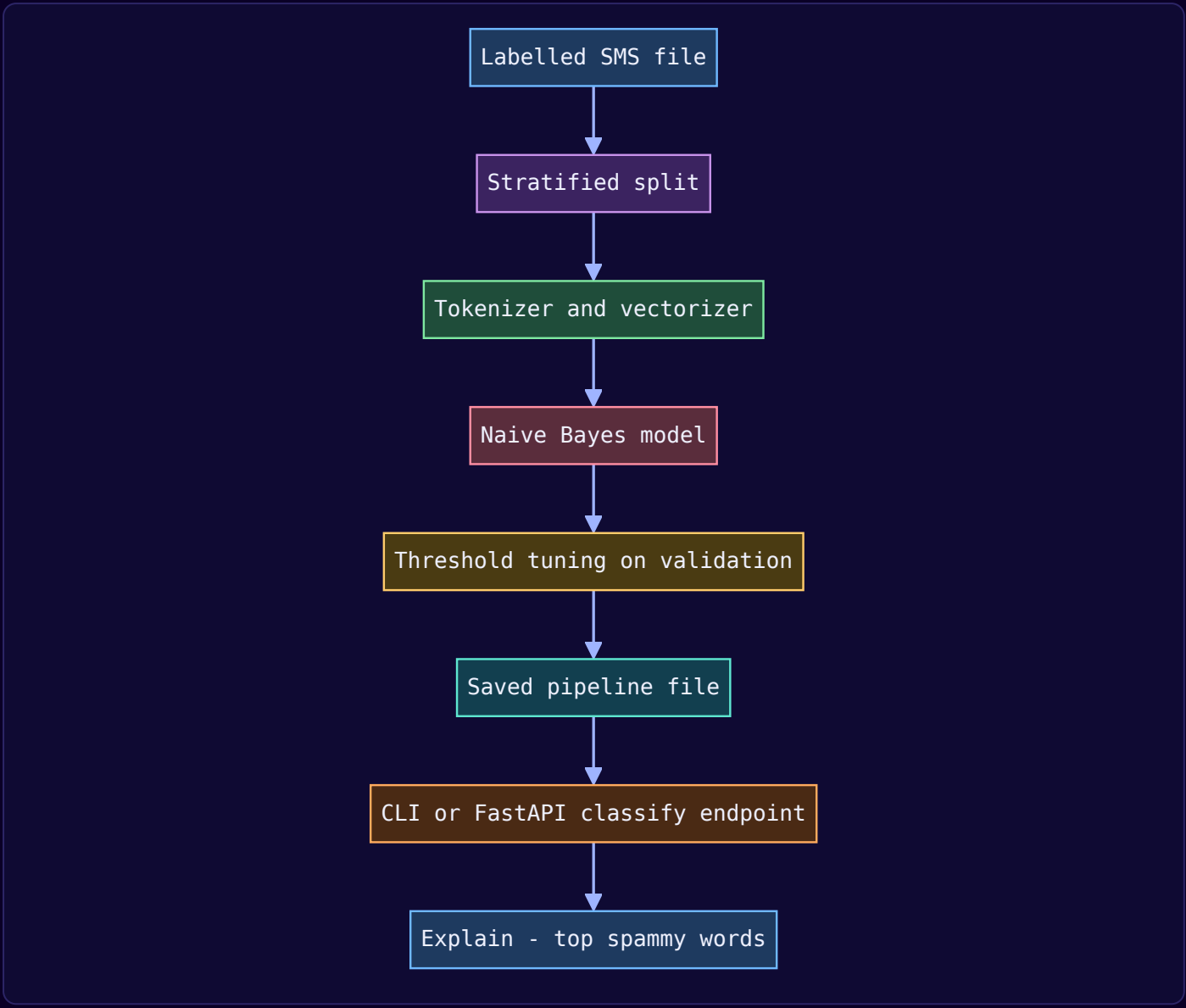
Functional

- Tokenise, build a vocabulary, vectorise with counts and with TF-IDF (term frequency times inverse document frequency, which down-weights words that appear everywhere).
- Train multinomial Naive Bayes and compare with logistic regression and a linear SVM.
- Choose a decision threshold for a target false-positive rate (legitimate messages wrongly flagged).
- Expose `classify(text) -> {label, p_spam, top_words}`.

Non-functional

- Classification under 5 ms per message on a laptop.
- Model file under 5 MB.
- At most 1% of ham flagged as spam at the chosen threshold, on held-out data.

The architecture



- **Vectorizer**: `CountVectorizer` or `TfidfVectorizer` builds the vocabulary from training messages only.
- **Naive Bayes**: estimates, for each word, how often it appears in spam vs ham, with **Laplace smoothing** (add a small count to every word so unseen words do not give zero probability).
- **Threshold tuning**: picks the probability cut-off on a validation split, not the test split.

- **Explain**: the words with the largest log-probability ratio between classes show *why* a message was flagged.

Data model

Item	Type	Notes
`label`	ham/spam	About 13% spam — imbalanced
`text`	string	Short, informal, abbreviations and numbers
Vocabulary	dict word to index	Learned from training split only
Class priors	2 floats	Fraction of spam and ham
Word log-likelihoods	matrix 2 × V	The whole model

Tech stack

Component	Choice	Why
Text features	scikit-learn `CountVectorizer`, `TfidfVectorizer`	Standard, fast, sparse
Model	`MultinomialNB`, `ComplementNB`	Strong on short text, trains in milliseconds
Comparison	`LogisticRegression`, `LinearSVC`	Discriminative baselines
Serving	FastAPI (optional)	Tiny JSON endpoint

Milestones

1. **Explore.** Message length by class, most common words per class. *Done when* you can name three features a human would use to spot spam.
2. **First model.** Counts + `MultinomialNB`. *Done when* precision, recall and F1 for spam are printed for a stratified CV.
3. **Compare.** TF-IDF, character n-grams (sequences of 3–5 characters, robust to "fr33"), and two linear models. *Done when* a table of PR AUC (area under the precision–recall curve) per variant exists.
4. **Threshold.** Pick the cut-off giving at most 1% false positives on validation. *Done when* test-set FPR and recall at that threshold are reported.
5. **Ship.** Save the pipeline, build the classify tool with top-word explanations. *Done when* typing a message returns a label and three reasons.

Deep dive: the hard parts

- **Why "naive" still works.** The independence assumption is wrong ("free" and "prize" co-occur), so the probabilities are over-confident. But the *ranking* of messages is often good, which is what classification needs. Do not trust the raw probabilities without calibration.
- **Imbalance and the right metric.** With 13% spam, a model predicting "ham" always is 87% accurate. Use precision (of flagged messages, how many were spam), recall (of spam, how many were caught) and PR AUC.
- **Asymmetric costs.** Losing a real message is worse than seeing a spam one. Express this as a maximum false-positive rate and pick the threshold from the validation precision–recall curve.
- **Adversarial drift.** Spammers change spelling and switch to URLs and phone numbers. Character n-grams and features like "contains a URL" or "contains a long number" make the filter harder to evade.

Testing strategy

- Golden examples: a handful of obvious spam and ham must classify correctly after every retrain.
- Vectorizer test: a word not in the training vocabulary is ignored, not an error.
- Assert predicted probabilities lie in [0, 1] and that thresholds are monotonic (higher threshold, fewer flags).
- Latency test: 1,000 messages classified in under 5 seconds.

Trade-offs, scaling and failure modes

- Naive Bayes trains in one pass and updates incrementally (`partial_fit`), ideal for user feedback loops. Logistic regression usually wins slightly on accuracy.
- A tiny English dataset will not generalise to other languages or to modern scam formats.
- In production, the label "spam" is reported by users — noisy, delayed and biased toward obvious cases.

Stretch goals

- Online learning: accept "mark as spam" feedback and update with `partial_fit`.
- Compare against a small transformer fine-tuned on the same data (see Project 13).
- Add calibration and display "92% likely spam" honestly.

Skills practised

- Ch 2 — Math for ML: Bayes' rule, conditional probability, logs to avoid underflow.
- Ch 8 — k-NN, SVMs and Naive Bayes: multinomial NB, smoothing, linear SVMs for text.
- Ch 6 — Evaluation and Tuning: precision, recall, PR curves, thresholds.

In the interview

Q: Why does Naive Bayes work despite its false assumption? **A:** Classification only needs the correct class to score highest. The independence assumption distorts the probabilities but often preserves the ranking, and with few parameters it needs little data.

Q: What is Laplace smoothing for? **A:** Without it, a word never seen in spam gets probability zero and wipes out the whole product. Adding a pseudo-count keeps every probability positive.

Q: How do you set the threshold? **A:** From business cost: decide the maximum tolerable rate of legitimate messages blocked, then choose the threshold on a validation set that meets it with the highest recall.

TL;DR

Bag-of-words plus Naive Bayes is a fast, strong text baseline. The real work is choosing the right metric for an imbalanced, asymmetric-cost problem and setting the threshold deliberately.

4. Customer segmentation with k-means

Level: Beginner · **Time:** 1 weekend · **Prerequisites:** Ch 4, 9

What it is

An **unsupervised** project (no labels) that groups an online shop's customers by behaviour. You compute **RFM** features for each customer — **R**ecency (days since last purchase), **F**requency (number of orders), **M**onetary value (total spend) — scale them, and cluster with **k-means** (which places k centre points and assigns each customer to the nearest). Then you name the segments in business language: "loyal big spenders", "lapsed one-timers".

The problem it solves

Marketing to every customer the same way wastes money. Segments let a business send a win-back offer to lapsed customers and early access to loyal ones. Because there is no ground truth, the hard part is deciding whether a clustering is *useful*, not just mathematically tidy.

Real-world examples

- The UCI Online Retail dataset holds about 540,000 transactions from a UK-based online gift retailer between December 2010 and December 2011 — the standard dataset for RFM exercises.
- The scikit-learn clustering guide compares k-means with DBSCAN, Gaussian mixtures and hierarchical methods and explains when each shape assumption holds.

Requirements

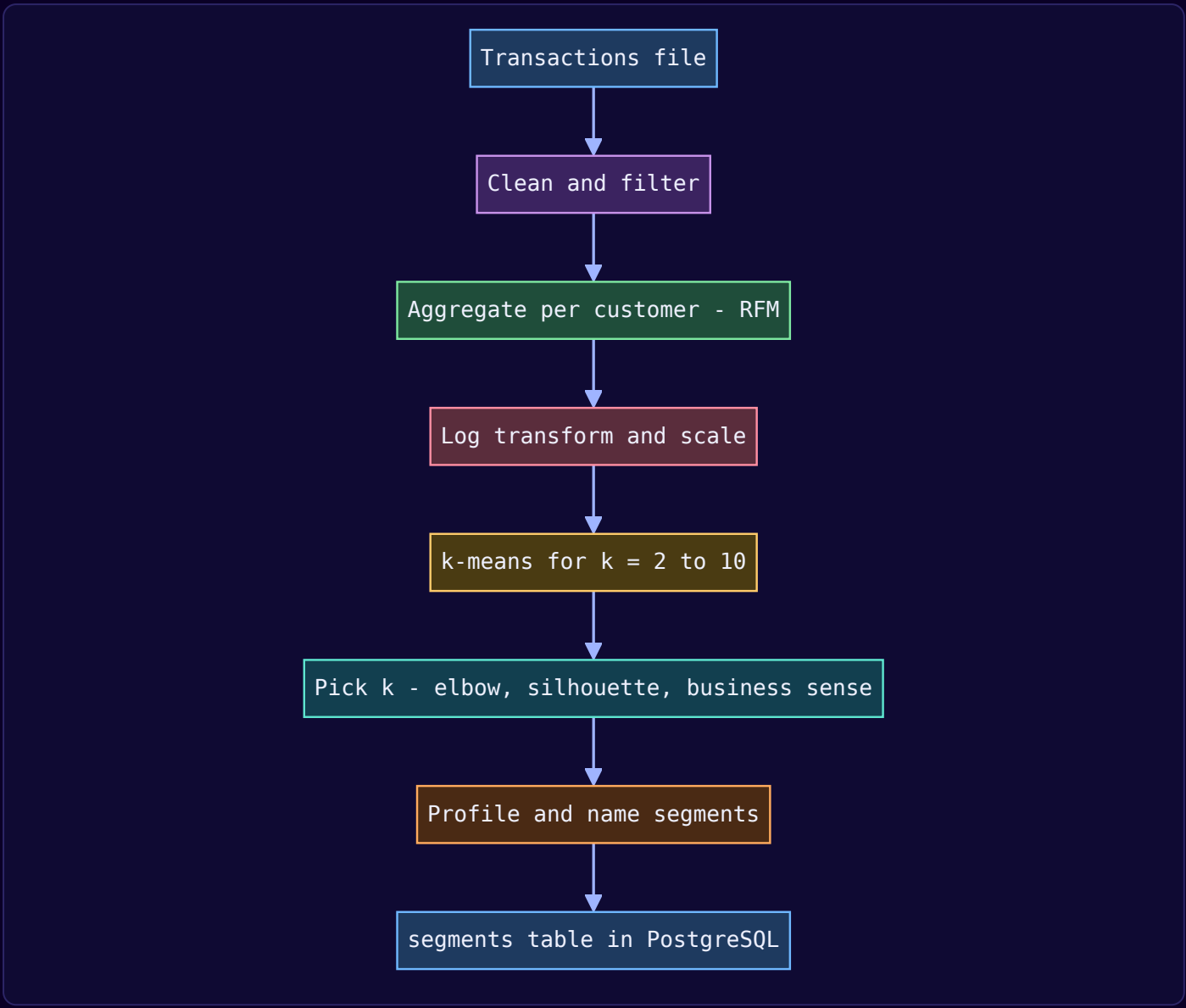
Functional

- Clean transactions: drop cancellations (invoice numbers starting with "C"), missing customer IDs and non-positive quantities.
- Aggregate to one row per customer with R, F, M and a few extras (average basket value, number of distinct products).
- Choose k with the elbow method and silhouette score; compare with a Gaussian mixture.
- Produce a segment profile table and a 2-D plot (PCA).

Non-functional

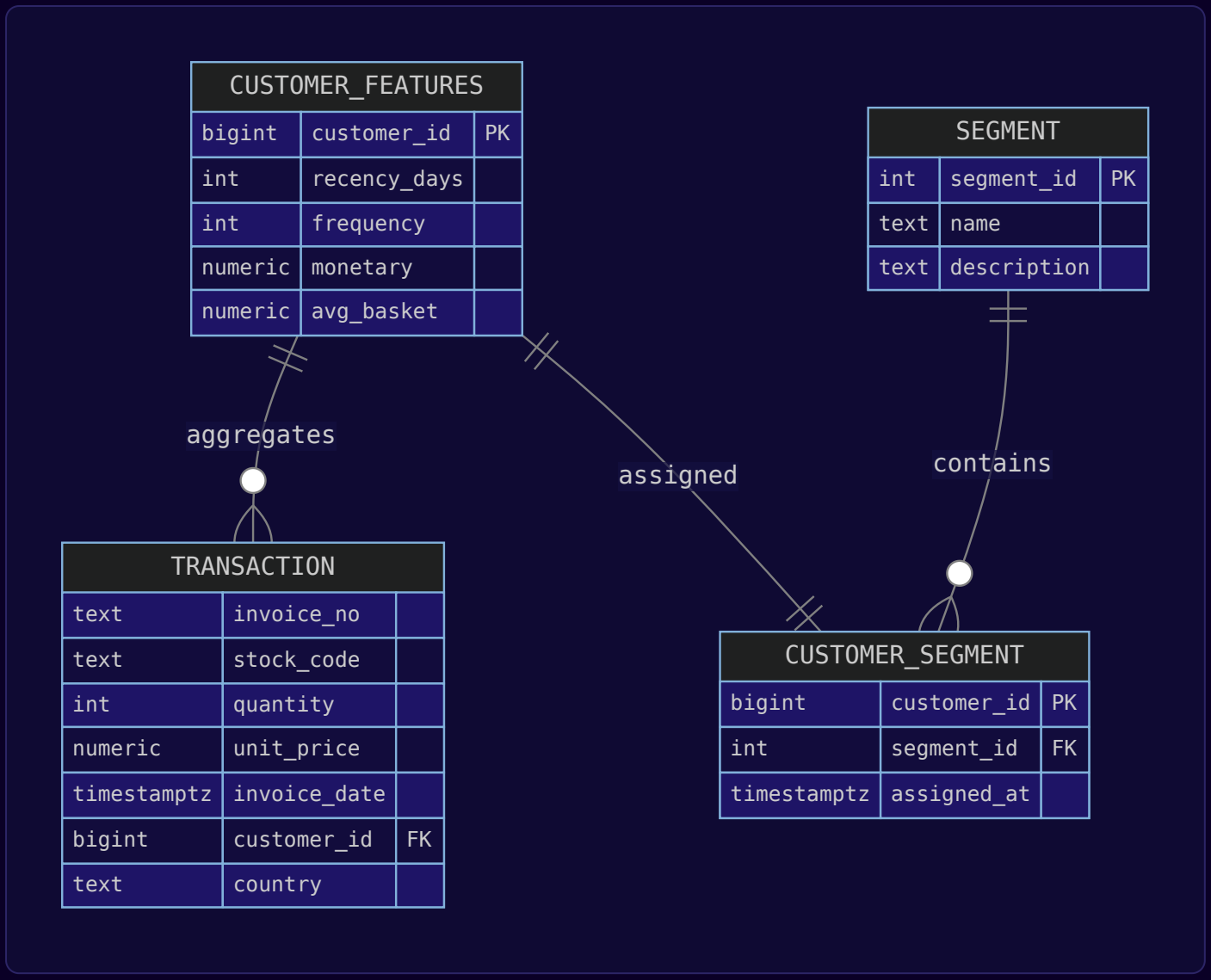
- Aggregation of the full file in under 30 seconds (polars or vectorised pandas).
- Segments stable: re-running with different seeds changes fewer than 5% of assignments.
- Segment table readable by a non-technical marketer.

The architecture



- **Clean**: removes returns and rows without a customer, which otherwise create negative spend.
- **Log and scale**: spend and frequency are heavily skewed. k-means uses Euclidean distance, so unscaled, the monetary column would decide everything.
- **Pick k**: the elbow plot shows where adding clusters stops reducing within-cluster spread much. The silhouette score (how much closer each point is to its own cluster than to the next) gives a second opinion.
- **Profile**: medians of R, F, M per cluster turn numbers into names.

Data model



Tech stack

Component	Choice	Why
Wrangling	polars	Fast group-by over half a million rows
Clustering	scikit-learn `KMeans`, `GaussianMixture`	Standard, quick
Visuals	matplotlib, PCA	2-D view of clusters
Storage	PostgreSQL	Marketing tools can query segments

Milestones

1. **Clean.** *Done when* total revenue after cleaning is positive and you can state how many rows each rule removed.
2. **RFM table.** *Done when* one row per customer exists with no negative values, and you have histograms before and after log transform.
3. **Cluster.** Elbow and silhouette for k = 2 to 10. *Done when* you justify one k in a sentence that mentions both the metric and the business.
4. **Profile.** *Done when* each segment has a name, a size and median R, F, M.
5. **Persist.** Write segments to PostgreSQL with an `assigned_at` timestamp. *Done when* a SQL query lists customers in a chosen segment.

Deep dive: the hard parts

- **Scaling decides the answer.** k-means minimises squared Euclidean distance. If monetary value ranges to tens of thousands and recency to a few hundred, clusters are just "spend bands". Log-transform skewed columns, then standardise.
- **k-means' shape assumption.** It prefers round, similarly sized clusters. A few huge wholesale customers form their own tiny cluster or drag a centroid. Consider capping outliers or treating them as a separate, rule-based segment.
- **Choosing k is a judgement.** There is rarely a sharp elbow. Pick the smallest k whose segments are distinct *and* actionable — four segments a team can act on beat nine nobody uses.
- **Stability.** Run k-means with several seeds (`n_init`) and on bootstrap samples. If customers jump between clusters, the structure is weak.

Testing strategy

- Unit test RFM aggregation on a hand-made five-transaction table with known answers.
- Assert recency is computed relative to a fixed snapshot date, not "now".
- Stability test: adjusted Rand index between two seeded runs above 0.9.
- Check every customer gets exactly one segment.

Trade-offs, scaling and failure modes

- k-means is fast (linear in rows) and easy to explain. Gaussian mixtures give soft memberships and elliptical clusters but are harder to explain.
- Segments go stale as behaviour changes. Re-run monthly, and track how many customers moved.
- Clusters are descriptive, not causal. "Big spenders" respond to offers is a hypothesis for an A/B test (Project 17), not a conclusion.

Stretch goals

- Add product-category shares as features and see whether new segments appear.
- Try DBSCAN or HDBSCAN to find outlier customers automatically.
- Predict each customer's next-90-day spend (customer lifetime value) and compare with the segments.

Skills practised

- Ch 4 — Data, EDA and Feature Engineering: cleaning, aggregation, log transforms, scaling.
- Ch 9 — Unsupervised Learning: k-means, elbow, silhouette, Gaussian mixtures, PCA, evaluation without labels.

In the interview

Q: How do you choose k? **A:** Elbow and silhouette for guidance, stability across seeds, then business usefulness. If the business can only run three campaigns, nine segments are not useful.

Q: Why scale before k-means? **A:** Because it uses distances. An unscaled large-range feature dominates, so clusters reflect units, not behaviour.

Q: How would you check segments are valuable? **A:** Use them: target one segment with an offer in a randomised test and compare response against a control group.

TL;DR

RFM features, log-scaled, clustered with k-means give marketers named, actionable segments — but scaling, outliers and the choice of k are judgements you must justify.

5. Handwritten digit recognizer

Level: Beginner · **Time:** 1–2 weekends · **Prerequisites:** Ch 10, 11

What it is

A neural network that reads a 28 × 28 pixel image of a handwritten digit and says which digit (0–9) it is. You start with logistic regression on flattened pixels, move to a **multi-layer perceptron** (MLP, layers of neurons fully connected to the previous layer), then a small **convolutional neural network** (CNN, which slides small learned filters over the image to detect strokes and shapes). Finally you build a canvas where you draw a digit and see the prediction live.

The problem it solves

Reading handwritten digits was one of the first commercial uses of neural networks (cheques and postal codes). For you, it is the cleanest way to learn the PyTorch training loop — data loaders, forward pass, loss, backpropagation, optimiser step — on a problem where you can see every mistake.

Real-world examples

- **MNIST**: 60,000 training and 10,000 test images, loaded in one line via `torchvision.datasets.MNIST`.
- LeNet-style CNNs were used to read handwritten digits on bank cheques in the 1990s, as described in LeCun et al.'s 1998 paper "Gradient-based learning applied to document recognition".

Requirements

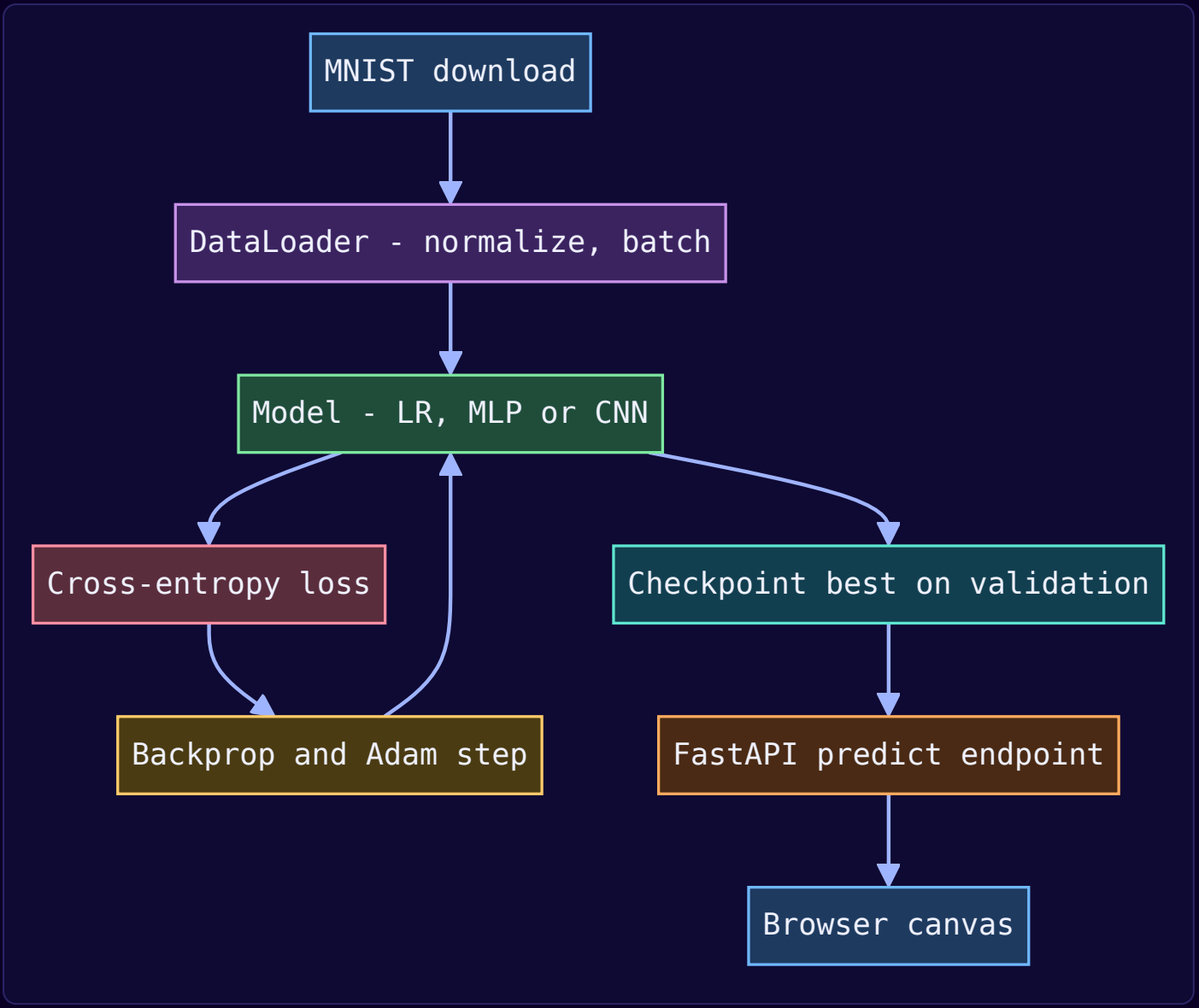
Functional

- Train three models: logistic regression, MLP, CNN. Compare test accuracy and parameter counts.
- Plot training and validation loss per epoch.
- Show a gallery of misclassified test images with predicted vs true labels.
- A small web page (or notebook widget) where you draw a digit and get the top-3 predictions.

Non-functional

- CNN trains on CPU in under 10 minutes.
- CNN test accuracy at least 99%.
- Inference under 10 ms per image.

The architecture



- **DataLoader**: yields shuffled mini-batches (for example 128 images) so the optimiser sees noisy but cheap gradient estimates.
- **Model**: CNN with two convolution + pooling blocks, then a fully connected layer and 10 outputs (logits — raw scores before softmax).
- **Cross-entropy loss**: penalises low probability on the correct digit.
- **Checkpoint**: save weights whenever validation accuracy improves (a 5,000-image slice of training data, not the test set).
- **Canvas**: the browser sends a 28 × 28 greyscale array to the API.

Data model

Item	Shape / type	Notes
Image tensor	float32 `[N, 1, 28, 28]`	Scaled to 0–1 then normalised by dataset mean and std
Label	int64 `[N]`	0–9
Logits	float32 `[N, 10]`	Softmax gives probabilities
Checkpoint	`.pt` file	`state_dict` plus normalisation constants

Tech stack

Component	Choice	Why
Framework	PyTorch, torchvision	Clear training loop, MNIST built in
Baseline	scikit-learn `LogisticRegression`	Linear reference point
Serving	FastAPI	JSON in, probabilities out
Front end	HTML canvas + a few lines of JS	Draw and send pixels

Milestones

1. **Linear baseline.** *Done when* logistic regression test accuracy is recorded (low 90s percent).
2. **MLP in PyTorch.** Your own training loop. *Done when* loss curves for train and validation are plotted and accuracy beats the baseline.
3. **CNN.** *Done when* test accuracy is at least 99% and you can state its parameter count vs the MLP's.
4. **Error analysis.** *Done when* a grid of 25 misclassified digits is saved and you have grouped the errors (ambiguous, badly written, mislabelled).
5. **Interactive demo.** *Done when* digits you draw are recognised — and you have fixed the preprocessing so they look like MNIST.

Deep dive: the hard parts

- **Train/serve skew.** MNIST digits were scaled to fit a 20 × 20 box (keeping their aspect ratio), then placed in the 28 × 28 frame so their centre of mass sits in the middle, white on black. Your canvas digit is probably off-centre, thick, and black on white. The model will fail until you apply the same preprocessing: invert, crop to the bounding box, resize, centre. This is the beginner version of a problem that ruins production systems.
- **Why convolutions win.** An MLP treats each pixel independently, so a digit shifted by two pixels looks new. Convolutions share weights across positions and pooling adds tolerance to small shifts, so the CNN needs fewer parameters and generalises better.
- **Overfitting signals.** Training loss keeps falling while validation loss rises. Use dropout (randomly zeroing activations during training), data augmentation (small rotations and shifts) and early stopping.
- `model.eval()`. Dropout and batch norm behave differently at inference. Forgetting to switch modes gives noisy predictions.

Testing strategy

- Overfit-one-batch test: the model must reach near-zero loss on a single batch. If it cannot, the loop is broken.
- Shape tests for each layer with a dummy tensor.
- Preprocessing test: an MNIST test image passed through the canvas preprocessing comes out unchanged (or nearly).
- API test: a known test image returns the right digit.

Trade-offs, scaling and failure modes

- Bigger CNNs add fractions of a percent on MNIST. Beyond about 99.3%, remaining errors are often ambiguous or mislabelled digits.
- MNIST is too easy to reveal modelling skill. Re-run on Fashion-MNIST to see a harder distribution.
- Confidently wrong predictions on non-digits (a drawn letter "A" gets 98% "4"). Softmax always picks something — consider a "not a digit" threshold.

Stretch goals

- Data augmentation and a learning-rate schedule. Measure each one's effect.
- Export to ONNX and run inference in the browser.
- Visualise first-layer filters and feature maps.

Skills practised

- Ch 10 — Neural Networks and Deep Learning: MLPs, cross-entropy, backpropagation, Adam, dropout, the PyTorch loop.
- Ch 11 — CNNs and Computer Vision: convolution, pooling, a small CNN, augmentation.

In the interview

Q: Why does a CNN beat an MLP on images? **A:** Local connectivity and weight sharing exploit the fact that nearby pixels are related and patterns can appear anywhere. That means far fewer parameters and built-in tolerance to small shifts.

Q: Your model is 99% accurate on the test set but fails on user drawings. Why? **A:** Train/serve skew: the user input is preprocessed differently from training data. Apply exactly the same normalisation, centring and scaling at serving time.

Q: How do you know your training loop is correct? **A:** Overfit a single small batch to near-zero loss. Then check that train and validation curves behave sensibly.

TL;DR

MNIST teaches the full deep-learning loop and the first lesson of production ML: a model is only as good as the match between training data and what it sees at serving time.

6. Credit-default scorecard

Level: Beginner · **Time:** 1–2 weekends · **Prerequisites:** Ch 5, 6

What it is

A model that estimates the probability that a credit-card customer will miss next month's payment, presented as a **scorecard**: a points table a loan officer could read ("utilisation above 80%: −40 points"). Under the hood it is **logistic regression** — a linear model whose output passes through the sigmoid function to give a probability — on carefully binned features. The focus is on well-calibrated probabilities and explanations, not on squeezing out accuracy.

The problem it solves

Lenders must price risk, set credit limits and explain decisions to customers and regulators. A black-box model with a slightly better AUC is often less useful than a transparent one whose every point can be justified. This project teaches you to build the transparent one properly, and to check that "20% risk" really means about one in five default.

Real-world examples

- The UCI Default of Credit Card Clients dataset has 30,000 Taiwanese card holders from 2005 with demographics, credit limits, six months of repayment status, bill and payment amounts, and a default flag.
- The scikit-learn probability calibration guide explains reliability diagrams and the tools (`CalibratedClassifierCV`, `calibration_curve`) you will use to validate the probabilities.

Requirements

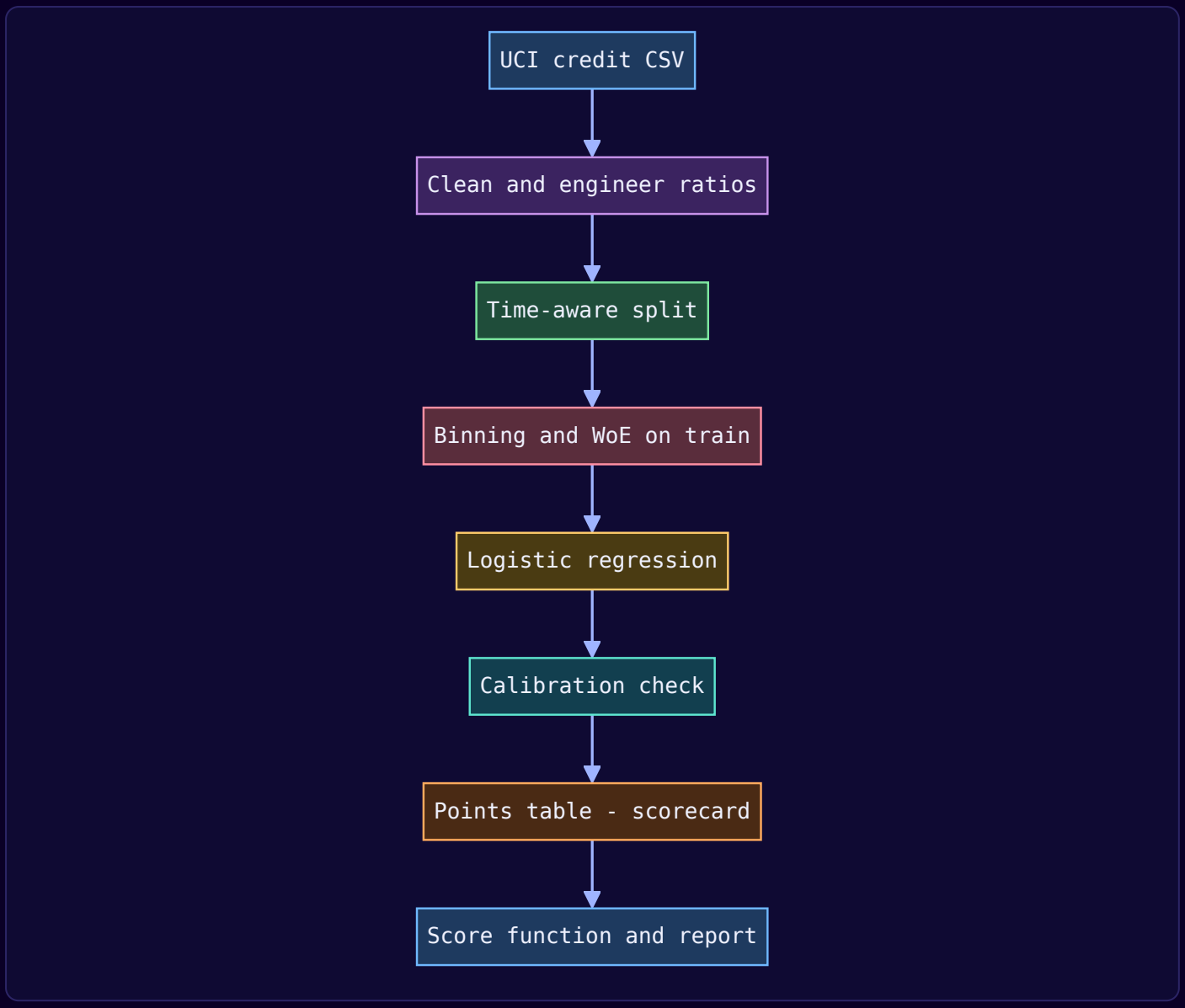
Functional

- Features: utilisation (bill ÷ limit), months of delinquency, payment-to-bill ratio, trend of bills.
- Bin numeric features into a handful of ranges and compute **WoE** (weight of evidence: the log of the ratio of good to bad customers in each bin) and **IV** (information value: a summary of how much a feature separates good from bad).
- Fit logistic regression on WoE features; convert coefficients to points.
- Report ROC AUC, a calibration plot and the **Brier score** (mean squared error of predicted probabilities).

Non-functional

- Every feature's points must change monotonically with risk where business logic expects it (more missed payments, never fewer points of risk).
- Scorecard printable on one page.
- Scoring one applicant under 1 ms.

The architecture



- **Binning and WoE**: learned from training rows only. WoE turns any feature, including categories, into a single number on the log-odds scale, which is exactly the scale logistic regression works in.
- **Logistic regression**: with WoE inputs, coefficients should all share the same sign and sit near 1 in size (with WoE defined as log of good over bad and the target "default", they come out near −1). A coefficient with the opposite sign warns of collinearity.
- **Points table**: a standard rescaling, for example 600 points at odds of 50:1 and 20 more points every time the odds double.

Data model

Item	Type	Notes
`default_next_month`	0/1 target	About 22% positives
`LIMIT_BAL`	numeric	Credit limit
`PAY_0`, `PAY_2` … `PAY_6`	int	Repayment status by month, −2 to 8 (there is no `PAY_1`)
`BILL_AMT1..6`, `PAY_AMT1..6`	numeric	Monthly bills and payments
Bin table	feature, range, WoE, points	The deliverable scorecard

Tech stack

Component	Choice	Why
Data	pandas	Small table
Model	scikit-learn `LogisticRegression`	Transparent, calibrated by design
Binning	your own code (or a scorecard library if it supports Python 3.14)	Learn the mechanics first
Comparison	`HistGradientBoostingClassifier`	Measures what transparency costs

Milestones

1. **Baseline.** Plain logistic regression on raw scaled features. *Done when* CV ROC AUC and Brier score are recorded.
2. **WoE binning.** *Done when* every feature has a bin table with WoE and IV, and you dropped features with IV close to 0.
3. **Scorecard.** *Done when* you can score a hand-made applicant by adding points from the table and get the same probability as the model.
4. **Calibration and comparison.** *Done when* a reliability diagram is plotted and you know how much AUC gradient boosting adds.
5. **Report.** *Done when* a one-page scorecard plus a "reasons for decline" function (the three features costing the most points) exist.

Deep dive: the hard parts

- **Monotonic bins.** Raw bins often zig-zag (risk goes up, down, up with income) because of noise. Merge adjacent bins until WoE is monotonic, otherwise the scorecard rewards nonsense and is hard to defend.
- **Calibration vs discrimination.** AUC measures ranking. A model can rank perfectly yet say 5% when the truth is 30%. Lending decisions use the probability directly (expected loss = probability × exposure), so check calibration with a reliability diagram.
- **Reject inference.** Real lenders only see outcomes for approved applicants, so the training data is biased. This dataset does not have that problem, but you should be able to explain it.
- **Fairness.** The dataset includes sex, marital status and age. In many jurisdictions some of these cannot be used for credit decisions. Train with and without them and compare outcomes by group (preview of Project 33).

Testing strategy

- Unit test the WoE function on a tiny table with known counts.
- Points-sum test: summed points convert back to the model's probability within rounding.
- Monotonicity test for each binned feature.
- Calibration test: on held-out data, each probability decile's observed default rate lies within a tolerance of its mean prediction.

Trade-offs, scaling and failure modes

- Scorecards give up a little accuracy for explainability, stability and easy audit. Boosting may add a few AUC points but needs SHAP-style explanations (Project 11).
- Economic shifts move default rates. Monitor the **population stability index** (how much the score distribution has shifted) and recalibrate.
- Bins learned on small groups are noisy. Require a minimum share (say 5%) of rows per bin.

Stretch goals

- Add a monotonic-constrained gradient boosting model and compare explanations.
- Simulate a cut-off policy: approval rate vs expected bad rate curve.
- Compute population stability index between two time slices.

Skills practised

- Ch 5 — Linear and Logistic Regression: logistic regression, log-odds, reading coefficients.
- Ch 6 — Evaluation and Tuning: ROC AUC, calibration, Brier score, thresholds.
- Ch 19 — Responsible ML: protected attributes and explanations for adverse decisions.

In the interview

Q: Why would a bank prefer logistic regression to gradient boosting? **A:** Transparency, stable behaviour, monotonic and explainable effects, easy regulatory review and "reasons for decline". The accuracy gap is often small on tabular credit data.

Q: What is weight of evidence? **A:** For each bin, the log of the share of good customers divided by the share of bad ones. It puts every feature on the log-odds scale and handles categories and outliers.

Q: How do you check probabilities are trustworthy? **A:** A reliability diagram and Brier score on held-out data. Fix miscalibration with Platt scaling or isotonic regression.

TL;DR

A credit scorecard is logistic regression made readable: monotonic WoE bins, points that add up to a calibrated probability, and reasons a customer can be told.

7. Tumour diagnosis with k-NN and SVM

Level: Beginner · **Time:** 1 weekend · **Prerequisites:** Ch 6, 8

What it is

A classifier that labels breast-tumour cell measurements as **malignant** or **benign**, using two classic algorithms: **k-nearest neighbours** (k-NN: classify a sample by majority vote of the k most similar training samples) and the **support vector machine** (SVM: find the boundary with the widest margin between classes, optionally in a transformed feature space via a **kernel**). The emphasis is on scaling, choosing hyperparameters and picking a decision threshold when one kind of mistake is far worse.

The problem it solves

Screening tools help doctors prioritise. Missing a malignant case (a **false negative**) is much worse than a false alarm, so the model must be tuned for high **recall** (sensitivity). The project teaches you how to make that trade-off explicit and why a model that is "98% accurate" can still be unsafe.

Real-world examples

- The Breast Cancer Wisconsin (Diagnostic) dataset has 569 samples with 30 features computed from digitised images of cell nuclei (radius, texture, concavity …). It also ships inside scikit-learn as `load_breast_cancer`.
- Clinical screening models are reported with sensitivity and specificity at a fixed operating point, which is exactly what you will report.

Requirements

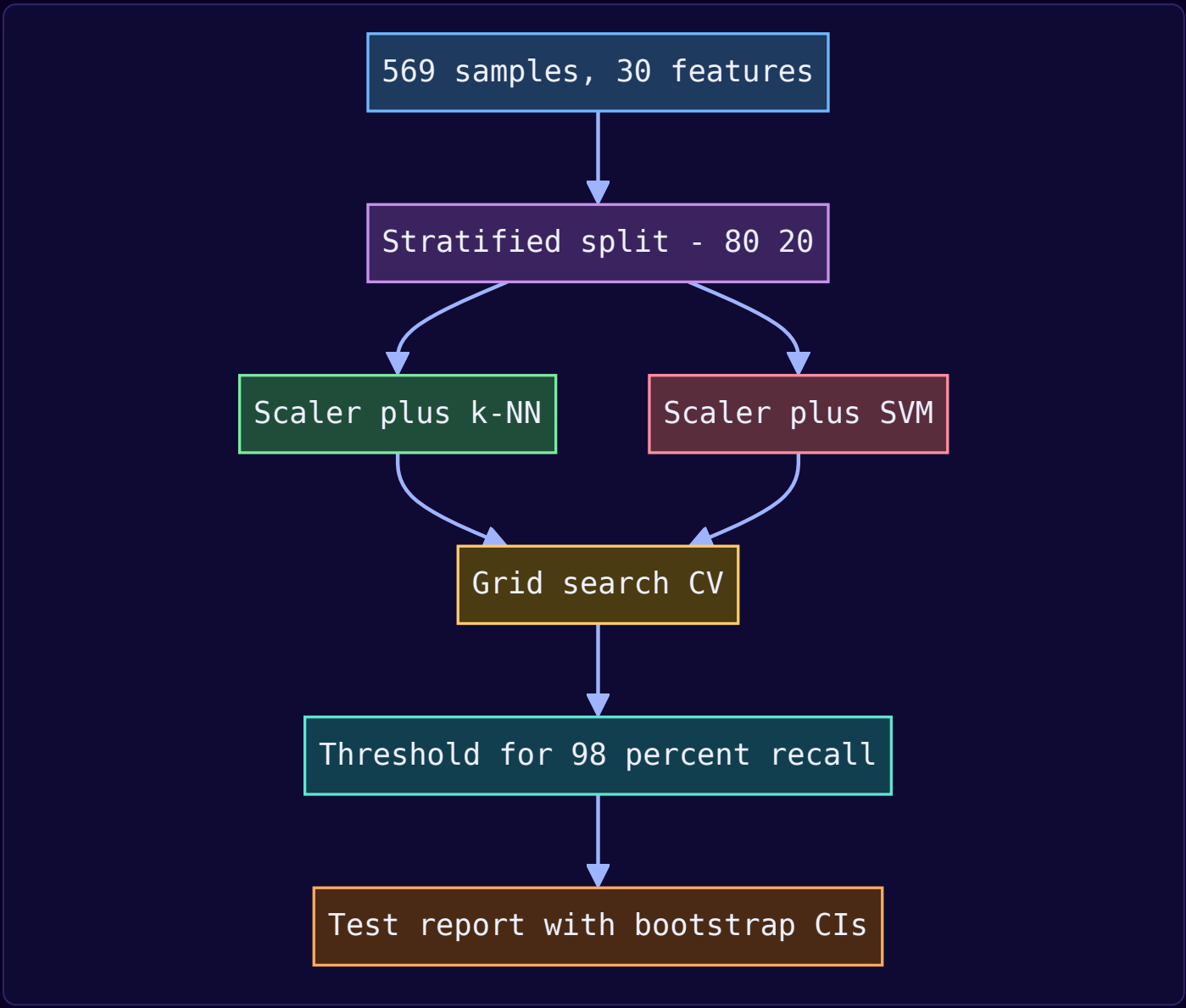
Functional

- Pipelines for k-NN and SVM (linear and RBF kernel), each with `StandardScaler`.
- Grid-search k, C (penalty for margin violations) and gamma (how far one training point's influence reaches) with stratified CV.
- Choose a threshold that gives at least 98% recall on validation, then report precision at that recall.
- Visualise the decision boundary in two PCA dimensions.

Non-functional

- Full grid search under 1 minute.
- Report 95% confidence intervals (bootstrap) for recall and precision.
- No preprocessing fitted outside CV.

The architecture



- **Scaler**: both algorithms use distances (k-NN directly, RBF SVM through the kernel), so features with large ranges, like area, would otherwise dominate texture.
- **Grid search**: tunes inside CV. For SVM, `probability=True` or `decision_function` scores give a continuous score for thresholding.
- **Bootstrap CIs**: resample the test set many times to show how uncertain the metrics are with only about 114 test samples.

Data model

Item	Type	Notes
`diagnosis`	M/B	212 malignant, 357 benign
Mean features (10)	float	radius, texture, perimeter, area, smoothness …
SE features (10)	float	Standard error of each
Worst features (10)	float	Mean of the three largest values
Model output	score	Distance to boundary or probability

Tech stack

Component	Choice	Why
Models	scikit-learn `KNeighborsClassifier`, `SVC`	Classic, well documented
Tuning	`GridSearchCV` with `StratifiedKFold`	Small grid, small data
Plots	matplotlib	PR curve, PCA boundary
Stats	numpy bootstrap	Confidence intervals

Milestones

1. **Unscaled vs scaled k-NN.** *Done when* you show the CV accuracy jump from scaling and explain it.
2. **Tune k-NN and SVM.** *Done when* best hyperparameters and CV ROC AUC for both are in a table.
3. **Threshold.** *Done when* you can state precision at 98% recall on validation.
4. **Test and uncertainty.** *Done when* test recall and precision with bootstrap 95% intervals are reported.
5. **Visualise.** *Done when* a 2-D PCA plot shows the SVM boundary and support vectors.

Deep dive: the hard parts

- **Correlated features.** Radius, perimeter and area are almost perfectly correlated, so they triple-count "size" in k-NN distances. Try PCA or dropping duplicates and see whether k-NN improves.
- **The C–gamma interaction.** Large gamma makes each point's influence tiny, so the boundary wraps around individual points (overfitting). Large C refuses margin violations. Plot CV score as a heat map over both to see the ridge of good settings.
- **Thresholds on SVMs.** SVM outputs a signed distance, not a probability. `probability=True` fits an internal Platt scaling with its own CV. Thresholding the raw decision function is often simpler and equivalent for ranking.
- **Tiny test sets.** With about 40 malignant test cases, one miss changes recall by 2.5 points. Say so, and show intervals.

Testing strategy

- Pipeline test: scaler statistics come only from training folds (check `mean_` against a manual calculation).
- Reproducibility: same seeds give identical grid results.
- Threshold test: recall at the chosen threshold on validation is at least 0.98.
- A "sanity" model (always benign) must score 0 recall.

Trade-offs, scaling and failure modes

- k-NN has no training cost but slow, memory-heavy prediction on big data. SVMs scale roughly quadratically in samples, fine here, painful above about 100,000 rows.
- Linear SVM is nearly as good here and far easier to explain. Prefer it if the gap is within noise.
- A model trained on one lab's imaging pipeline may not transfer to another lab. Never deploy medical models without external validation and regulatory review — this is a learning exercise.

Stretch goals

- Compare with logistic regression and a random forest under the same protocol.
- Use repeated stratified CV to compare models with a paired test.
- Add feature selection inside the pipeline and check whether recall holds.

Skills practised

- Ch 8 — k-NN, SVMs and Naive Bayes: distance-based learning, margins, kernels, C and gamma.
- Ch 6 — Evaluation and Tuning: recall vs precision, thresholds, grid search, confidence intervals.
- Ch 9 — Unsupervised Learning: PCA for visualisation.

In the interview

Q: Why must you scale features for k-NN and SVM? **A:** Both rely on distances. Without scaling, the feature with the largest numeric range dominates the distance and the others are ignored.

Q: What do C and gamma do in an RBF SVM? **A:** C trades margin width for training errors. Gamma sets how local each support vector's influence is. High values of either lead to overfitting.

Q: Accuracy is 97%. Is the model good enough? **A:** Not by itself. For diagnosis, recall on the malignant class at a chosen operating point matters, along with its uncertainty and external validation.

TL;DR

Scale, tune C and gamma or k inside CV, and choose the threshold from the cost of a missed malignancy — then admit how uncertain a 114-sample test is.

8. Bike-share demand with tree ensembles

Level: Beginner · **Time:** 1–2 weekends · **Prerequisites:** Ch 4, 7

What it is

A regression model that predicts how many bikes will be rented in a city's bike-share system each hour, from the time, the season, the weather and whether it is a working day. You compare a single **decision tree** (a flowchart of yes/no questions learned from data) with a **random forest** (many trees on random subsets, averaged) and **gradient boosting** (trees added one at a time, each fixing the previous ones' errors).

The problem it solves

Bike-share operators must move bikes between stations before the morning rush and schedule maintenance on quiet days. An hourly demand forecast drives both. Demand depends on interactions — rain hurts far more at 5 pm on a weekday than at 3 am — which is exactly what tree ensembles capture without manual feature crosses.

Real-world examples

- The UCI Bike Sharing dataset has two years (2011–2012) of hourly and daily rental counts from Capital Bikeshare in Washington, D.C., joined with weather and holidays.
- The scikit-learn HistGradientBoostingRegressor docs show native categorical support and Poisson loss, both useful here.

Requirements

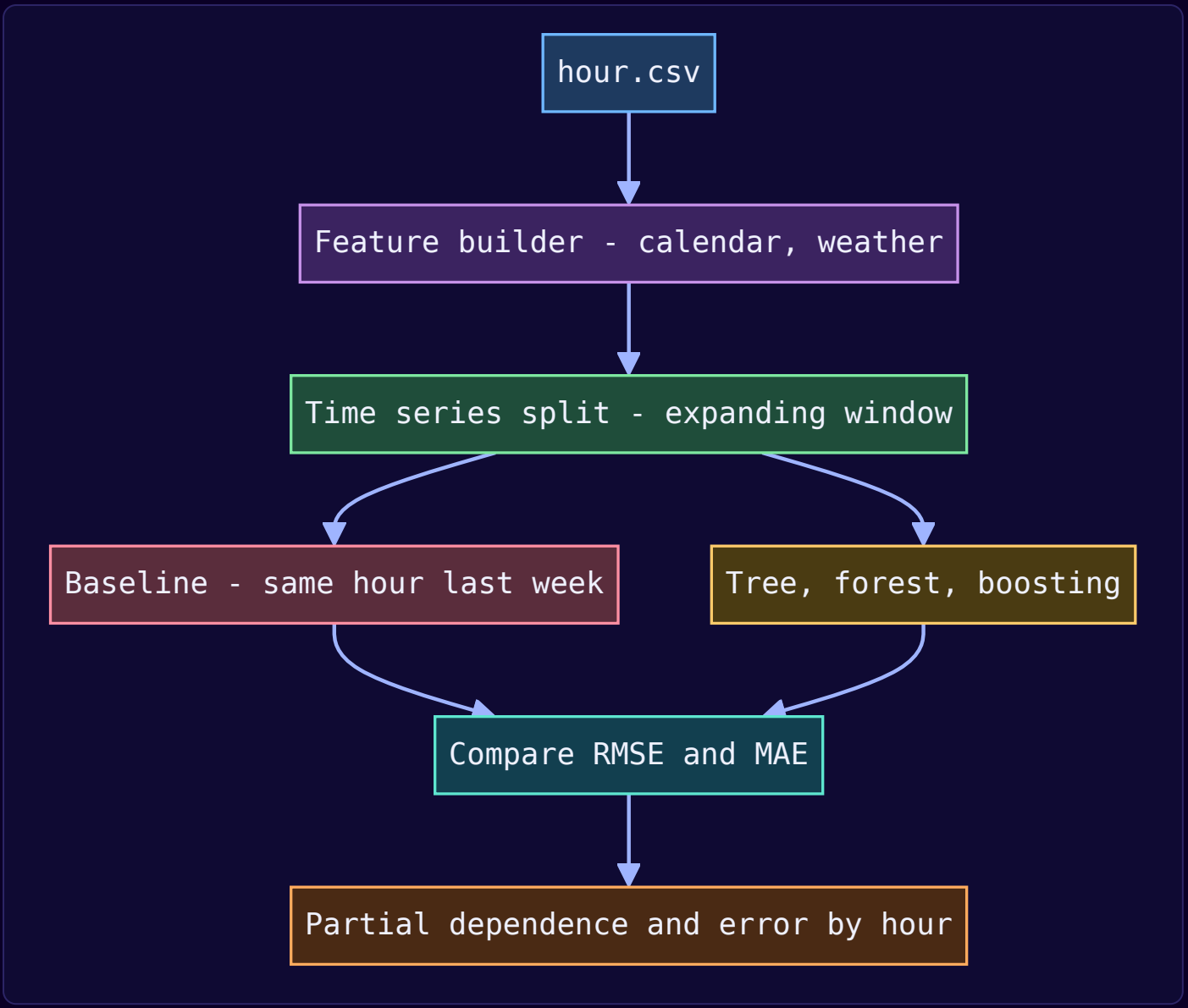
Functional

- Features: hour, weekday, month, holiday, working day, weather situation, temperature, humidity, wind speed.
- Models: seasonal-naive baseline (same hour last week), decision tree, random forest, gradient boosting.
- **Time-based** validation: train on the past, validate on later months.
- Report RMSE and MAE, and plot predicted vs actual for a sample week.

Non-functional

- Train all models in under 2 minutes.
- No future information in any feature.
- Predictions never negative.

The architecture



- **Feature builder**: cyclical encodings for hour (sine and cosine) are optional for trees but needed for linear models.
- **Expanding-window split**: `TimeSeriesSplit` trains on months 1..k and validates on month k+1, mimicking real forecasting.
- **Partial dependence plots**: show how predicted demand changes with temperature or hour while averaging over other features.

Data model

Column	Type	Notes
`dteday`, `hr`	date, int	Timestamp
`season`, `mnth`, `weekday`, `holiday`, `workingday`	category/int	Calendar
`weathersit`	1–4	Clear to heavy rain
`temp`, `atemp`, `hum`, `windspeed`	float, normalised	Weather
`casual`, `registered`, `cnt`	int	Target is `cnt` (do not use the other two as features!)

Tech stack

Component	Choice	Why
Data	pandas or polars	Simple joins and date parts
Models	scikit-learn trees and ensembles	Good defaults, `TimeSeriesSplit`
Explainability	`PartialDependenceDisplay`	Understand effects
Optional	LightGBM	Faster boosting, same ideas

Milestones

1. **Explore.** *Done when* you have plots of demand by hour for working days vs weekends and can describe the two commuter peaks.
2. **Baseline and split.** *Done when* the seasonal-naive baseline is scored on an expanding-window split.
3. **Single tree.** *Done when* you show how max depth changes train vs validation error (the bias–variance trade-off in one plot).
4. **Ensembles.** *Done when* random forest and boosting are tuned and both beat the baseline.
5. **Explain.** *Done when* partial dependence plots for temperature and hour are saved and error by hour is plotted.

Deep dive: the hard parts

- **The leak in plain sight.** `casual + registered = cnt`. Including either as a feature gives near-perfect scores and a useless model. Always ask "would I know this at prediction time?"
- **Random splits lie for time data.** A random split puts tomorrow in training and today in validation, so the model interpolates. Validate on later periods. Expect higher error, and believe it.
- **Count targets.** Demand is a non-negative count with variance growing with the mean. Gradient boosting with Poisson loss (`loss="poisson"`) respects this and avoids negative predictions.
- **Bagging vs boosting.** Random forests reduce variance by averaging deep trees and are hard to overfit. Boosting reduces bias by adding shallow trees and needs a learning rate and early stopping.

Testing strategy

- Feature-availability test: assert no column whose name contains `casual` or `registered` reaches the model.
- Split test: every validation timestamp is later than every training timestamp in each fold.
- Non-negativity test on predictions.
- Regression test: saved model's validation RMSE does not worsen by more than 5% after code changes.

Trade-offs, scaling and failure modes

- Trees cannot extrapolate: if next summer is hotter than any training day, predictions plateau. Linear terms or recent-trend features help.

- Weather features in training are *observed*. At serving time you have *forecasts*, which are noisier. Evaluate with forecast-quality inputs if you can.
- Demand grew from 2011 to 2012, so a model trained only on 2011 under-predicts. Add a trend feature or retrain often.

Stretch goals

- Add lag features (demand 24 and 168 hours ago) and compare.
- Predict casual and registered separately and sum them.
- Quantile predictions for staffing decisions (the 90th percentile of demand).

Skills practised

- Ch 7 — Decision Trees and Ensembles: trees, bagging, random forests, gradient boosting.
- Ch 4 — Data, EDA and Feature Engineering: calendar features, leakage.
- Ch 6 — Evaluation and Tuning: time-based validation, bias–variance.

In the interview

Q: Random forest vs gradient boosting? **A:** Forests average many deep, decorrelated trees to cut variance — robust with little tuning. Boosting fits shallow trees sequentially to residuals to cut bias — usually more accurate but needs learning rate, depth and early stopping.

Q: Why not a random train/test split here? **A:** The data is ordered in time. Random splits let the model see the future and overstate accuracy. Use forward-chaining splits.

Q: What is a partial dependence plot? **A:** The model's average prediction as one feature varies, averaging over the others. It shows the shape of an effect but can mislead when features are strongly correlated.

TL;DR

Tree ensembles capture hour × weather interactions for free. Validate forward in time, drop the target's own components, and use a count-aware loss.

9. Your first model API (wine quality)

Level: Beginner · **Time:** 1–2 weekends · **Prerequisites:** Ch 3, 6, 16

What it is

A small but complete **ML service**: train a model that predicts wine quality from lab measurements, track every experiment in **MLflow** (a tool that records parameters, metrics and model files for each training run), register the best model, and serve it through a **FastAPI** web endpoint that validates input and logs every prediction to PostgreSQL. The model is simple on purpose; the point is the plumbing around it.

The problem it solves

Most models die in notebooks. To be useful, a model must be reproducible (you can rebuild it), versioned (you know which one is live), callable (other software can use it) and observable (you can see what it is predicting). This project gives you a template for all four that you will reuse in many later capstones.

Real-world examples

- The UCI Wine Quality dataset has 1,599 red and 4,898 white Portuguese "vinho verde" samples with 11 physicochemical measurements and a sensory quality score.
- The MLflow documentation covers tracking, model packaging and the model registry.
- The FastAPI documentation shows typed request models with automatic validation and interactive API docs.

Requirements

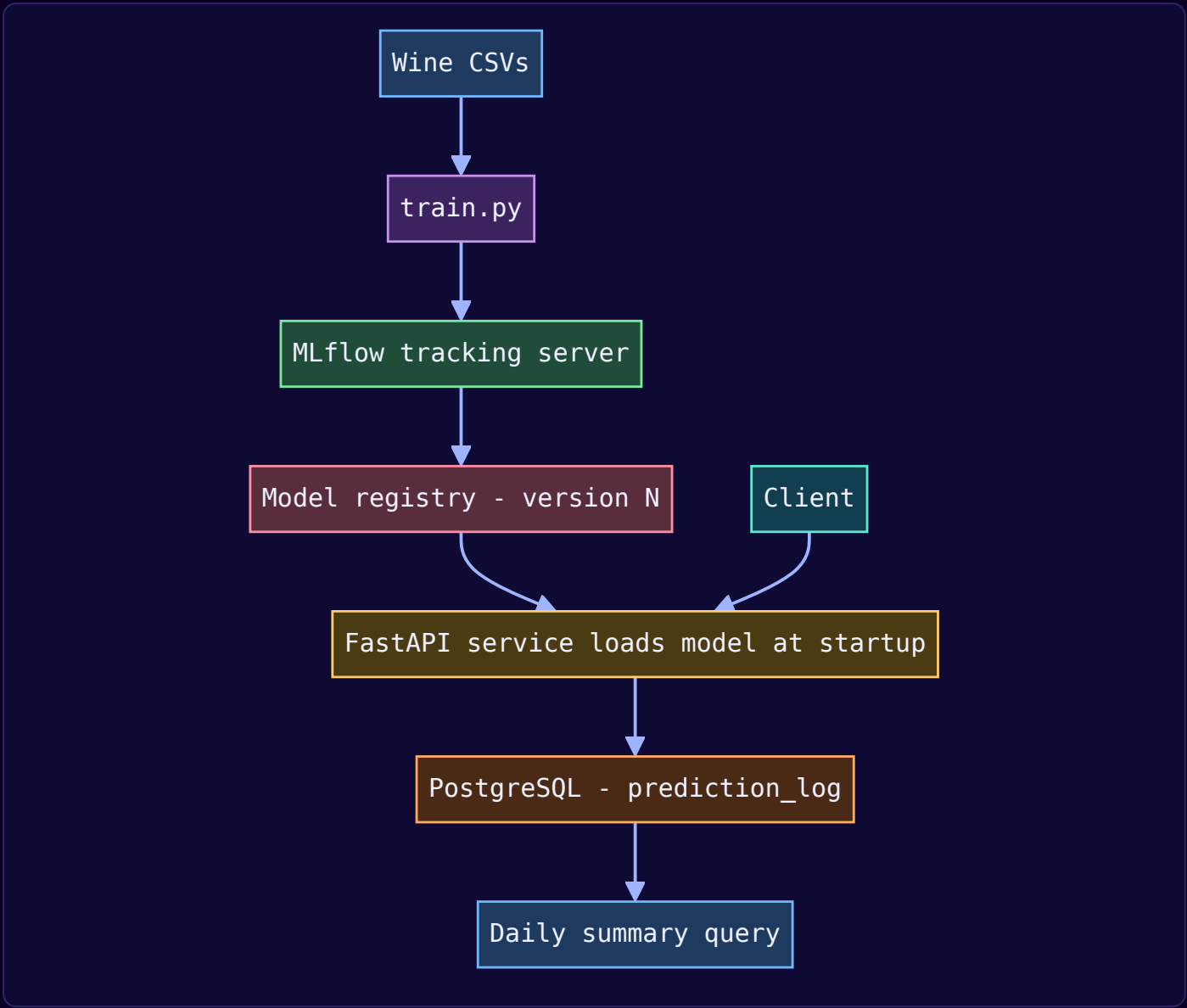
Functional

- Train at least three models (linear, random forest, gradient boosting), logging parameters, metrics and the model artefact to MLflow.
- Register the best run as a named model version.
- `POST /predict` accepts the 11 measurements plus wine colour and returns predicted quality and model version.
- `GET /health` and `GET /model-info` endpoints.
- Log each request, prediction, model version and latency to PostgreSQL.

Non-functional

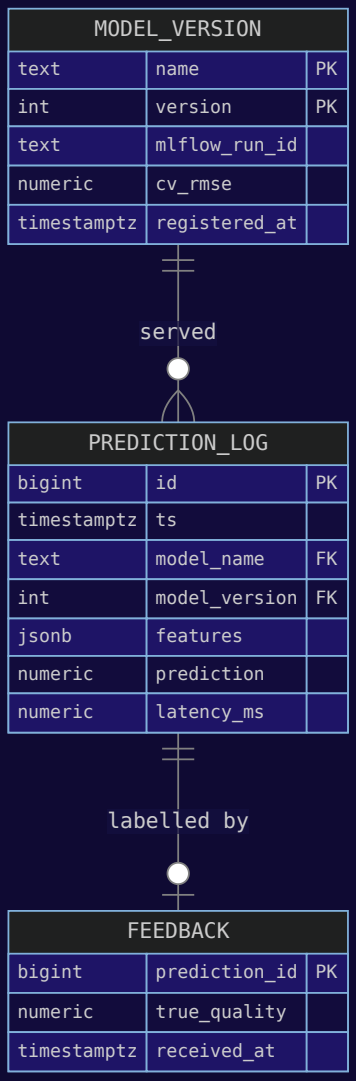
- p95 (95th-percentile) latency under 50 ms for a single prediction on a laptop.
- Invalid input (negative pH, missing field) rejected with HTTP 422 and a clear message.
- Container image builds and starts with one command.

The architecture



- **train.py**: one script, seeds fixed, logs everything to MLflow.
- **Registry**: holds versions with aliases such as `champion`. The service loads "the champion", not a file path.
- **FastAPI service**: Pydantic models define the input schema; the model is loaded once at startup, not per request.
- **prediction_log**: lets you audit and later monitor inputs and outputs (the start of Project 31).

Data model



Tech stack

Component	Choice	Why
Training	scikit-learn	Simple tabular models
Tracking and registry	MLflow	Runs, artefacts, versions, aliases
API	FastAPI + Pydantic	Typed validation, auto docs
Logging store	PostgreSQL	Durable, queryable prediction log
Packaging	Docker	Same environment everywhere

Milestones

1. **Tracked training.** *Done when* three runs appear in the MLflow UI with parameters, CV RMSE and a saved model.
2. **Registry.** *Done when* the best run is registered and tagged as champion.
3. **API.** *Done when* `curl` to `/predict` returns a prediction and model version, and bad input returns 422.
4. **Logging.** *Done when* every call writes one row to `prediction_log`, and a SQL query shows prediction counts per day.
5. **Container and load test.** *Done when* `docker compose up` starts API, MLflow and PostgreSQL, and a load test reports p95 latency.

Deep dive: the hard parts

- **Regression or classification?** Quality is an integer from 0 to 10 but most wines score 5–7. Treating it as regression is simple; treating it as ordinal classification is more faithful. Pick one, justify it, and report MAE either way.
- **Same preprocessing everywhere.** Save the entire scikit-learn pipeline (scaler + model) as one artefact. If the API re-implements scaling by hand, train/serve skew is only a matter of time.
- **Model loading and versions.** Load the model at startup and expose its version in every response. To roll out a new version, change the registry alias and restart (or hot-reload) — never overwrite files in place.
- **Logging without slowing requests.** Writing to PostgreSQL synchronously adds milliseconds. Use FastAPI background tasks or a small queue, and make the endpoint still respond if the log write fails.

Testing strategy

- Unit test the Pydantic schema with valid and invalid payloads.
- API tests with FastAPI's `TestClient`: health, predict, bad input.
- Parity test: the API's prediction equals the training pipeline's prediction for the same row.
- Load test (a few hundred requests) checks p95 latency.

Trade-offs, scaling and failure modes

- One process with the model in memory is fine for tens of requests per second. For more, run several workers behind a load balancer or batch requests.
- If the registry is down at startup, the service should fail fast rather than serve nothing silently.
- Logged features may include personal data in real systems — decide retention and access up front.

Stretch goals

- Add a `/predict/batch` endpoint for many rows at once.
- Shadow mode: load two versions, serve one, log both predictions.
- A Grafana or simple HTML dashboard over `prediction_log`.

Skills practised

- Ch 3 — The Python ML Stack: environments, the estimator API, saving models.
- Ch 6 — Evaluation and Tuning: choosing and comparing models.
- Ch 16 — ML in Production: serving, model registry, logging, train/serve skew.

In the interview

Q: What does a model registry give you over saving files? **A:** Versioning with lineage back to the training run, stage or alias management, and a single source of truth for which model is live, which makes rollback a metadata change.

Q: How do you avoid train/serve skew in a simple service? **A:** Serialise the full preprocessing + model pipeline as one artefact, validate inputs against a schema, and run a parity test between offline and online predictions.

Q: What would you log for each prediction? **A:** Timestamp, model version, input features (or a reference), output, latency and a request ID to join later labels. That enables debugging, monitoring and retraining.

TL;DR

Wrap even a simple model in tracking, a registry, a validated API and a prediction log — that skeleton is what turns a notebook into a service.

10. Book "more like this" finder

Level: Beginner · **Time:** 1 weekend · **Prerequisites:** Ch 8, 13

What it is

A **content-based recommender**: given a book you liked, it finds books whose plot summaries are most similar. Each summary becomes a vector (first with TF-IDF word weights, later with pre-trained sentence embeddings), and similarity is the **cosine** of the angle between vectors (1 means pointing the same way, 0 means unrelated). No user data is needed, which makes it the natural first recommender and the fix for the "cold start" problem of new items.

The problem it solves

A new book has no ratings, so collaborative methods (Project 14) cannot recommend it. Content similarity can, from day one. It also gives explainable results ("both mention dragons, a heist and a city under siege"), which users trust.

Real-world examples

- The CMU Book Summary Dataset contains plot summaries for 16,559 books extracted from Wikipedia, with author, title, date and genre metadata.
- Sentence-BERT showed how to produce sentence embeddings where cosine similarity reflects meaning — your upgrade path from TF-IDF.

Requirements

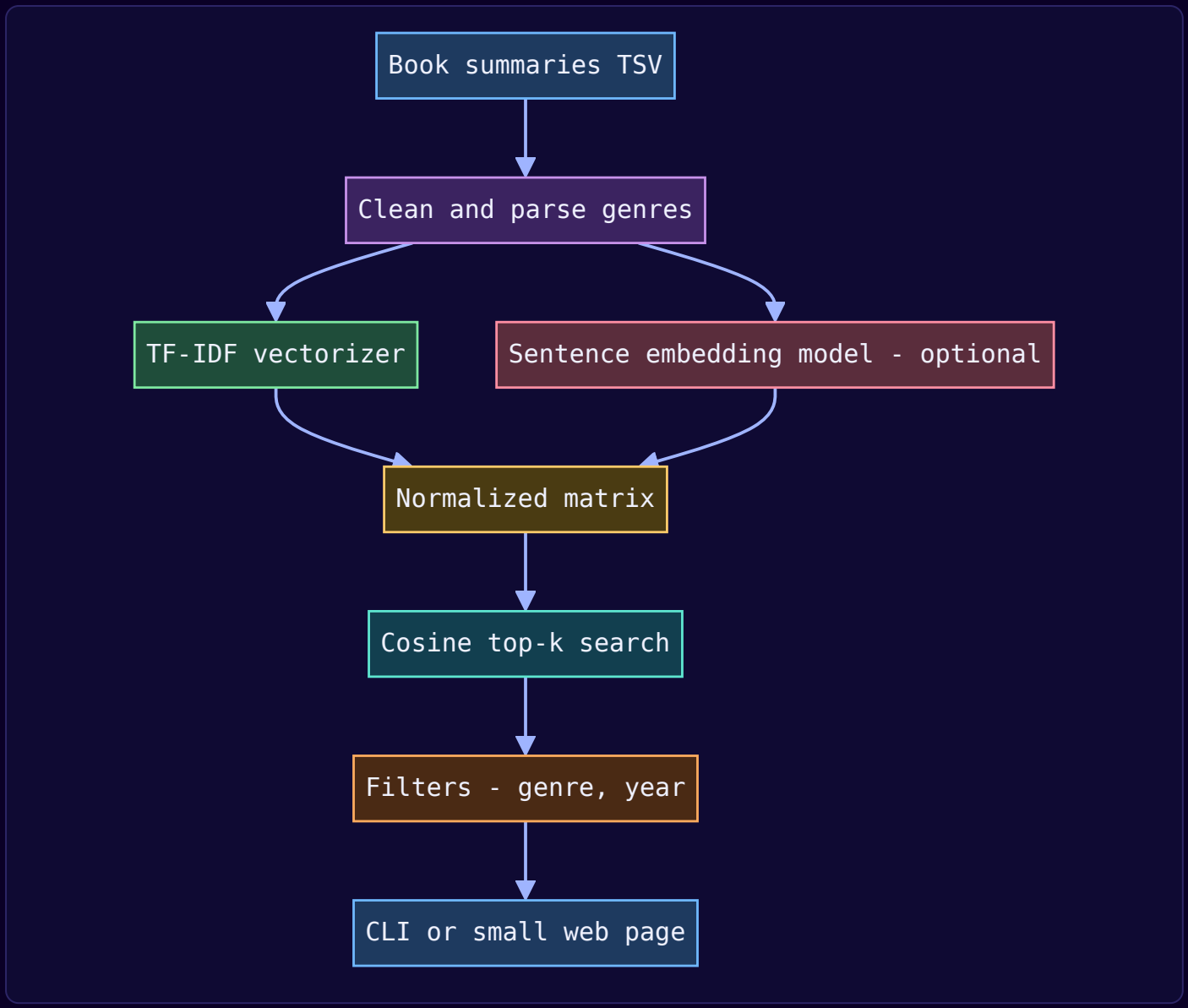
Functional

- Clean summaries; build TF-IDF vectors with unigrams and bigrams, removing stop words.
- `similar(book_id, k=10)` returns the top k books with similarity scores and shared top terms.
- Optional filters: same genre, published within N years.
- Compare TF-IDF results with a sentence-embedding model on a hand-labelled test set.

Non-functional

- Query under 50 ms for 16,000 books (brute-force cosine is fine at this size).
- Index build under 1 minute.
- Results stable across runs.

The architecture



- **TF-IDF**: rare words that are frequent in one summary ("hobbit") get high weight; common words ("the", "story") get low weight.
- **Normalized matrix**: L2-normalising each row makes cosine similarity a simple dot product.
- **Top-k search**: one sparse matrix–vector product plus `argpartition`.
- **Filters**: applied after scoring, with a larger k so you still return enough results.

Data model

Item	Type	Notes
`book_id`	int	Wikipedia ID
`title`, `author`, `pub_date`	text	Metadata, some missing
`genres`	JSON string	Parse into a list
`summary`	text	Hundreds of words each
Vectors	sparse matrix N × V or dense N × 384	Row-normalised

Tech stack

Component	Choice	Why
Vectors	scikit-learn `TfidfVectorizer`	Sparse, fast, explainable
Embeddings (stretch)	a sentence-transformers model	Captures meaning beyond shared words
Search	numpy / scipy sparse	Brute force fits in memory
UI	FastAPI + HTML, or a CLI	Quick demo

Milestones

1. **Parse.** *Done when* every book has an ID, title, genre list and non-empty summary.
2. **TF-IDF search.** *Done when* `similar()` returns 10 neighbours for a famous book and they look plausible.
3. **Explanations.** *Done when* each result lists the 5 terms contributing most to the similarity.
4. **Evaluation set.** Pick 30 books and mark which of their top 10 are "good" recommendations. *Done when* you can report precision at 10.
5. **Embeddings.** *Done when* you have compared TF-IDF and embeddings on the same evaluation set.

Deep dive: the hard parts

- **Names dominate TF-IDF.** Character names are rare and therefore heavily weighted, so sequels match each other perfectly but nothing else does. Consider removing proper nouns or capping per-term weight, and decide whether "same series" is a feature or a bug.
- **Evaluation without users.** There are no clicks. A small hand-labelled set and genre overlap as a weak proxy are honest substitutes; say clearly that they are proxies.
- **Lexical vs semantic similarity.** TF-IDF matches words; embeddings match meaning ("murder" ≈ "killing"). Embeddings can also match on style or length in surprising ways — inspect results, do not just trust metrics.
- **Diversity.** The top 10 may all be by the same author. Add a simple re-rank that limits each author to two results.

Testing strategy

- A book's most similar item (excluding itself) must not be itself; self-similarity must be 1.0.
- Similarity is symmetric: `sim(a, b) == sim(b, a)`.
- Filter test: genre-filtered results all have that genre.
- Golden test: a sequel appears in the top 10 for its first book.

Trade-offs, scaling and failure modes

- Brute force is O(N) per query — fine for thousands, too slow for millions. Then use an approximate nearest-neighbour index (Project 18).
- Content-based systems over-specialise: users see more of the same. Collaborative signals add serendipity.
- Summary quality varies; a two-line summary gives a poor vector.

Stretch goals

- Combine TF-IDF and embedding scores with a weighted sum and tune the weight on your labelled set.
- Add a "because you liked X and Y" mode that averages two book vectors.
- Store vectors in PostgreSQL with pgvector and query from SQL.

Skills practised

- Ch 8 — k-NN, SVMs and Naive Bayes: nearest-neighbour search and distance choices.
- Ch 13 — Recommender Systems and Embeddings: content-based filtering, cosine similarity, cold start, embeddings.
- Ch 12 — Sequences, RNNs and Attention: tokenisation and word/sentence embeddings.

In the interview

Q: When would you choose content-based over collaborative filtering? **A:** For new items or users with no interactions, for niche catalogues with little traffic, and when explanations matter. Collaborative filtering wins when interaction data is rich.

Q: Why cosine similarity rather than Euclidean distance? **A:** Cosine ignores vector length, so a long summary and a short one about the same topic still match. With normalised vectors the two are equivalent rankings.

Q: How would you evaluate without user data? **A:** A small human-labelled relevance set, proxy labels such as genre overlap, and later an online test measuring click-through on recommendations.

TL;DR

Turn text into vectors, rank by cosine similarity, explain with shared terms — the simplest recommender, and the one that works for brand-new items.

11. Churn model with gradient boosting and SHAP

Level: Intermediate · **Time:** 2 weeks · **Prerequisites:** Ch 6, 7, 19

What it is

A model that predicts which subscribers of a telecom company are likely to cancel (**churn**) in the near future, built with **gradient-boosted trees** and explained with **SHAP** (SHapley Additive exPlanations — a method that splits each prediction into contributions from each feature, so they add up to the model's output). The output is a weekly ranked list for the retention team, each customer with their top reasons.

The problem it solves

Winning a new customer usually costs more than keeping one. A retention team can call or discount only a limited number of people per week, so they need a ranked list of who is at risk *and* why (a price complaint needs a different offer from a network-quality complaint). Accuracy without reasons is hard to act on; reasons without accuracy waste the budget.

Real-world examples

- The IBM sample Telco Customer Churn dataset on Kaggle has 7,043 customers with contract type, tenure, services, charges and a churn flag.
- Lundberg and Lee's paper A Unified Approach to Interpreting Model Predictions introduced SHAP; the SHAP documentation covers the fast TreeExplainer used for boosted trees.

Requirements

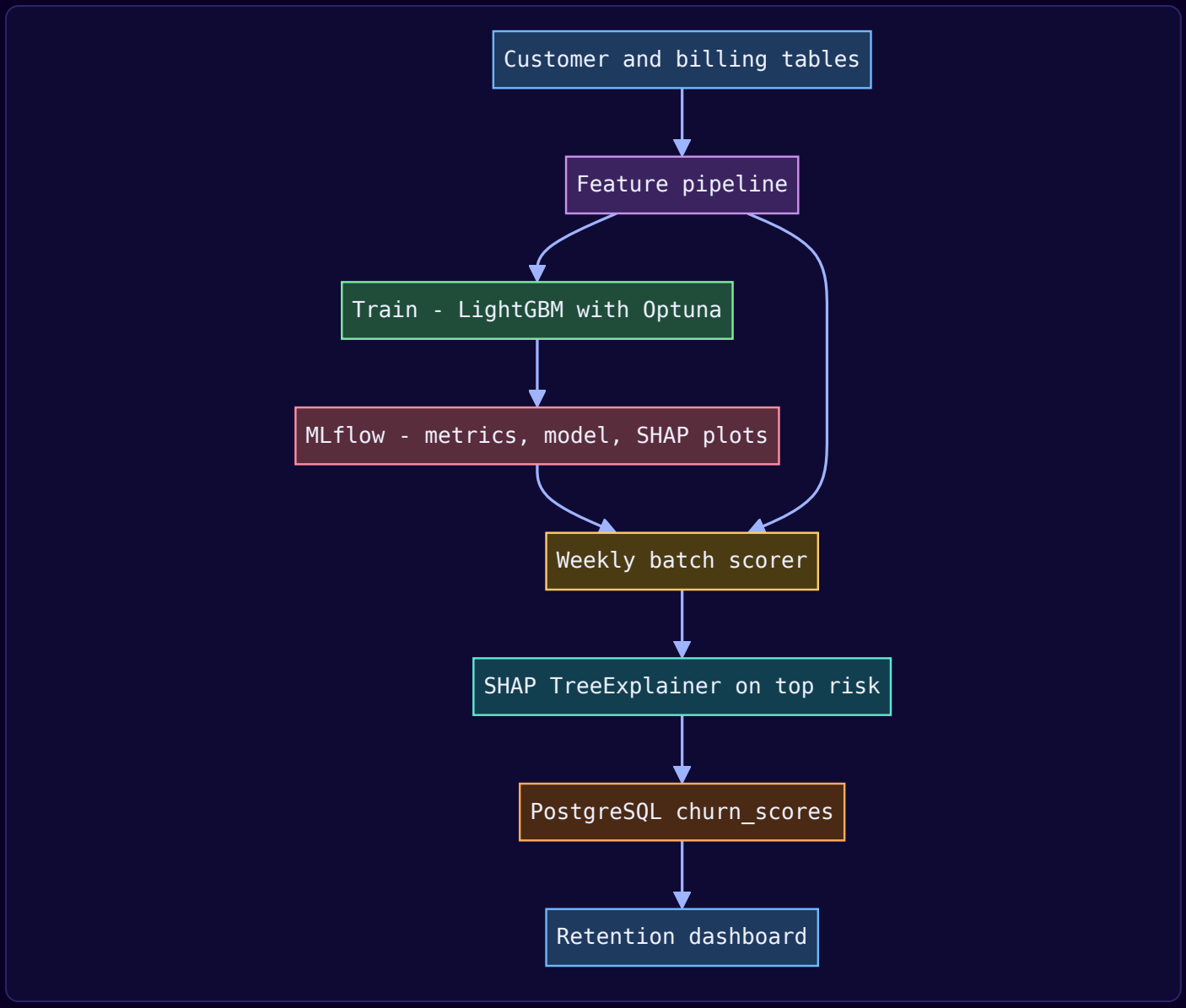
Functional

- Feature pipeline: tenure buckets, contract type, payment method, number of add-on services, charges per month of tenure.
- Models: logistic regression baseline, LightGBM (or scikit-learn `HistGradientBoostingClassifier`), with tuned hyperparameters.
- Metrics: ROC AUC, PR AUC, **lift at top 10%** (how many times more churners are in the top decile than average), and calibration.
- Global explanations (SHAP summary) and per-customer top three reasons.
- A batch job writing `churn_scores` to PostgreSQL weekly.

Non-functional

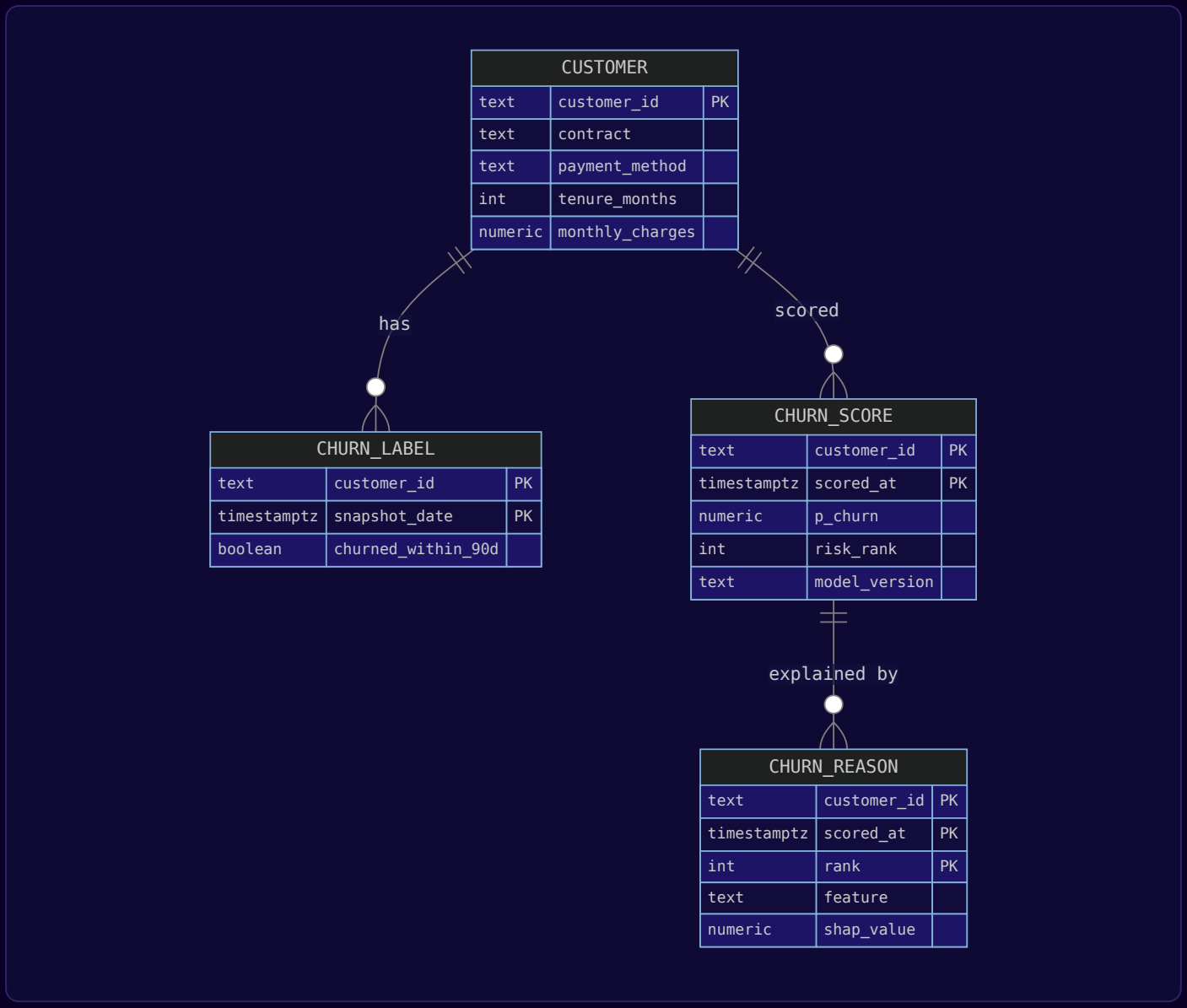
- Scoring 1 million customers under 5 minutes on a laptop-class machine.
- Explanations computed for the top 10,000 risk customers in under 2 minutes.
- Model and SHAP versions recorded with each score.

The architecture



- **Feature pipeline**: a single function used for training and scoring; in real life it reads from a snapshot as of a cut-off date.
- **Optuna**: a hyperparameter search library that tries promising settings first (Optuna).
- **Batch scorer**: scores all active customers, sorts by risk.
- **TreeExplainer**: exact SHAP values for tree models, fast enough for thousands of rows.

Data model



Tech stack

Component	Choice	Why
Model	LightGBM or scikit-learn HistGB	Strong on tabular data, handles categories
Tuning	Optuna	Efficient search, pruning
Explanations	SHAP (check Python 3.14 wheels, else use a 3.13 env)	Consistent additive attributions
Tracking	MLflow	Runs, plots, model versions
Storage	PostgreSQL	Scores and reasons for the dashboard

Milestones

1. **Label definition and baseline.** *Done when* you have written down exactly what "churn" means (window and snapshot date) and logistic regression has CV ROC AUC, PR AUC and lift.
2. **Boosting.** *Done when* a tuned boosted model beats the baseline on PR AUC and lift at 10%, with early stopping.
3. **Explanations.** *Done when* a SHAP summary plot exists and you can describe the top three global drivers in plain words.
4. **Reasons.** *Done when* each top-risk customer has three human-readable reasons ("month-to-month contract", "fibre optic with no tech support").

5. **Batch job.** *Done when* a scheduled script writes scores and reasons to PostgreSQL, tagged with the model version.

Deep dive: the hard parts

- **Defining the label in time.** The Kaggle file is a single snapshot, so "churn" already happened. In a real system you build features as of date T and label "cancelled between T and T + 90 days". Features computed after T (for example, "number of support calls this month" when the month includes the cancellation call) leak the answer.
- **Ranking metrics for a budget.** The team calls the top N, so what matters is precision and lift in the top slice, not overall accuracy. Report a cumulative gains curve.
- **SHAP is about the model, not causation.** A high SHAP value for "month-to-month contract" means the model relies on it, not that switching a customer to a yearly contract will keep them. Retention offers need an uplift model (Project 23) or an experiment.
- **Correlated features split credit.** Tenure and total charges are strongly correlated, so SHAP may spread credit between them unpredictably. Group related features when presenting reasons.

Testing strategy

- Point-in-time test: every feature value for a training row has a timestamp before its snapshot date.
- SHAP additivity check: base value + sum of SHAP values equals the model's raw output (log-odds) for sampled rows.
- Batch scorer test on a fixture of 100 customers: row counts, no nulls, ranks are a permutation.
- Drift check: compare this week's score distribution with last week's.

Trade-offs, scaling and failure modes

- Logistic regression is easier to defend; boosting typically ranks better. Keep both and report the gap.
- Scores influence who gets offers, and offers change who churns — future labels are contaminated by the intervention. Keep a random holdout group that receives no offers.
- SHAP on millions of rows is slow; explain only those you will act on.

Stretch goals

- Survival analysis: predict *when* a customer will churn with a Cox model or a survival forest.
- Cost-based threshold: maximise expected profit given offer cost and customer value.
- A small Streamlit or HTML dashboard of top-risk customers and their reasons.

Skills practised

- Ch 7 — Decision Trees and Ensembles: gradient boosting, early stopping, feature importance.
- Ch 6 — Evaluation and Tuning: PR AUC, lift, calibration, hyperparameter search.
- Ch 19 — Responsible ML: explainability with SHAP and its limits.
- Ch 16 — ML in Production: batch inference and point-in-time features.

In the interview

Q: How do you define churn for training? **A:** Pick a snapshot date T, build features only from data before T, and label whether the customer cancels in a fixed window after T. Use several snapshots for more training rows, and validate on a later snapshot.

Q: What do SHAP values tell you? **A:** How much each feature pushed this prediction above or below the average prediction, adding up exactly to the output. They explain the model's reasoning, not the causal effect of changing the feature.

Q: The team can call 500 customers a week. Which metric do you optimise? **A:** Precision or expected saved revenue in the top 500 — a ranking metric at a fixed budget — plus calibration if offers depend on the probability.

TL;DR

Churn modelling is mostly about a leak-free, point-in-time label and a ranking metric that matches the retention budget; SHAP turns scores into reasons, not causes.

12. Pet-breed classifier with transfer learning

Level: Intermediate · **Time:** 2 weeks · **Prerequisites:** Ch 10, 11

What it is

An image classifier that recognises 37 breeds of cats and dogs from photos, built by **transfer learning**: you take a CNN pre-trained on ImageNet (millions of general photos), replace its final layer with a new 37-way layer, and **fine-tune** (continue training) on a few thousand pet photos. You then ship it as an API with a confidence threshold and an "I'm not sure" answer.

The problem it solves

Training a strong image model from scratch needs huge labelled datasets and lots of GPU time. Most real projects — product photos, defects, medical images — have only thousands of labelled images. Transfer learning reuses general visual features (edges, textures, fur, eyes) and makes these projects feasible on one GPU or even a CPU.

Real-world examples

- The Oxford-IIIT Pet Dataset has 37 categories with roughly 200 images each, plus breed, head bounding box and pixel-level segmentation annotations.
- The official PyTorch transfer learning tutorial walks through feature extraction vs fine-tuning on a small dataset.
- ResNet introduced residual connections and remains a common transfer-learning backbone.

Requirements

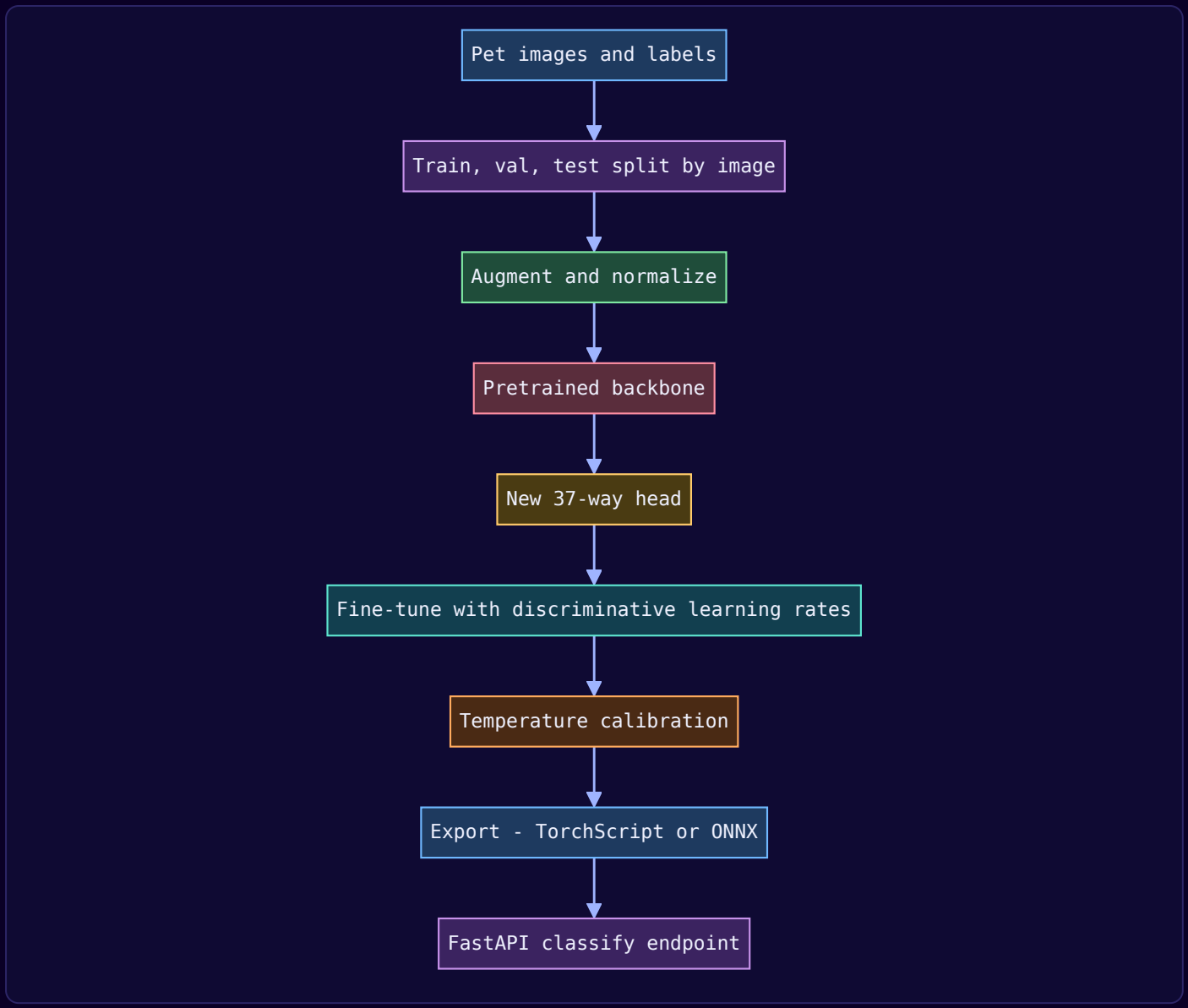
Functional

- Train with three strategies: frozen backbone (only the new head learns), full fine-tuning, and gradual unfreezing.
- Data augmentation: random resized crops, flips, colour jitter.
- Metrics: top-1 and top-5 accuracy, per-class accuracy, confusion matrix of the most confused breeds.
- `POST /classify` with an image returns top-3 breeds and probabilities, or "unsure" below a threshold.

Non-functional

- Top-1 test accuracy above 85% with a ResNet-50-class backbone (good fine-tuning typically exceeds this).
- CPU inference under 150 ms per image at 224 × 224.
- Model artefact under 100 MB.

The architecture



- **Pretrained backbone**: a torchvision ResNet or EfficientNet with ImageNet weights.
- **Discriminative learning rates**: small rates for early layers (general features you want to keep), larger for later layers and the new head.
- **Temperature calibration**: divide logits by a single learned number so that probabilities match observed accuracy, which makes the "unsure" threshold meaningful.
- **Export**: a format that loads without your training code.

Data model

Item	Type	Notes
Image file	JPEG, variable size	Resize then crop to 224 × 224
Breed label	int 0–36	From the file name
Species	cat/dog	Useful for error analysis
Normalisation	ImageNet mean and std	Must match the pre-training
Checkpoint	state dict + class names + temperature	One artefact

Tech stack

Component	Choice	Why
Framework	PyTorch, torchvision	Pretrained weights, transforms
Tracking	MLflow	Compare strategies
Serving	FastAPI with `UploadFile`	Simple multipart upload
Export	TorchScript or ONNX Runtime	Fast CPU inference

Milestones

1. **Feature extraction.** Freeze the backbone, train the head. *Done when* you have a validation accuracy after a few epochs.
2. **Fine-tuning.** Unfreeze, lower the learning rate, add augmentation. *Done when* validation accuracy improves and curves show no runaway overfitting.
3. **Error analysis.** *Done when* you list the five most-confused breed pairs and inspect example images for each.
4. **Calibration and threshold.** *Done when* expected calibration error is reported before and after temperature scaling and the "unsure" threshold is chosen.
5. **Serve.** *Done when* uploading a photo of a pet returns top-3 breeds within the latency target.

Deep dive: the hard parts

- **Catastrophic forgetting.** A high learning rate on the whole network destroys useful pre-trained features in the first few steps. Train the head first, then unfreeze with a much lower rate, or use a warm-up.
- **Batch norm with small batches.** Batch normalisation layers keep running statistics. With tiny batches during fine-tuning these statistics become noisy; consider freezing batch-norm statistics.
- **Fine-grained confusions.** Similar breeds (for example, two terriers) are genuinely hard. Higher input resolution and longer fine-tuning help more than a bigger head.
- **Out-of-distribution inputs.** A photo of a rabbit still gets a confident breed. Calibrated max probability is a weak but cheap detector; mention stronger options (an "other" class, energy scores).

Testing strategy

- Transform test: train and inference preprocessing produce identical tensors for the same image (except random augmentation).
- Overfit-one-batch test.
- Golden image test after export: TorchScript/ONNX outputs match PyTorch within 1e-4.
- API test with a non-image file returns a 400 error.

Trade-offs, scaling and failure modes

- Bigger backbones raise accuracy and latency. For mobile, MobileNet or EfficientNet-B0 class models trade a few points for speed.
- Dataset bias: pet photos are well-lit, centred and often from breeders' sites. User photos are blurry and off-centre.
- Label noise in public datasets caps achievable accuracy; inspect top-loss images.

Stretch goals

- Compare a CNN backbone with a vision transformer (ViT).
- Grad-CAM heat maps to show which pixels drove a prediction.
- Quantise the model to int8 and measure the accuracy/latency trade.

Skills practised

- Ch 11 — CNNs and Computer Vision: architectures, transfer learning, augmentation.
- Ch 10 — Neural Networks and Deep Learning: optimisers, learning-rate schedules, regularisation.
- Ch 6 — Evaluation and Tuning: calibration, confusion matrices.

In the interview

Q: Feature extraction or fine-tuning? **A:** With very little data or a target domain close to ImageNet, freeze most layers. With more data or a different domain, fine-tune more layers at lower learning rates.

Q: Why calibrate a classifier you will threshold? **A:** Deep nets tend to be over-confident. If 0.9 does not mean 90% correct, the threshold for "unsure" is arbitrary. Temperature scaling fixes this cheaply without changing rankings.

Q: How do you handle a class the model has never seen? **A:** Detect out-of-distribution inputs with confidence or energy scores, add an explicit "other" class with diverse negatives, and route low-confidence cases to a human.

TL;DR

Start from a pre-trained backbone, train the new head, then fine-tune gently with augmentation; calibrate before you threshold.

13. Movie-review sentiment, TF-IDF to transformer

Level: Intermediate · **Time:** 2–3 weeks · **Prerequisites:** Ch 8, 12

What it is

A sentiment classifier for movie reviews (positive or negative) built in three generations: a **TF-IDF + logistic regression** baseline, an **LSTM** (a recurrent network that reads text word by word and keeps a memory), and a fine-tuned **transformer** (a pre-trained language model such as BERT that uses attention to look at all words at once). You compare accuracy, cost and robustness, and serve the winner.

The problem it solves

Companies want to know how people feel about products from reviews, tickets and social posts at a scale no human can read. The project also teaches an important professional habit: measure how much a large model really buys you over a cheap baseline before paying its inference bill.

Real-world examples

- The Large Movie Review Dataset (Maas et al., 2011) has 50,000 IMDB reviews split evenly into training and test, also on Hugging Face as stanfordnlp/imdb.
- BERT showed that fine-tuning one pre-trained transformer beats task-specific architectures on many text classification benchmarks; the Hugging Face Transformers docs show the fine-tuning workflow.

Requirements

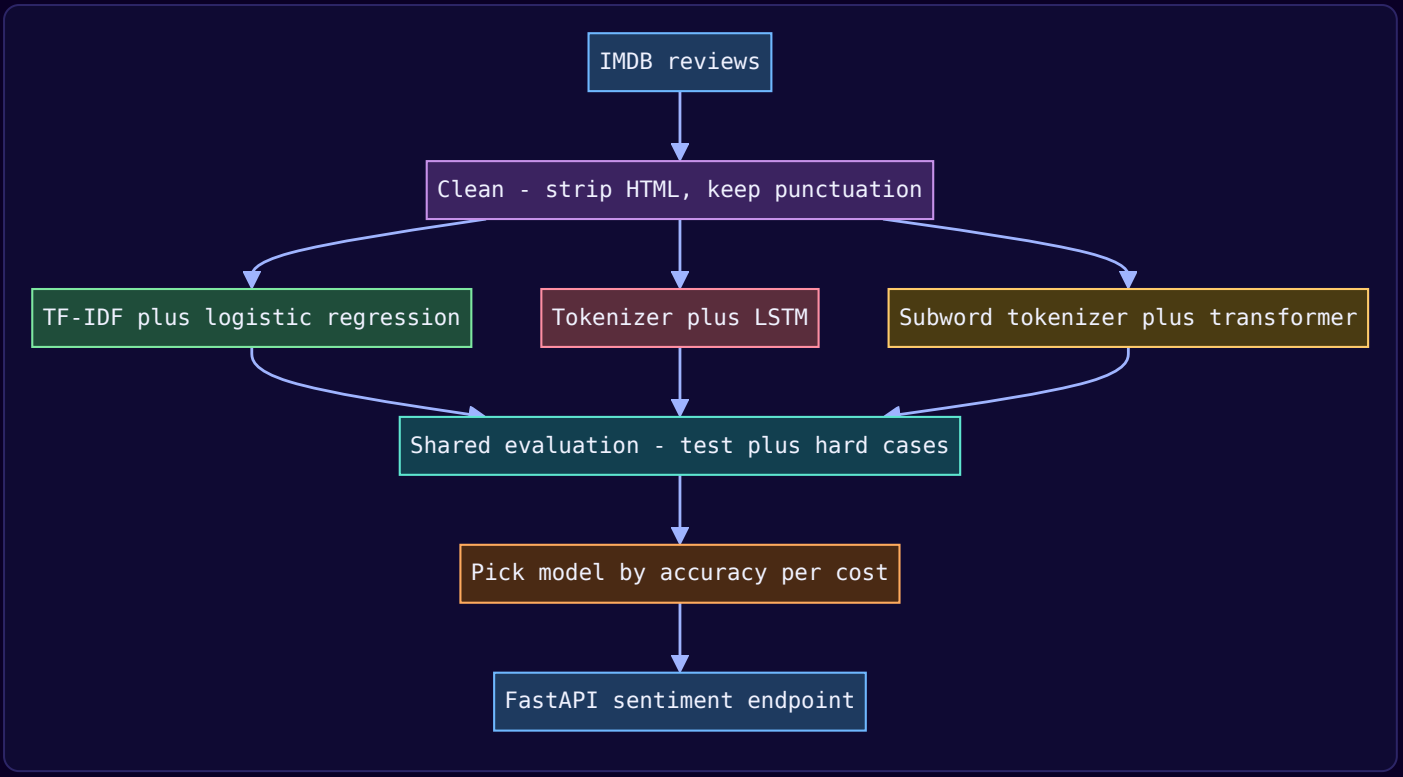
Functional

- Three models trained on the same split: TF-IDF + logistic regression, embedding + LSTM, fine-tuned small transformer (for example DistilBERT).
- Accuracy, F1 and a curated "hard cases" set: negation ("not bad at all"), sarcasm, mixed reviews.
- Error analysis by review length.
- API returning label, probability and model name.

Non-functional

- Baseline trains in under 1 minute on CPU.
- Transformer inference under 100 ms per review on CPU at up to 256 tokens (or document the GPU requirement).
- Report cost per 1,000 predictions for each model.

The architecture



- **Clean**: reviews contain `<br />` tags; keep punctuation because "!!!" and "?" carry sentiment.
- **Tokenizers**: word-level for TF-IDF and LSTM; subword (WordPiece) for the transformer, which splits rare words into known pieces.
- **Shared evaluation**: every model sees exactly the same test rows and hard cases.

Data model

Item	Type	Notes
Review text	string	25k train, 25k test, balanced
Label	0/1	Ratings ≤ 4 negative, ≥ 7 positive (neutral excluded)
Token IDs	int tensor `[B, T]`	Truncated or padded to T tokens
Attention mask	0/1 tensor	Marks padding for the transformer
Hard cases	small CSV	Hand-written, with expected labels

Tech stack

Component	Choice	Why
Baseline	scikit-learn	Fast, strong
LSTM	PyTorch `nn.LSTM`	Learn sequence modelling
Transformer	Hugging Face Transformers + PyTorch	Pre-trained checkpoints, tokenizers
Tracking	MLflow	Compare three families
Serving	FastAPI	One endpoint, swappable model

Milestones

1. **Baseline.** *Done when* TF-IDF + logistic regression reports test accuracy (high 80s percent is typical) and its top positive and negative words.
2. **LSTM.** *Done when* it trains with padding handled correctly and you can say whether it beats the baseline.

3. **Transformer fine-tune.** Two or three epochs, small learning rate, warm-up. *Done when* test accuracy is recorded along with training time and hardware.
4. **Hard cases and error analysis.** *Done when* each model's score on hard cases is in a table and you have read 30 errors of the best model.
5. **Serve and cost.** *Done when* the API runs and you have a cost-per-1,000 estimate for each model.

Deep dive: the hard parts

- **Long reviews.** Many reviews exceed 512 tokens, the usual transformer limit. Truncating to the first 256 tokens loses the verdict that often comes at the end. Try "head + tail" (first 128 and last 128 tokens) and compare.
- **Negation and context.** Bag-of-words sees "not" and "good" separately. Bigrams help ("not_good"). Transformers handle this naturally because attention relates words to each other.
- **Fine-tuning stability.** Transformers fine-tuned with too high a learning rate can collapse to predicting one class. Use about 2e-5, warm-up, and several seeds.
- **Domain shift.** A model trained on movie reviews misreads product reviews ("unpredictable" is praise for a plot and a complaint about a car's brakes). Test on a few out-of-domain examples to show this.

Testing strategy

- Tokenizer round trip on a sample: padding and truncation lengths correct.
- Invariance tests: changing an actor's name must not flip the label. Directional tests: adding "terrible" should lower positivity.
- Model parity: exported model matches in-training predictions.
- API latency test on a 1,000-word review.

Trade-offs, scaling and failure modes

- The transformer usually wins by a few points but can cost far more per prediction than the baseline. For high-volume, low-stakes uses, the baseline may be the right choice.
- Distillation (training a small model to mimic a big one) recovers much of the gap cheaply.
- Sentiment models learn biases from text, for example associating some identity terms with negativity (see Project 25).

Stretch goals

- Distil the transformer into the LSTM or a smaller transformer.
- Aspect-based sentiment: separate scores for acting, plot and music.
- Attention or integrated-gradient visualisations for explanations.

Skills practised

- Ch 8 — k-NN, SVMs and Naive Bayes: linear text classifiers.
- Ch 12 — Sequences, RNNs and Attention: tokenisation, LSTMs, attention, transformers.
- Ch 10 — Neural Networks and Deep Learning: learning-rate schedules, fine-tuning habits.

In the interview

Q: Why start with TF-IDF and logistic regression? **A:** It is fast, strong and interpretable, and it sets the bar. If a transformer adds only a point, its cost may not be justified.

Q: How do you handle documents longer than the model's context? **A:** Truncate smartly (head + tail), split into chunks and aggregate predictions, or use a long-context model. Measure which works on your data.

Q: How would you test a sentiment model beyond accuracy? **A:** Behavioural tests: invariance (names, typos), directional changes (adding negative words) and minimum functionality (simple negation), plus slices by length and domain.

TL;DR

Climb from TF-IDF to LSTM to a fine-tuned transformer on one shared evaluation, then choose by accuracy per unit of cost, not accuracy alone.

14. Movie recommender with matrix factorisation

Level: Intermediate · **Time:** 2–3 weeks · **Prerequisites:** Ch 6, 13

What it is

A **collaborative filtering** recommender: it learns from the pattern of who watched or rated what, not from movie content. **Matrix factorisation** (MF) represents every user and every movie as a short vector (an **embedding**, say 64 numbers) so that the dot product of a user vector and a movie vector predicts how much the user will like the movie. You train it on MovieLens, evaluate it as a ranking problem, and serve top-N recommendations.

The problem it solves

Catalogues are too big to browse. Users who liked the same movies tend to like the same new movies, even when the movies share no obvious attributes. MF compresses millions of sparse interactions into embeddings that capture taste — the core idea behind the Netflix Prize era and still the backbone of many retrieval systems.

Real-world examples

- MovieLens datasets from GroupLens range from 100,000 ratings to the 32M release with 32 million ratings.
- Koren, Bell and Volinsky's article Matrix Factorization Techniques for Recommender Systems (IEEE Computer, 2009) describes the biased MF and implicit-feedback ideas used by Netflix Prize winners.

Requirements

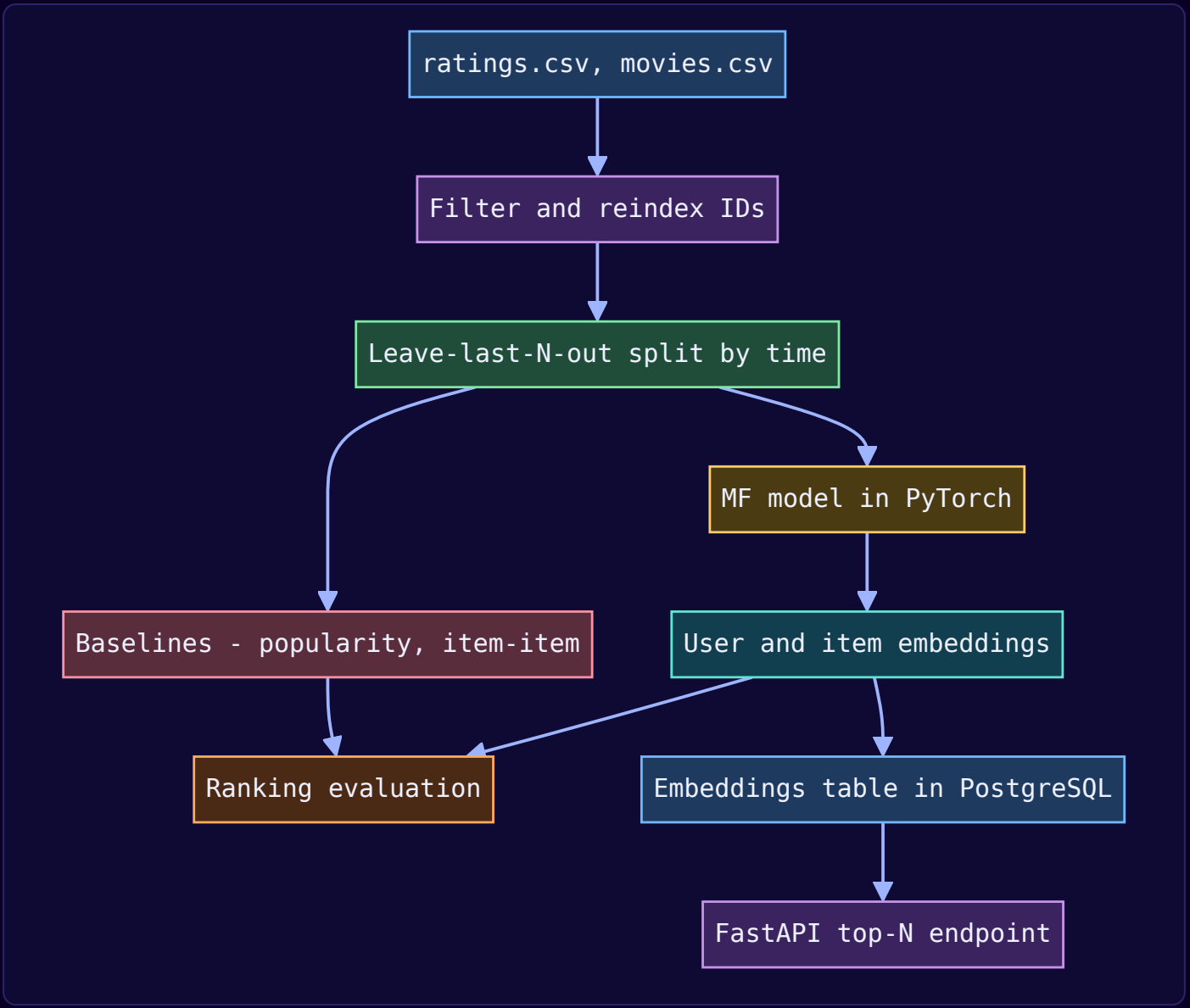
Functional

- Baselines: global popularity and item–item cosine similarity.
- MF trained two ways: explicit ratings with squared loss plus user and item biases, and implicit feedback (watched or not) with negative sampling and BPR (Bayesian personalised ranking: a pairwise loss that prefers watched over unwatched items).
- Time-based split: last N interactions per user for testing.
- Metrics: RMSE for ratings; Recall@20, NDCG@20 (normalised discounted cumulative gain: rewards relevant items more when ranked higher) and catalogue coverage for ranking.
- `GET /recommendations/{user_id}` excluding already-seen movies.

Non-functional

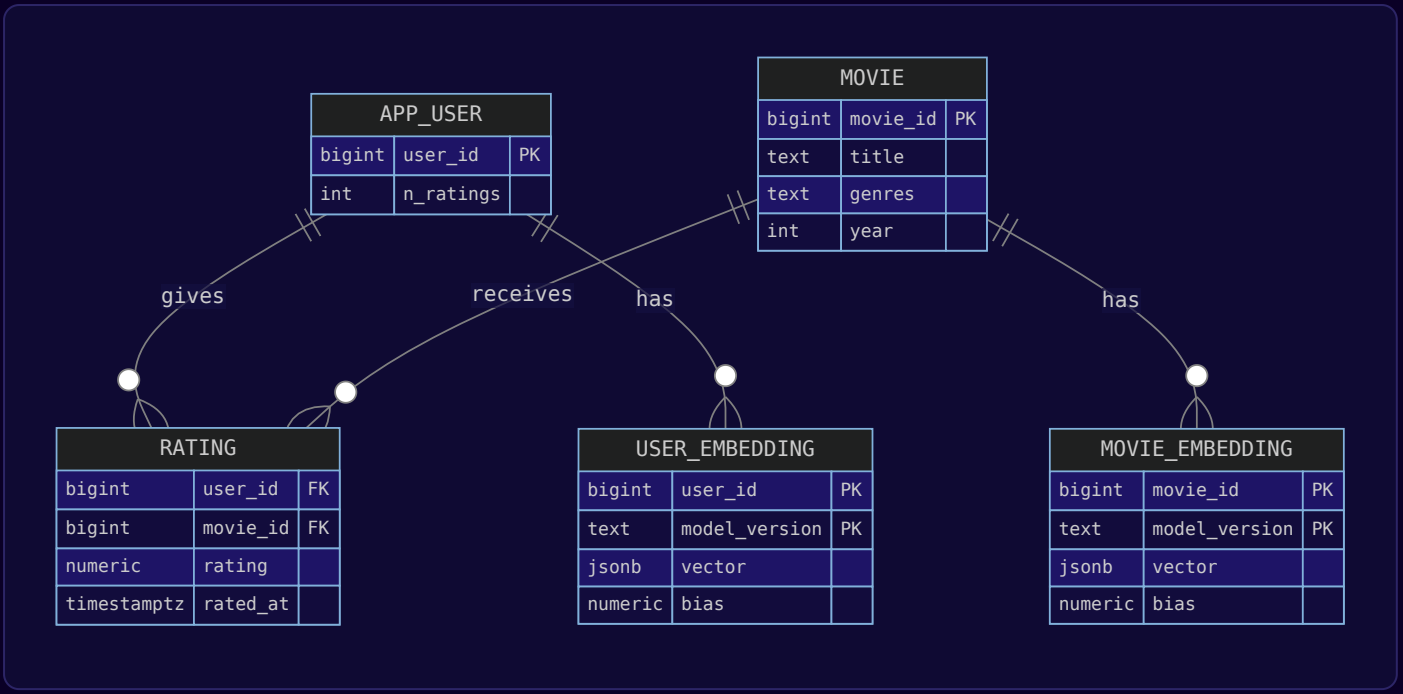
- Training on MovieLens 25M/32M under 30 minutes on one machine (GPU optional).
- Recommendation request under 50 ms for one user.
- Embeddings and model versioned in MLflow.

The architecture



- **Reindex IDs**: map raw IDs to dense integers so embedding tables are compact.
- **MF model**: two `nn.Embedding` tables plus biases; prediction = dot product + biases.
- **Ranking evaluation**: for each test user, score all unseen movies, take top 20, check how many test movies appear.
- **Serving**: pre-compute embeddings; at request time, one matrix–vector product over all movies, then mask seen items.

Data model



Tech stack

Component	Choice	Why
Data	polars	Fast loading of tens of millions of rows
Model	PyTorch embeddings	Flexible losses (MSE, BPR)
Alternative	an ALS implementation (if it supports Python 3.14)	Classic implicit-feedback solver
Storage	PostgreSQL (pgvector optional)	Persist embeddings
Serving	FastAPI	Top-N endpoint

Milestones

1. **Data and split.** *Done when* each user's last 5 ratings by time are held out and you report users, items and density.
2. **Baselines.** *Done when* popularity and item–item Recall@20 and NDCG@20 are recorded (popularity is embarrassingly strong).
3. **Explicit MF.** *Done when* RMSE beats a user+item bias-only model and you can show the effect of embedding size and weight decay.
4. **Implicit MF with BPR.** *Done when* Recall@20 beats both baselines and coverage is reported.
5. **Serve.** *Done when* the API returns 20 unseen movies for a user in under 50 ms, with a popularity fallback for unknown users.

Deep dive: the hard parts

- **Ratings vs ranking.** A model that predicts ratings well (low RMSE) can still recommend poorly, because RMSE is dominated by movies users chose to rate. What users experience is the top of a ranked list over *all* movies — evaluate that.
- **Negative sampling.** With implicit data you only see positives. Treat random unwatched movies as negatives; sampling proportional to popularity makes negatives harder and the model better at separating popular items.
- **Popularity bias and coverage.** MF tends to recommend the same blockbusters to everyone. Track catalogue coverage and the share of long-tail items; penalise popularity or re-rank for diversity.
- **Cold start.** New users and new movies have no embeddings. Fall back to popularity for users and content similarity (Project 10) for new movies, or add genre features to the model.

Testing strategy

- Split test: no test interaction appears in training; all test timestamps are after each user's training ones.
- Metric unit tests on tiny hand-computed rankings (Recall@k, NDCG@k).
- Seen-item filter test: no recommended movie was rated by the user in training.
- Determinism test: same seed, same embeddings.

Trade-offs, scaling and failure modes

- Full scoring over 80,000+ movies per request is fine on CPU; over millions of items you need approximate nearest-neighbour search (Project 26).
- Embeddings drift between retrains; never mix user vectors from one version with item vectors from another.
- Offline metrics on historical data reward recommending what users found anyway; only an online test measures what recommendations change.

Stretch goals

- Add side features (genres, year) to make a hybrid model.
- Train a two-tower model (Project 26 preview) and compare.
- Explain recommendations by nearest watched movies in embedding space.

Skills practised

- Ch 13 — Recommender Systems and Embeddings: collaborative filtering, matrix factorisation, implicit feedback, cold start, evaluation.
- Ch 6 — Evaluation and Tuning: ranking metrics (Recall@k, NDCG).
- Ch 2 — Math for ML: dot products, low-rank approximation, SVD.

In the interview

Q: Explicit vs implicit feedback? **A:** Explicit feedback is ratings — clear but rare and biased toward what people chose to rate. Implicit feedback is clicks or watches — abundant but only positive and noisy. Most production systems train on implicit data with sampled negatives.

Q: Why can popularity be a hard baseline to beat? **A:** Because interactions are concentrated on popular items, and offline metrics reward predicting them. Personalisation must beat it on recall and also add coverage and diversity.

Q: How do you handle a brand-new movie? **A:** Use content features (genres, text, cast) to produce an initial embedding or similarity, give it some exploration traffic, and switch to collaborative signals once interactions arrive.

TL;DR

Matrix factorisation learns user and item embeddings from interactions; evaluate it as a ranking problem against a strong popularity baseline, and plan for cold start.

15. Retail demand forecasting

Level: Intermediate · **Time:** 2–3 weeks · **Prerequisites:** Ch 6, 7, 12

What it is

A system that forecasts daily unit sales for thousands of products in many stores for the next 28 days, using calendar effects, prices, promotions and events. You compare classic **seasonal-naive** and exponential smoothing baselines with a **global gradient-boosting model** (one model trained across all products using lag features) and optionally a neural forecaster. Forecasts include **prediction intervals** (a range the true value should fall in, say, 90% of the time).

The problem it solves

Retailers order stock weeks ahead. Too little means empty shelves and lost sales; too much means waste and tied-up cash. Forecasts at product-store level drive replenishment, and the uncertainty matters as much as the point forecast because safety stock is set from it.

Real-world examples

- The Kaggle M5 Forecasting — Accuracy competition provides hierarchical daily Walmart sales for 3,049 products across 10 stores in three US states, with prices and event calendars.
- Amazon's DeepAR paper describes training one recurrent network across many related time series to produce probabilistic forecasts.

Requirements

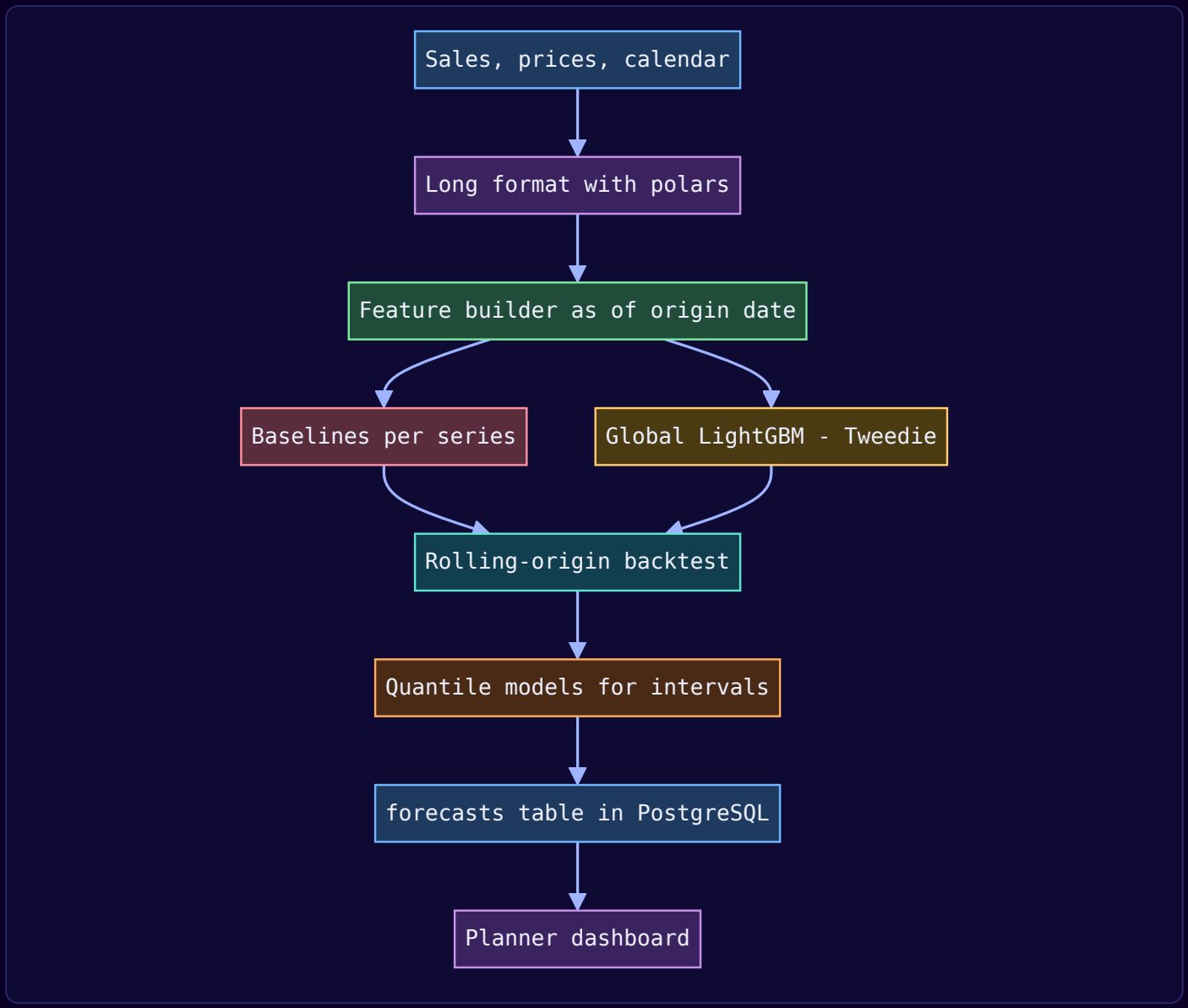
Functional

- Reshape the wide sales table into long format (item, store, date, units).
- Features: lags (7, 14, 28 days), rolling means, day of week, month, events, SNAP days, price and price change.
- Models: seasonal-naive, exponential smoothing per series, global LightGBM with Tweedie loss, optional DeepAR-style RNN.
- **Backtesting**: several forecast origins, each predicting 28 days ahead.
- Metrics: RMSE, MAE, a scaled error (for example RMSSE, error relative to a naive forecast) and interval coverage.

Non-functional

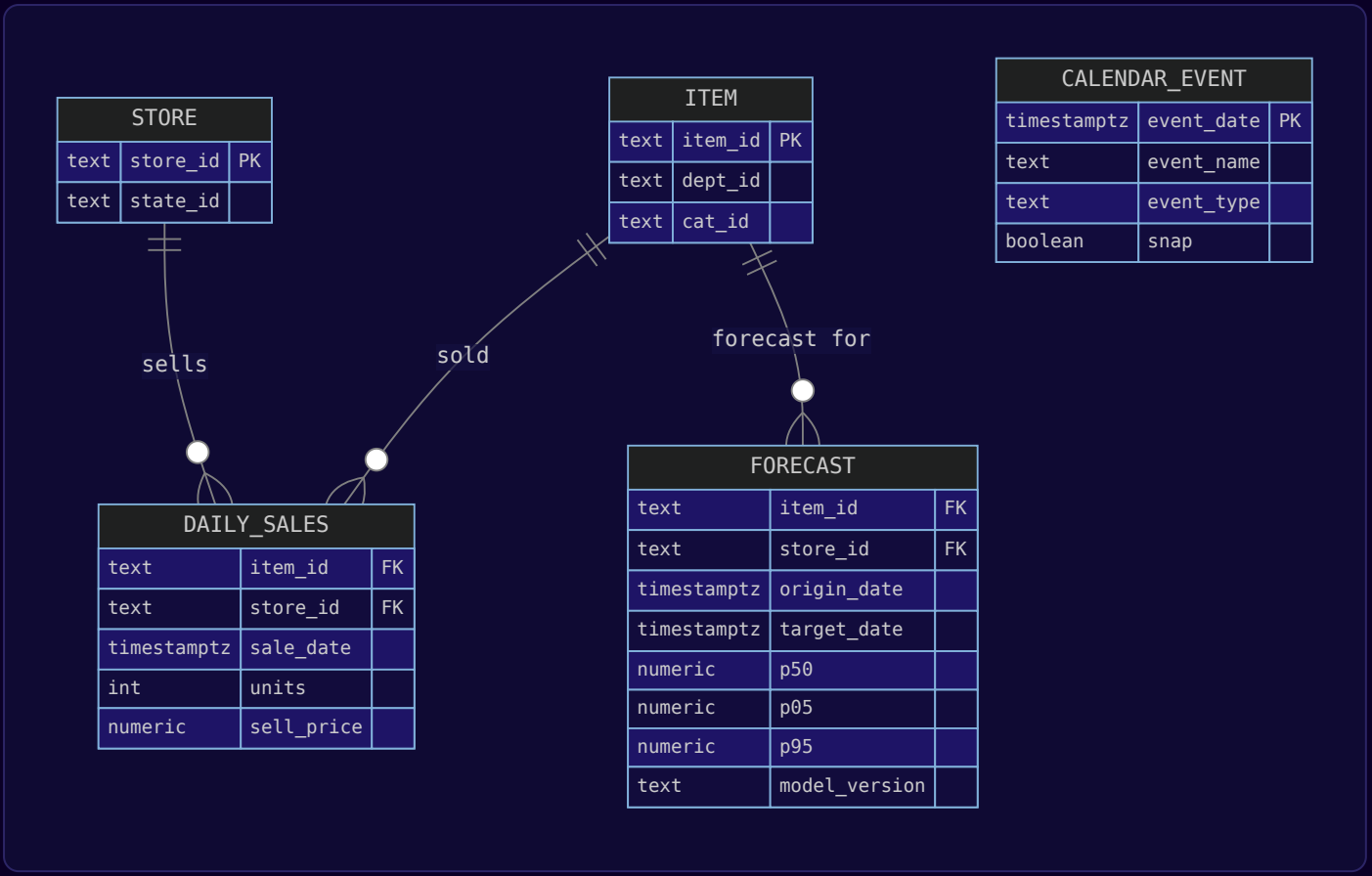
- Full backtest of the global model under 1 hour on a laptop with 16 GB RAM.
- Forecast file for all series written in under 10 minutes.
- No feature uses information after the forecast origin.

The architecture



- **Feature builder as of origin**: computes every feature using only data up to the forecast origin, so backtests mimic real use.
- **Global model**: one model over all series shares strength — slow-moving items learn from similar items.
- **Tweedie loss**: suits non-negative targets with many zeros and occasional spikes.
- **Quantile models**: train for the 5th and 95th percentiles to get a 90% interval.

Data model



Tech stack

Component	Choice	Why
Data	polars	Tens of millions of long-format rows
Baselines	statsmodels (exponential smoothing)	Standard classical methods
Global model	LightGBM	Tweedie and quantile losses, fast
Neural (stretch)	PyTorch	DeepAR-style RNN
Storage	PostgreSQL	Forecasts by origin and version

Milestones

1. **Reshape and explore.** *Done when* you can show weekly seasonality, the share of zero-sales days, and an event effect.
2. **Baselines and backtest harness.** *Done when* seasonal-naive is scored on three rolling origins with RMSSE.
3. **Global model.** *Done when* LightGBM beats seasonal-naive on the backtest at store and total level.
4. **Intervals.** *Done when* 90% intervals cover between 85% and 95% of actuals in the backtest.
5. **Publish.** *Done when* forecasts for all series are written with origin date and model version, and a dashboard query shows one item's forecast vs actuals.

Deep dive: the hard parts

- **Recursive vs direct multi-step.** Lag-7 is unknown when forecasting 14 days ahead. Either use only lags at least as long as the horizon (direct: lag-28 and above for a 28-day horizon), train one model per horizon, or feed predictions back in (recursive, which compounds errors). Pick one and be consistent between training and forecasting.
- **Intermittent demand.** Many products sell zero most days. Squared error rewards predicting tiny fractional values; Tweedie or Poisson losses and scaled metrics behave better. Report accuracy separately for fast and slow movers.

- **Backtesting correctly.** A single train/test split hides variance. Use several origins (for example the last three 28-day windows) and average. Never compute rolling features across the origin.
- **Hierarchy coherence.** Item forecasts should add up to store and state totals. Simple bottom-up summing is a start; reconciliation methods adjust levels to be consistent.

Testing strategy

- Leakage test: for a fixed origin, recomputing features after deleting all data after the origin gives identical feature values.
- Backtest harness test on a synthetic series with known seasonality: seasonal-naive should be near perfect.
- Interval test: quantile 5 ≤ median ≤ quantile 95 for every row.
- Volume test: the forecast table has exactly series × horizon rows per origin.

Trade-offs, scaling and failure modes

- Per-series models are interpretable but need history; global models share information and scale better.
- Promotions and stock-outs distort history: zero sales during a stock-out is not zero demand. Flag or impute those days.
- Structural breaks (a pandemic, a new competitor) invalidate history. Monitor forecast error weekly and retrain.

Stretch goals

- DeepAR-style probabilistic RNN and comparison with LightGBM quantiles.
- Hierarchical reconciliation.
- A replenishment simulation that turns forecasts into order quantities and measures stock-outs vs waste.

Skills practised

- Ch 12 — Sequences, RNNs and Attention: time-series forecasting, lags, sequence models.
- Ch 7 — Decision Trees and Ensembles: gradient boosting with custom losses.
- Ch 6 — Evaluation and Tuning: time-aware validation, scaled metrics.

In the interview

Q: How do you validate a forecasting model? **A:** Rolling-origin backtesting: repeatedly train up to a date, forecast the horizon, score, move the origin forward. All features must be computed as of each origin.

Q: Why one global model rather than one per product? **A:** It shares patterns across series, handles short or sparse histories, and is one artefact to maintain. Per-series models can still win for a few long, stable, high-volume series.

Q: What loss for sparse count demand? **A:** Poisson or Tweedie, which fit non-negative skewed data with many zeros, plus quantile losses when you need intervals for safety stock.

TL;DR

Demand forecasting is backtesting discipline: features as of each origin, a seasonal-naive bar to beat, count-aware losses, and intervals that actually cover.

16. Metric anomaly detection

Level: Intermediate · **Time:** 2 weeks · **Prerequisites:** Ch 9, 12

What it is

A service that watches streams of operational metrics (CPU usage, request latency, sales per minute) and raises an alert when a value is **anomalous** — unusual given recent history and the usual daily or weekly pattern. You implement and compare three families: statistical (rolling z-score and seasonal residuals), tree-based **isolation forest** (which isolates points with random splits; anomalies are isolated in fewer splits), and a forecasting approach (alert when the actual value falls far outside the predicted interval).

The problem it solves

Teams have thousands of metrics and cannot set a hand-tuned threshold for each. Static thresholds also fail on seasonal metrics (low traffic at 3 am is normal; the same value at noon is an outage). Learned detectors flag what is unusual for *this* metric at *this* time, reducing both missed incidents and alert fatigue.

Real-world examples

- The Numenta Anomaly Benchmark provides 58 labelled real and artificial time series (server metrics, traffic, tweets volume) with a scoring method that rewards early detection; its design is explained in Lavin and Ahmad, 2015.
- scikit-learn's IsolationForest documents the algorithm and its contamination parameter.

Requirements

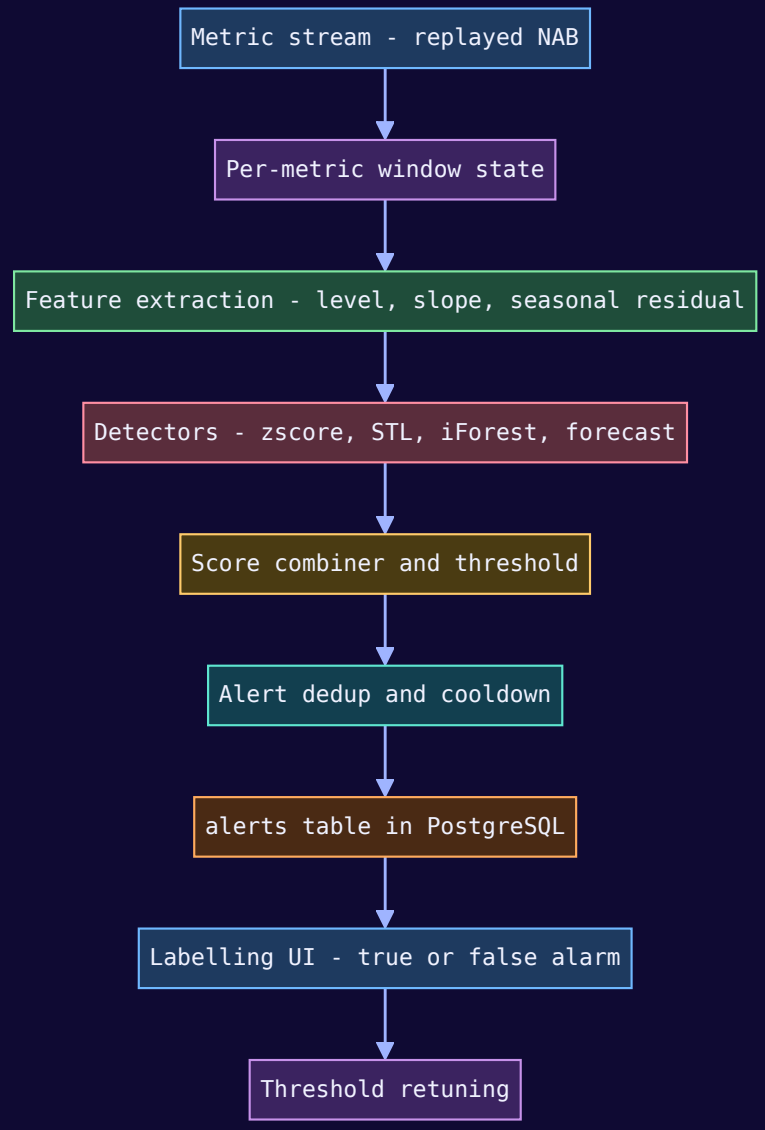
Functional

- Ingest NAB series and replay them as a stream, point by point.
- Detectors: rolling z-score, seasonal decomposition residual (STL), isolation forest on windowed features, forecast-interval detector.
- Each detector outputs an anomaly score per point; a threshold converts it to alerts.
- Alert deduplication: one alert per incident, not one per point.
- Evaluate with NAB-style windows: precision, recall and detection delay.

Non-functional

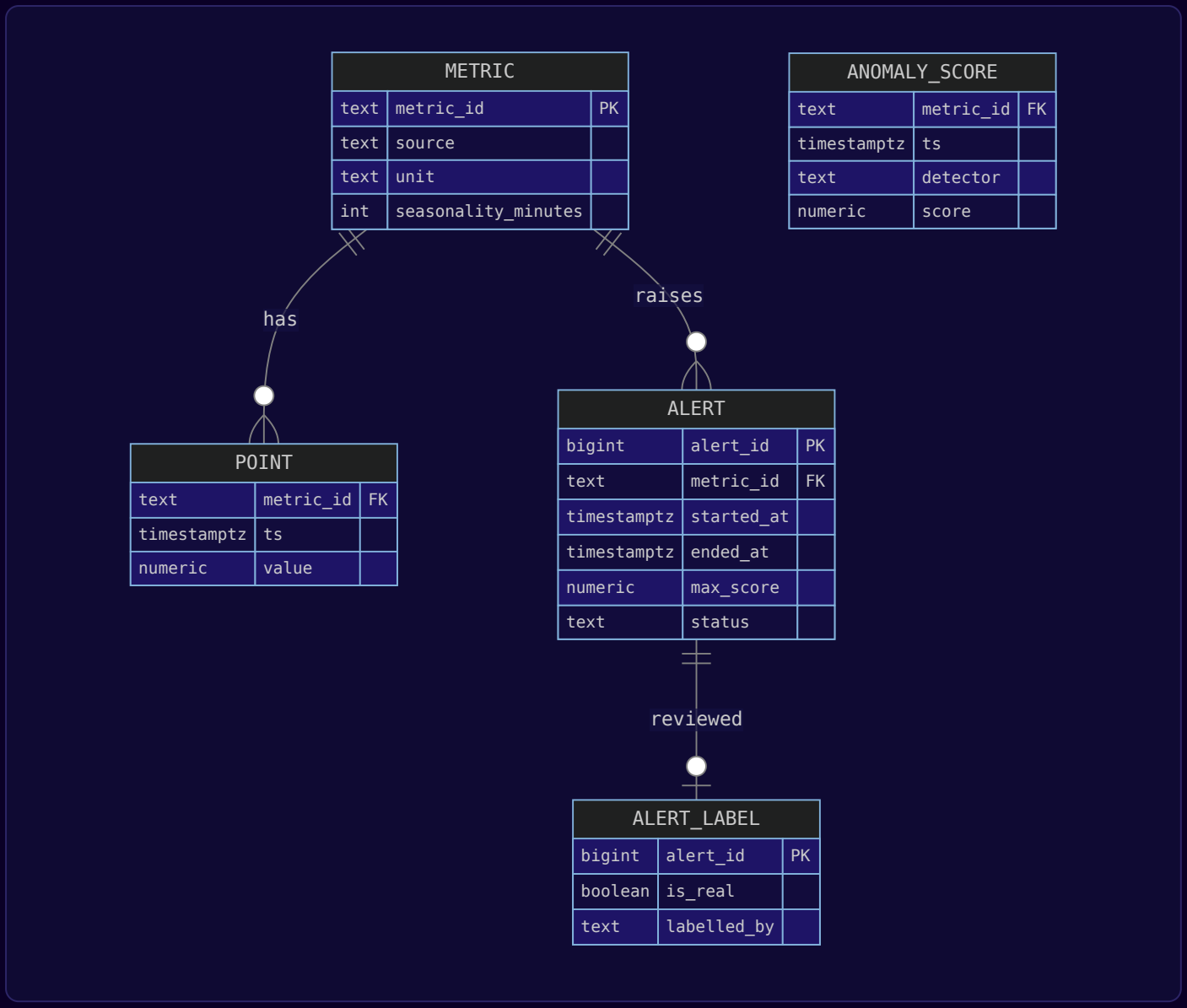
- Process 10,000 metrics at one point per minute each on one machine (about 170 points per second).
- Detection decision under 100 ms per point.
- Detector state per metric under 100 KB.

The architecture



- **Window state**: a ring buffer of the last N points per metric plus running statistics.
- **Detectors**: run in parallel; each is a small class with `update(value)` and `score()`.
- **Combiner**: start with "max of normalised scores"; later learn weights from labelled alerts.
- **Dedup and cooldown**: once an alert fires, suppress further alerts for the same metric for a cooldown period.
- **Labelling UI**: operators mark alerts as real or noise, giving labels for tuning.

Data model



Tech stack

Component	Choice	Why
Stream replay	Python generator or Kafka (stretch)	Simulate real-time arrival
Statistics	numpy, statsmodels STL	Seasonal decomposition
ML detector	scikit-learn IsolationForest	Unsupervised, fast
Storage	PostgreSQL	Alerts and labels
UI	FastAPI + simple HTML	Review alerts

Milestones

1. **Replay and baseline.** *Done when* a rolling z-score detector runs over all NAB series and you have precision and recall at a chosen threshold.
2. **Seasonality.** *Done when* an STL-residual detector reduces false positives on clearly seasonal series and you can show one example.
3. **Isolation forest.** *Done when* windowed features feed an isolation forest retrained periodically per metric, and its scores are compared.
4. **Alerting logic.** *Done when* dedup and cooldown turn point scores into incident alerts and the alert count per series is sensible.

5. **Feedback loop.** *Done when* labels from the UI are used to retune per-metric thresholds and precision improves.

Deep dive: the hard parts

- **No labels, or few.** Real anomalies are rare and unlabelled. Use NAB labels for development, then rely on operator feedback in production. Choose thresholds per metric by a target alert budget (for example, at most one alert per metric per week) when labels are missing.
- **Contamination of the baseline.** If the training window includes an incident, the detector learns the incident as normal. Use robust statistics (median and MAD, median absolute deviation) and exclude flagged periods from refits.
- **Concept drift vs anomaly.** A permanent level shift after a deployment is an anomaly at first and the new normal afterwards. Let the detector adapt after a configurable time, and record the change as an annotation.
- **Point vs contextual vs collective anomalies.** A single spike, a normal value at the wrong time, and a run of slightly odd values need different detectors. This is why you combine several.

Testing strategy

- Synthetic tests: inject spikes, level shifts and variance changes into clean seasonal series; assert each detector catches its target type.
- Cold-start test: detectors behave (no alerts) until enough history exists.
- Replay determinism: replaying the same series twice gives identical alerts.
- Load test: 10,000 metrics at the target rate with latency measured.

Trade-offs, scaling and failure modes

- Simple statistical detectors are cheap, explainable and often competitive; ML detectors catch subtler patterns but are harder to explain to an on-call engineer at 3 am.
- Alert fatigue is the main failure mode: too many false alarms and people ignore all alerts. Optimise precision at an acceptable recall.
- Correlated metrics fire together during one incident. Group alerts by service or dependency graph.

Stretch goals

- Multivariate detection: an autoencoder over related metrics, alerting on reconstruction error.
- Root-cause hints: rank which metrics moved first.
- Real streaming with Kafka and a consumer group per metric shard.

Skills practised

- Ch 9 — Unsupervised Learning: anomaly detection, isolation forest, evaluation without labels.
- Ch 12 — Sequences, RNNs and Attention: time-series features and forecasting.
- Ch 16 — ML in Production: monitoring, alerting, feedback loops.

In the interview

Q: How do you detect anomalies in a seasonal metric? **A:** Remove the expected seasonal pattern (decomposition or a forecast) and score the residual, so a value is compared with what is normal for that time.

Q: How do you choose a threshold without labels? **A:** From an alert budget per metric, robust statistics on the residual distribution, and operator feedback over time.

Q: How does isolation forest work? **A:** It builds random trees with random splits. Anomalies are few and different, so they get isolated near the root; the average path length becomes the anomaly score.

TL;DR

Score residuals against what is normal for this metric at this time, combine simple detectors, and judge success by alert precision and detection delay.

17. A/B test analysis toolkit

Level: Intermediate · **Time:** 2 weeks · **Prerequisites:** Ch 2, 16

What it is

A small library and report generator that analyses **A/B tests** (randomised controlled experiments where users are split between a control and a new variant). It computes the effect on key metrics with confidence intervals, checks experiment health, applies **CUPED** (a variance-reduction method that adjusts each user's metric using their pre-experiment behaviour) and produces a one-page decision report.

The problem it solves

Most ML models are launched through an A/B test, because offline metrics do not guarantee online impact. Analysing tests badly — peeking every day, ignoring a broken randomisation, testing twenty metrics and celebrating the one that moved — leads to shipping changes that do nothing or harm. A toolkit makes the correct analysis the easy default.

Real-world examples

- Deng, Xu, Kohavi and Walker's paper Improving the Sensitivity of Online Controlled Experiments by Utilizing Pre-Experiment Data (WSDM 2013) introduced CUPED at Microsoft Bing.
- statsmodels provides the t-tests, proportion tests and power calculations you will build on.

Requirements

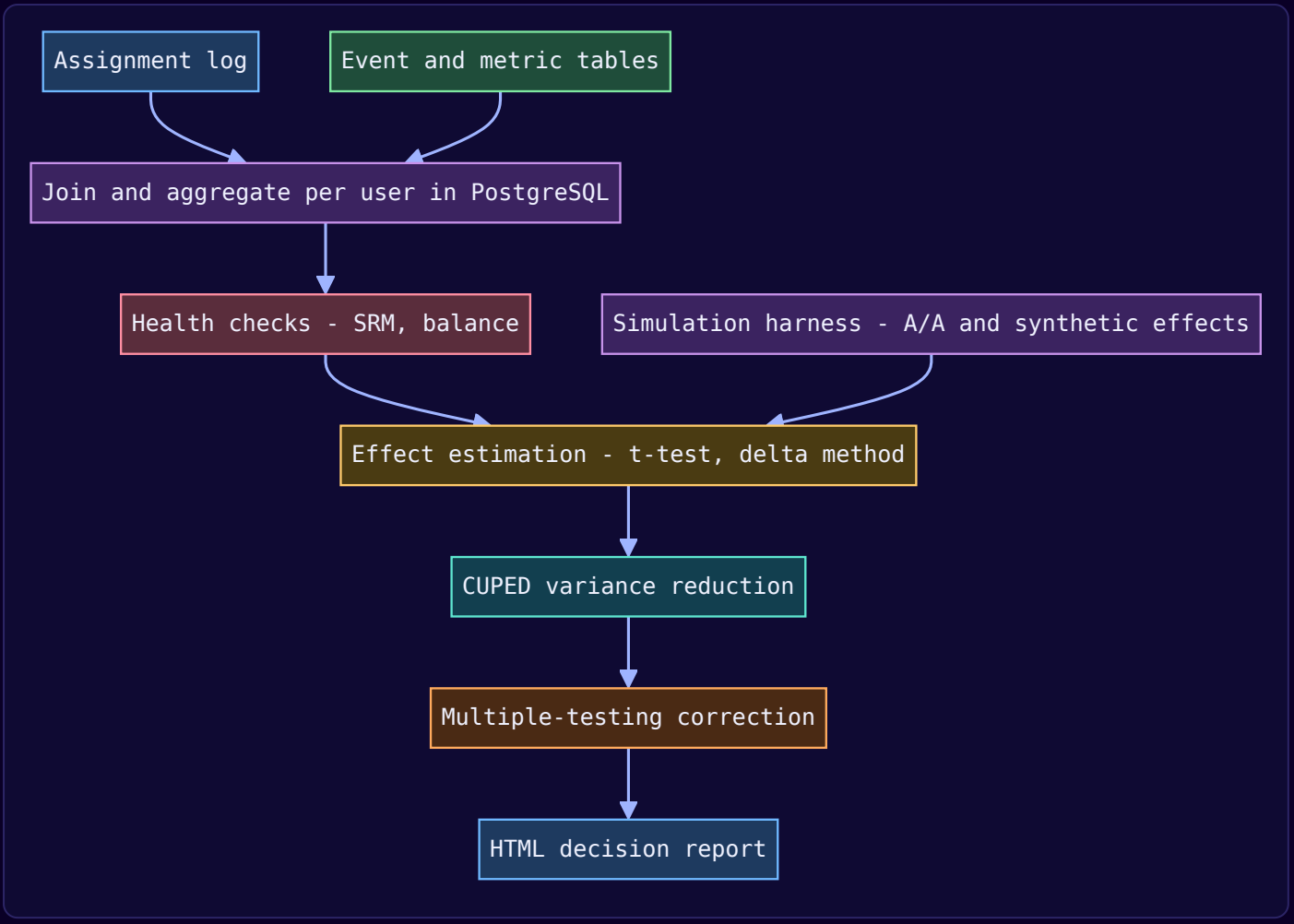
Functional

- Inputs: an assignment table (user, variant, timestamp) and a metrics table (user, metric values, pre-period values).
- Health checks: **sample ratio mismatch** (SRM — the observed split differs from the planned one more than chance allows), pre-period balance.
- Effects: difference in means and relative lift with 95% confidence intervals, for means and proportions; delta method for ratio metrics (for example clicks per session).
- CUPED adjustment and the resulting variance reduction.
- Power and sample-size calculator.
- Multiple-testing correction across secondary metrics.
- HTML report: decision summary, guardrail metrics, charts.

Non-functional

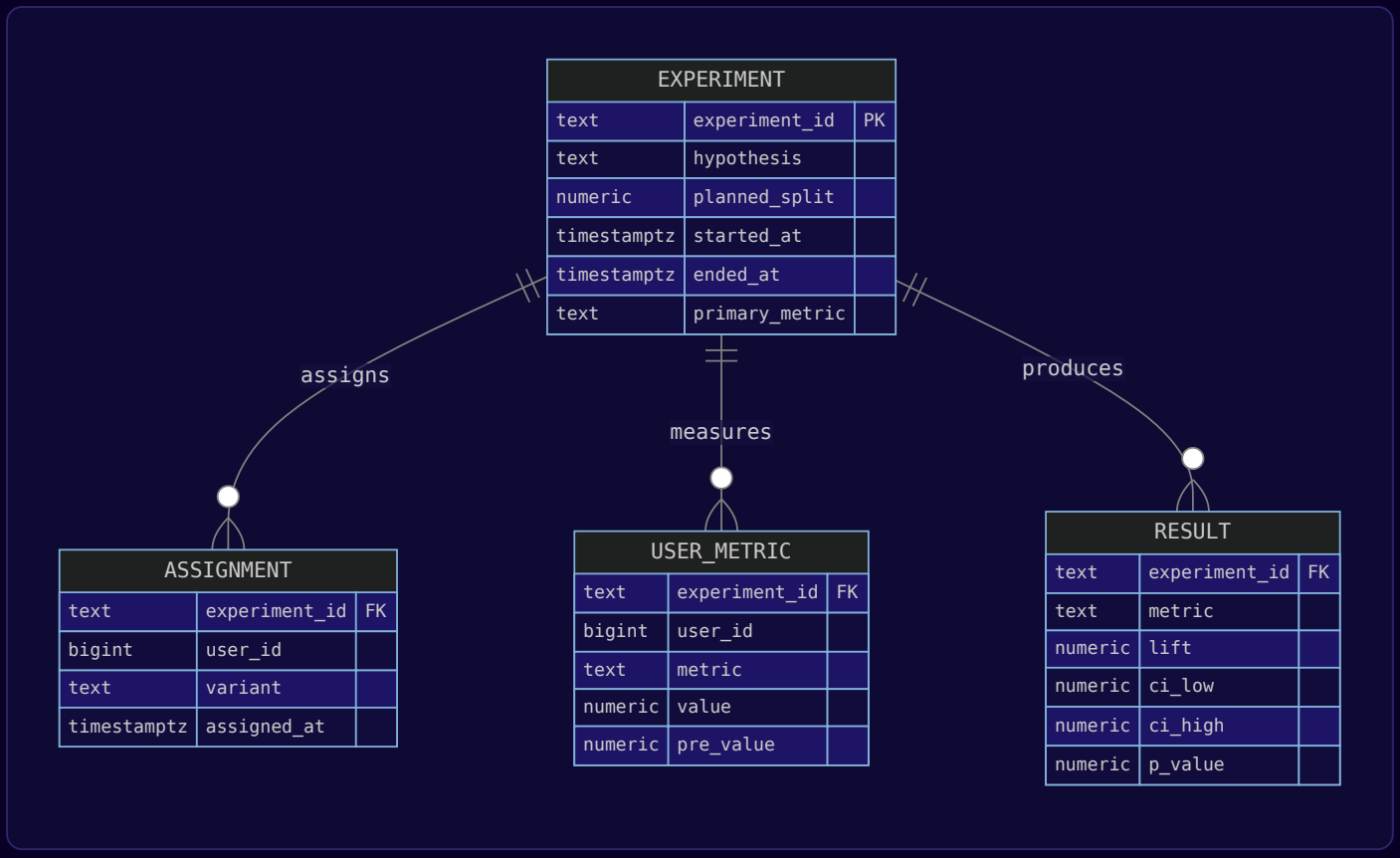
- Analyse 10 million users × 20 metrics in under 2 minutes.
- Every number reproducible from SQL inputs.
- Validated by simulation: false-positive rate close to 5% on A/A tests.

The architecture



- **Per-user aggregation**: the unit of analysis must match the unit of randomisation (users), otherwise sessions from the same user are wrongly treated as independent.
- **Health checks**: an SRM failure means the result cannot be trusted, whatever the p-value.
- **CUPED**: subtract $\theta \times$ (pre-period metric − its mean) from each user's metric, with θ estimated by regression. The mean effect is unchanged; the variance shrinks.
- **Simulation harness**: runs thousands of fake experiments to prove the false-positive rate and power are what the theory says.

Data model



Tech stack

Component	Choice	Why
Aggregation	PostgreSQL SQL	Close to the data, auditable
Stats	numpy, scipy, statsmodels	Tests, power, corrections
Data frames	polars	Fast per-user tables
Report	Jinja2 HTML + matplotlib	Shareable output

Milestones

1. **Synthetic data generator.** *Done when* you can create experiments with known true effect, metric skew and pre-period correlation.
2. **Core estimates.** *Done when* difference in means with confidence intervals matches statsmodels on test data, and A/A simulations show about 5% false positives.
3. **Health checks.** *Done when* an injected 50.5/49.5 split on a million users fails the SRM check.
4. **CUPED and ratio metrics.** *Done when* CUPED shows variance reduction on correlated data and delta-method intervals have correct coverage in simulation.
5. **Report.** *Done when* one command turns two tables into an HTML report with a clear ship / do not ship / inconclusive verdict.

Deep dive: the hard parts

- **Peeking.** Checking significance daily and stopping when $p < 0.05$ inflates false positives far above 5%. Fix the sample size in advance with a power calculation, or use a sequential method designed for continuous monitoring. Demonstrate the inflation with your simulator.
- **Ratio metrics.** Click-through rate per session has a random denominator, and sessions from one user are correlated. The delta method gives a correct variance for ratios of per-user sums; naive per-session t-tests are over-confident.

- **SRM.** A chi-square test on assignment counts. Causes include bots filtered in one arm, crashes in the new variant, or redirect delays. When SRM fails, debug instead of analysing.
- **Novelty and network effects.** Users may click a new feature because it is new; effects fade. In social products, treated users influence control users, biasing results; cluster randomisation helps.

Testing strategy

- Simulation tests are the core: A/A false-positive rate within 4–6% over 2,000 runs; power matches the calculator for a planted effect.
- Unit tests against known textbook examples for each test.
- CUPED unbiasedness: average estimated effect equals planted effect over many simulations.
- SQL tests: per-user aggregation on a small fixture.

Trade-offs, scaling and failure modes

- Bigger samples detect smaller effects but take longer; CUPED often gives a sizeable variance cut when pre-period behaviour predicts the metric well, which shortens tests.
- Too many metrics without correction guarantees false discoveries. Name one primary metric up front; treat others as guardrails or exploratory.
- Statistically significant is not practically significant. Report effect size and confidence interval against a minimum worthwhile effect.

Stretch goals

- Sequential testing (always-valid confidence intervals) with a simulation proving the error rate.
- Bayesian analysis with a prior and posterior probability of improvement.
- Heterogeneous effects by segment, with a warning about multiple comparisons.

Skills practised

- Ch 2 — Math for ML: hypothesis tests, confidence intervals, variance.
- Ch 16 — ML in Production: online experiments, guardrails, launch decisions.
- Ch 15 — ML System Design Framework: offline vs online metrics.

In the interview

Q: What is sample ratio mismatch and why does it matter? **A:** The observed split between arms differs from the designed split beyond chance. It signals a bug in assignment or logging, which can bias every metric, so the test should not be trusted until fixed.

Q: How does CUPED reduce variance? **A:** It regresses the metric on a pre-experiment covariate and removes the part explained by it. Randomisation keeps the covariate balanced, so the effect estimate is unbiased, but the leftover noise is smaller.

Q: Why not stop the test as soon as it's significant? **A:** Repeated looks inflate the false-positive rate. Pre-commit to a sample size or use sequential methods.

TL;DR

Check health before effects, analyse at the randomisation unit, use CUPED and the delta method, never peek naively — and prove all of it with simulations.

18. Semantic document search

Level: Intermediate · **Time:** 2 weeks · **Prerequisites:** Ch 12, 13

What it is

A search engine that finds documents by meaning rather than exact words. Each document chunk is turned into an **embedding** by a pre-trained sentence-embedding model; a query is embedded the same way, and the nearest chunks by cosine similarity are returned. You combine this **dense retrieval** with classic keyword search (**BM25**, a

ranking formula based on term frequency and rarity) in a **hybrid** system and measure the gain with standard IR metrics.

The problem it solves

Keyword search fails when users and documents use different words ("cancel subscription" vs "terminate membership"). Embeddings match synonyms and paraphrases. But dense retrieval misses exact identifiers (error codes, product names) that keyword search nails — hence hybrid search. This is also the retrieval half of question-answering systems (Project 38).

Real-world examples

- Sentence-BERT introduced siamese fine-tuning so that cosine similarity of sentence embeddings reflects semantic similarity.
- The BEIR benchmark evaluates retrieval models across many domains and shows that BM25 remains a strong zero-shot baseline.
- pgvector adds vector columns and approximate nearest-neighbour indexes (HNSW, IVFFlat) to PostgreSQL; FAISS is a dedicated similarity-search library.

Requirements

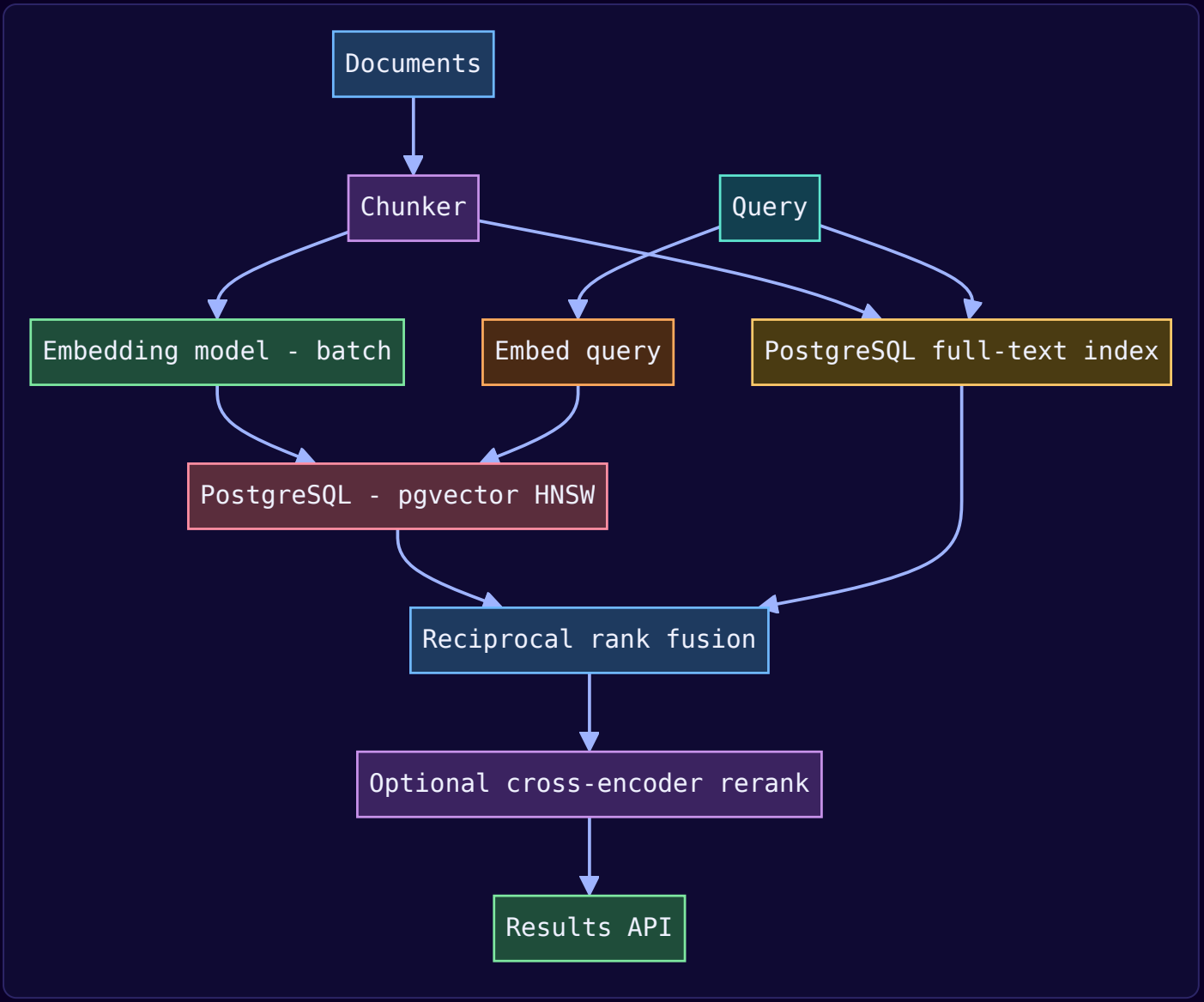
Functional

- Ingest a document collection (a BEIR dataset such as SciFact or FiQA, which come with queries and relevance labels).
- Chunk documents (for example 200–300 words with overlap), embed with a sentence-transformer model.
- Index in PostgreSQL with pgvector (HNSW) and with full-text search for BM25-style ranking.
- Hybrid fusion with **reciprocal rank fusion** (RRF: sum of 1/(k + rank) across result lists).
- Metrics: Recall@10, nDCG@10, MRR (mean reciprocal rank of the first relevant hit).
- `GET /search?q=` returning ranked chunks with highlights.

Non-functional

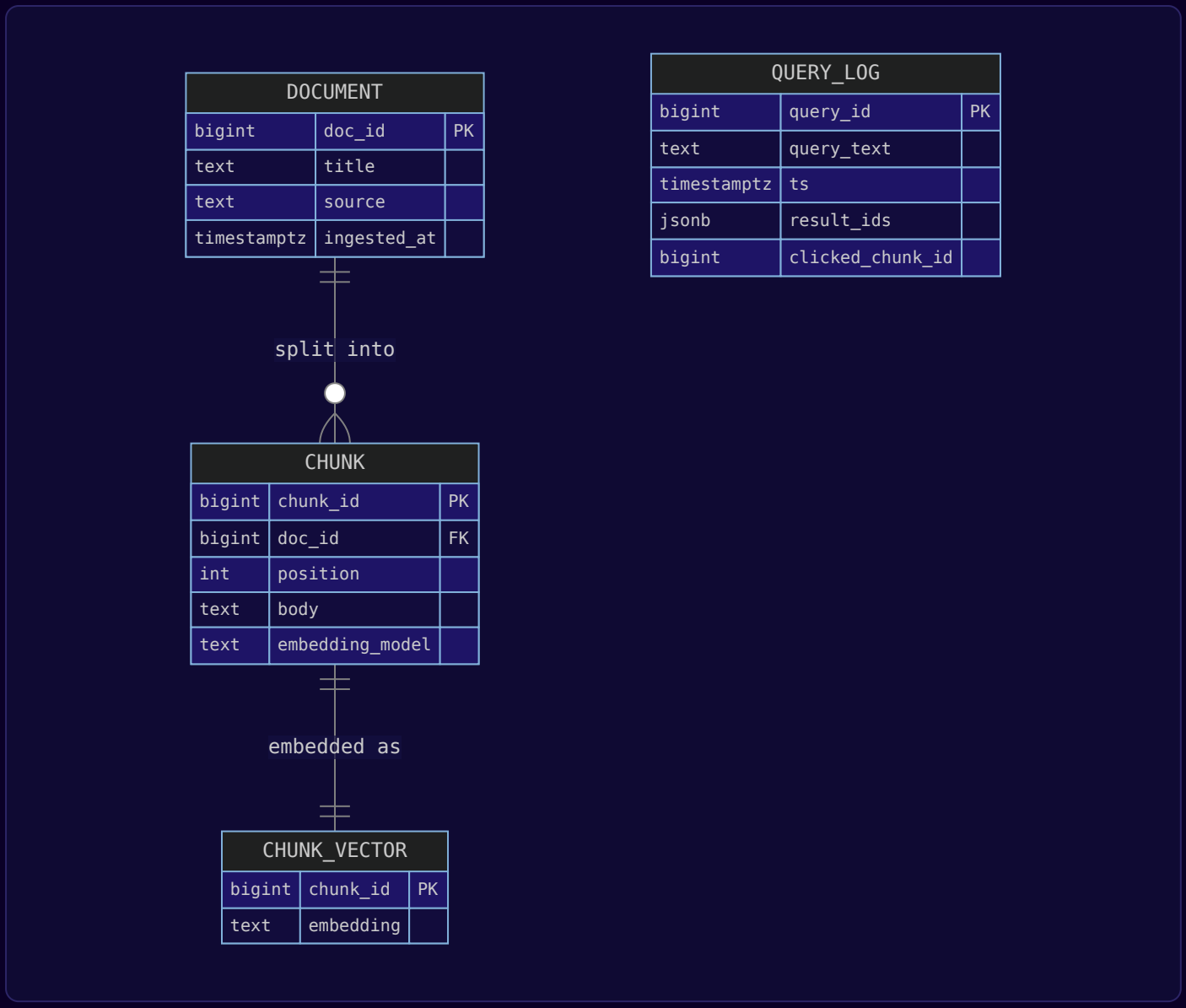
- p95 query latency under 100 ms for 1 million chunks (ANN index, CPU).
- Re-indexing 100,000 chunks under 30 minutes on CPU (faster with GPU).
- Recall of the ANN index at least 95% of exact search.

The architecture



- **Chunker**: long documents are split so each vector represents one idea; overlap avoids cutting a key sentence in half.
- **HNSW index**: a graph-based approximate nearest-neighbour index — fast, with a recall vs speed knob (`ef_search`).
- **Full-text index**: PostgreSQL `tsvector` with ranking, standing in for BM25.
- **Fusion**: RRF needs no score calibration between the two systems.
- **Cross-encoder rerank** (stretch): a model that reads query and chunk together for a more accurate score on the top 50.

Data model



(In SQL the `embedding` column has type `vector(384)` from pgvector; the diagram shows it as text only because diagram types must be single words.)

Tech stack

Component	Choice	Why
Embeddings	sentence-transformers model on PyTorch	Strong general-purpose embeddings
Vector index	PostgreSQL + pgvector (HNSW)	One database for text, vectors, metadata
Keyword search	PostgreSQL full-text search	No extra service
Alternative index	FAISS	Benchmark against pgvector
API	FastAPI	Search endpoint

Milestones

1. **Keyword baseline.** *Done when* full-text search over a BEIR dataset reports nDCG@10 and Recall@10.
2. **Dense retrieval.** *Done when* embedding search (exact, brute force) is scored on the same queries.
3. **ANN index.** *Done when* HNSW recall vs exact search and latency at several `ef_search` settings are plotted.
4. **Hybrid.** *Done when* RRF fusion is scored and beats (or ties) the better single method; note which query types each wins.

5. **API and logging.** *Done when* the endpoint returns highlighted results and logs queries and clicks.

Deep dive: the hard parts

- **Chunking decides quality.** Too large and a vector blurs several topics; too small and context is lost. Evaluate 2–3 chunk sizes on the benchmark rather than guessing.
- **Domain mismatch.** General embedding models may do poorly on legal or medical text, sometimes below BM25 — BEIR shows this pattern across domains. Hybrid search is the safety net; domain fine-tuning with query–document pairs is the upgrade.
- **ANN trade-offs.** HNSW gives high recall at low latency but uses memory and builds slowly; IVFFlat builds faster with lower recall. Filtering (for example by document type) interacts with ANN: filtering after search can return too few results.
- **Evaluation without labels in production.** Use clicks and dwell time as weak labels, but beware position bias — users click what is on top.

Testing strategy

- Metric unit tests on hand-built rankings.
- Index parity: for 100 random queries, ANN top-10 overlaps exact top-10 by at least 95%.
- Re-embedding test: the stored `embedding_model` matches the model used for queries (mixing models silently breaks search).
- Latency test at 1 million chunks.

Trade-offs, scaling and failure modes

- pgvector keeps everything in one database — simple operations, transactional updates. Dedicated vector databases scale further and offer more index types.
- Changing the embedding model requires re-embedding everything; version vectors by model.
- Dense search can return confident nonsense for out-of-domain queries; show scores and a "no good match" threshold.

Stretch goals

- Cross-encoder reranking of the top 50.
- Fine-tune the embedding model on in-domain pairs.
- Multilingual search with a multilingual embedding model.

Skills practised

- Ch 12 — Sequences, RNNs and Attention: transformers, embeddings.
- Ch 13 — Recommender Systems and Embeddings: embedding similarity, ANN search.
- Ch 6 — Evaluation and Tuning: ranking metrics (nDCG, MRR, Recall@k).
- Ch 17 — ML Case Studies I: hybrid lexical and semantic retrieval in search design.

In the interview

Q: Dense vs sparse retrieval? **A:** Sparse (BM25) matches exact terms — great for names and codes, robust out of domain. Dense matches meaning — great for paraphrase. Hybrid fusion usually beats both.

Q: How does HNSW trade recall for speed? **A:** It searches a layered proximity graph greedily; the size of the candidate list during search (`ef_search`) controls how much of the graph is explored. Bigger is slower but more accurate.

Q: How would you evaluate search quality? **A:** Offline with labelled query–document relevance (nDCG, Recall@k), online with click-through, reformulation rate and success rate, correcting for position bias.

TL;DR

Embed chunks, index them with an ANN index, fuse with keyword search, and measure with nDCG on labelled queries — chunking and hybrid fusion matter more than the model.

19. Pedestrian detector by fine-tuning

Level: Intermediate · **Time:** 2–3 weeks · **Prerequisites:** Ch 11

What it is

An **object detector**: given an image, it draws a box around every pedestrian and gives a confidence for each. Unlike classification (one label per image), detection must find a variable number of objects and localise them. You fine-tune a pre-trained detector (Faster R-CNN, which first proposes regions then classifies and refines them) on a small pedestrian dataset, evaluate with **mAP** (mean average precision over IoU thresholds) and serve it.

The problem it solves

Counting people in a shop, blurring faces for privacy, and driver assistance all start with detection. Labelling boxes is expensive, so real projects start from a detector pre-trained on a large dataset like COCO and fine-tune on a few hundred domain images.

Real-world examples

- The Penn-Fudan Pedestrian Database has 170 images with 345 labelled pedestrians, including segmentation masks.
- The PyTorch TorchVision object detection fine-tuning tutorial fine-tunes Mask R-CNN on Penn-Fudan.
- COCO is the standard large detection dataset and the source of most pre-trained detector weights and the COCO-style mAP metric.

Requirements

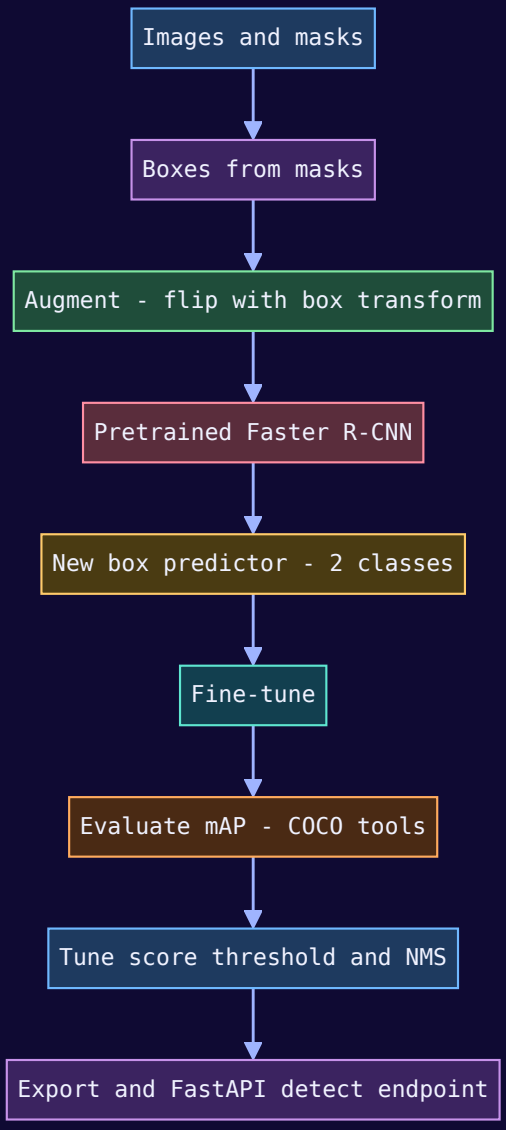
Functional

- Dataset class returning image and target (boxes, labels) with augmentations that move boxes correctly.
- Fine-tune a torchvision Faster R-CNN with a pre-trained backbone; replace the box predictor for 2 classes (background, person).
- Metrics: mAP@0.5 and COCO-style mAP@[0.5:0.95]; precision–recall curve.
- **Non-maximum suppression** (NMS: drop overlapping boxes that describe the same person) tuned on validation.
- `POST /detect` returns boxes and scores; a visualisation endpoint draws them.

Non-functional

- Training under 1 hour on a single consumer GPU (CPU possible but slow — note it).
- CPU inference under 1 second per 800-pixel image; GPU under 100 ms.
- mAP@0.5 above the pre-trained model's zero-shot person detection on this dataset.

The architecture



- **Boxes from masks**: Penn-Fudan gives instance masks; the bounding box is the min/max of each mask's pixels.
- **Augment**: a horizontal flip must also flip box x-coordinates. Colour jitter needs no box change.
- **Box predictor**: the final classification and box-regression layer is replaced; the region proposal network and backbone are reused.
- **Evaluate**: IoU (intersection over union of predicted and true boxes) decides whether a detection matches a ground-truth person.

Data model

Item	Type	Notes
Image	uint8 `[3, H, W]`	Various sizes
`boxes`	float `[N, 4]`	x1, y1, x2, y2 in pixels
`labels`	int64 `[N]`	1 = person
`iscrowd`, `area`	per box	Needed by COCO evaluation
Prediction	boxes, scores, labels	After NMS and threshold

Tech stack

Component	Choice	Why
Detector	torchvision Faster R-CNN	Pre-trained, well documented
Evaluation	COCO-style mAP (pycocotools or torchmetrics)	Standard metric
Training	PyTorch, mixed precision on GPU	Speed
Serving	FastAPI	Image upload, JSON boxes

Milestones

1. **Dataset.** *Done when* drawing boxes from your dataset class over 10 images shows correct boxes after augmentation.
2. **Zero-shot baseline.** *Done when* the COCO-pre-trained detector's person class is evaluated on your test split.
3. **Fine-tune.** *Done when* mAP@0.5 and mAP@[0.5:0.95] beat the baseline.
4. **Thresholds.** *Done when* score threshold and NMS IoU are tuned on validation and the PR curve is plotted.
5. **Serve.** *Done when* uploading a street photo returns boxes drawn correctly within the latency target.

Deep dive: the hard parts

- **Box transforms.** Every geometric augmentation must transform boxes identically, and boxes that leave the image must be clipped or dropped. A silent bug here trains the model on wrong targets. Visual checks are the best test.
- **mAP explained.** For each class, rank detections by score, mark each as a true positive if it matches an unmatched ground-truth box with IoU ≥ threshold, compute the precision–recall curve and its area. COCO averages this over IoU thresholds 0.5 to 0.95, rewarding tight boxes.
- **Small data, big model.** 170 images is tiny. Keep most of the backbone frozen early, use a low learning rate and strong augmentation, and watch validation mAP.
- **Crowds and occlusion.** Overlapping pedestrians challenge NMS: too aggressive and you merge two people, too lenient and one person gets two boxes.

Testing strategy

- Visual regression: a fixed set of images with expected box counts.
- Augmentation unit test: flipping twice returns original boxes.
- Metric test: perfect predictions give mAP = 1.0, no predictions give 0.
- API test with an image containing no people returns an empty list.

Trade-offs, scaling and failure modes

- Two-stage detectors (Faster R-CNN) are accurate; one-stage detectors (RetinaNet, which introduced focal loss for class imbalance, and YOLO-style models) are faster. Choose by latency budget.
- Detection quality drops at night, in rain, at unusual camera angles. Collect slice-specific test sets.
- People-detection systems raise privacy concerns. Process on device or blur faces when possible (Ch 18's street-imagery case study).

Stretch goals

- Instance segmentation with Mask R-CNN on the same data.
- A one-stage detector comparison on accuracy vs speed.
- Track pedestrians across video frames (detection + simple IoU tracking).

Skills practised

- Ch 11 — CNNs and Computer Vision: detection, IoU, NMS, transfer learning, augmentation.
- Ch 6 — Evaluation and Tuning: precision–recall curves and average precision.
- Ch 18 — ML Case Studies II: privacy blurring of street imagery.

In the interview

Q: What is mAP? **A:** Average precision per class (area under the precision–recall curve where a detection counts as correct above an IoU threshold), averaged over classes and, in COCO style, over IoU thresholds 0.5 to 0.95.

Q: What does non-maximum suppression do? **A:** It keeps the highest-scoring box and removes others overlapping it above an IoU threshold, so each object gets one box.

Q: One-stage vs two-stage detectors? **A:** Two-stage detectors propose regions then classify them — more accurate, slower. One-stage detectors predict boxes densely in one pass — faster, and with focal loss they close much of the accuracy gap.

TL;DR

Fine-tune a pre-trained detector, transform boxes with every augmentation, and judge it by mAP with tuned thresholds and NMS.

20. Headline-testing bandit

Level: Intermediate · **Time:** 2 weeks · **Prerequisites:** Ch 14, 16

What it is

A **multi-armed bandit** service that chooses which of several headlines (or thumbnails, or button texts) to show each visitor, learning from clicks as it goes. Unlike an A/B test, which splits traffic evenly until the end, a bandit shifts traffic toward the winners while still **exploring** the others. You implement epsilon-greedy, UCB (upper confidence bound) and Thompson sampling, then a **contextual bandit** (LinUCB) that personalises by visitor features.

The problem it solves

News articles live for hours, not weeks. A classic A/B test on headlines would end after the article is stale, and half the traffic would see the worse headline. Bandits reduce **regret** (clicks lost compared with always showing the best option) by adapting during the test.

Real-world examples

- Li, Chu, Langford and Schapire's A Contextual-Bandit Approach to Personalized News Article Recommendation introduced LinUCB on the Yahoo! Front Page Today Module and showed how to evaluate bandits offline from logged random traffic.
- Netflix's Artwork Personalization at Netflix post describes choosing the image shown for a title per member with contextual bandits.

Requirements

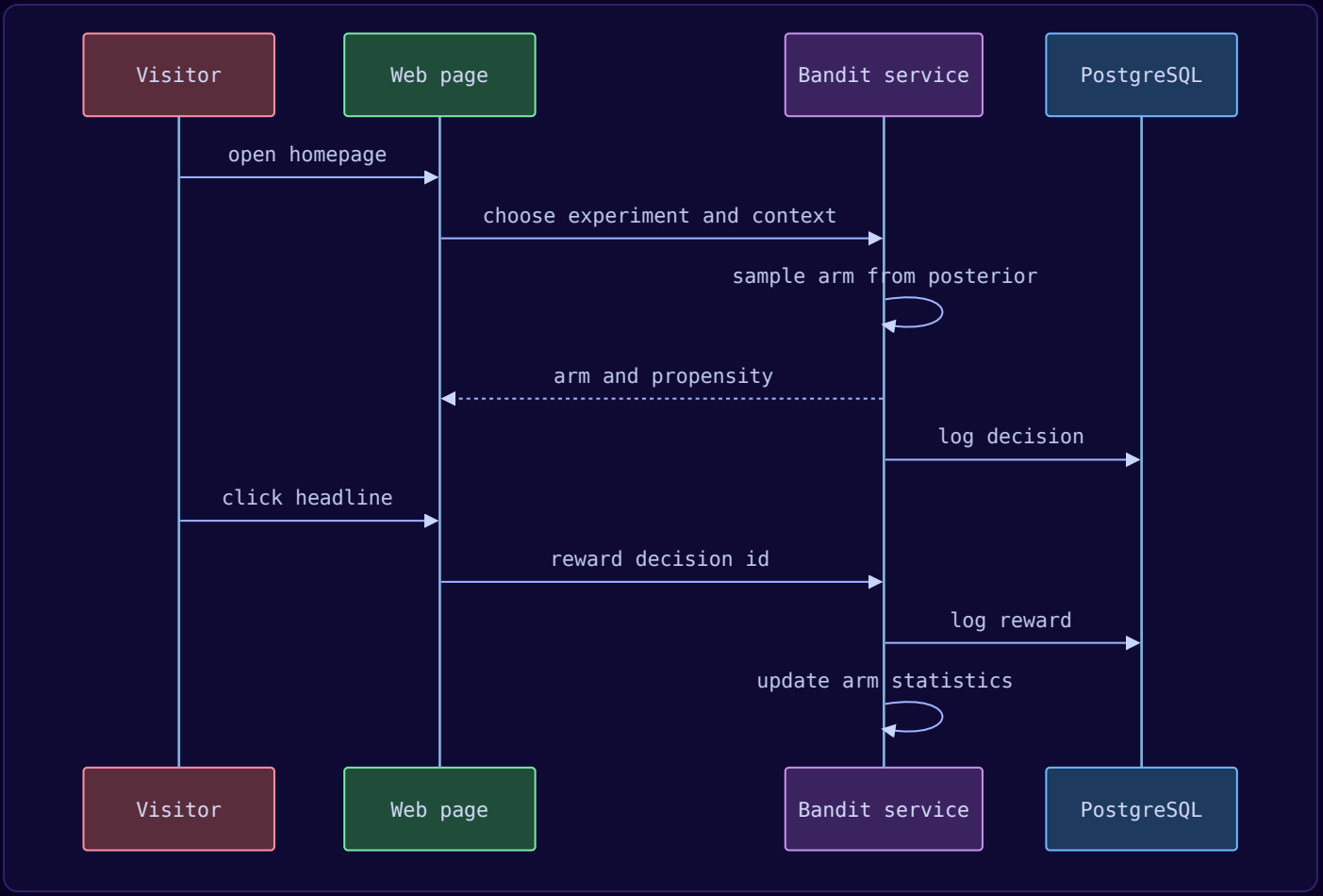
Functional

- Simulator: each headline has a hidden click probability (optionally depending on visitor features and time).
- Policies: uniform random (the A/B test), epsilon-greedy, UCB1, Thompson sampling (Beta-Bernoulli), LinUCB.
- Service: `GET /choose?experiment=…&context=…` returns an arm and a **propensity** (the probability the policy chose it); `POST /reward` records a click.
- **Off-policy evaluation**: estimate a new policy's click rate from logs of an old one using inverse propensity scoring (IPS).
- Dashboard of traffic share and cumulative regret.

Non-functional

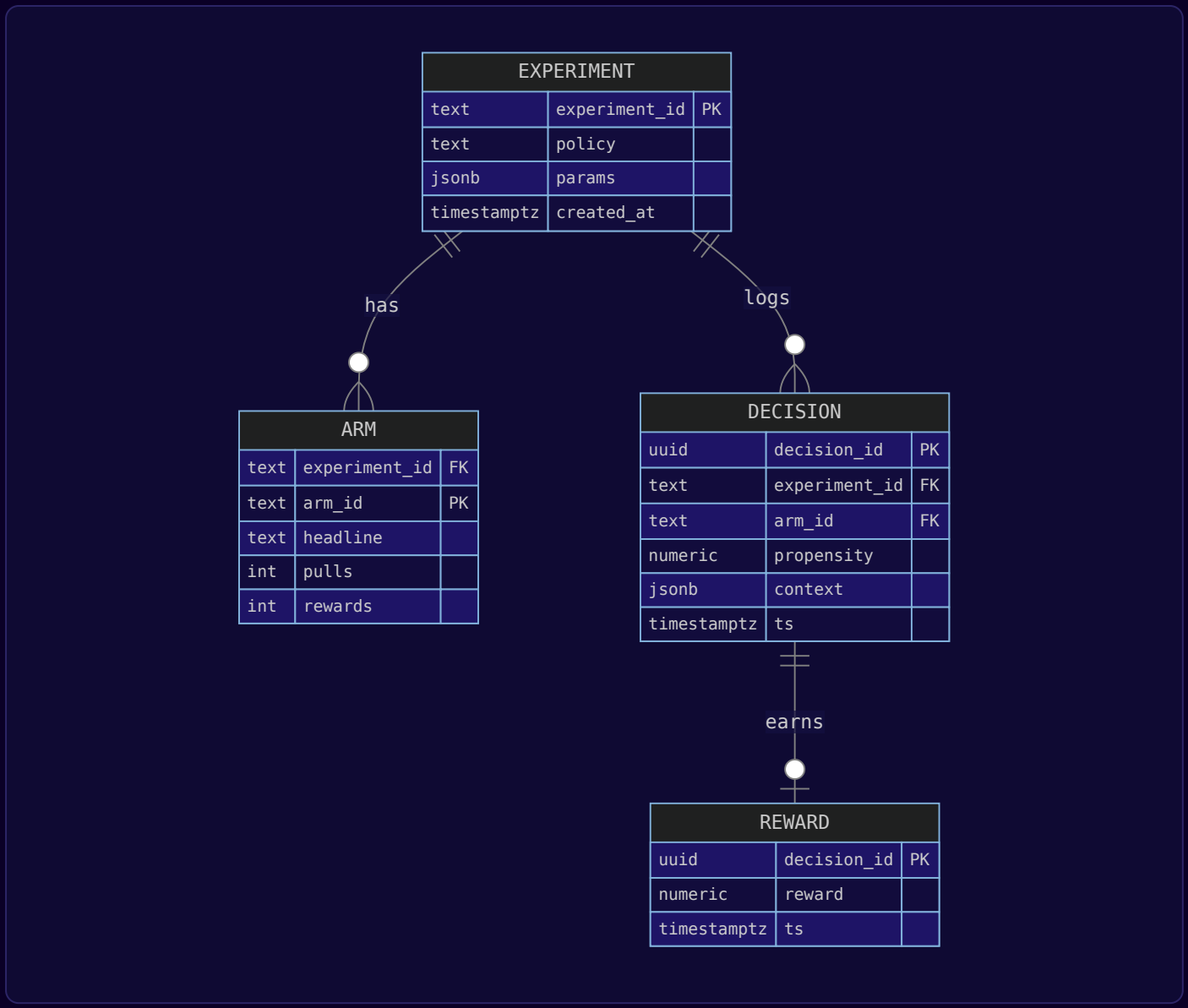
- `choose` under 5 ms p99.
- Handle 2,000 decisions per second per experiment on one instance.
- Rewards may arrive late or never; the service must not block.

The architecture



- **Choose**: reads in-memory arm statistics, applies the policy, returns the arm with its propensity.
- **Log decision**: every decision is stored with context and propensity — essential for off-policy evaluation later.
- **Reward**: joined to the decision by ID; clicks after a timeout window count as no click.
- **Update**: Beta posterior counts for Thompson, or LinUCB matrices per arm.

Data model



Tech stack

Component	Choice	Why
Service	FastAPI	Low-latency JSON
State	in-memory + periodic snapshot to PostgreSQL	Fast reads, durable logs
Math	numpy	Beta sampling, LinUCB matrix updates
Simulator	Python	Reproducible experiments
Evaluation	your own IPS code	Learn the mechanics

Milestones

1. **Simulator and policies.** *Done when* cumulative regret curves for all non-contextual policies over 100 simulated runs are plotted, with Thompson and UCB clearly below uniform random.
2. **Service.** *Done when* choose and reward endpoints work and decisions are logged with propensities.
3. **Delayed rewards.** *Done when* simulated clicks arriving 0–10 minutes later are handled, and you show the effect of delay on regret.
4. **Contextual bandit.** *Done when* LinUCB beats Thompson in a simulator where different visitor segments prefer different headlines.

5. **Off-policy evaluation.** *Done when* IPS estimates of a new policy from uniform-random logs match its true simulated click rate within the confidence band.

Deep dive: the hard parts

- **Exploration strategies.** Epsilon-greedy explores blindly at a fixed rate. UCB adds an optimism bonus that shrinks with more data. Thompson sampling draws from each arm's posterior and picks the best draw, naturally exploring uncertain arms. Thompson is simple and robust in practice.
- **Propensities make logs reusable.** If you log the probability of each choice, you can later evaluate any other policy by reweighting rewards (IPS). Deterministic policies have propensity 0 or 1 for unchosen arms — so keep some randomness if you want to evaluate alternatives.
- **Non-stationarity.** Click rates change through the day and as news ages. Use discounted counts or sliding windows so old evidence fades.
- **Bandit vs A/B test.** Bandits minimise regret during the test; A/B tests give clean, unbiased effect estimates for long-term decisions. Use bandits for short-lived, many-option choices; A/B tests for launches.

Testing strategy

- Statistical tests in the simulator: with one clearly best arm, its traffic share exceeds 80% after N decisions in at least 95% of runs.
- Propensity tests: for each policy, reported propensities sum to 1 over arms.
- Idempotency: posting the same reward twice does not double count.
- Load test: 2,000 choose calls per second within latency.

Trade-offs, scaling and failure modes

- Fast convergence can lock in an early lucky arm; keep a minimum exploration floor.
- Clicks reward clickbait. Add guardrails such as dwell time or bounce rate, or use them as the reward.
- In multi-instance deployments, arm statistics must be shared (database or cache) or periodically merged.

Stretch goals

- Replay Yahoo-style logged data offline using the replay method from the LinUCB paper.
- Doubly robust off-policy estimation (combine a reward model with IPS).
- Use dwell time as a continuous reward with a Gaussian posterior.

Skills practised

- Ch 14 — Reinforcement Learning: bandits, exploration vs exploitation, UCB, Thompson sampling.
- Ch 16 — ML in Production: online experiments, logging, delayed feedback.
- Ch 2 — Math for ML: Beta distributions, Bayesian updating.

In the interview

Q: When would you use a bandit instead of an A/B test? **A:** When options are short-lived or numerous and you care about reward during the test more than a precise effect estimate — headlines, thumbnails, promotions.

Q: How does Thompson sampling work? **A:** Keep a posterior over each arm's reward rate, sample one value per arm, show the arm with the highest sample, update with the outcome. Uncertain arms sometimes sample high, so they get explored.

Q: Why log propensities? **A:** So you can evaluate new policies offline with inverse propensity scoring, correcting for the old policy's choices.

TL;DR

A bandit trades a clean experiment for less regret; log propensities so your traffic remains useful evidence, and watch for non-stationarity and clickbait.

21. Predictive maintenance (remaining useful life)

Level: Intermediate · **Time:** 2–3 weeks · **Prerequisites:** Ch 7, 12

What it is

A model that reads multivariate sensor histories from jet engines and predicts **remaining useful life** (RUL): how many more operating cycles each engine can run before failure. You compare gradient boosting on hand-crafted window features with a sequence model (an LSTM or a 1-D CNN over the last N cycles), and turn predictions into a maintenance schedule.

The problem it solves

Maintaining machines too early wastes parts and downtime; too late causes failures, which can be dangerous and expensive. Predicting RUL lets operators plan maintenance "just in time". The same pattern applies to wind turbines, trains, pumps and data-centre disks.

Real-world examples

- NASA's C-MAPSS jet engine simulated data contains four sub-datasets (FD001–FD004) of run-to-failure trajectories with 3 operational settings and 21 sensor channels per cycle, with test trajectories cut off before failure.
- The LightGBM documentation covers the gradient-boosting library used for the feature-based model.

Requirements

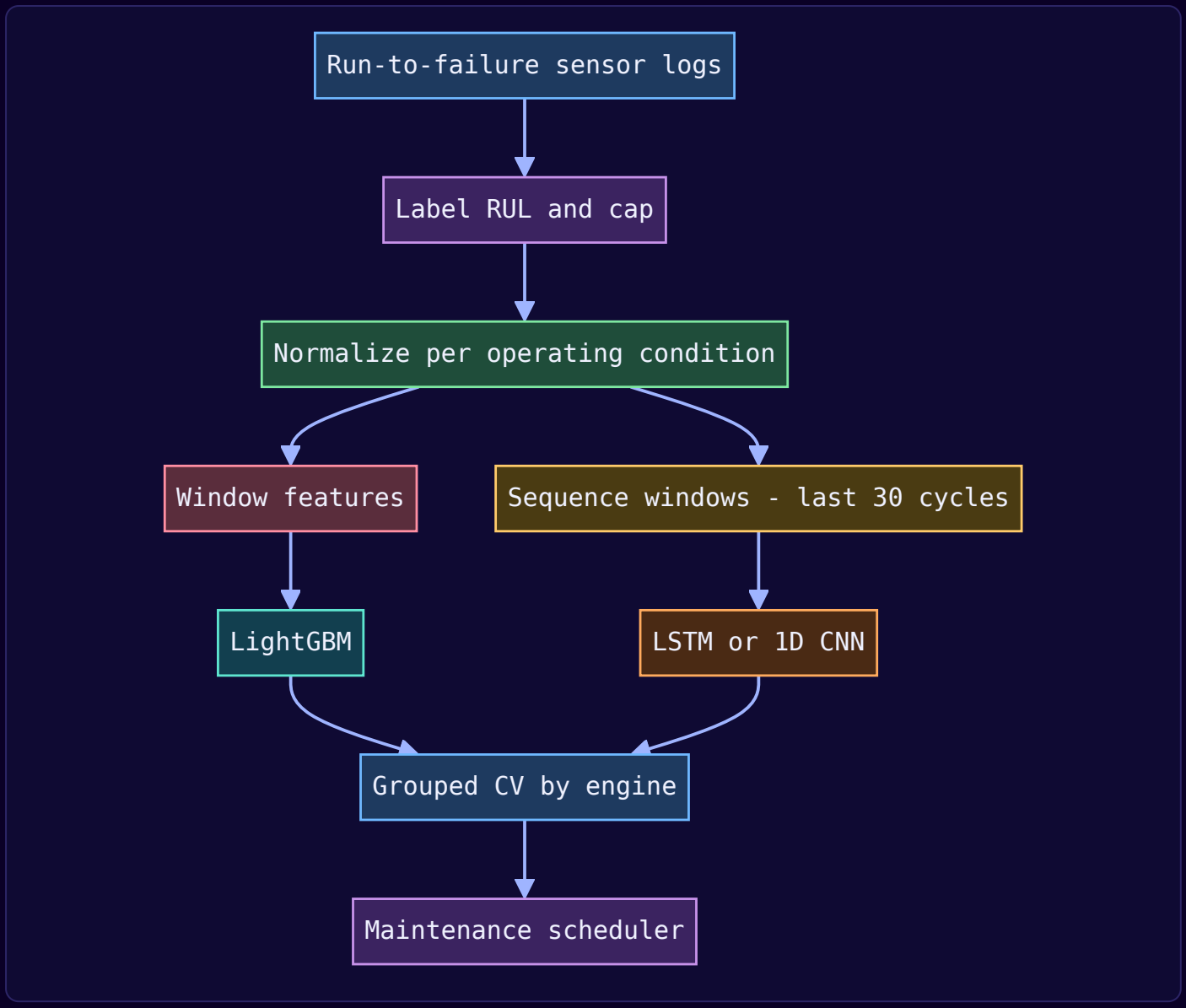
Functional

- Build RUL labels for training engines (cycles until their final cycle), capped at a ceiling (for example 125) because early life is uninformative.
- Features: rolling means, slopes and standard deviations of each sensor over several windows; normalisation per operating condition for FD002/FD004.
- Models: linear baseline, LightGBM, LSTM or 1-D CNN on windows of the last 30 cycles.
- Metrics: RMSE and the asymmetric score used with C-MAPSS that penalises late predictions (predicting more life than remains) more than early ones.
- A scheduler that flags engines whose predicted RUL falls below a maintenance lead time.

Non-functional

- Validation split by **engine**, never by row.
- Training for all models under 30 minutes on CPU.
- Predictions with an uncertainty estimate (quantiles or an ensemble spread).

The architecture



- **Cap RUL**: early in life all engines look healthy, so labels like "250 cycles left" are impossible to learn; a piecewise-linear cap focuses the model on degradation.
- **Normalise per operating condition**: in FD002/FD004, sensors jump with altitude and throttle settings; cluster the settings and standardise within each cluster.
- **Grouped CV**: all rows of one engine are in the same fold.
- **Scheduler**: converts RUL and its lower quantile into "service within N cycles".

Data model

Item	Type	Notes
`unit`	int	Engine ID
`cycle`	int	Time step
`setting_1..3`	float	Operating condition
`s1..s21`	float	Sensors; several are constant and can be dropped
`rul`	int (train only)	Max cycle − current cycle, capped
Prediction	float + quantiles	Per engine at its last observed cycle

Tech stack

Component	Choice	Why
Data	pandas or polars	Window aggregations
Feature model	LightGBM	Strong on engineered features
Sequence model	PyTorch LSTM / Conv1d	Learns temporal patterns directly
Tracking	MLflow	Compare feature vs sequence approaches

Milestones

1. **Explore.** *Done when* you plot sensor trajectories for a few engines and identify constant and informative sensors.
2. **Labels and baseline.** *Done when* capped RUL labels exist and a linear model is scored with engine-grouped CV.
3. **Feature model.** *Done when* LightGBM on window features beats the baseline on RMSE and the asymmetric score.
4. **Sequence model.** *Done when* an LSTM or 1-D CNN is trained on windows and compared fairly.
5. **Scheduling.** *Done when* a simple policy (service when the 10th-percentile RUL falls below lead time) is simulated on test engines, reporting failures avoided vs early maintenance waste.

Deep dive: the hard parts

- **Row leakage across engines.** Consecutive cycles of one engine are nearly identical. A random row split puts cycle 100 in training and cycle 101 in validation, so the model memorises engines. Always group by engine.
- **Asymmetric costs.** Predicting 20 cycles left when only 10 remain risks failure; predicting 10 when 20 remain only wastes a little. Train with a quantile loss or bias toward early predictions, and report the asymmetric score.
- **Operating regimes.** In multi-condition datasets, raw sensor values reflect flight conditions more than wear. Per-regime normalisation is often the single biggest gain.
- **Test trajectories are truncated.** The test set gives partial histories; predict RUL at the last available cycle. Short histories carry little degradation signal, so uncertainty should be larger.

Testing strategy

- Label test: RUL decreases by 1 each cycle and is 0 (or capped value logic) at the final training cycle.
- Split test: no engine appears in both train and validation within a fold.
- Window test: no window uses cycles after the prediction point.
- Quantile sanity: lower quantile ≤ median ≤ upper quantile.

Trade-offs, scaling and failure modes

- Hand-crafted features plus boosting are fast and interpretable; sequence models learn features but need more tuning and data.
- Simulated data is cleaner than reality, where sensors drift, fail and get recalibrated. Add missing-data handling before trusting it on real fleets.
- Real failures are rare, so real labels are scarce; censored data (machines maintained before failure) needs survival-analysis methods.

Stretch goals

- Survival model (for example a Weibull time-to-event model) treating maintained units as censored.
- Transformer encoder over cycle windows.
- An anomaly-based early-warning detector (Project 16) as a complement.

Skills practised

- Ch 12 — Sequences, RNNs and Attention: LSTMs, windowing, time-series features.
- Ch 7 — Decision Trees and Ensembles: gradient boosting on engineered features.
- Ch 6 — Evaluation and Tuning: grouped CV, asymmetric costs, quantiles.

In the interview

Q: Why cap RUL labels? **A:** Healthy engines look alike regardless of how much life remains, so large RUL values are unlearnable noise. Capping focuses the model on the degradation phase where predictions matter.

Q: How do you split the data? **A:** By machine. Rows from the same machine are highly correlated, so a row split leaks identity and inflates scores.

Q: How do you turn RUL into decisions? **A:** Use a conservative quantile against the maintenance lead time, and tune the policy by simulating the cost of failures vs early maintenance.

TL;DR

Group by engine, cap labels, normalise per regime, and use a conservative quantile — RUL prediction is about safe decisions, not just low RMSE.

22. Retraining pipeline with registry and batch scoring

Level: Intermediate · **Time:** 2–3 weeks · **Prerequisites:** Ch 16

What it is

An automated **training pipeline**: on a schedule (or when new data lands), it extracts a fresh training snapshot from PostgreSQL, validates the data, trains a candidate model, compares it with the current production model on the same holdout, registers it in MLflow if it wins, and runs a **batch scoring** job that writes predictions back to the database. The model can be any tabular one from earlier projects; Project 11's churn model is a good choice.

The problem it solves

Models decay as the world changes, and manual retraining is error-prone: someone forgets a preprocessing step, trains on a broken data extract, or deploys a worse model. A pipeline makes retraining repeatable, checked and auditable — the "Ops" in MLOps.

Real-world examples

- Sculley et al.'s Hidden Technical Debt in Machine Learning Systems (NeurIPS 2015) explains why glue code, pipeline jungles and unchecked data dependencies dominate real ML maintenance.
- The MLflow Model Registry documents versioning, aliases and lineage back to training runs.
- Uber's Michelangelo post describes a platform that standardised the manage-data, train, evaluate, deploy, predict and monitor workflow across teams.

Requirements

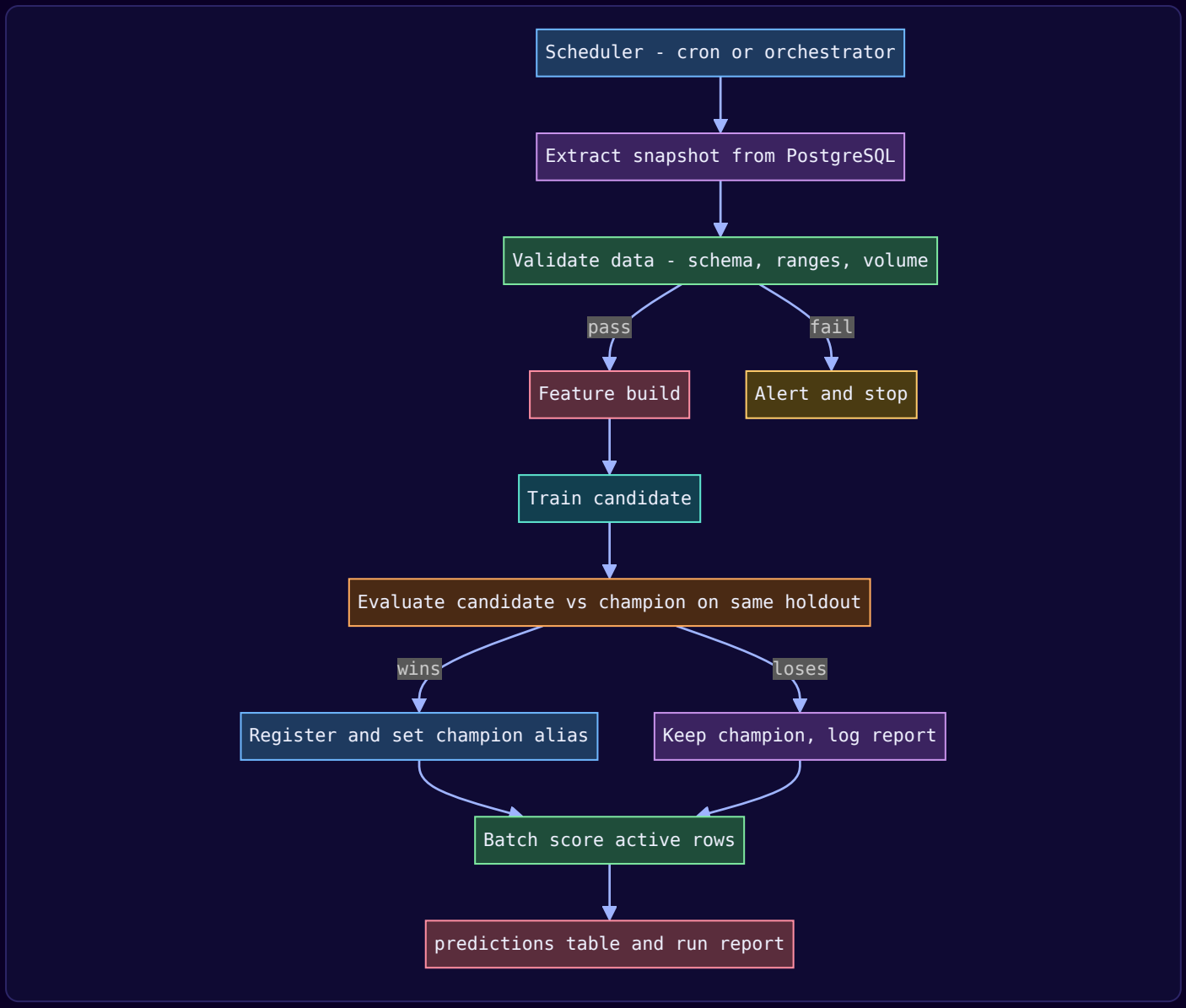
Functional

- Steps: extract snapshot → validate data → build features → train → evaluate vs champion → register → batch score → publish report.
- **Data validation**: schema, null rates, value ranges, row-count change vs last run; fail the pipeline on violation.
- **Promotion gate**: candidate must beat the champion on the primary metric by a margin and not regress on guardrail slices.
- Every artefact (data snapshot hash, code version, parameters, metrics) recorded.
- Rollback: one command restores the previous champion.

Non-functional

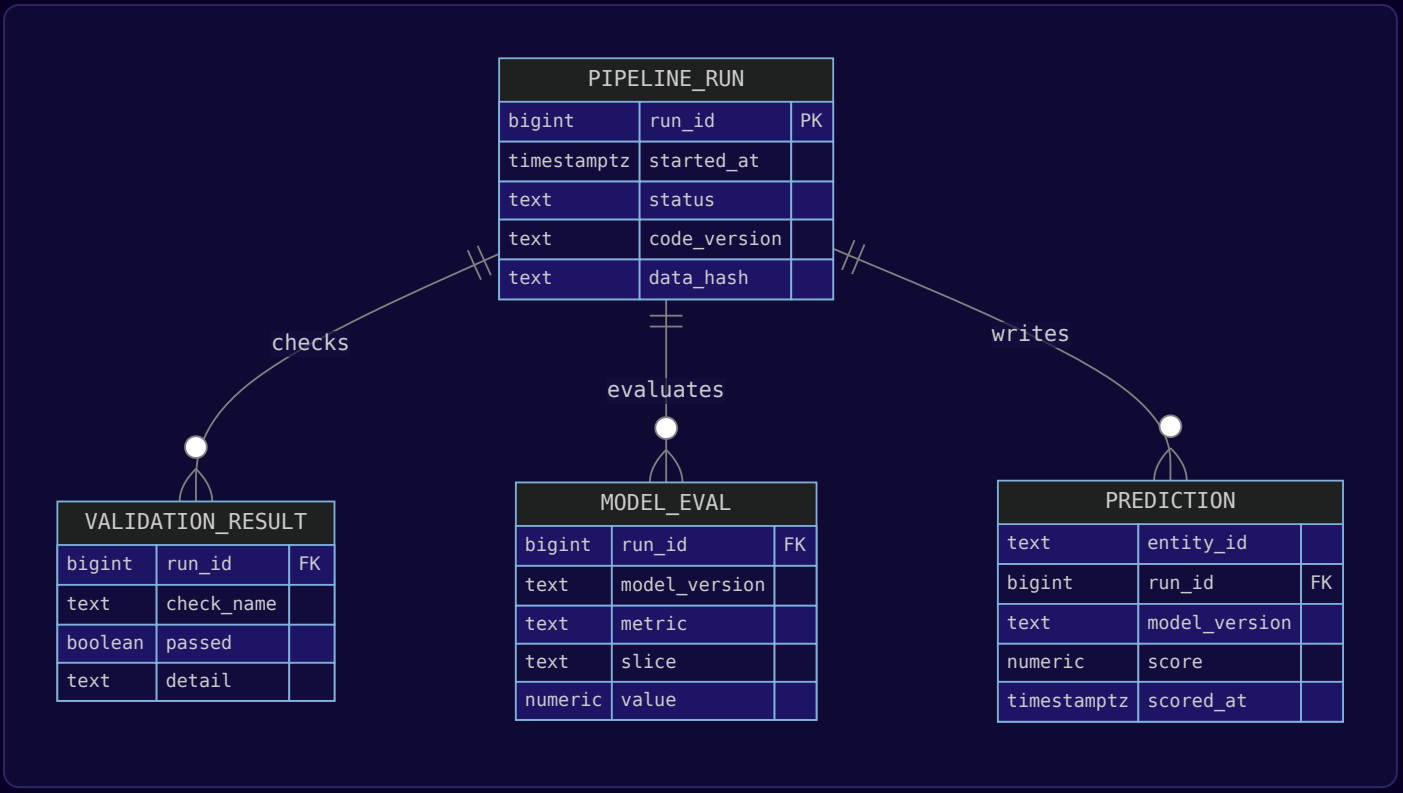
- Idempotent: re-running a step with the same inputs produces the same outputs.
- End-to-end run under 1 hour for 1 million rows.
- Failure notifications within 5 minutes.

The architecture



- **Scheduler**: cron is enough to start; an orchestrator (Airflow, Prefect, Dagster) adds retries, dependencies and a UI.
- **Validate**: catches the most common production failure — broken upstream data — before it trains a broken model.
- **Evaluate vs champion**: both models scored on the same, newest holdout, so the comparison is fair.
- **Batch score**: always uses whatever model holds the champion alias.

Data model



Tech stack

Component	Choice	Why
Orchestration	cron to start, then Prefect, Dagster or Airflow	Retries, lineage, UI
Data	PostgreSQL + polars	Source and fast transforms
Validation	your own checks, or a validation library	Fail fast on bad data
Registry	MLflow	Versions, aliases, lineage
Packaging	Docker	Same environment every run

Milestones

1. **Script the steps.** *Done when* each step is a function with explicit inputs and outputs, runnable alone.
2. **Validation.** *Done when* corrupting the input (nulls in a key column, a halved row count) fails the run with a clear message.
3. **Champion/challenger.** *Done when* a deliberately worse candidate is rejected and a better one is promoted automatically.
4. **Batch scoring.** *Done when* predictions are written with run ID and model version, and re-running does not duplicate rows.
5. **Orchestrate and alert.** *Done when* the pipeline runs on a schedule, retries a flaky step, and notifies on failure.

Deep dive: the hard parts

- **Fair comparison.** Comparing a new model's score on new data with the old model's score from months ago is meaningless. Re-score the champion on the same new holdout. Add slice checks so a better average does not hide a regression for one region or customer type.
- **Reproducibility.** Store a hash of the training snapshot (or the snapshot itself), the code commit and the environment. Without these, "why did the model change?" is unanswerable.
- **Idempotency.** Batch scoring should upsert by `(entity_id, run_id)` so a retried job does not double-write. Steps should check whether their output already exists.

- **Feedback loops.** If predictions drive actions that change outcomes (for example retention offers), the next training set is biased. Keep a random holdout population untouched by the model.

Testing strategy

- Unit tests for each validation check with good and bad fixtures.
- Integration test: the whole pipeline on a 1,000-row fixture database in CI, in under 2 minutes.
- Promotion gate test with synthetic champion and challenger metrics.
- Rollback drill: switch alias back and confirm the next batch uses the old model.

Trade-offs, scaling and failure modes

- Scheduled retraining is simple; trigger-based retraining (on drift or new data) saves compute but needs monitoring (Project 31).
- Automatic promotion is fast but risky; many teams require human approval for the final step at first.
- The most common failures are upstream: schema changes, late data, duplicated rows. Validation is your cheapest insurance.

Stretch goals

- Data drift report comparing the new snapshot with the training snapshot.
- Canary scoring: score 5% with the challenger first and compare distributions.
- Train on a cluster with Ray or Spark for larger data.

Skills practised

- Ch 16 — ML in Production: pipelines, registries, batch inference, retraining, monitoring.
- Ch 6 — Evaluation and Tuning: fair model comparison, slice metrics.
- Ch 15 — ML System Design Framework: the deployment and monitoring steps.

In the interview

Q: How do you decide whether to promote a retrained model? **A:** Score champion and challenger on the same recent holdout, require a meaningful improvement on the primary metric with no guardrail or slice regressions, then roll out gradually with monitoring.

Q: What should a training pipeline validate? **A:** Schema, nulls, ranges, category sets, row counts and label distribution versus the previous run, and freshness of the source tables.

Q: How do you make a model reproducible? **A:** Version the data snapshot, code, environment and parameters, and fix random seeds — all recorded with the registered model.

TL;DR

Automate extract, validate, train, compare against the champion, register and batch-score — with data checks first and reproducibility everywhere.

23. Uplift model for a marketing campaign

Level: Intermediate · **Time:** 2 weeks · **Prerequisites:** Ch 6, 7

What it is

An **uplift model** (also called a treatment-effect model) predicts how much a marketing action *changes* each customer's behaviour, not just whether they will buy. It estimates, per customer, P(buy | treated) − P(buy | not treated), using data from a randomised campaign. You build two-model (T-learner), class-transformation and tree-based approaches, and evaluate with **Qini curves** (how much extra conversion you get by targeting customers in order of predicted uplift).

The problem it solves

A response model targets people likely to buy — but many of them would have bought anyway, so the discount is wasted. Some customers even react badly to contact ("sleeping dogs"). Uplift modelling targets the **persuadables**: people who buy *because* of the campaign. It directly optimises campaign ROI.

Real-world examples

- The Criteo Uplift Prediction Dataset contains millions of users from incrementality tests with a treatment flag, 12 anonymised features, and visit and conversion labels.
- The scikit-uplift documentation describes meta-learners and uplift metrics such as Qini and uplift@k.

Requirements

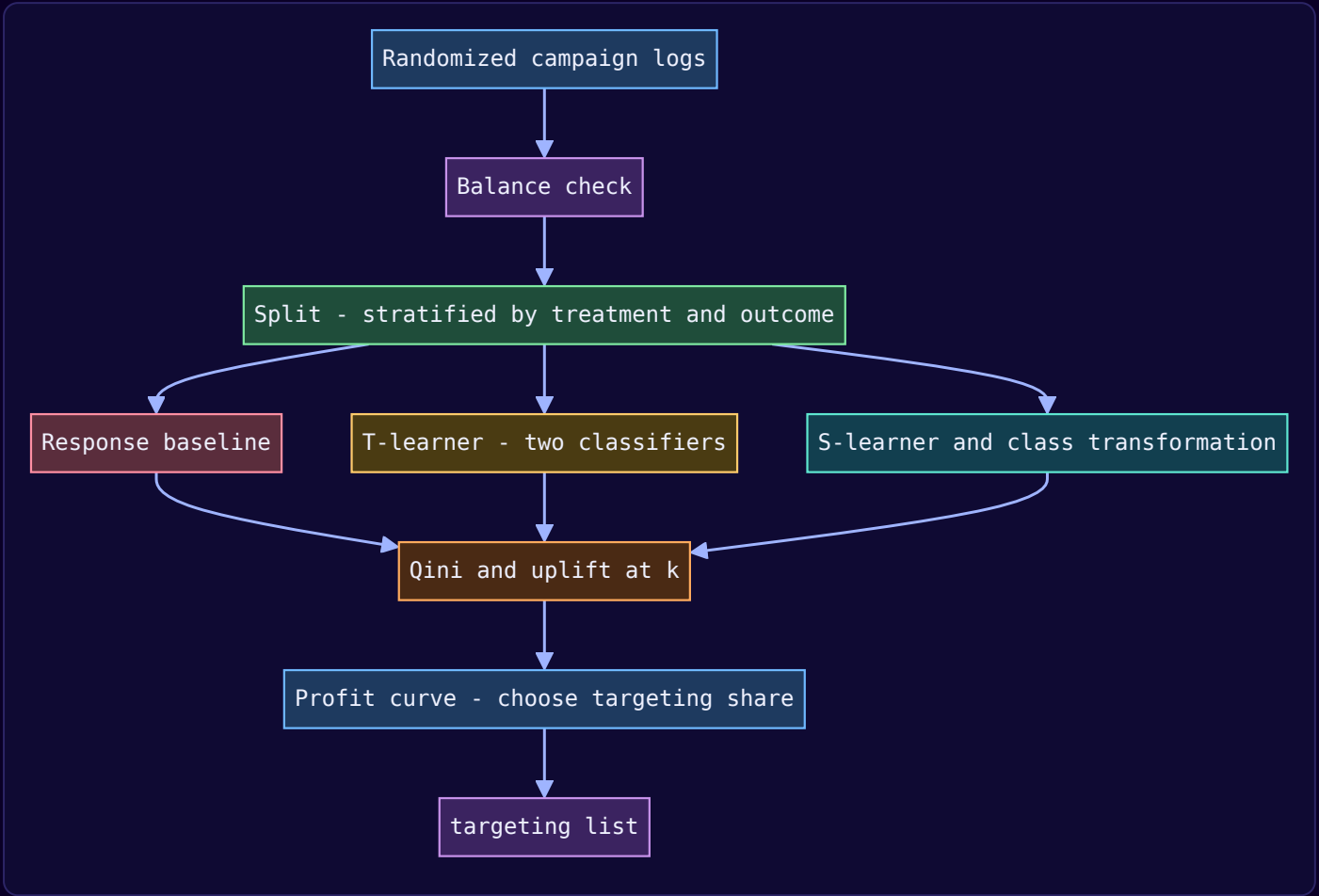
Functional

- Verify randomisation: treatment and control have similar feature distributions.
- Models: response-only baseline (ignores treatment), T-learner (two separate models), S-learner (one model with treatment as a feature), class transformation.
- Metrics: Qini curve and Qini coefficient, uplift at top 10% and 30%.
- Targeting policy: choose the share of customers to treat that maximises expected profit given offer cost and conversion value.

Non-functional

- Handle about 14 million rows (sample to 2 million for fast iteration).
- All evaluation on a held-out randomised test split.
- Report confidence intervals via bootstrap for top-k uplift.

The architecture



- **Balance check**: if treatment was not random, the whole method breaks.
- **T-learner**: model A learns conversion among treated, model B among control; uplift = A(x) − B(x).

- **S-learner**: one model with a treatment flag; uplift = f(x, 1) − f(x, 0).
- **Class transformation**: a trick that turns uplift into a single classification problem when treatment is 50/50 (reweight otherwise).
- **Profit curve**: expected incremental revenue minus cost as a function of how many top-uplift customers you treat.

Data model

Item	Type	Notes
`f0..f11`	float	Anonymised features
`treatment`	0/1	Randomly assigned, heavily treated share
`visit`, `conversion`	0/1	Outcomes; conversion is very rare
`exposure`	0/1	Whether the ad was actually shown
Predicted uplift	float	Can be negative

Tech stack

Component	Choice	Why
Data	polars	Millions of rows
Models	LightGBM / scikit-learn classifiers	Base learners for meta-learners
Uplift tools	scikit-uplift (if it supports Python 3.14), or your own Qini code	Metrics and baselines
Plots	matplotlib	Qini and profit curves

Milestones

1. **Explore and balance.** *Done when* treatment share, outcome rates per arm, and a standardised-difference table for features exist; the average treatment effect (ATE) is computed with a confidence interval.
2. **Response baseline.** *Done when* a plain conversion model's Qini curve is plotted — and you can explain why it is weak at finding uplift.
3. **Meta-learners.** *Done when* T- and S-learners and class transformation are evaluated on Qini and uplift@k.
4. **Policy.** *Done when* a profit curve picks the treatment share for an assumed offer cost and conversion value.
5. **Report.** *Done when* a one-page summary says whom to target, the expected incremental conversions and the uncertainty.

Deep dive: the hard parts

- **You never observe the label.** For any one customer you see only the treated *or* the untreated outcome. Uplift can only be evaluated on groups: rank customers by predicted uplift, then compare treated vs control conversion within each bucket. That is what the Qini curve does.
- **Rare outcomes and noise.** Conversion rates are tiny, so the difference between two tiny rates is very noisy. Use visit as an auxiliary outcome, large buckets, and bootstrap intervals.
- **Imbalanced treatment.** When most users are treated, the control model in a T-learner has fewer positives; class transformation needs reweighting by the propensity of treatment.
- **Negative uplift.** Some customers are less likely to convert when treated. A good policy *excludes* them, which a response model never would.

Testing strategy

- Synthetic data with planted heterogeneous effects: each method must recover the ranking of true uplift (Spearman correlation reported).
- Qini implementation test: a random model's Qini curve is close to the diagonal; an oracle's is above all others.
- Balance test fails on a deliberately non-random split.

- Reproducibility of bootstrap intervals with fixed seeds.

Trade-offs, scaling and failure modes

- Meta-learners are simple and reuse any classifier; specialised uplift trees split directly on treatment-effect differences but are less mature in tooling.
- Uplift models need randomised data; on observational data you must adjust for confounding, which is much harder.
- Uplift changes over time and with offer type; re-run randomised tests periodically to refresh labels.

Stretch goals

- X-learner and causal forests; compare with T/S-learners.
- Multi-treatment uplift (two offer sizes).
- Combine with Project 11: target churn-risk customers by uplift, not risk.

Skills practised

- Ch 6 — Evaluation and Tuning: ranking metrics, bootstrap intervals, cost-based thresholds.
- Ch 7 — Decision Trees and Ensembles: boosted base learners.
- Ch 2 — Math for ML: conditional probability, randomisation, confidence intervals.
- Ch 16 — ML in Production: experiments and their use for modelling.

In the interview

Q: Why not just target customers most likely to buy? **A:** Many would buy anyway, so the offer is wasted on them, and some react negatively. Uplift targets people whose behaviour the offer changes.

Q: How do you evaluate an uplift model? **A:** On randomised holdout data, rank by predicted uplift and compute the observed treated-minus-control difference cumulatively — the Qini or uplift curve — since individual effects are never observed.

Q: What is a T-learner? **A:** Two outcome models, one trained on treated and one on control users; their difference is the uplift estimate.

TL;DR

Uplift models estimate the effect of acting, not the chance of an outcome; they need randomised data and are judged by Qini curves and profit, never by accuracy.

24. Spoken keyword spotter

Level: Intermediate · **Time:** 2–3 weeks · **Prerequisites:** Ch 11, 12

What it is

A small neural network that listens to one-second audio clips and recognises a fixed vocabulary of spoken commands ("yes", "no", "up", "down", "stop", "go" …) or says "unknown" or "silence". Audio is converted to a **log-mel spectrogram** (an image-like grid of frequency energy over time, on a scale similar to human hearing), then classified by a compact CNN. The goal is high accuracy in a model small enough for a phone or microcontroller.

The problem it solves

Voice assistants listen for a wake word on-device, constantly, without sending audio to the cloud — for privacy, battery life and latency. That requires a tiny, always-on model that rarely triggers falsely. Keyword spotting is the standard teaching problem for this kind of **edge ML**.

Real-world examples

- Warden's Speech Commands dataset (Google) provides one-second utterances of short words, designed specifically for keyword spotting; version 2 has 105,829 clips of 35 words from 2,618 speakers.

- torchaudio provides loaders and spectrogram transforms used in this project.

Requirements

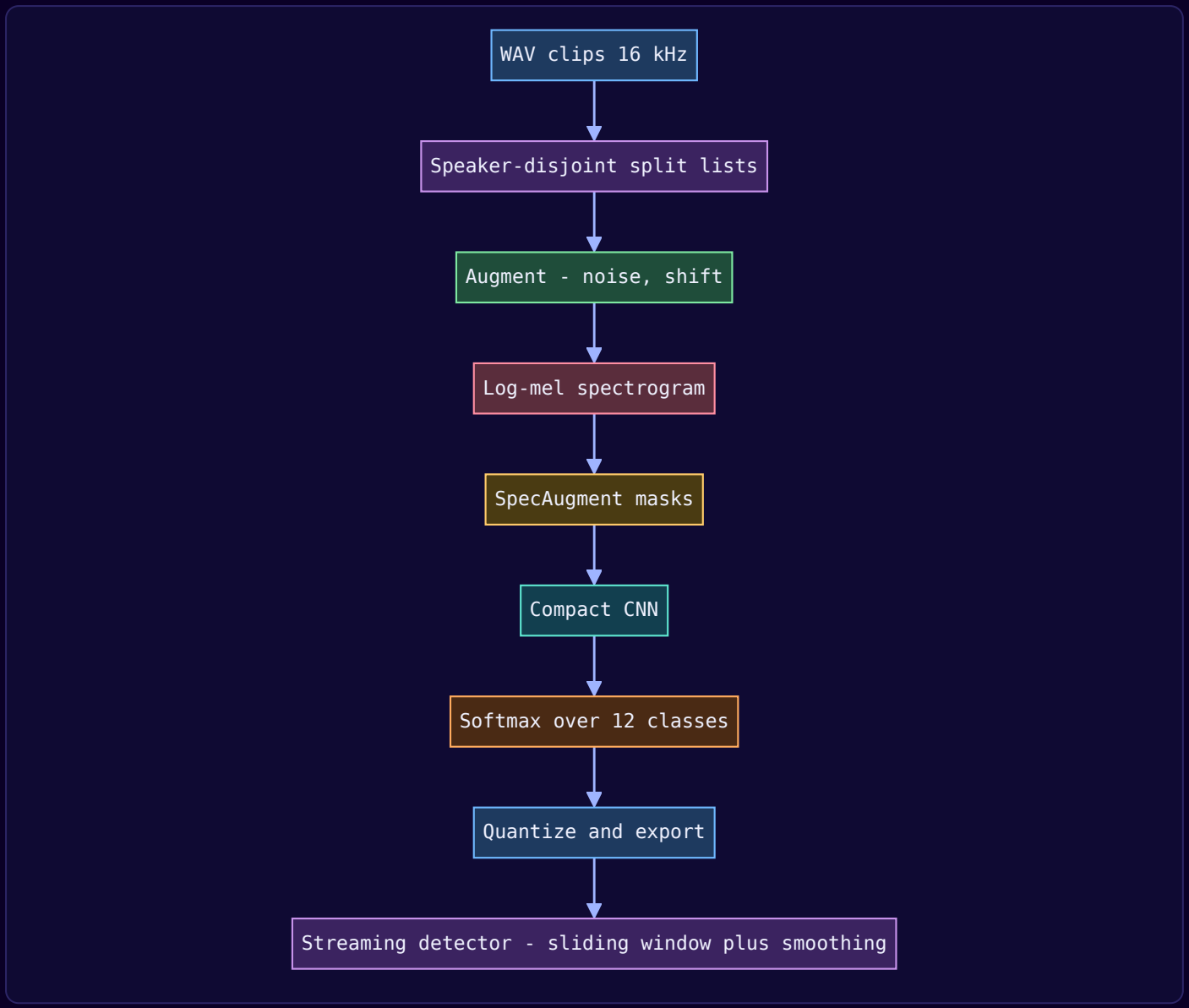
Functional

- Use the official train/validation/test split lists (they keep each speaker in one split).
- Features: log-mel spectrograms (for example 40 mel bands, 25 ms windows, 10 ms hop).
- Augmentation: background noise mixing, time shift, SpecAugment-style masking (blank out random time and frequency bands).
- Model: a small CNN (depthwise-separable convolutions to cut parameters).
- Classes: 10 keywords + unknown + silence.
- Streaming demo: run on a microphone stream with a sliding window and smoothing.

Non-functional

- Model under 500,000 parameters and under 2 MB after int8 quantisation.
- Test accuracy above 90% on the 12-class task.
- Streaming inference under 20 ms per window on a laptop CPU.
- False triggers under 1 per hour of background audio in the demo (measured on noise recordings).

The architecture



- **Speaker-disjoint split**: the same person never appears in both train and test, so the model must generalise to new voices.

- **Log-mel**: compresses the waveform into a 40 × 98 grid for a one-second clip.
- **Compact CNN**: depthwise-separable convolutions split a standard convolution into cheaper per-channel and 1 × 1 parts.
- **Streaming detector**: runs every 100 ms over the last second, averages recent probabilities, and requires a confidence threshold to trigger.

Data model

Item	Type	Notes
Audio clip	int16 PCM, 16,000 samples	Pad or trim to 1 s
Label	12 classes	10 keywords, unknown (other words), silence (noise clips)
Speaker ID	hash in file name	For splitting
Spectrogram	float `[1, 40, 98]`	Normalised per dataset
Model	`.pt` and quantised export	Plus class list

Tech stack

Component	Choice	Why
Audio	torchaudio	Loading, mel transforms
Model	PyTorch	Small CNN, quantisation tools
Export	ONNX or TorchScript	Run without training code
Demo	`sounddevice` or browser audio + FastAPI	Live microphone

Milestones

1. **Data pipeline.** *Done when* you can play a clip, view its spectrogram, and class counts per split match the official lists.
2. **Baseline CNN.** *Done when* a simple CNN reaches a recorded test accuracy.
3. **Augmentation.** *Done when* noise mixing and SpecAugment improve test accuracy or robustness on a noisy test copy.
4. **Make it small.** *Done when* a depthwise-separable model meets the parameter budget and int8 quantisation costs less than 1 accuracy point.
5. **Streaming demo.** *Done when* saying keywords to a microphone triggers correctly and an hour of background audio produces few false triggers.

Deep dive: the hard parts

- **Unknown and silence classes.** Real-world audio is mostly not your keywords. Without strong unknown and silence classes, the model triggers on everything. Balance these classes deliberately and measure false triggers per hour, not just accuracy.
- **Speaker leakage.** Random splits put the same voice in train and test. Use the dataset's validation and testing lists, which are hashed by speaker.
- **Streaming vs clips.** Training clips are centred; in streaming, a word may fall across window boundaries. Time-shift augmentation and posterior smoothing over several windows handle this.
- **Quantisation.** Converting weights and activations to 8-bit integers shrinks the model about 4× and speeds it up, but needs calibration data and can hurt small models. Quantisation-aware training helps if post-training quantisation loses too much.

Testing strategy

- Feature parity: spectrograms computed in training and in the streaming demo match for the same clip.
- Split test: no speaker hash in more than one split.
- Robustness test set: test clips mixed with noise at several signal-to-noise ratios; plot accuracy vs SNR.

- Exported model parity with PyTorch outputs.

Trade-offs, scaling and failure modes

- Bigger models gain accuracy but drain battery on always-on hardware. Two-stage designs use a tiny always-on detector and a bigger verifier.
- Accents, children's voices and far-field microphones are under-represented; collect slice test sets.
- Always-listening devices raise privacy concerns; keep processing on device and be transparent about it.

Stretch goals

- Custom wake word with few-shot fine-tuning from a handful of your own recordings.
- Train a keyword-spotting transformer or attention-pooled CRNN and compare size vs accuracy.
- Deploy to a microcontroller toolchain.

Skills practised

- Ch 11 — CNNs and Computer Vision: CNNs on 2-D inputs, augmentation, efficient architectures.
- Ch 12 — Sequences, RNNs and Attention: sequential signals and streaming inference.
- Ch 16 — ML in Production: model compression and edge deployment.

In the interview

Q: Why log-mel spectrograms? **A:** They capture how energy is distributed across frequencies over time, on a perceptual scale and with compressed dynamic range, turning audio into an image-like input that CNNs handle well.

Q: How do you measure a wake-word detector? **A:** False rejection rate on keywords and false acceptances per hour on background audio, at the operating threshold — not just clip accuracy.

Q: How do you shrink a model for a device? **A:** Efficient architectures (depthwise-separable convolutions), pruning, quantisation, and distillation from a bigger teacher.

TL;DR

Spectrogram plus compact CNN, speaker-disjoint splits, strong unknown/silence classes and quantisation — measured in false triggers per hour, not just accuracy.

25. Toxic-comment classifier with bias evaluation

Level: Intermediate · **Time:** 2–3 weeks · **Prerequisites:** Ch 12, 19

What it is

A **multi-label** text classifier (each comment can have several labels at once) that scores online comments for toxicity, insults, threats, obscenity and identity attacks. You build a TF-IDF baseline and a fine-tuned transformer, then audit the model for **unintended bias**: whether innocent comments that mention identity groups ("I am a gay man") get unfairly high toxicity scores.

The problem it solves

Platforms need help moderating huge comment volumes. But a toxicity model trained on real data can learn that identity terms co-occur with abuse and then penalise the very communities it should protect. This project teaches you to measure that harm with per-subgroup metrics, not just overall AUC.

Real-world examples

- Kaggle's Toxic Comment Classification Challenge (Jigsaw) provides Wikipedia talk-page comments with six labels: toxic, severe toxic, obscene, threat, insult, identity hate.
- The follow-up Jigsaw Unintended Bias in Toxicity Classification competition added identity annotations and a bias-aware metric that blends overall AUC with subgroup, BPSN and BNSP AUCs.

Requirements

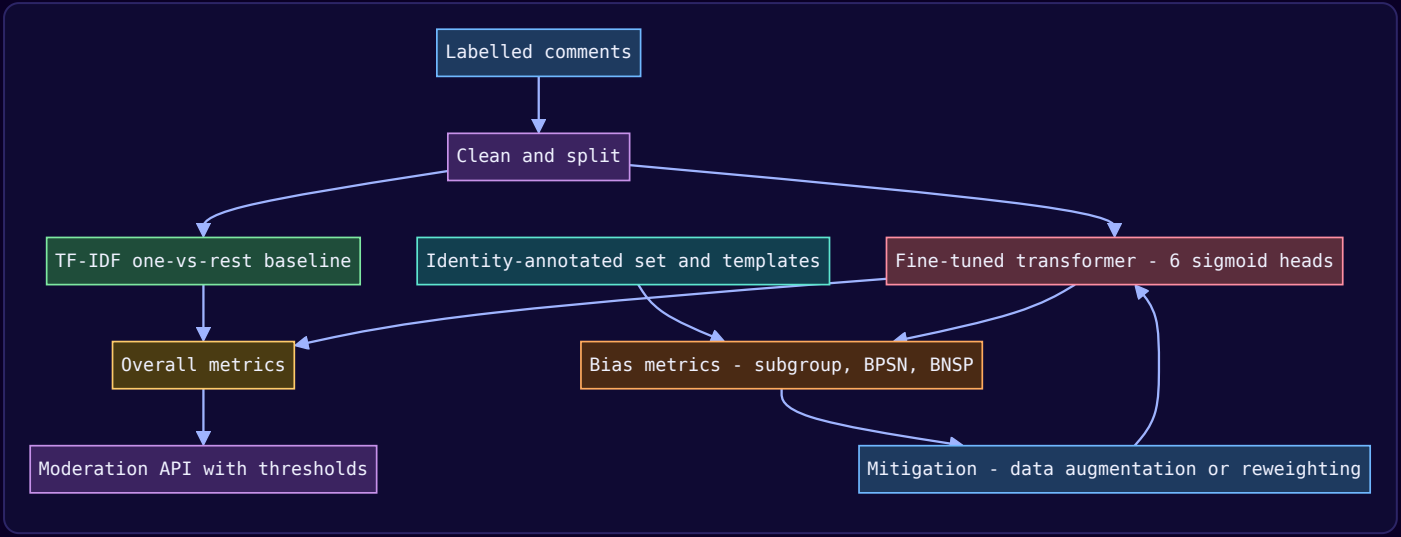
Functional

- Multi-label models: one-vs-rest logistic regression on TF-IDF (word + character n-grams), and a fine-tuned transformer with six sigmoid outputs.
- Metrics: mean column-wise ROC AUC; per-label PR AUC.
- Bias audit on identity-annotated data: **subgroup AUC** (AUC within comments mentioning a group), **BPSN** (background positive, subgroup negative: can the model separate toxic comments *without* the identity from non-toxic ones *with* it?) and **BNSP** (the reverse).
- A synthetic template test set ("I am a [identity] person") with all non-toxic labels.
- Mitigation experiment and before/after report.

Non-functional

- Transformer inference under 50 ms per comment on GPU (document CPU latency).
- Audit report reproducible from a single script.
- Thresholds chosen per moderation use (auto-remove vs send to human review).

The architecture



- **Six sigmoid heads**: independent probabilities per label, trained with binary cross-entropy, since labels overlap.
- **Bias metrics**: computed per identity subgroup; the worst subgroup matters more than the average.
- **Mitigation**: add non-toxic examples mentioning identities, reweight training examples, or both — then re-audit.
- **Moderation API**: returns per-label scores; the calling product decides actions.

Data model

Item	Type	Notes
`comment_text`	string	Wikipedia talk-page comments
Six labels	0/1 each	Heavily imbalanced, overlapping
Identity columns (bias set)	fraction of annotators	e.g. mentions of gender, religion, sexuality, race
Template set	generated text	All labelled non-toxic
Scores	float × 6	Sigmoid outputs

Tech stack

Component	Choice	Why
Baseline	scikit-learn TF-IDF + logistic regression	Strong and explainable
Transformer	Hugging Face Transformers on PyTorch	Fine-tuning
Fairness metrics	your own subgroup AUC code (Fairlearn for group metrics)	Learn the definitions
API	FastAPI	Score endpoint

Milestones

1. **Baseline.** *Done when* mean column-wise ROC AUC is reported and the top n-grams for each label are listed.
2. **Transformer.** *Done when* a fine-tuned model beats the baseline, with per-label PR AUC (threats and identity hate are rare and hard).
3. **Bias audit.** *Done when* subgroup AUC, BPSN and BNSP are tabulated per identity and the template test shows score distributions per identity term.
4. **Mitigation.** *Done when* one mitigation is applied and the worst-subgroup metric improves without a large drop in overall AUC.
5. **Policy and API.** *Done when* thresholds for "auto-hide" and "send to review" are chosen from precision targets and documented.

Deep dive: the hard parts

- **Where the bias comes from.** In the training data, identity terms appear disproportionately in toxic comments, so the model learns the shortcut "identity word → toxic". Low BPSN AUC for a subgroup is the signature: non-toxic comments mentioning that group score higher than toxic ones that do not.
- **Multi-label imbalance.** Severe toxic and threat are very rare. Report PR AUC per label; consider per-label thresholds and class weights.
- **Label subjectivity.** Annotators disagree. Labels are fractions in the bias dataset; treat them as soft targets or threshold at 0.5, and accept a ceiling on achievable accuracy.
- **Adversarial spelling.** Users write "f*ck" or insert spaces. Character n-grams and subword tokenisers help; still test with obfuscated variants.

Testing strategy

- Template tests: average toxicity on "I am a [X] person" stays below a threshold for every identity.
- Invariance tests: swapping one identity term for another changes scores by less than a tolerance.
- Directional tests: adding an explicit insult raises the insult score.
- Regression suite of known false positives from earlier audits.

Trade-offs, scaling and failure modes

- Removing identity terms from inputs hides the symptom but also blinds the model to real identity attacks. Fix data, not inputs.
- Auto-removal with an imperfect model silences people; use human review for borderline scores.
- Language and dialect bias: comments in some dialects may be flagged more often. Audit by dialect if you can.

Stretch goals

- Multilingual toxicity with a multilingual transformer.
- Calibrate per label and show reliability diagrams per subgroup.
- An appeals simulator: estimate how many wrongful removals per day per group at your thresholds.

Skills practised

- Ch 12 — Sequences, RNNs and Attention: transformers, tokenisation, fine-tuning.
- Ch 19 — Responsible ML: bias sources, subgroup metrics, mitigation.
- Ch 18 — ML Case Studies II: harmful-content moderation design.

In the interview

Q: What is BPSN AUC? **A:** AUC on a mix of toxic comments that do not mention the identity and non-toxic comments that do. Low values mean the model confuses mentions of the identity with toxicity.

Q: How would you reduce identity bias? **A:** Add or up-weight non-toxic examples mentioning identities, balance training data, and re-audit with subgroup metrics; avoid simply masking identity words.

Q: Why is overall AUC not enough? **A:** Errors concentrated on small groups barely move the average but cause real harm. Report per-subgroup and worst-group metrics.

TL;DR

A toxicity model is only as good as its worst subgroup: measure subgroup, BPSN and BNSP AUC, mitigate with better data, and keep humans in the loop for borderline cases.

26. Two-tower retrieval and ranking recommender

Level: Advanced · **Time:** 4–6 weeks · **Prerequisites:** Ch 13, 15, 17

What it is

A production-style recommender with the industry-standard **funnel**: a **two-tower retrieval model** (one neural network embeds the user and their recent history, another embeds each item; the dot product scores relevance) narrows millions of items to a few hundred candidates via an **approximate nearest-neighbour** (ANN) index, then a heavier **ranking model** with rich features orders those candidates, and a **re-ranker** applies business rules (diversity, freshness, already-seen filtering). You build all three stages, an offline evaluation, and an online serving path.

The problem it solves

You cannot run a heavy model on every item for every request: millions of items × thousands of requests per second is far too much compute. The funnel splits the work — cheap and broad first, expensive and precise last — so you get both coverage and accuracy within a latency budget of a couple of hundred milliseconds.

Real-world examples

- Covington, Adams and Sargin's Deep Neural Networks for YouTube Recommendations (RecSys 2016) describes the candidate generation + ranking split, training on watch history and weighting by watch time.
- Yi et al.'s Sampling-Bias-Corrected Neural Modeling for Large Corpus Item Recommendations (RecSys 2019) explains in-batch negatives in two-tower models and how to correct their popularity bias.
- Meta's Scaling the Instagram Explore recommendations system describes two-tower retrieval feeding multi-stage ranking.

Requirements

Functional

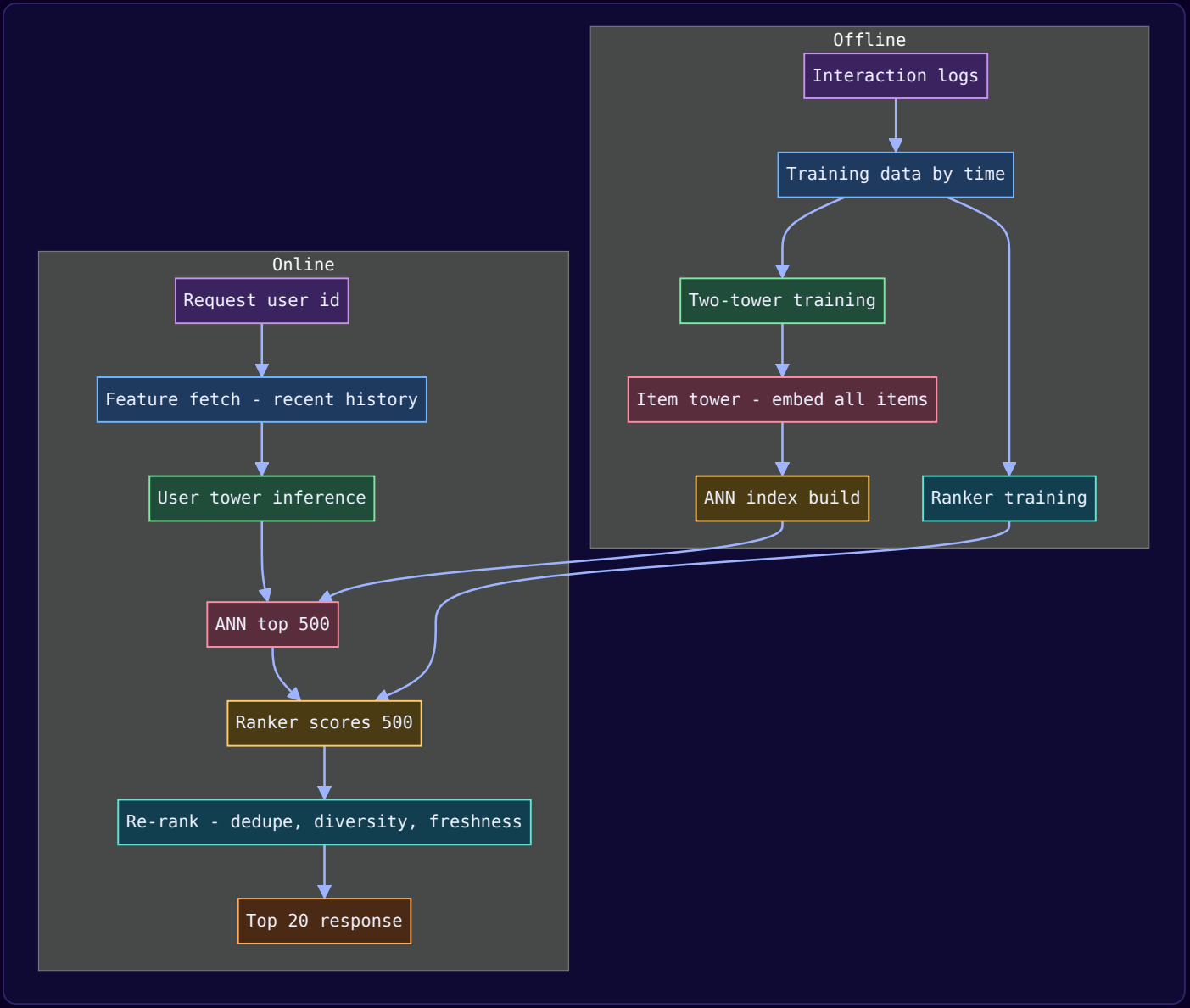
- Data: MovieLens 32M (or KuaiRec for a dense, video-style dataset), treated as implicit feedback, split by time.
- Retrieval: two-tower model with user features (history, recency-weighted genres) and item features (ID, genres, year, tag embeddings), trained with in-batch softmax and logQ correction.
- ANN index (FAISS or pgvector HNSW) over item embeddings; retrieve top 500.
- Ranking: gradient-boosted or deep model with user–item cross features, predicting "will watch and rate ≥ 4".
- Re-rank: remove seen items, cap items per genre, mix in fresh items.
- Offline metrics: Recall@500 for retrieval, NDCG@20 for the full pipeline; compare with popularity and MF (Project 14).
- `GET /feed/{user_id}` returning 20 items with stage-by-stage debug info.

Non-functional

- End-to-end latency under 150 ms p95 (retrieval under 20 ms, ranking 500 candidates under 80 ms).

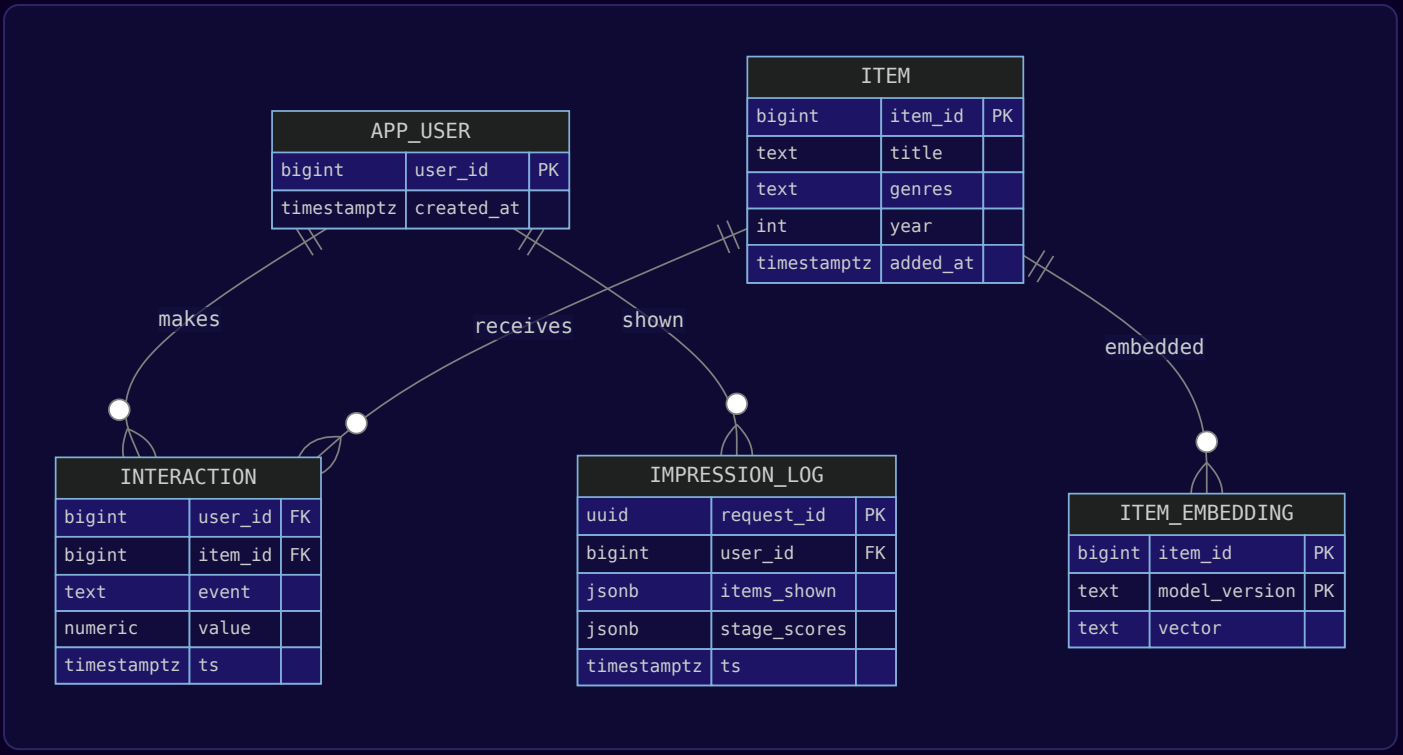
- Item embeddings refreshed daily; new items indexed within an hour.
- Serve 200 requests per second on one machine in a load test.

The architecture



- **Two-tower training**: positives are (user context, next watched item); negatives are other items in the same batch. The logQ correction subtracts the log of each item's sampling probability so popular items are not over-penalised.
- **Item tower offline**: item embeddings depend only on item features, so they are pre-computed and indexed.
- **User tower online**: runs per request on the latest history, so recommendations react to what the user just did.
- **Ranker**: sees features the towers cannot combine, such as "how many movies by this director has the user watched".
- **Re-rank**: deterministic rules that encode product policy.

Data model



Tech stack

Component	Choice	Why
Models	PyTorch	Two towers, custom losses
Ranker	LightGBM (lambdarank or binary)	Strong, fast on 500 candidates
ANN	FAISS or pgvector HNSW	Millisecond top-k
Features	PostgreSQL + in-memory cache	Recent history lookup
Serving	FastAPI	One endpoint, stage timings
Tracking	MLflow	Versions for both models and the index

Milestones

1. **Data and baselines.** *Done when* a time-based split exists and popularity and MF recall/NDCG numbers are recorded.
2. **Two-tower retrieval.** *Done when* Recall@500 beats MF retrieval and you have measured the effect of logQ correction.
3. **ANN index.** *Done when* ANN recall vs exact search is above 95% at under 20 ms.
4. **Ranker.** *Done when* NDCG@20 of retrieval + ranking beats retrieval alone.
5. **Serving and re-rank.** *Done when* the endpoint returns 20 items within latency, logs impressions with stage scores, and re-rank rules are unit-tested.
6. **Load test and freshness.** *Done when* 200 RPS holds p95 under 150 ms and a newly added item appears in the index within an hour.

Deep dive: the hard parts

- **In-batch negatives and popularity.** A frequently watched item turns up in many training batches, so it is drafted as a negative for other users again and again, and its scores get pushed down. The logQ correction compensates; without it, retrieval under-recommends popular items in a way that hurts online metrics.
- **Stage alignment.** Retrieval is trained on "watched", the ranker on "rated highly"; if objectives diverge too much, retrieval hands the ranker candidates it does not like. Measure the ranker's NDCG restricted to retrieved candidates, not on a random sample.

- **Training–serving consistency.** The user tower must see the same history window and feature definitions online as offline. Compute both from one feature function and log served features to train on.
- **Feedback loops and position bias.** Users click what you show and what is on top. Training on your own impressions reinforces the current model. Add exploration slots and position as a training-only feature (set to a constant at serving).

Testing strategy

- Offline evaluation harness with fixed splits so every model change is comparable.
- Contract tests between towers and index: embedding dimension, normalisation, model version must match.
- Re-rank rule tests (seen items never shown, genre cap respected).
- Load tests with realistic user mixes (heavy and new users).
- Shadow test: a new model scores live requests without serving, comparing distributions.

Trade-offs, scaling and failure modes

- Two-tower models cannot model fine user–item interactions (late interaction) — that is the ranker's job. More candidates help the ranker but cost latency.
- Index refresh vs freshness: rebuilding HNSW for millions of items is slow; add new items incrementally and rebuild nightly.
- If the ANN service or feature store fails, fall back to cached recommendations or popularity, never an empty page.
- Napkin math: 10 million items × 64 floats × 4 bytes is about 2.6 GB of vectors — fits in RAM on one machine, before index overhead.

Stretch goals

- Multiple retrieval sources (two-tower, co-watch item-to-item, trending) merged before ranking.
- Multi-task ranker (watch, rate highly, finish) with a weighted value formula (see Project 27).
- An A/B test plan with guardrails (Project 17).

Skills practised

- Ch 13 — Recommender Systems and Embeddings: two-tower retrieval, the funnel, ANN search, cold start.
- Ch 15 — ML System Design Framework: scoping, metrics, serving architecture.
- Ch 17 — ML Case Studies I: the video recommendation case study.
- Ch 16 — ML in Production: feature consistency, shadow testing, fallbacks.

In the interview

Q: Why separate retrieval and ranking? **A:** Compute budget: retrieval must scan millions of items, so it uses a model whose item side can be pre-computed and searched with ANN. Ranking sees only hundreds of items, so it can afford rich cross features and heavy models.

Q: How do you train a two-tower model without explicit negatives? **A:** Use in-batch negatives (other users' positives in the batch) with a softmax loss, plus a sampling-bias correction so popular items are not unfairly penalised; optionally add hard negatives.

Q: How do you handle new items? **A:** The item tower uses content features, so a new item gets an embedding immediately and can be added to the index; reserve some exploration traffic so it collects interactions.

Q: What offline metrics would you use per stage? **A:** Recall@K for retrieval (did the good items survive?), NDCG@K or MAP for ranking, plus coverage and diversity for the final list.

TL;DR

Retrieval (two towers + ANN) for recall, a feature-rich ranker for precision, rules for policy — each stage evaluated on its own job and served within one latency budget.

27. Social feed ranking with a multi-task model

Level: Advanced · **Time:** 4–6 weeks · **Prerequisites:** Ch 15, 17

What it is

A ranking system for a social news feed: from all recent posts by friends, groups and followed pages (the **candidate inventory**), it predicts several engagement probabilities for each post — like, comment, share, long dwell, hide — with one **multi-task neural network** (a shared body with one output head per task), combines them with a **value formula** into a single score, then applies integrity and diversity rules. You build it on a simulated social network, because real feed logs are not public.

The problem it solves

Sorting by time buries good posts from close friends under floods of low-value content. Optimising only for clicks rewards clickbait and outrage. A multi-task ranker lets product owners choose explicitly what a "good" feed means — weighting comments and shares between friends higher than passive likes, penalising predicted hides — and change that definition without retraining.

Real-world examples

- Meta's How machine learning powers Facebook's News Feed ranking algorithm describes the inventory, per-action predictions from multitask models, a linear combination into one value, and passes that apply diversity rules.
- Meta's Instagram Explore post describes a multi-task multi-label ranker and a weighted value model including negative actions such as "see less".
- X (Twitter) open-sourced its home timeline recommendation code as the-algorithm, showing candidate sources, a heavy ranker and heuristic filters.

Requirements

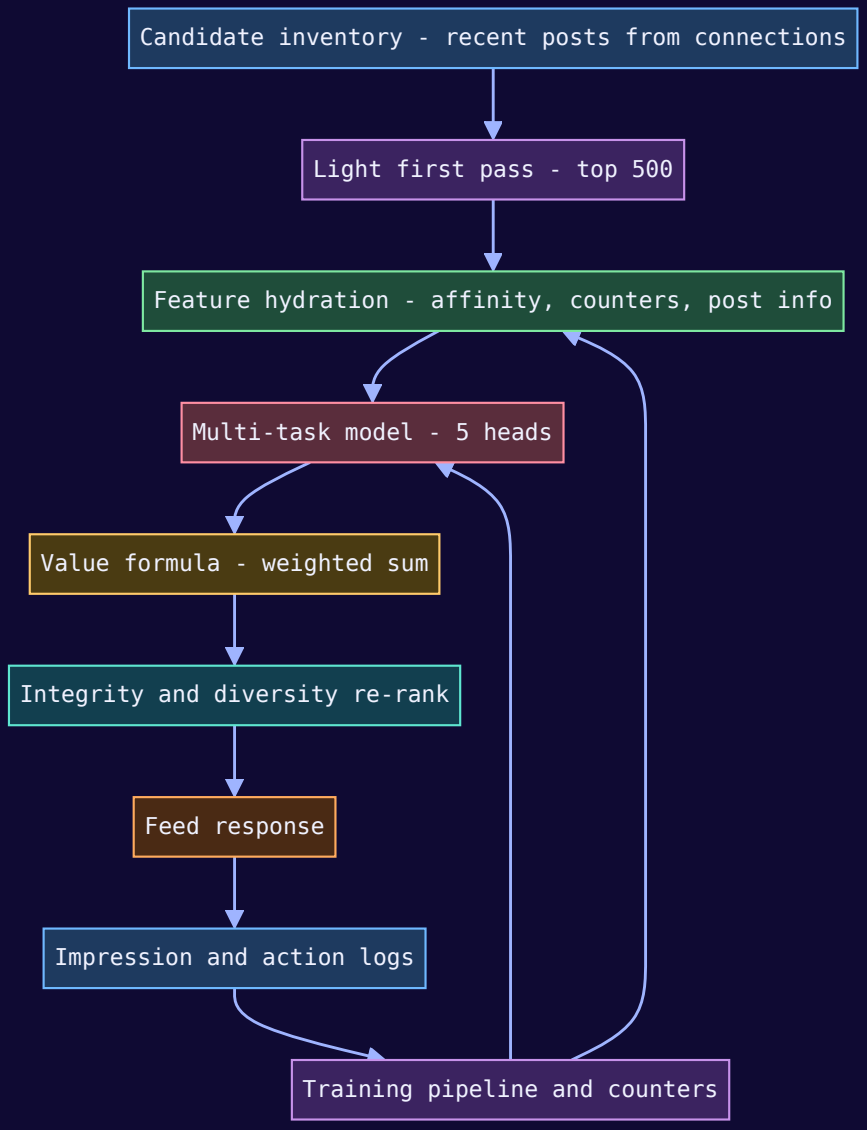
Functional

- Simulator: users with interest vectors and friendship graph, authors posting content with topic vectors, engagement probabilities that depend on affinity, topic match, freshness and content quality; logged impressions with position.
- Features: viewer–author affinity (past interactions), post age, post type, early engagement counts, viewer's historical action rates, topic similarity.
- Model: shared MLP + 5 task heads (like, comment, share, dwell > 10 s, hide), trained with summed weighted binary cross-entropy.
- Value formula: $\Sigma\, w_i \times P(\text{action}_i)$, with configurable weights and a negative weight for hide.
- Re-rank: author diversity (no more than 2 posts in a row from one author), integrity down-ranking for flagged posts.
- Offline metrics per task (AUC, calibration) and NDCG on value; online metrics in the simulator (sessions, comments, hides).

Non-functional

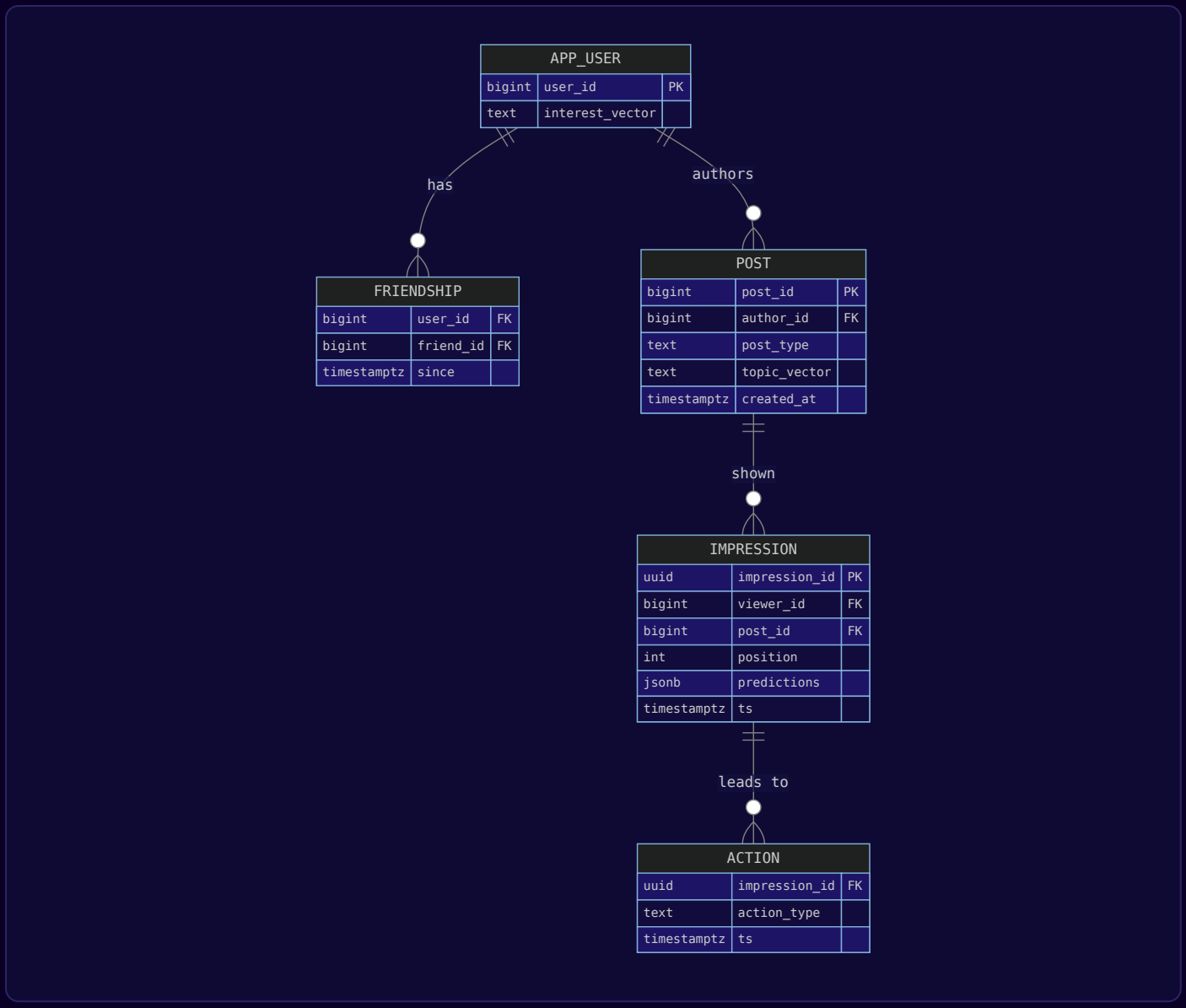
- Rank 1,000 candidates in under 50 ms on CPU.
- Feature freshness: engagement counters at most 1 minute stale.
- Weights changeable at runtime without redeploying the model.

The architecture



- **Candidate inventory**: unseen posts from the last few days from connections.
- **Light first pass**: a cheap model or heuristic that cuts thousands to hundreds.
- **Feature hydration**: fetches per-post and per-viewer features; counters come from a streaming aggregator.
- **Multi-task model**: one forward pass yields five probabilities; shared layers let rare tasks (share) learn from common ones (like).
- **Value formula**: owned by product; changes are tested with A/B experiments.

Data model



Tech stack

Component	Choice	Why
Simulator	Python + numpy	Controllable ground truth
Model	PyTorch multi-task MLP	Shared body, many heads
Counters	Redis or PostgreSQL with periodic aggregation	Fresh engagement features
Serving	FastAPI	Ranking endpoint
Config	weights in a config table	Change without redeploy

Milestones

1. **Simulator.** *Done when* it produces a month of impressions and actions with realistic rates (likes common, shares rare) and known ground-truth probabilities.
2. **Single-task baselines.** *Done when* separate logistic models per action report AUC and calibration.
3. **Multi-task model.** *Done when* one shared model matches or beats single-task models, especially on rare tasks.
4. **Value formula and re-rank.** *Done when* changing weights visibly changes the feed and diversity rules hold.
5. **Online simulation.** *Done when* an A/B test inside the simulator compares chronological, like-only and multi-task value feeds on comments, hides and session length.

Deep dive: the hard parts

- **Task imbalance and loss weights.** Likes vastly outnumber shares. Unweighted summed losses let the like task dominate shared layers. Tune task weights, or use per-task normalisation, and check every head's calibration.
- **Calibration matters for combination.** The value formula adds probabilities. If one head is over-confident, it silently dominates the score. Calibrate each head on held-out data before combining.
- **Position bias.** Posts at the top get more engagement regardless of quality. Log position, include it as a training feature, set it to a fixed value at serving — or estimate propensities with randomised positions in a small slice of traffic.
- **Freshness of counters.** Early engagement ("10 comments in 5 minutes") is a powerful feature, but if training used final counts while serving sees live counts, you have skew. Log the features as served and train on those.

Testing strategy

- Simulator ground-truth tests: model probabilities correlate strongly with the simulator's true probabilities.
- Per-head calibration tests (expected calibration error below a tolerance).
- Re-rank rule tests with adversarial candidate lists (ten posts by one author).
- Feature logging test: served features are stored and match those used to train.

Trade-offs, scaling and failure modes

- Engagement objectives can amplify sensational content; negative-feedback heads (hide, report) and integrity classifiers are the counterweight.
- More tasks increase model capacity needs and the risk of negative transfer (one task hurting another); monitor each head.
- Napkin math: 500 candidates × 1 forward pass of a small MLP is a few milliseconds on CPU; feature hydration (network calls) usually dominates latency, so batch fetches.

Stretch goals

- Mixture-of-experts multi-task architecture (separate expert layers with task-specific gates).
- Long-term value: predict next-day return and add it to the formula.
- Off-policy evaluation of new weightings from logged randomised traffic.

Skills practised

- Ch 17 — ML Case Studies I: feed ranking, multi-task heads, value formulas, integrity.
- Ch 15 — ML System Design Framework: framing the objective, offline vs online metrics.
- Ch 10 — Neural Networks and Deep Learning: multi-task losses.
- Ch 6 — Evaluation and Tuning: calibration.

In the interview

Q: Why multi-task instead of separate models? **A:** One model and one feature fetch for all predictions, and rare tasks benefit from representations learned on common ones. The cost is tuning loss weights and watching for negative transfer.

Q: How do you combine the predictions? **A:** A weighted sum of calibrated probabilities, including negative weights for hides or reports. Weights encode product values and are tuned with online experiments.

Q: How do you avoid optimising for clickbait? **A:** Include negative-feedback predictions and integrity signals, weight meaningful interactions higher, and use long-term metrics (retention, surveys) as guardrails.

TL;DR

Predict many engagement types with one calibrated multi-task model, combine them with a product-owned value formula, and correct for position bias and stale counters.

28. Real-time fraud detection with streaming features

Level: Advanced · **Time:** 4–6 weeks · **Prerequisites:** Ch 16, 18

What it is

A system that scores every card payment for fraud risk in real time — while the customer waits at checkout — and decides to approve, challenge (for example, ask for 3-D Secure verification, an extra check by the card issuer such as a code sent to the cardholder's phone) or block. Its power comes from **streaming features** (velocity counters such as "number of transactions on this card in the last 10 minutes" or "distinct cards used from this device today") computed continuously from an event stream, combined with a gradient-boosted model and a rules layer.

The problem it solves

Fraudsters move fast: a stolen card is used many times in minutes. Features computed in a nightly batch are useless by then. The system must compute fresh aggregates, score in tens of milliseconds, cope with extreme class imbalance and with labels (chargebacks) that arrive weeks later.

Real-world examples

- Stripe's How we built it: Stripe Radar describes scoring each payment in under 100 ms using a large set of transaction characteristics and their move from an XGBoost + DNN ensemble to a DNN-only model.
- The Kaggle IEEE-CIS Fraud Detection dataset (from Vesta) has hundreds of transaction and identity features with timestamps, suitable for velocity features.
- The Credit Card Fraud Detection dataset (ULB) has 284,807 transactions with 492 frauds, a classic imbalance benchmark.

Requirements

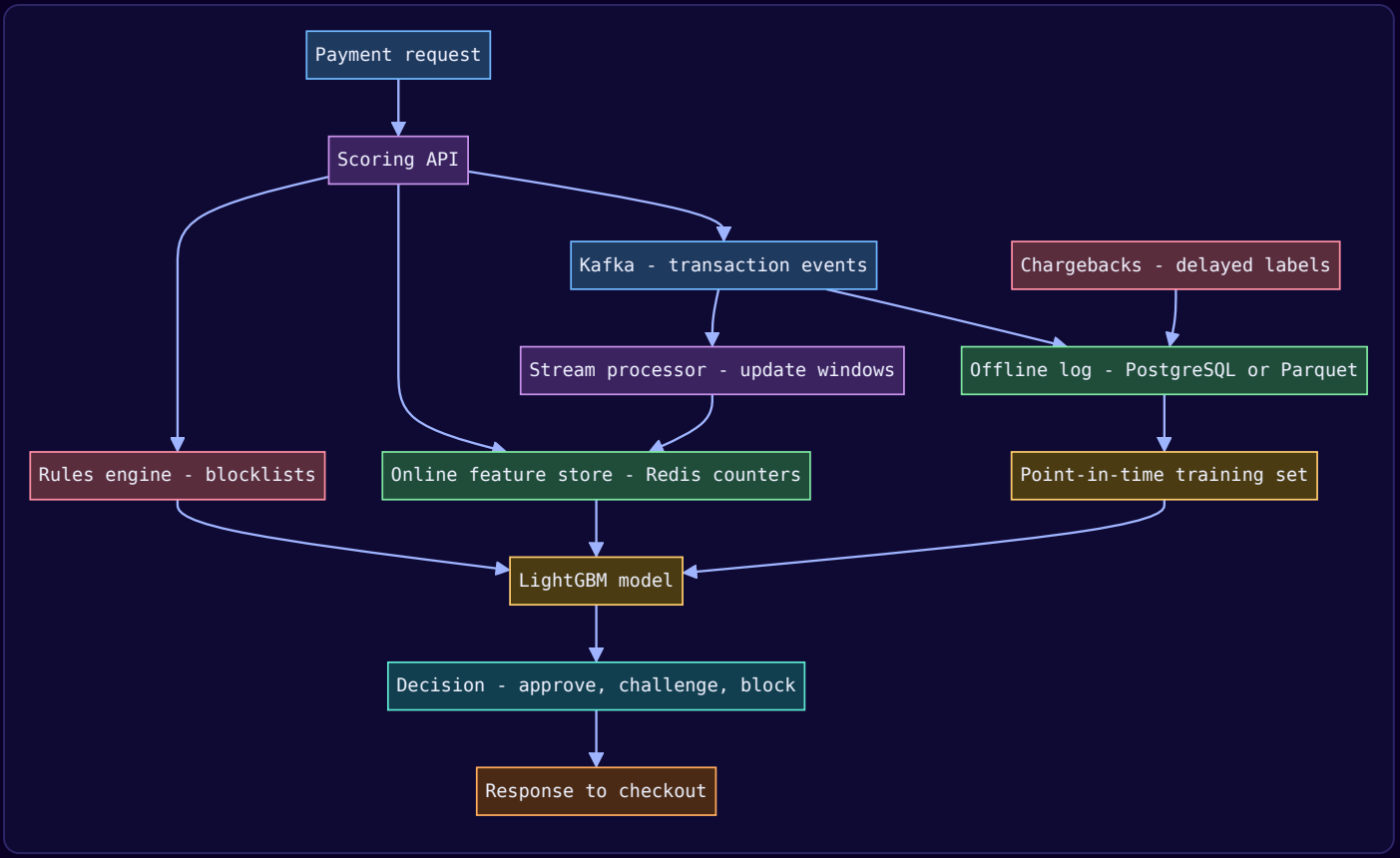
Functional

- Event replay: stream IEEE-CIS transactions in timestamp order into Kafka (or a Python queue to start).
- Streaming feature service: sliding-window counts and sums per card, email domain, device and address (1 min, 1 h, 24 h), plus "first time seen" flags.
- Model: LightGBM trained on **point-in-time** features (exactly as they would have been at the transaction moment).
- Decision layer: score → approve / challenge / block thresholds; hard rules (blocklists) applied first.
- Delayed labels: simulate chargebacks arriving 30–60 days later; retrain only on matured data.
- Analyst queue: challenged/blocked transactions with explanations.

Non-functional

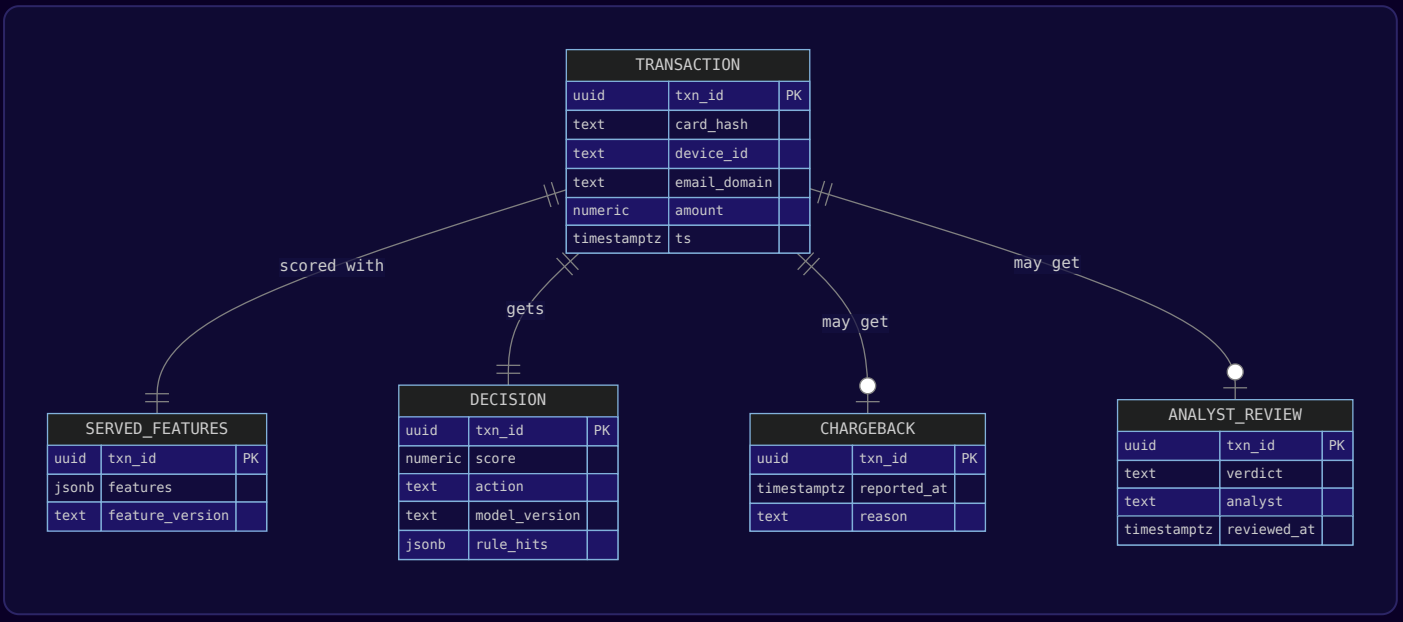
- p99 scoring latency under 50 ms including feature lookup.
- Throughput 1,000 transactions per second on one machine in the replay test.
- Feature values online and offline identical for the same transaction (tested).
- False-positive budget: block at most 0.1% of legitimate transactions.

The architecture



- **Scoring API**: one synchronous request per payment; reads features, runs rules and model, returns a decision.
- **Stream processor**: consumes events and updates sliding-window counters keyed by card, device, email, IP.
- **Online feature store**: low-latency key–value reads (Redis) of the latest counters.
- **Offline log**: every event and the features *as served*, joined later with chargebacks to make training data.
- **Point-in-time training set**: features reconstructed as of each transaction's timestamp — never using later events.

Data model



Tech stack

Component	Choice	Why
Event bus	Kafka	Durable, ordered, replayable stream
Stream processing	Python consumer (Faust-style) or Flink	Window aggregations
Online store	Redis	Sub-millisecond counter reads
Model	LightGBM	Strong on tabular, fast inference
API	FastAPI	Scoring endpoint
Offline store	PostgreSQL or Parquet + polars	Training data

Milestones

1. **Offline model.** *Done when* a LightGBM model on IEEE-CIS with a time-based split reports PR AUC and recall at 0.1% false-positive rate.
2. **Point-in-time velocity features.** *Done when* offline window features are built with strict as-of logic and improve PR AUC.
3. **Streaming pipeline.** *Done when* replaying events through Kafka updates Redis counters, and online values equal offline values for a sample of 10,000 transactions.
4. **Scoring service.** *Done when* the API meets the latency target at 1,000 TPS and logs served features and decisions.
5. **Delayed labels and retraining.** *Done when* a simulated 45-day label delay is handled and you report how performance decays between retrains.
6. **Analyst loop.** *Done when* challenged transactions appear in a queue with top reasons, and analyst verdicts flow into training.

Deep dive: the hard parts

- **Training/serving skew in features.** Offline, it is tempting to compute "transactions in the last hour" with a SQL window that includes the current transaction or events that arrived late. Online, the counter may not have processed the previous event yet. Define windows precisely (exclusive of the current event), log served features, and train on logged features once you have them.
- **Label delay and selection bias.** Chargebacks arrive weeks later, and blocked transactions never get labels at all (you never learn whether they were fraud). Retrain on matured windows, and send a small random sample of would-be-blocked transactions to challenge instead, so you can still learn.
- **Extreme imbalance.** Fraud may be one in a thousand. Optimise and report PR AUC and recall at a fixed false-positive rate, which maps to customer friction. Class weights or down-sampling the majority (with recalibration) speed training.
- **Adversaries adapt.** Once a pattern is blocked, fraudsters change it. Monitor feature drift and the share of fraud caught by rules vs model, and retrain often.

Testing strategy

- Online/offline parity test for every feature on replayed data.
- Window boundary tests: an event exactly at the window edge counts as specified.
- Latency and throughput tests with realistic key distributions (hot keys: one card with many attempts).
- Backtest: replay a month, apply decisions, compute fraud caught and legitimate blocked.
- Chaos test: Redis unavailable → API falls back to a model without velocity features or to rules, never times out.

Trade-offs, scaling and failure modes

- Approve-by-default on failure keeps customers happy but lets fraud through during outages; block-by-default does the opposite. Decide per merchant risk and document it.
- More windows and keys mean more memory: 100 million cards × 10 counters × 8 bytes is about 8 GB before overhead — plan for a Redis cluster.

- Gradient boosting is fast and strong; deep models (as at Stripe) handle more raw features and scale with data but need more infrastructure.

Stretch goals

- Graph features: number of cards linked to a device or address, connected-component size.
- Cost-sensitive thresholds per merchant category.
- Shadow-deploy a new model and compare decisions before switching.

Skills practised

- Ch 16 — ML in Production: feature stores, online vs offline features, training/serving skew, delayed labels.
- Ch 18 — ML Case Studies II: the fraud detection case study.
- Ch 7 — Decision Trees and Ensembles: gradient boosting on imbalanced data.
- Ch 6 — Evaluation and Tuning: PR AUC, thresholds at a fixed FPR.

In the interview

Q: How do you guarantee features are the same online and offline? **A:** Define each feature once, compute it with the same logic in both paths (or log online values and train on them), use point-in-time joins offline, and run parity tests on replayed data.

Q: How do you handle labels arriving 60 days late? **A:** Train only on matured data, use faster proxy labels (analyst verdicts, customer disputes) for monitoring, and account for the gap when measuring drift.

Q: How do you learn from transactions you blocked? **A:** You can't directly — they have no outcome. Route a small random fraction to challenge instead of block, and use those outcomes (with inverse propensity weighting) for training and evaluation.

TL;DR

Real-time fraud is a feature-freshness problem: streaming velocity counters, point-in-time training data, sub-50 ms scoring, and a plan for delayed and missing labels.

29. Search ranking with learning to rank

Level: Advanced · **Time:** 4–5 weeks · **Prerequisites:** Ch 6, 7, 17

What it is

A **learning-to-rank** (LTR) system: for each query, it orders candidate documents so the most relevant appear first. You train pointwise (predict each document's relevance), pairwise (predict which of two documents is better) and listwise models — in particular **LambdaMART**, gradient-boosted trees optimised for NDCG — on a public benchmark, then build a two-stage search service (BM25 retrieval — a classic keyword-match score based on term frequency and rarity — followed by LTR re-ranking) with click logging and click-bias correction.

The problem it solves

Search quality decides whether users find what they need. Hand-tuned scoring formulas break down once there are dozens of signals (text match, freshness, popularity, personalisation). LTR learns how to weigh them from labelled judgements or clicks, and optimises the metric that matters — quality at the top of the list.

Real-world examples

- Microsoft's MSLR-WEB30K dataset has more than 30,000 queries with 136 features per query–URL pair and relevance labels from 0 to 4, split into five folds (also in TensorFlow Datasets).
- Haldar et al.'s Applying Deep Learning to Airbnb Search tells how Airbnb moved from gradient-boosted ranking to neural networks, with candid lessons on what failed.
- The LightGBM documentation includes the `lambdarank` objective used here.

Requirements

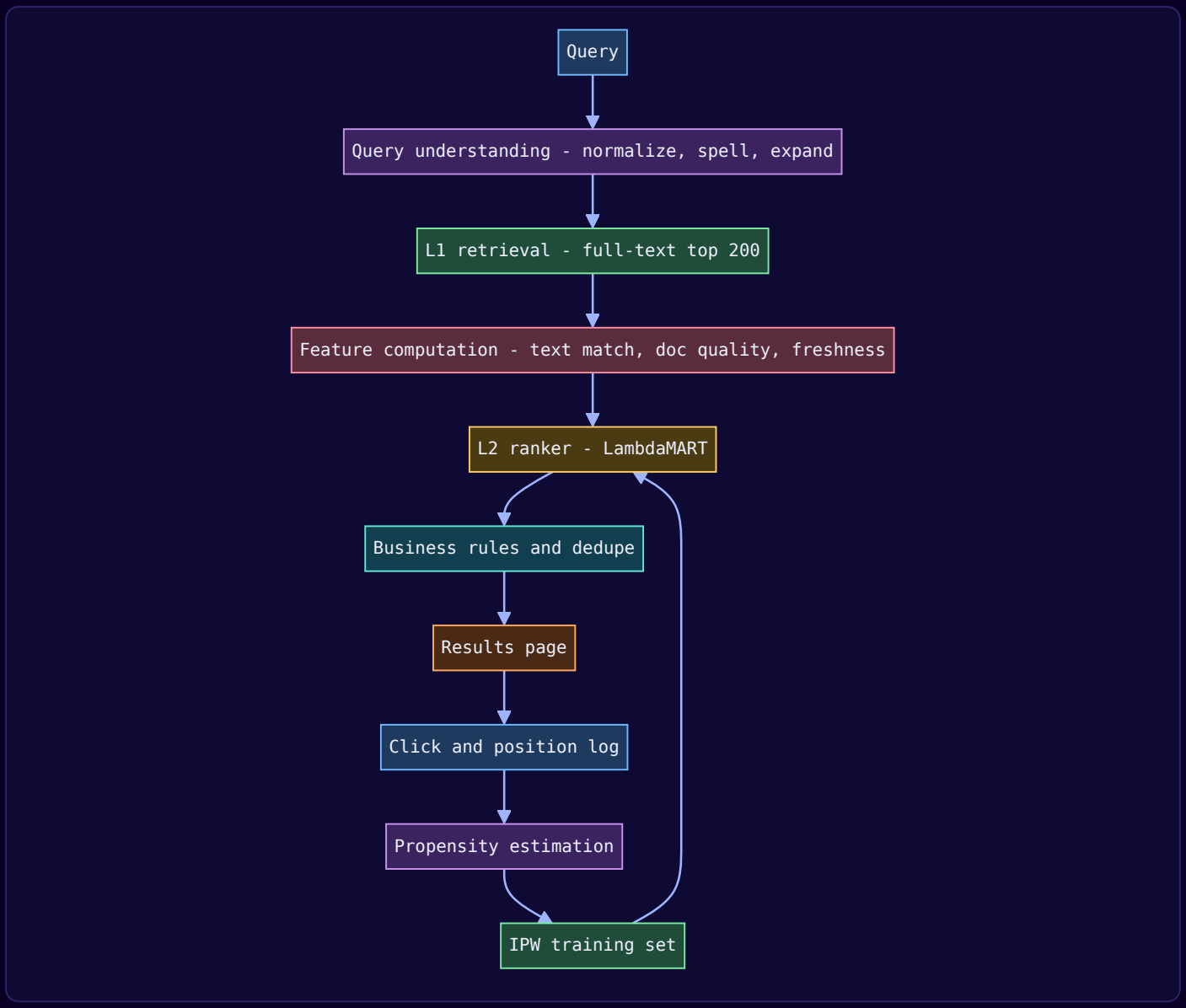
Functional

- Offline: train pointwise regression, pairwise (RankNet-style) and LambdaMART on MSLR-WEB10K/30K folds; report NDCG@1, @5, @10.
- Feature analysis: importance and ablations by feature group.
- Online prototype: index a document collection (for example an MS MARCO passage subset) in PostgreSQL full-text search; BM25-style top 200 → LTR re-rank top 20.
- Click simulation with **position bias** (users are more likely to examine top results) and an inverse-propensity-weighted (IPW) LTR model trained on simulated clicks.

Non-functional

- Re-ranking 200 candidates in under 30 ms.
- End-to-end query under 100 ms p95.
- Every query logs candidates, features, positions and clicks for training.

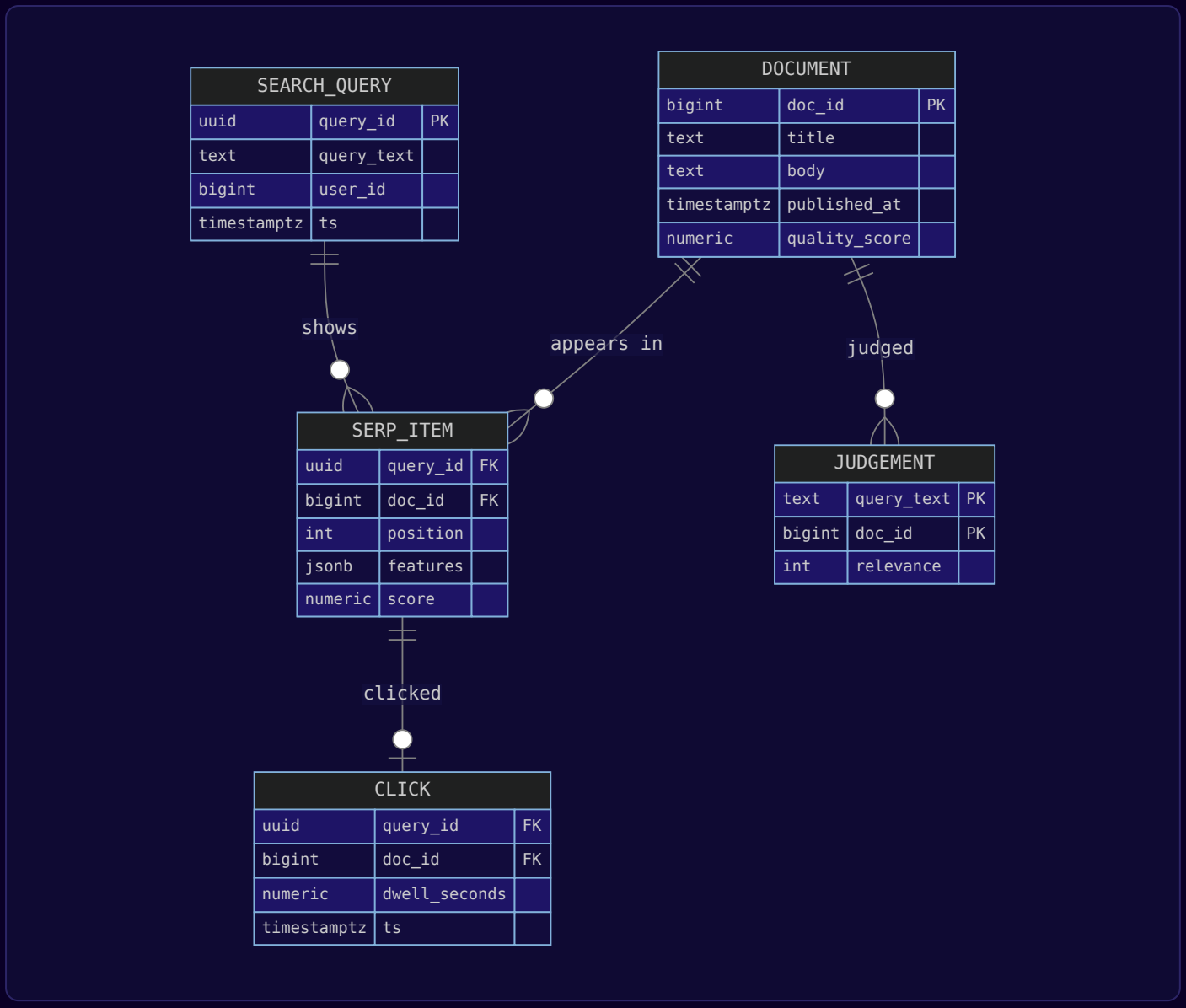
The architecture



- **Query understanding**: lowercasing, spelling correction, synonym expansion.
- **L1 retrieval**: cheap and recall-oriented.
- **Feature computation**: per query–document pair (BM25 score per field, term overlap, document length, click-through history, age).
- **L2 ranker**: LambdaMART on those features.

- **Propensity estimation**: estimate how likely each position is examined (for example, with result randomisation on a small traffic slice), then weight clicks by the inverse.

Data model



Tech stack

Component	Choice	Why
Ranker	LightGBM `lambdarank`	Standard LambdaMART implementation
Neural comparison	PyTorch (pairwise/listwise losses)	Learn the losses
Retrieval	PostgreSQL full-text search	Simple L1
Service	FastAPI	Search endpoint
Data	polars	Large LETOR-format files

Milestones

1. **Benchmark baselines.** *Done when* a single feature (BM25 column) and pointwise regression NDCG@10 are recorded on MSLR fold 1.
2. **LambdaMART.** *Done when* it beats pointwise on all five folds with mean ± std reported.
3. **Neural ranker.** *Done when* a pairwise/listwise MLP is compared fairly (feature normalisation matters).
4. **Online two-stage service.** *Done when* real queries return re-ranked results under 100 ms with full logging.

5. **Click bias.** *Done when* a click simulator with position bias shows that naive click training underperforms IPW training on true relevance.

Deep dive: the hard parts

- **Why LambdaMART works.** NDCG is not differentiable — it depends on sorting. LambdaRank defines gradients ("lambdas") for each document pair proportional to how much swapping them would change NDCG, so training focuses on mistakes near the top. MART (gradient-boosted trees) fits those gradients.
- **Query-grouped data.** LTR training data is grouped by query; splits must be by query, and the `group` array must match row order exactly. A mismatch silently produces nonsense.
- **Clicks are biased labels.** Users click top results more because they look at them more, and they cannot click what is not shown. Training on raw clicks teaches the model to copy the old ranking. Position-based propensity models and IPW reduce this bias.
- **Feature scaling for neural rankers.** MSLR features have wildly different ranges; trees do not care, neural nets do. Log-transform and standardise per feature; Airbnb's paper describes similar lessons about normalisation.

Testing strategy

- Metric tests: NDCG on hand-computed rankings, including ties and all-zero relevance queries.
- Group integrity test: sum of group sizes equals row count; every query in exactly one split.
- Simulator test: with no position bias, naive and IPW training converge to the same ranking.
- Latency test for feature computation + ranking on 200 candidates.

Trade-offs, scaling and failure modes

- GBDT rankers are strong on hand-crafted features and cheap to serve; neural rankers can use raw text (cross-encoders) but cost more per candidate.
- L1 recall limits everything: a perfect ranker cannot rank a document retrieval missed. Measure L1 recall separately.
- Popularity features create rich-get-richer loops; add freshness and exploration.

Stretch goals

- Add a dense retrieval channel (Project 18) to L1 and measure recall.
- Cross-encoder re-ranker on the top 20 after LambdaMART.
- Interleaving experiments (mixing two rankers' results in one list) as a fast online comparison.

Skills practised

- Ch 17 — ML Case Studies I: search ranking, L1/L2 rankers, NDCG, click bias.
- Ch 7 — Decision Trees and Ensembles: gradient boosting with custom objectives.
- Ch 6 — Evaluation and Tuning: ranking metrics, grouped CV.
- Ch 13 — Recommender Systems and Embeddings: retrieval funnels.

In the interview

Q: Pointwise, pairwise or listwise? **A:** Pointwise predicts relevance per item — simple but ignores order. Pairwise learns which of two items is better. Listwise (LambdaMART, softmax losses) optimises a list metric directly and usually wins on NDCG.

Q: Why can't you just train on clicks? **A:** Clicks depend on position and on what was shown, so they reflect the old ranker as much as relevance. Correct with propensity weighting, randomisation or click models.

Q: How would you evaluate a new ranker before a full A/B test? **A:** Offline NDCG on human judgements and IPW-corrected click data, then interleaving on a small slice of traffic, then an A/B test on success metrics.

TL;DR

Two-stage search with a LambdaMART re-ranker, query-grouped evaluation by NDCG, and click labels corrected for position bias.

30. Ad click-through-rate prediction

Level: Advanced · **Time:** 4–5 weeks · **Prerequisites:** Ch 6, 13, 18

What it is

A model that predicts the probability a user clicks an ad (**CTR**, click-through rate), trained on billions-scale-style sparse data: dozens of categorical features with millions of distinct values (user IDs, site IDs, ad IDs). You progress from logistic regression with the **hashing trick** (map any category string to one of N buckets with a hash function) to a deep **feature-interaction** model (DeepFM or DCN), with a strong focus on **calibration**, because the predicted probability directly sets the price in an ad auction.

The problem it solves

Ad systems rank ads by expected value: bid × P(click). If P(click) is too high, advertisers overpay and leave; too low, the platform loses revenue. Small relative improvements in log loss are worth a lot at scale, so CTR models are some of the most heavily optimised in industry.

Real-world examples

- He et al.'s Practical Lessons from Predicting Clicks on Ads at Facebook (2014) combined boosted trees as feature transformers with logistic regression and stressed data freshness and calibration, using normalised entropy as the metric.
- The Criteo Display Advertising Challenge dataset has 13 integer and 26 hashed categorical features over about a week of traffic; Avazu is a smaller mobile-ads alternative.
- DCN and DCN V2 (Google) learn explicit feature crosses; DeepFM combines factorisation machines with a deep network.

Requirements

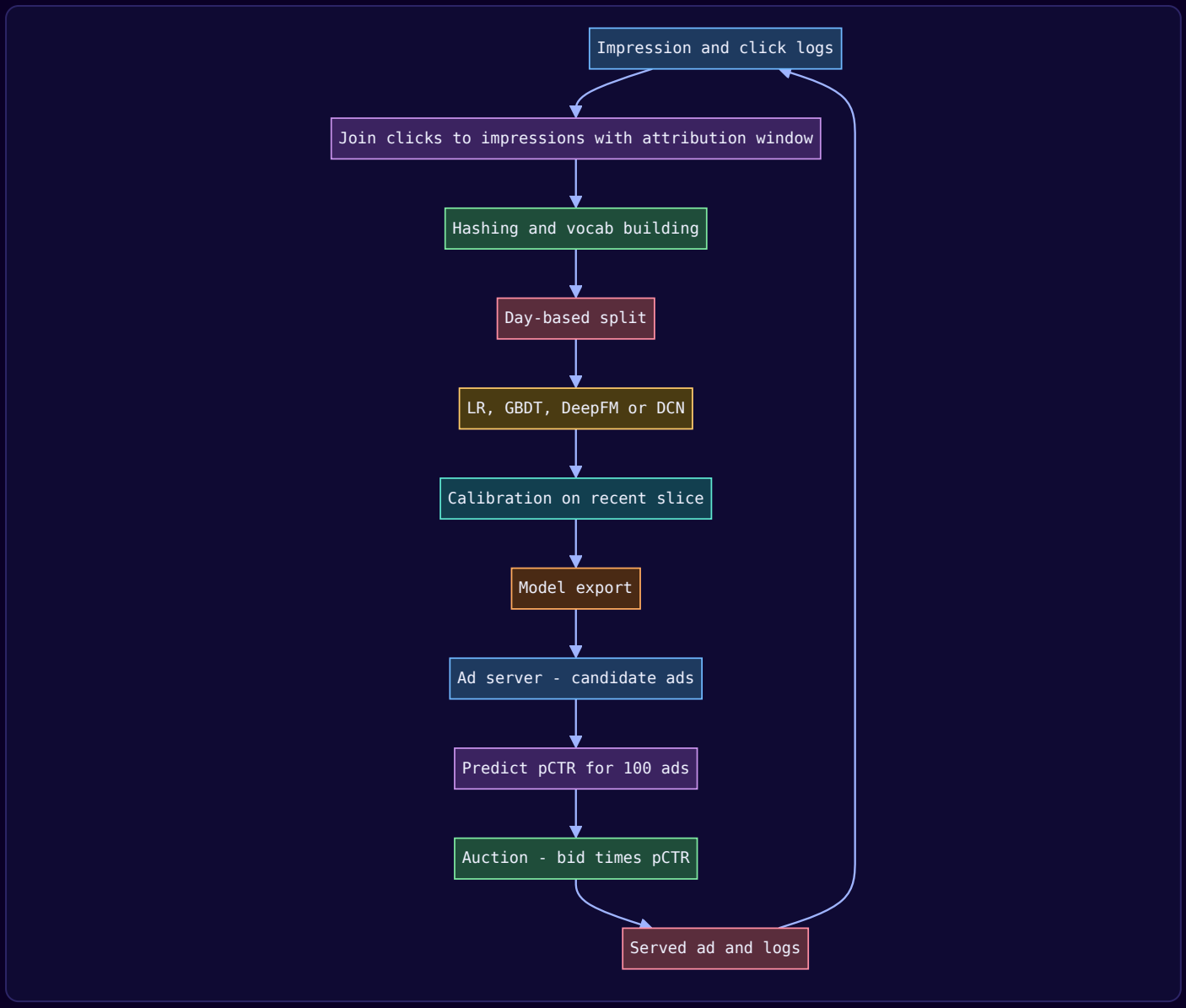
Functional

- Preprocess Criteo: log-transform integers, hash categoricals (with frequency thresholds), split by day (train on earlier days, test on the last).
- Models: logistic regression with hashed one-hot features, GBDT, DeepFM or DCN V2 in PyTorch.
- Metrics: **log loss** (penalises confident wrong probabilities), **normalised entropy** (log loss divided by the log loss of always predicting the average CTR), ROC AUC, and calibration (predicted vs observed CTR by bucket and overall).
- Calibration layer (isotonic or Platt) fitted on a recent slice.
- Serving: score a batch of 100 candidate ads in under 10 ms; an auction simulator that ranks by bid × pCTR.

Non-functional

- Train on at least 50 million rows with streaming data loading in bounded memory.
- Embedding tables sized to fit in 16 GB RAM.
- Daily incremental training from yesterday's data.

The architecture



- **Join with attribution window**: a click counts if it arrives within, say, 30 minutes of the impression; late clicks are missed labels.
- **Hashing**: fixed-size embedding tables even as new IDs appear, at the cost of some collisions.
- **Deep model**: embeddings for each field, an explicit cross network (or FM part) plus an MLP.
- **Calibration**: adjusts for down-sampling of negatives and drift since training.
- **Auction**: expected value ranking; pCTR errors become price errors.

Data model

Item	Type	Notes
`label`	0/1 click	CTR typically a few percent in Criteo
`I1..I13`	integer counts	Many missing; log-transform
`C1..C26`	hashed categorical	Millions of distinct values
Embedding tables	float `[buckets, d]` per field	The bulk of model size
Served log	impression ID, features, pCTR, bid, price	Training and auditing

Tech stack

Component	Choice	Why
Data loading	polars / PyArrow streaming	Larger-than-memory files
Baselines	scikit-learn `SGDClassifier`, LightGBM	Hashed LR and GBDT
Deep model	PyTorch (`nn.EmbeddingBag`)	DeepFM / DCN V2
Calibration	scikit-learn `IsotonicRegression`	Simple, effective
Serving	FastAPI + ONNX Runtime	Fast batch scoring

Milestones

1. **Data pipeline.** *Done when* a streaming loader processes the training days with bounded memory and a day-based split.
2. **Baselines.** *Done when* hashed LR and GBDT report log loss, normalised entropy and AUC on the test day.
3. **Deep model.** *Done when* DeepFM or DCN V2 improves log loss over both baselines (even a small relative gain is meaningful here).
4. **Calibration.** *Done when* predicted vs observed CTR by decile is plotted before and after calibration, and overall predicted CTR is within 1–2% of observed.
5. **Serving and auction.** *Done when* 100 ads score in under 10 ms and the auction simulator shows how miscalibration shifts revenue.

Deep dive: the hard parts

- **Negative down-sampling and recalibration.** Clicks are rare, so teams often keep all positives and a fraction w of negatives. Predictions are then inflated; correct with q = p / (p + (1 − p) / w). Forgetting this is a classic and expensive bug.
- **Freshness.** Ads, campaigns and user interests change daily. The Facebook paper found that model staleness costs measurable accuracy; plan incremental training and measure decay by training on day 1 and testing on days 2, 3, 4.
- **High-cardinality features.** Most IDs appear once or twice. Use frequency thresholds (map rare values to an "other" bucket), hashing, and regularisation on embeddings. Field-aware interactions capture "this user on this site" patterns.
- **Metric choice.** AUC ignores calibration; log loss and normalised entropy reward both ranking and calibration. A model with better AUC and worse calibration can lose money in the auction.

Testing strategy

- Down-sampling correction test on synthetic data with known CTR.
- Hash stability test: the same raw value maps to the same bucket in training and serving (same hash function and seed — Python's built-in `hash()` for strings is randomised per process, so do not use it).
- Calibration test on the latest day before every deployment.
- Latency test with batches of 100 ads.

Trade-offs, scaling and failure modes

- Bigger embedding tables reduce collisions but cost memory: 26 fields × 1 million buckets × 16 dims × 4 bytes is about 1.7 GB.
- Feature crosses explode in count; deep interaction models learn them but are harder to debug than LR.
- Feedback loops: the model decides which ads get shown, so ads it under-rates never get data. Reserve exploration traffic for new ads.

Stretch goals

- Multi-task prediction of click and conversion (with conversion delays).
- Online learning with FTRL-style logistic regression on a stream.
- Privacy: train without user IDs and measure the loss.

Skills practised

- Ch 18 — ML Case Studies II: ad click prediction, calibration, auctions.
- Ch 13 — Recommender Systems and Embeddings: embeddings for sparse IDs, feature interactions.
- Ch 6 — Evaluation and Tuning: log loss, calibration.
- Ch 16 — ML in Production: freshness and continual training.

In the interview

Q: Why does calibration matter so much for ads? **A:** The auction ranks and prices by bid × pCTR. Miscalibrated probabilities misprice every ad, hurting advertisers or the platform even if the ranking is good.

Q: How do you handle millions of categorical values? **A:** Hashing into fixed buckets, frequency thresholds for rare values, learned embeddings with regularisation, and models that learn interactions (FM, DCN).

Q: You down-sampled negatives by 10×. What changes? **A:** Predicted probabilities are inflated; apply the correction $q = p / (p + (1 - p)/w)$ with $w = 0.1$, then verify calibration on un-sampled data.

TL;DR

CTR prediction is sparse features, embeddings and crosses — but calibrated probabilities and fresh models are what make it profitable.

31. Feature store, model serving and drift monitoring

Level: Advanced · **Time:** 5–8 weeks · **Prerequisites:** Ch 16

What it is

A small **ML platform** — your own mini Michelangelo — that other projects can plug into. It has three parts: a **feature store** (one place to define features, compute them in batch and streaming, store them offline for training and online for serving, with point-in-time correct training-set generation), a **model serving** layer (versioned models behind a common API with canary rollout and shadow mode), and a **monitoring** service (data drift, prediction drift, latency, and performance once labels arrive, with alerts).

The problem it solves

Every team that ships models rebuilds the same plumbing, and each rebuild introduces the same bugs: features computed differently in training and serving, models deployed without a rollback path, silent degradation nobody notices for months. A platform turns these into solved problems so data scientists can focus on models, and gives the organisation one consistent view of what is running.

Real-world examples

- Uber's Meet Michelangelo post describes a shared platform covering data management (including a shared feature store), training, evaluation, deployment, prediction and monitoring.
- Feast is an open-source feature store with offline and online stores and point-in-time joins.
- Evidently is an open-source library for data and model drift reports and test suites.

Requirements

Functional

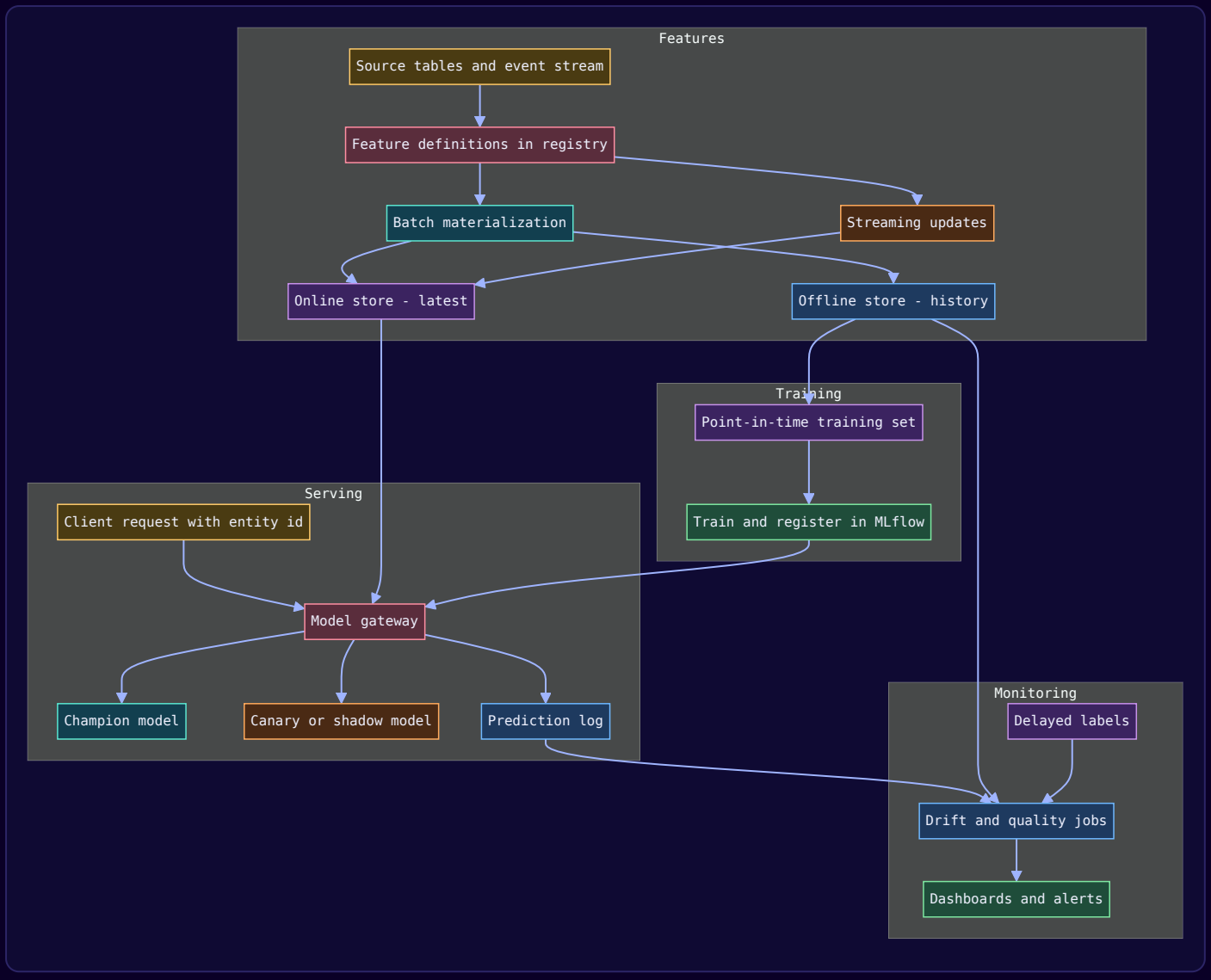
- **Feature registry**: features declared in code (entity, source, transformation, owner, freshness SLA) and listed in a catalogue.
- **Offline store**: historical feature values with timestamps (PostgreSQL or Parquet); `get_training_data(entity_df)` performs point-in-time joins.
- **Online store**: latest values per entity (Redis or PostgreSQL), filled by a materialisation job and optionally a stream processor.
- **Serving**: `POST /v1/models/{name}/predict` fetches online features by entity ID, calls the model, logs inputs and outputs. Supports canary (x% traffic to a new version) and shadow (new version scores but is not returned).

- **Monitoring**: daily and hourly drift metrics per feature (PSI — population stability index — and the KS statistic, the largest gap between two cumulative distributions), prediction distribution drift, missing-value rates, latency percentiles; delayed-label performance; alerts by threshold.
- Plug in at least two earlier projects (for example churn from Project 11 and fraud from Project 28).

Non-functional

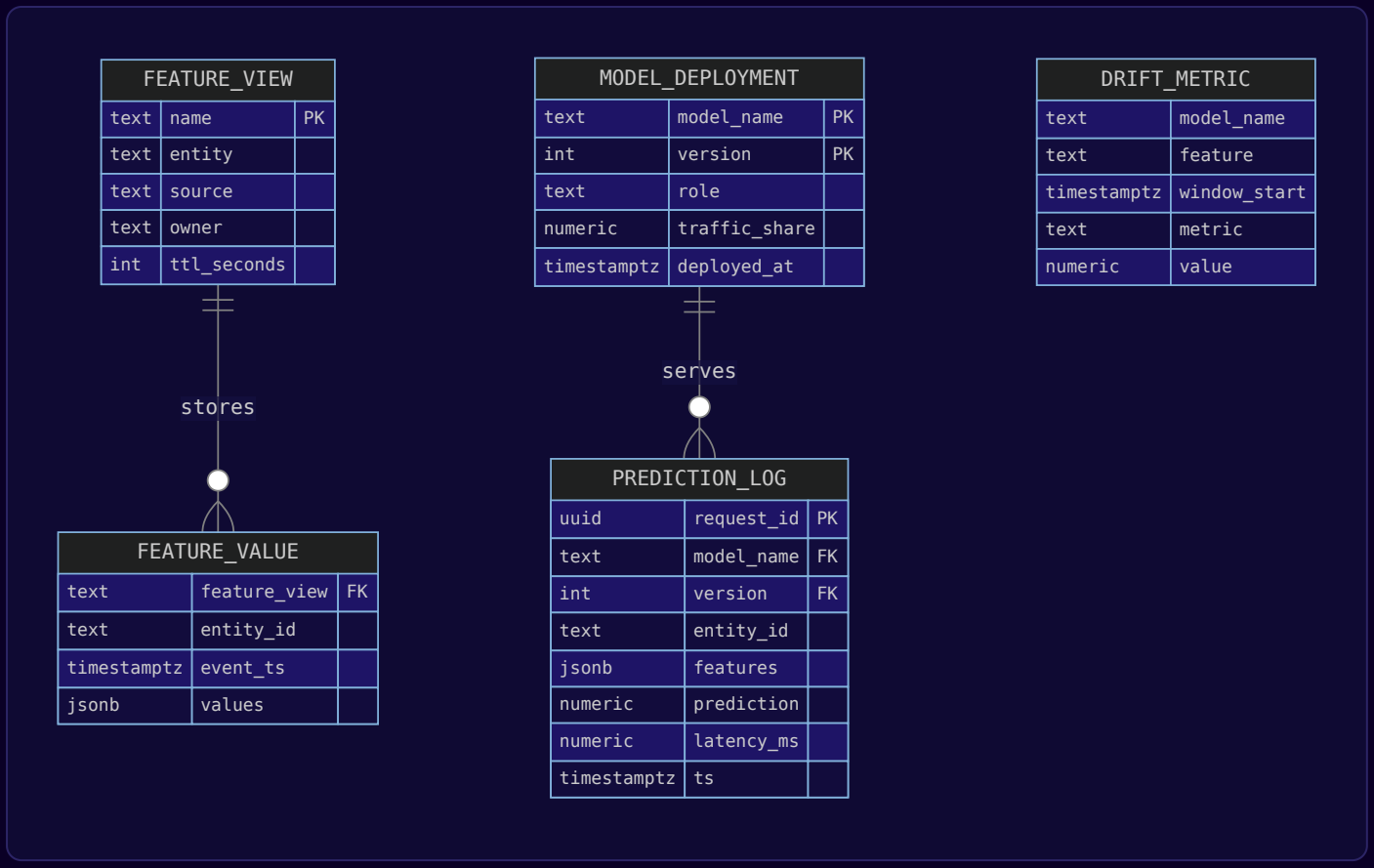
- Online feature fetch p99 under 10 ms for 20 features.
- Serving overhead (excluding model) under 5 ms.
- Offline/online parity: the same entity and timestamp give the same values.
- Monitoring jobs process 10 million prediction logs per day.

The architecture



- **Feature definitions**: Python objects or YAML describing entity keys, the SQL or function, and the timestamp column. One definition feeds both stores.
- **Materialisation**: copies the newest values to the online store on a schedule; streaming updates handle features needing seconds of freshness.
- **Point-in-time training set**: for each (entity, event time) row, join the most recent feature value *at or before* that time.
- **Model gateway**: resolves model version, fetches features, routes canary/shadow traffic, logs everything.
- **Drift jobs**: compare recent feature and prediction distributions with the training reference.

Data model



Tech stack

Component	Choice	Why
Feature store	Feast, or your own minimal version first	Learn the mechanics, then compare
Offline store	PostgreSQL or Parquet + polars	Point-in-time joins
Online store	Redis	Low-latency key–value reads
Registry and models	MLflow	Versions and aliases
Gateway	FastAPI	Routing, logging
Monitoring	Evidently or your own PSI/KS jobs	Drift reports
Scheduling	Prefect, Dagster or cron	Materialisation and monitoring jobs

Milestones

1. **Minimal feature store.** *Done when* features are declared once, materialised offline and online, and `get_training_data` passes a point-in-time correctness test.
2. **Gateway.** *Done when* a registered model serves through `/predict` by entity ID with online features, and logs every request.
3. **Rollouts.** *Done when* canary (5% traffic) and shadow modes work and a one-command rollback restores the previous version.
4. **Monitoring.** *Done when* daily drift (PSI per feature, prediction drift) runs, and an injected drift (shifted feature) triggers an alert.
5. **Delayed-label performance.** *Done when* labels joined to prediction logs produce weekly AUC or RMSE per model version.
6. **Onboard two models.** *Done when* two different earlier projects run through the platform without custom plumbing.

Deep dive: the hard parts

- **Point-in-time joins.** A naive join of labels with the latest feature values leaks the future (a customer's "total spend" includes purchases after the label date). The correct join takes, for each training row, the latest feature value with timestamp ≤ the row's event time — an "as-of" join (`ASOF JOIN` in some engines; a lateral subquery or `DISTINCT ON` in PostgreSQL). Test it with hand-made timelines.
- **Freshness vs cost.** Batch materialisation every hour is cheap; streaming updates are fresh but complex. Give each feature a freshness SLA and choose per feature. Monitor feature age at serving time.
- **Drift is not always a problem.** PSI flags any distribution change, including harmless seasonal ones. Prioritise drift on important features (by model importance), and confirm with performance metrics when labels arrive. Alert on sustained, large drift; report the rest.
- **Canary vs shadow.** Shadow tests risk-free (outputs not used) but cannot measure user impact. Canary measures impact on a small slice but exposes real users. Use shadow first for correctness and latency, canary for metrics.

Testing strategy

- Point-in-time test: synthetic entity with values changing at known times; training rows at various times must pick the right values.
- Parity test: online and offline values for 1,000 random (entity, now) pairs match.
- Gateway tests: routing proportions within tolerance over 10,000 requests; shadow responses never returned.
- Drift test: PSI near 0 for identical distributions, above threshold for a shifted one.
- Failure tests: online store down → gateway uses defaults or fails clearly, per model policy.

Trade-offs, scaling and failure modes

- Build vs buy: managed feature stores and serving platforms save time but add cost and lock-in; a small team may need only an offline store plus a cache.
- A central platform becomes a dependency for every model — its outages are everyone's outages. Design graceful degradation and per-model fallbacks.
- Logging all features for all predictions is expensive at scale; sample logs for high-volume models, but keep full logs for those that need retraining on served features.

Stretch goals

- On-demand features computed at request time from request data (for example, distance between user and store).
- Automated retraining triggered by sustained drift (with the pipeline from Project 22).
- A feature-usage catalogue showing which models use which features, to manage deprecation.

Skills practised

- Ch 16 — ML in Production: feature stores, serving, canary and shadow, drift, MLOps.
- Ch 15 — ML System Design Framework: deployment and monitoring steps.
- Ch 6 — Evaluation and Tuning: performance tracking over time.

In the interview

Q: What problem does a feature store solve? **A:** Consistent feature definitions for training and serving, point-in-time correct training data, low-latency online access, and reuse of features across teams.

Q: How do you detect model degradation before labels arrive? **A:** Monitor input drift (PSI, KS), prediction distribution drift, missing values and upstream freshness; confirm with performance once labels arrive.

Q: Canary or shadow deployment? **A:** Shadow first to validate correctness and latency on real traffic with no user impact, then canary to measure business metrics on a small slice before full rollout.

TL;DR

A platform makes features consistent (one definition, point-in-time joins), deployments safe (registry, shadow, canary, rollback) and degradation visible (drift and delayed-label monitoring).

32. Reinforcement-learning agent: DQN to PPO

Level: Advanced · **Time:** 3–5 weeks · **Prerequisites:** Ch 14

What it is

An agent that learns to act by trial and error in simulated environments. You implement **tabular Q-learning** on a small grid world, then **DQN** (deep Q-network: a neural network estimates the value of each action, trained from a replay buffer with a slowly updated target network) on CartPole and LunarLander, then **PPO** (proximal policy optimisation: a policy-gradient method that limits how far each update can move the policy) — and compare learning curves, stability and sample efficiency across random seeds.

The problem it solves

Some problems are sequences of decisions with delayed rewards — games, robotics, resource scheduling, recommendation sequences. Supervised learning needs the right answer per step; reinforcement learning (RL) learns from rewards alone. Implementing the core algorithms yourself is the only way to understand why RL is notoriously sensitive to hyperparameters and seeds.

Real-world examples

- Mnih et al.'s Human-level control through deep reinforcement learning (Nature, 2015) introduced DQN with experience replay and target networks on Atari games.
- Schulman et al.'s Proximal Policy Optimization Algorithms (2017) introduced PPO's clipped objective, now a default policy-gradient method.
- Gymnasium (the maintained successor to OpenAI Gym) provides the environments; Stable-Baselines3 provides reference implementations to compare against.

Requirements

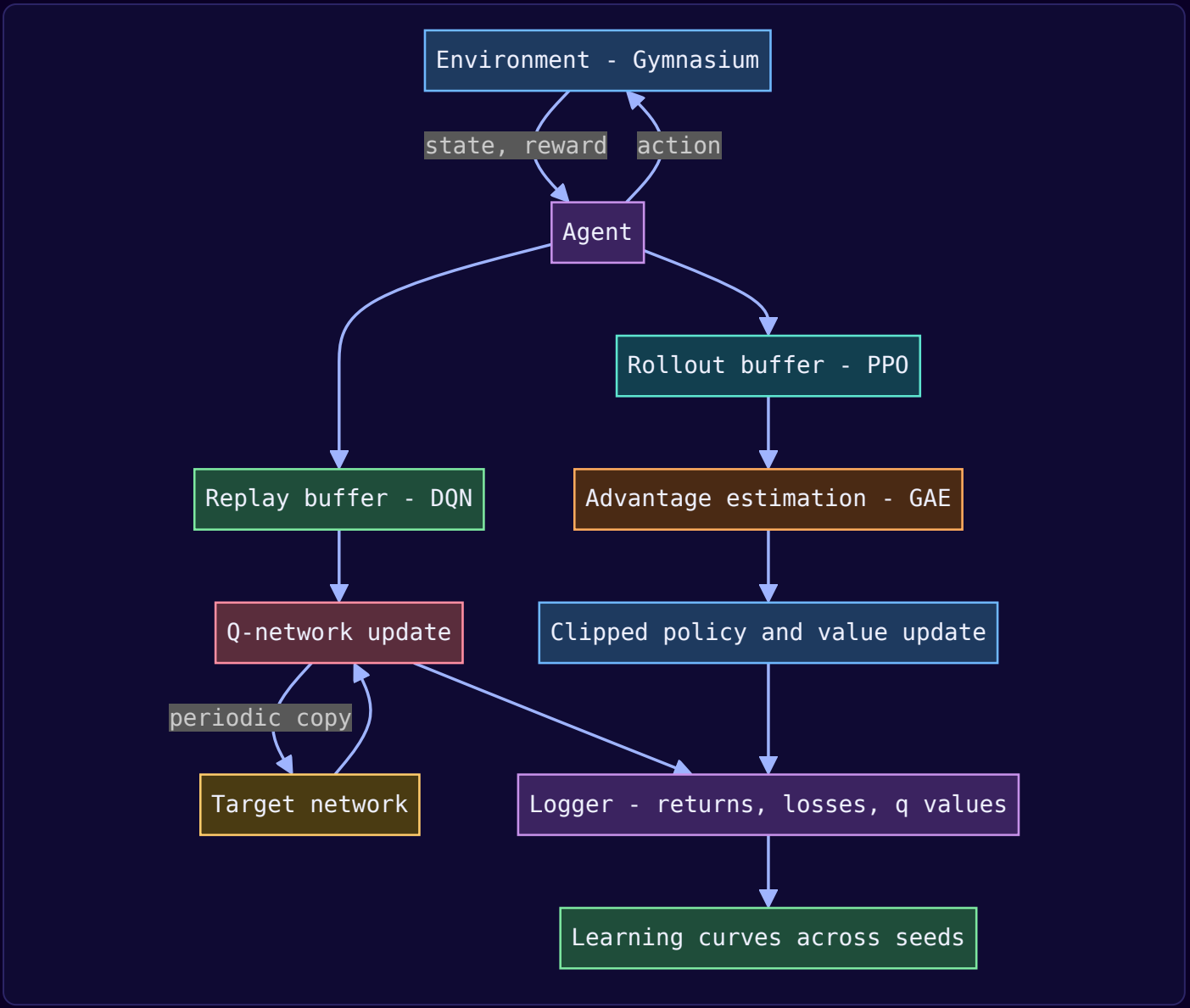
Functional

- Tabular Q-learning and SARSA on FrozenLake or a custom grid world.
- DQN from scratch: replay buffer, target network, epsilon schedule, Huber loss; solve CartPole and then LunarLander.
- PPO from scratch: actor-critic network, generalised advantage estimation (GAE), clipped objective, multiple epochs per batch.
- Experiment runner: N seeds per configuration, learning curves with mean and confidence bands.
- Evaluation: average return over 100 evaluation episodes with exploration off; videos of trained agents.
- Comparison against Stable-Baselines3 on the same environments.

Non-functional

- CartPole solved (average return near the 500 cap) within 100,000 environment steps by DQN on CPU.
- All runs reproducible from a config file and seed.
- At least 5 seeds per reported result.

The architecture



- **Replay buffer**: stores past transitions; sampling randomly breaks the correlation between consecutive steps.
- **Target network**: a lagged copy of the Q-network used to compute targets, which stabilises training.
- **Rollout buffer**: PPO collects a batch of on-policy experience, computes advantages, updates for a few epochs, then discards it.
- **GAE**: blends short- and long-horizon estimates of how much better an action was than average.
- **Logger**: tracks not only returns but Q-value magnitudes, entropy and KL divergence — the early warnings of instability.

Data model

Item	Type	Notes
Transition	(s, a, r, s', done)	Replay buffer entries
Q-network	MLP state → action values	Plus target copy
Policy	MLP state → action distribution	PPO actor
Value function	MLP state → scalar	PPO critic
Run record	config, seed, curves	One row per run in MLflow

Tech stack

Component	Choice	Why
Environments	Gymnasium (Box2D extra for LunarLander)	Standard API
Networks	PyTorch	Small MLPs
Reference	Stable-Baselines3 (check Python 3.14 support)	Sanity comparison
Tracking	MLflow or TensorBoard	Curves across seeds

Milestones

1. **Tabular.** *Done when* Q-learning and SARSA solve a grid world and you can show the learned value table and policy.
2. **DQN on CartPole.** *Done when* it reaches near-max return over 5 seeds, and an ablation without target network or replay shows the instability.
3. **DQN on LunarLander.** *Done when* average return passes 200 (the usual "solved" level) on most seeds.
4. **PPO.** *Done when* PPO solves CartPole and LunarLander and you compare sample efficiency and wall-clock time with DQN.
5. **Reference check.** *Done when* your implementations' curves are in the same range as Stable-Baselines3 with matched hyperparameters.

Deep dive: the hard parts

- **The deadly triad.** Function approximation + bootstrapping (targets from your own estimates) + off-policy learning can diverge. Replay buffers and target networks are DQN's mitigations. Double DQN reduces the over-estimation from taking a max over noisy estimates.
- **Seeds and variance.** RL results vary wildly between seeds. One good curve proves nothing; report mean and confidence bands over at least 5 seeds and never tune on one seed.
- **PPO's clip and details.** The clipped ratio stops the new policy from moving too far from the old one in one update. Small details — advantage normalisation, learning-rate annealing, value-loss clipping, entropy bonus, observation normalisation — matter a lot. Implement them one at a time and measure.
- **Reward design.** The agent optimises exactly what you reward. In custom environments, badly shaped rewards produce surprising "hacks" (spinning in circles to collect a small bonus). Watch the videos, not just the curves.

Testing strategy

- Unit tests: Bellman target computation on a hand-made batch; GAE against a hand-computed short trajectory.
- Sanity environments: a one-step bandit-like environment where the optimal action is known; agents must find it.
- Determinism: same seed and config give identical first 1,000 steps.
- Evaluation separation: evaluation episodes use a separate environment instance with exploration off.

Trade-offs, scaling and failure modes

- DQN (off-policy) reuses data and is sample-efficient for discrete actions; PPO (on-policy) is more stable, handles continuous actions, but needs more environment steps.
- Simulators are cheap; real-world RL is expensive and risky, so production systems often use offline RL or bandits instead.
- Over-fitting to the environment: an agent may fail with small changes (different gravity). Test on perturbed environments.

Stretch goals

- Continuous control (for example, Pendulum or LunarLanderContinuous) with PPO or SAC.
- Distributed rollouts with vectorised environments.
- A custom environment from a real problem (inventory reorder policy) and comparison with a heuristic.

Skills practised

- Ch 14 — Reinforcement Learning: MDPs, Q-learning, SARSA, DQN, policy gradients, actor-critic.
- Ch 10 — Neural Networks and Deep Learning: training stability, optimisers.
- Ch 6 — Evaluation and Tuning: reporting variance across runs.

In the interview

Q: Why does DQN need a replay buffer and a target network? **A:** The replay buffer breaks correlations between consecutive samples and reuses data; the target network keeps training targets stable instead of chasing a moving estimate.

Q: On-policy vs off-policy? **A:** On-policy methods (PPO, SARSA) learn from data generated by the current policy; off-policy methods (Q-learning, DQN) can learn from data from any policy, enabling replay and logged data.

Q: What does PPO's clipping do? **A:** It limits the change in action probabilities per update, preventing destructive large policy steps while allowing several epochs of updates on each batch.

TL;DR

Implement Q-learning, DQN and PPO yourself, report results across seeds, and learn why replay, target networks and clipping exist.

33. Fairness audit of an income model

Level: Advanced · **Time:** 3–4 weeks · **Prerequisites:** Ch 6, 19

What it is

An end-to-end **fairness audit** of a model that predicts whether a person's income exceeds a threshold — a stand-in for lending, hiring or benefits decisions. You train a strong model on US census data, measure its behaviour across groups (race, sex, age band) with several **fairness metrics**, test **mitigations** before, during and after training, and write a **model card** (a short document stating what the model is for, how it was evaluated, and its limitations).

The problem it solves

Models trained on historical data reproduce historical disparities, and an overall accuracy number hides who bears the errors. Regulators, auditors and affected people increasingly expect evidence that a model has been examined. This project teaches you to measure, explain and trade off fairness criteria — knowing that several reasonable definitions cannot all be satisfied at once.

Real-world examples

- Ding, Hardt, Miller and Schmidt's Retiring Adult: New Datasets for Fair Machine Learning introduced Folktables, which builds prediction tasks (such as ACSIncome) from US Census American Community Survey data by state and year.
- Fairlearn provides group metrics (`MetricFrame`) and mitigation algorithms (exponentiated gradient reductions, threshold optimisation).
- ProPublica's Machine Bias investigation of the COMPAS risk score sparked the debate about which fairness criterion should apply.

Requirements

Functional

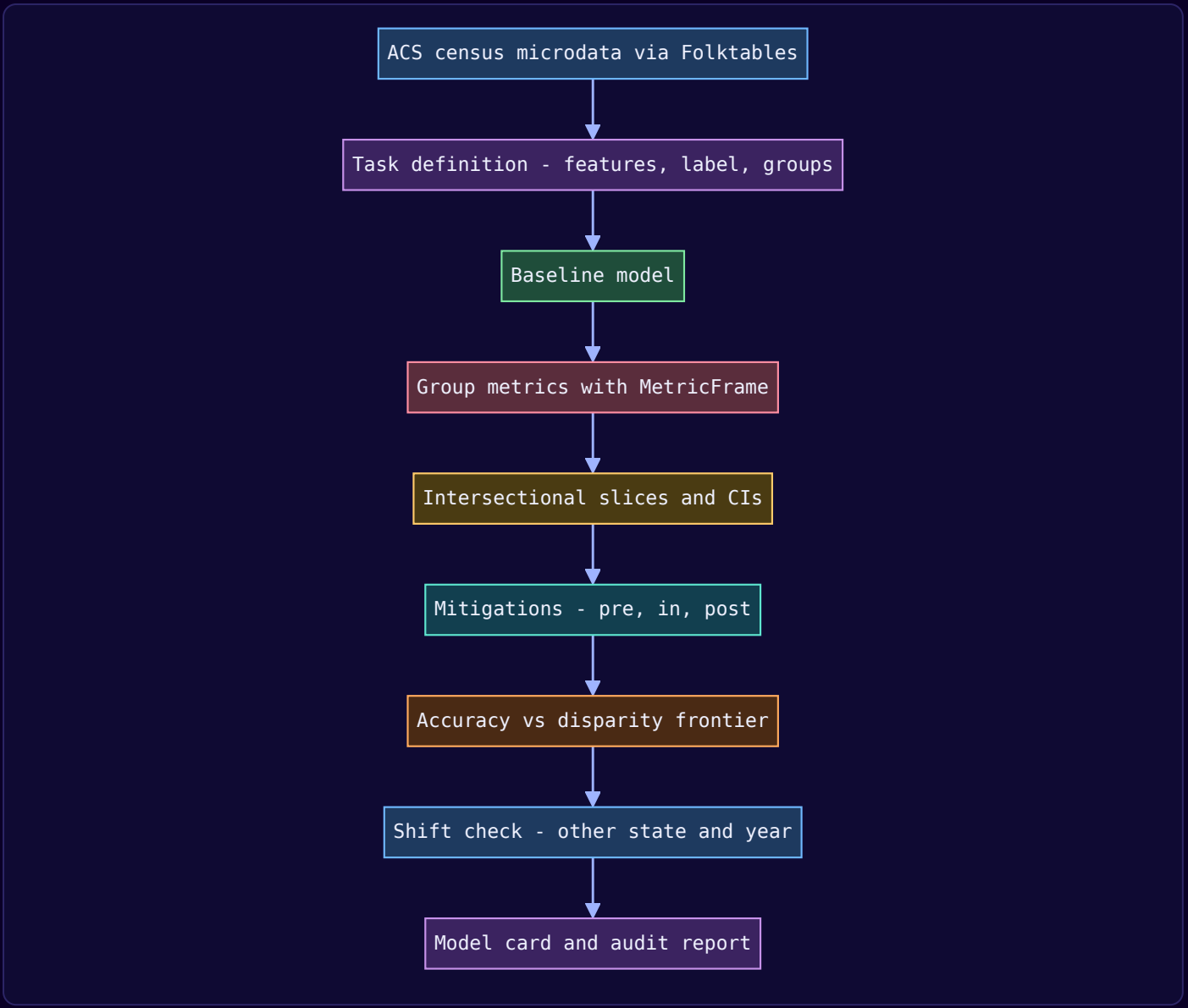
- Data: Folktables ACSIncome for at least two states and two years (to study shift).
- Model: gradient boosting with and without sensitive attributes as inputs.
- Metrics per group: selection rate, true-positive rate, false-positive rate, precision, calibration; disparities as differences and ratios (demographic parity difference, equalised-odds difference).
- Intersectional slices (for example race × sex) with sample sizes and confidence intervals.

- Mitigations: reweighting (pre-processing), exponentiated-gradient reduction (in-processing), group-specific thresholds (post-processing).
- Trade-off plot: accuracy vs disparity for each mitigation.
- Model card and an audit report.

Non-functional

- All metrics with bootstrap 95% confidence intervals; slices under a minimum size flagged rather than over-interpreted.
- Audit reproducible from one command.
- Report readable by a non-technical reviewer in under 10 minutes.

The architecture



- **Task definition**: which features are allowed, what the label means and which attributes are audited (and whether they are model inputs).
- **MetricFrame**: computes any metric per group and the disparities between groups.
- **Mitigations**: pre (change data weights), in (constrain training), post (change thresholds per group).
- **Shift check**: fairness measured in one state and year may not hold in another.
- **Model card**: intended use, out-of-scope uses, metrics by group, limitations, ethical considerations.

Data model

Item	Type	Notes
Features	age, education, occupation, hours, class of worker, marital status, place of birth …	From ACS person records
Sensitive attributes	race, sex, age band	Audited; may or may not be inputs
Label	income above threshold	Binary
Group metrics table	group, metric, value, CI, n	Core audit output
Mitigation results	method, accuracy, disparity metrics	For the frontier plot

Tech stack

Component	Choice	Why
Data	Folktables + pandas	Reproducible census tasks
Model	LightGBM or scikit-learn HistGB	Strong baseline
Fairness	Fairlearn	Metrics and mitigations
Explanations	SHAP (check Python 3.14 wheels)	Which features drive group gaps
Report	Jinja2 HTML or Markdown	Model card

Milestones

1. **Baseline and overall metrics.** *Done when* accuracy, AUC and calibration are reported for one state and year.
2. **Group audit.** *Done when* selection rate, TPR, FPR and calibration per group and intersection are tabulated with CIs.
3. **Diagnose.** *Done when* you can explain *where* gaps come from (base-rate differences, feature proxies, sample size) with evidence.
4. **Mitigate.** *Done when* three mitigations are compared on an accuracy–disparity plot.
5. **Shift and report.** *Done when* metrics are re-measured on another state and year and the model card is written.

Deep dive: the hard parts

- **Incompatible definitions.** Demographic parity (equal selection rates), equalised odds (equal TPR and FPR) and calibration within groups cannot generally all hold when base rates differ between groups. You must choose based on the decision context and say why — this is a value judgement, not a technical one.
- **Removing the attribute is not enough.** Dropping race from the inputs ("fairness through unawareness") leaves proxies such as place of birth or occupation. Measure outcomes by group regardless of inputs.
- **Small groups and uncertainty.** Intersectional groups can be small, so their metrics are noisy. Show confidence intervals and minimum-size rules; do not declare a disparity from 30 people.
- **Post-processing trade-offs.** Group-specific thresholds can equalise error rates but require using the sensitive attribute at decision time, which may be prohibited in some settings. Note legal context as a question for counsel, not something the model builder decides.

Testing strategy

- Metric tests: MetricFrame results match manual per-group calculations.
- Synthetic bias test: plant a known disparity in synthetic data and verify the audit detects it; plant none and verify it reports none (within CIs).
- Reproducibility: report numbers identical across runs with fixed seeds.
- Mitigation sanity: threshold optimiser achieves its target constraint on the data it was fitted to.

Trade-offs, scaling and failure modes

- Reducing disparity usually costs some overall accuracy; the frontier makes the trade explicit for decision-makers.
- Fairness achieved on training data can erode under distribution shift — re-audit on new data regularly.
- Audits can become box-ticking. The value is in the diagnosis and the decision record, not a single number.

Stretch goals

- Counterfactual analysis: change one attribute and its plausible downstream features, measure prediction changes.
- Audit a pre-trained text or image model with the same methodology.
- Add a monitoring job that recomputes group metrics monthly (with Project 31).

Skills practised

- Ch 19 — Responsible ML: fairness definitions, bias sources, mitigations, model cards.
- Ch 6 — Evaluation and Tuning: per-slice metrics, calibration, bootstrap CIs.
- Ch 7 — Decision Trees and Ensembles: the boosted baseline.

In the interview

Q: Name two fairness criteria and why they conflict. **A:** Demographic parity (equal positive rates) and equalised odds (equal error rates). When base rates differ between groups, a model that equalises error rates will have different positive rates, and vice versa.

Q: Is dropping the sensitive attribute enough? **A:** No. Other features act as proxies. You need the attribute (at least for auditing) to measure outcomes by group.

Q: How do you mitigate a disparity you found? **A:** Diagnose first (data, labels, proxies, sample size), then choose among reweighting, constrained training or threshold adjustment, comparing accuracy–disparity trade-offs and documenting the choice.

TL;DR

A fairness audit measures error and selection rates per group with uncertainty, explains the gaps, compares mitigations on an explicit trade-off, and documents the choice in a model card.

34. Visual search ("shop the look")

Level: Advanced · **Time:** 4–6 weeks · **Prerequisites:** Ch 11, 13, 18

What it is

A system where a user uploads a photo (or crops part of one) and gets visually similar products from a catalogue. Images are mapped to **embeddings** by a CNN or vision transformer fine-tuned with **metric learning** (training so that images of the same product are close and different products are far apart), indexed with ANN search, and optionally combined with text through a CLIP-style model so users can refine with words ("same but in red").

The problem it solves

Users often cannot describe what they want in words — a lamp in a photo, a jacket on a stranger. Keyword search fails; visual similarity succeeds. The same machinery powers duplicate detection, near-duplicate copyright checks and "more like this" for image-heavy catalogues.

Real-world examples

- Jing et al.'s Visual Search at Pinterest (KDD 2015) describes building visual search with deep features, distributed indexing and incremental updates at Pinterest scale.
- The Stanford Online Products dataset (about 120,000 images of about 22,600 products) is a standard metric-learning benchmark for product retrieval.

- CLIP (OpenAI, 2021) trained image and text encoders jointly, enabling text-to-image and image-to-image retrieval in a shared space.

Requirements

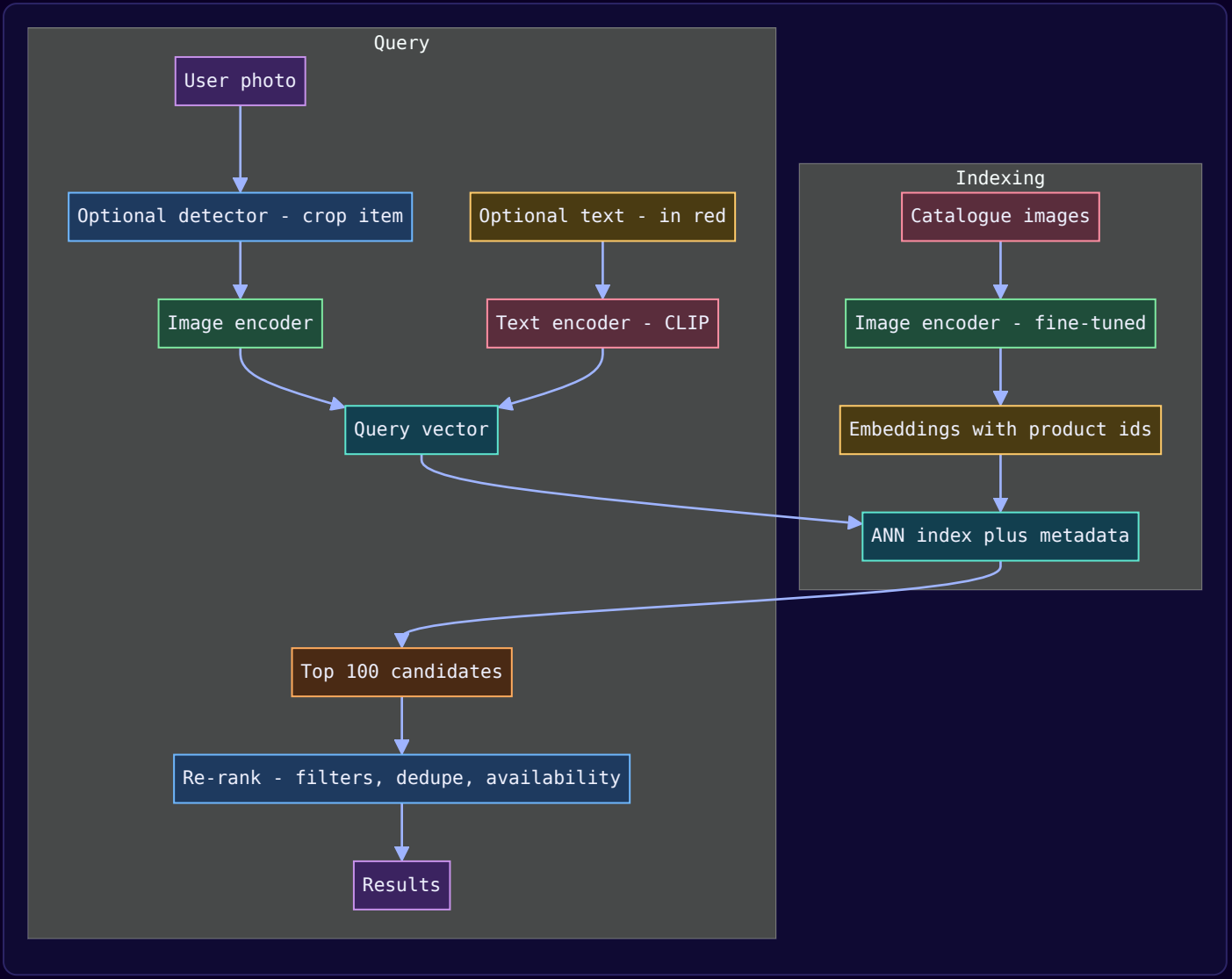
Functional

- Baseline: embeddings from an ImageNet-pre-trained backbone (penultimate layer), cosine search.
- Fine-tune with a metric-learning loss (triplet loss with hard-negative mining, or a proxy-based or contrastive loss) on Stanford Online Products, splitting by **product class** so test products are unseen.
- Metrics: Recall@1, @10, @100 (does a same-product image appear in the top k?).
- ANN index (FAISS or pgvector) with product metadata filters (category, price range).
- API: `POST /search/image` with optional crop box and text refinement; returns products with similarity scores.
- Optional object detection step to crop the main item from a busy photo (Project 19 skills).

Non-functional

- Query latency under 200 ms p95 including embedding on CPU (GPU optional).
- Index of 1 million items under 4 GB RAM (use reduced dimension or product quantisation if needed).
- New products searchable within an hour of upload.

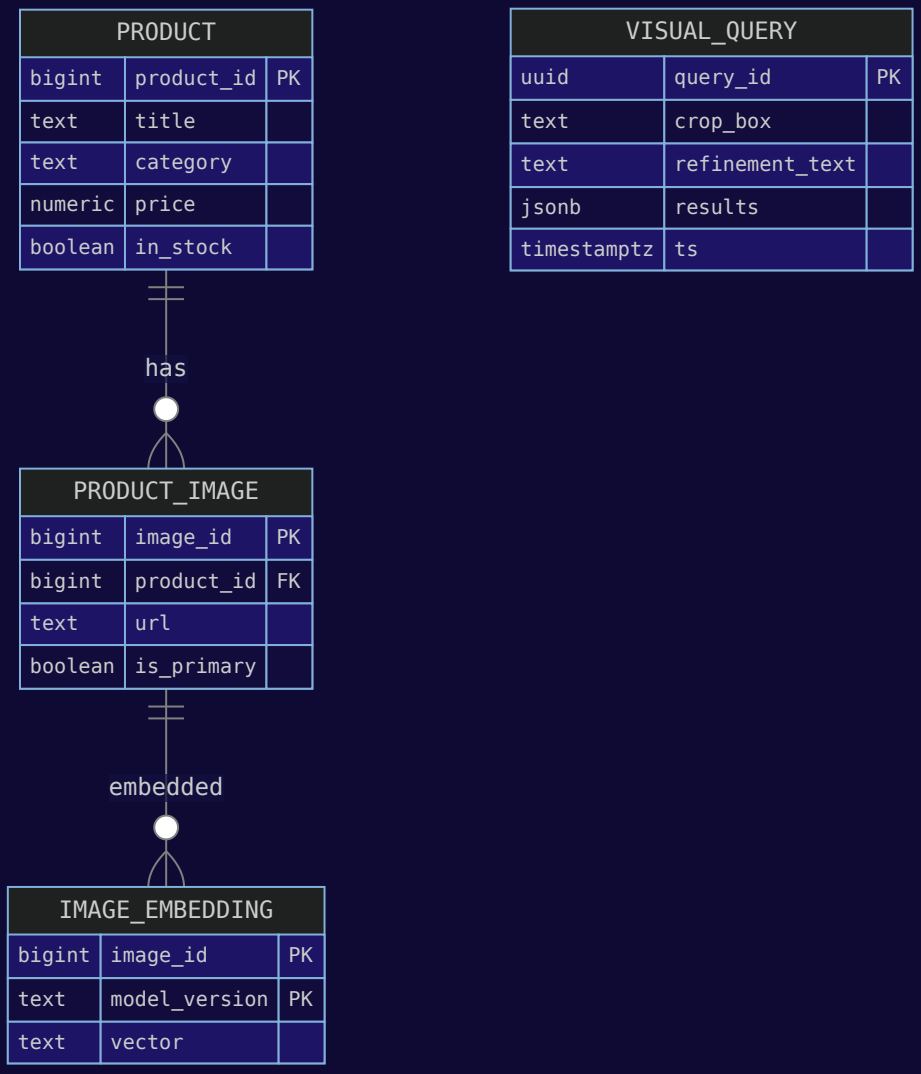
The architecture



- **Image encoder**: the same model and preprocessing for catalogue and queries — essential.
- **Detector crop**: user photos contain scenes; cropping the main item improves matches dramatically.
- **Text refinement**: with a joint image–text model, add a scaled text vector to the image vector.

- **Re-rank**: removes out-of-stock items and near-duplicate listings, applies filters.

Data model



Tech stack

Component	Choice	Why
Encoder	PyTorch + torchvision backbone or a ViT	Fine-tunable
Joint text–image	an open CLIP implementation	Text refinement
ANN	FAISS (or pgvector HNSW)	Fast similarity search
Detector	torchvision Faster R-CNN	Crop main item
API	FastAPI	Upload and search

Milestones

1. **Pre-trained baseline.** *Done when* Recall@1/10 on Stanford Online Products test classes is recorded with off-the-shelf embeddings.
2. **Metric learning.** *Done when* fine-tuning with hard-negative mining improves Recall@1 substantially on unseen products.
3. **Index.** *Done when* ANN recall vs exact search and memory usage are measured for full and compressed vectors.

4. **API with crops.** *Done when* uploading a cluttered photo, cropping and searching returns relevant products in the latency budget.
5. **Text refinement.** *Done when* "same but in red" style queries shift results measurably on a small hand-labelled set.

Deep dive: the hard parts

- **Mining hard negatives.** Random negatives are too easy, so the loss quickly goes to zero and learning stops. Mine negatives that the current model confuses (similar-looking different products) within each batch, but avoid the hardest ones early, which are often label noise.
- **Splitting by product.** The real task is finding products the model never saw during training. Split by product class, never by image, or results are inflated.
- **Domain gap between catalogue and user photos.** Catalogue images are clean, on white backgrounds; user photos are cluttered, angled, badly lit. Augment heavily, crop with a detector, and if possible collect "street-to-shop" pairs.
- **Compression.** 1 million × 512 floats × 4 bytes is about 2 GB. Product quantisation (store vectors as short codes) cuts memory several-fold at some recall cost — measure it.

Testing strategy

- Retrieval metric tests on a toy set with known matches.
- Preprocessing parity: catalogue and query pipelines produce identical embeddings for the same image.
- Index consistency: every in-stock product has an embedding for the current model version.
- Load test with image uploads of realistic sizes.

Trade-offs, scaling and failure modes

- Larger encoders improve recall but cost GPU time for both indexing and queries.
- Re-embedding the whole catalogue on each model change is expensive; version embeddings and migrate in the background.
- Visual similarity can surface counterfeits or inappropriate content; apply policy filters in re-ranking.

Stretch goals

- Multi-object queries: detect several items and search each.
- Learn from clicks: fine-tune with (query, clicked product) pairs.
- Duplicate-listing detection using the same embeddings.

Skills practised

- Ch 11 — CNNs and Computer Vision: backbones, transfer learning, detection.
- Ch 13 — Recommender Systems and Embeddings: embeddings, ANN search.
- Ch 18 — ML Case Studies II: the visual search case study.

In the interview

Q: How do you train image embeddings for retrieval? **A:** Start from a pre-trained backbone and fine-tune with a metric-learning loss (triplet, contrastive or proxy-based), mining hard negatives, evaluated with Recall@k on unseen products.

Q: How do you scale to a billion images? **A:** ANN indexes with compression (product quantisation), sharding by category or hash, distributed index builds and incremental updates for new items.

Q: User photos look very different from catalogue photos. What do you do? **A:** Crop with a detector, augment training images toward user-photo conditions, and fine-tune on street-to-shop pairs or click data.

TL;DR

Fine-tune an image encoder with metric learning, split by product, index with ANN, and close the gap between catalogue and user photos with crops and augmentation.

35. People You May Know (link prediction)

Level: Advanced · **Time:** 4–6 weeks · **Prerequisites:** Ch 13, 18

What it is

A friend-recommendation system for a social graph: predict which pairs of users who are not yet connected are likely to connect. **Candidate generation** uses graph structure (friends of friends, shared groups, embeddings from random walks), and a **ranker** scores each candidate pair with features such as number of mutual friends, **Adamic–Adar** score (mutual friends weighted by how selective each mutual friend is), and embedding similarity. You evaluate on future edges with a time-based split.

The problem it solves

Social networks grow when people find others they know. Searching manually is slow, and recommending random people is spam. The graph itself holds strong hints — people with many mutual friends usually know each other — but naive friends-of-friends lists are huge for popular users and must be ranked.

Real-world examples

- Grover and Leskovec's node2vec (KDD 2016) learns node embeddings from biased random walks and evaluates them on link prediction.
- Ying et al.'s PinSage (Pinterest, KDD 2018) applied graph convolutional networks to web-scale recommendation.
- The Open Graph Benchmark link prediction tasks (for example ogbl-collab, a time-evolving collaboration network) and the SNAP datasets provide real graphs with standard evaluation.

Requirements

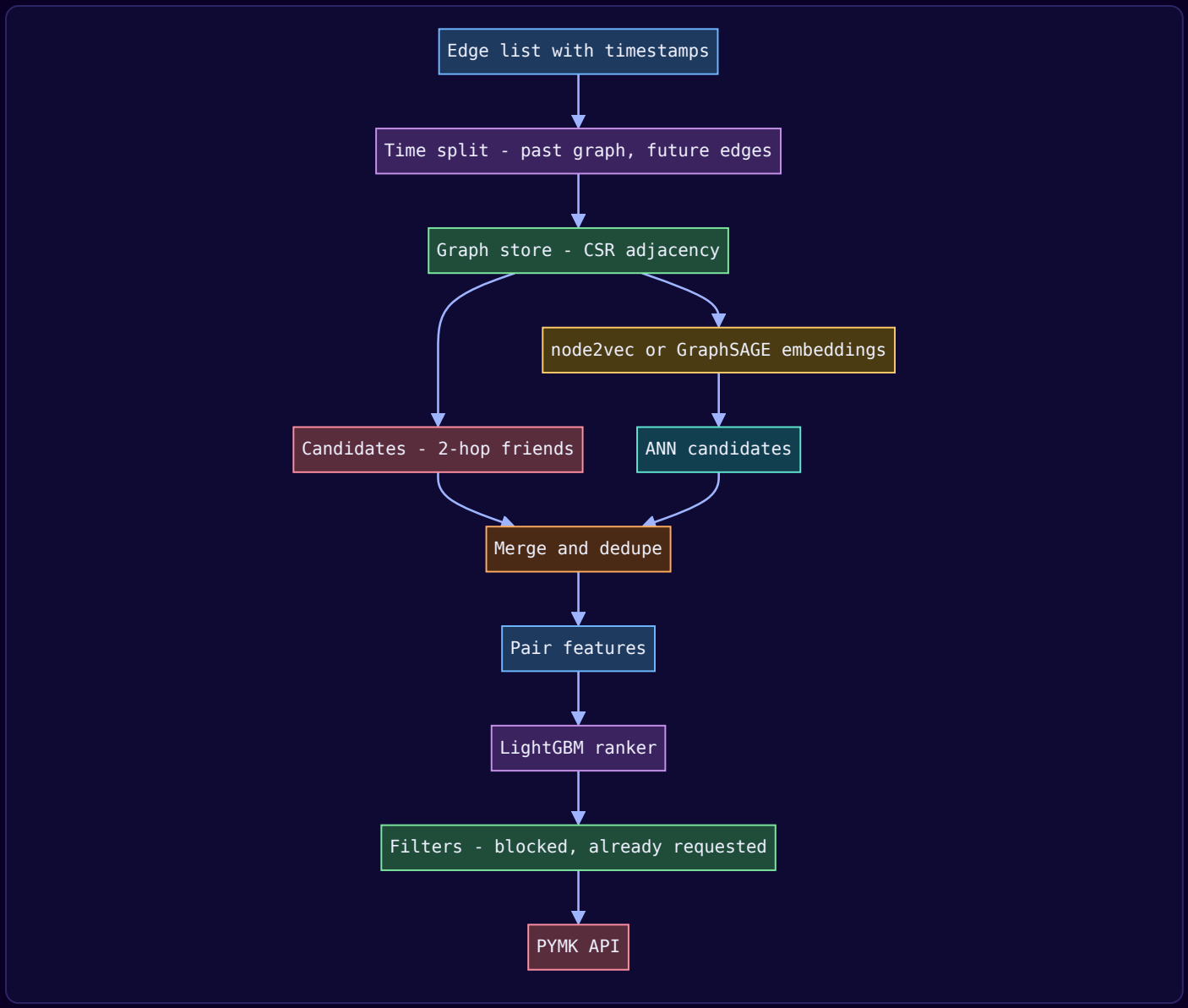
Functional

- Data: ogbl-collab (or a SNAP social graph with timestamps); split edges by time.
- Candidates: friends-of-friends (2-hop), capped per user; plus nearest neighbours in node2vec embedding space.
- Features per pair: mutual-friend count, Jaccard, Adamic–Adar, degrees, embedding cosine, shared attributes.
- Ranker: LightGBM (binary) on sampled candidate pairs; compare with heuristic-only ranking.
- Stretch model: a graph neural network (GraphSAGE) for embeddings.
- Metrics: Hits@50 (OGB style: the share of true future edges that score higher than all but at most 49 of a fixed set of negative pairs), Recall@k per user, and coverage.
- API: `GET /pymk/{user_id}` returning 20 suggestions with "N mutual friends" explanation.

Non-functional

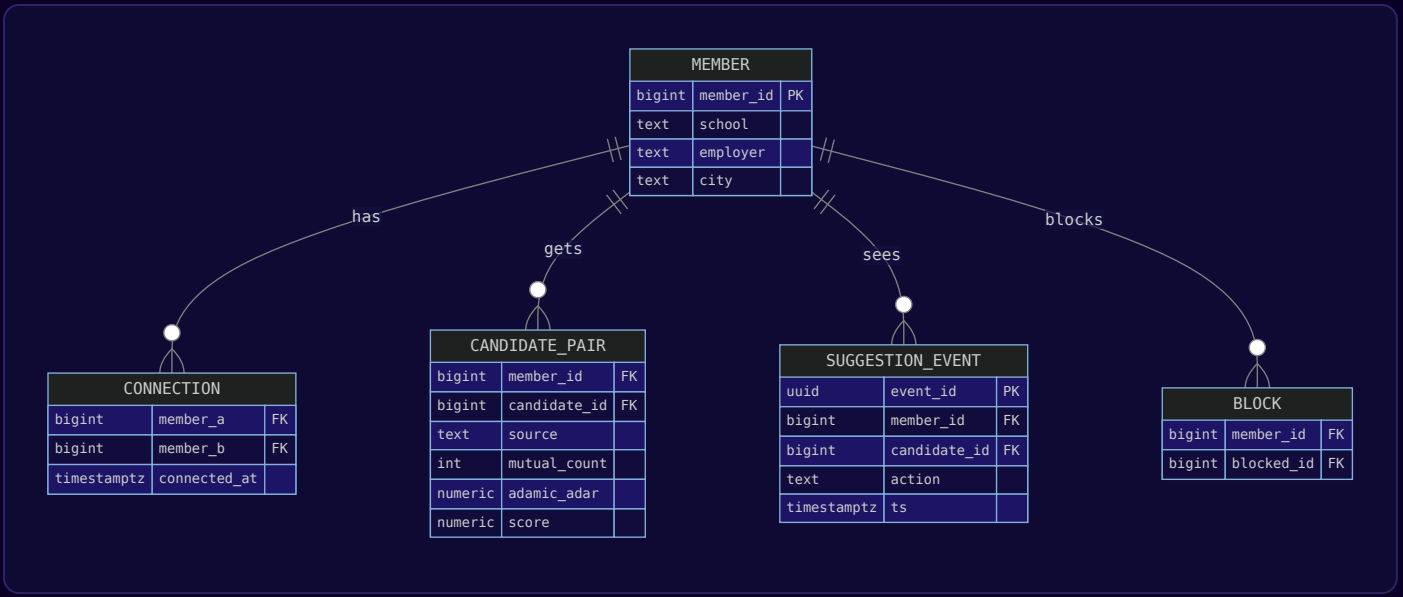
- Candidate generation for one user under 50 ms (precomputed for heavy users).
- Daily batch refresh of embeddings and candidates for the whole graph.
- Memory: adjacency of a 1-million-node graph held in a compressed sparse format.

The architecture



- **Time split**: build the graph from edges before a date; positives are edges formed after.
- **CSR adjacency**: compressed sparse row — a fast, compact way to store neighbour lists.
- **2-hop candidates**: friends of friends ranked by mutual count, capped (popular users have huge 2-hop neighbourhoods).
- **Embedding candidates**: find people who are structurally similar but not within 2 hops.
- **Filters**: never suggest blocked users, existing connections or pending requests.

Data model



Tech stack

Component	Choice	Why
Graph	scipy sparse CSR, NetworkX for small checks	Fast neighbour operations
Embeddings	PyTorch (node2vec), PyTorch Geometric for GNNs (check Python 3.14 support)	Learned structure
Ranker	LightGBM	Pair features
Storage	PostgreSQL	Members, edges, suggestions
API	FastAPI	Suggestions with explanations

Milestones

1. **Heuristic baseline.** *Done when* mutual-count and Adamic–Adar rankings are evaluated with Hits@50 on future edges.
2. **Embeddings.** *Done when* node2vec embeddings are trained on the past graph and cosine-similarity ranking is evaluated.
3. **Ranker.** *Done when* a LightGBM model on combined features beats each heuristic.
4. **Serving.** *Done when* the API returns suggestions under 50 ms with mutual-friend explanations and filters.
5. **GNN (stretch-level milestone).** *Done when* GraphSAGE embeddings are compared with node2vec as features.

Deep dive: the hard parts

- **Leakage through the graph.** If a future edge is present when computing features (mutual counts, embeddings), the model sees the answer. Build every feature from the past graph only, and remove the target edge when computing its own features.
- **Hub explosion.** A user with 5,000 friends whose friends also have thousands has millions of 2-hop candidates. Cap by sampling, by mutual count thresholds, or precompute with a batch job for heavy users.
- **Negative sampling.** Most pairs never connect. Sample negatives from the candidate set itself (hard negatives that share friends but did not connect), not random pairs, or the ranker learns something trivially easy.
- **Asymmetric harms.** Suggesting an ex-partner, an abuser or a patient to a doctor can cause real harm. Add privacy controls (opt-out from being suggested), sensitive-context filters and quick blocking.

Testing strategy

- Leakage test: no target edge exists in the training graph; features for a pair are unchanged whether or not its future edge exists in the database.
- Heuristic unit tests on a tiny hand-drawn graph with known Adamic–Adar values.

- Filter tests: existing connections and blocked users never appear.
- Performance test for a hub user.

Trade-offs, scaling and failure modes

- Heuristics are cheap, explainable and surprisingly strong; embeddings add reach beyond 2 hops; GNNs combine structure and attributes but are expensive to train at scale.
- Graph data changes constantly; nightly batch suggestions go stale for very active users — add a light online refresh after new connections.
- Engagement-optimised PYMK can amplify existing clusters and isolate others; track suggestion diversity.

Stretch goals

- Online update: after a user accepts a suggestion, refresh their candidates immediately.
- Multi-objective: predict "send request" and "request accepted" and combine.
- Explanations beyond mutual friends ("you both worked at X").

Skills practised

- Ch 18 — ML Case Studies II: the PYMK case study.
- Ch 13 — Recommender Systems and Embeddings: embeddings, candidate generation and ranking.
- Ch 7 — Decision Trees and Ensembles: pairwise feature ranking.
- Ch 19 — Responsible ML: privacy risks of recommendations.

In the interview

Q: What's the simplest strong baseline for friend recommendations? **A:** Friends of friends ranked by mutual count or Adamic–Adar. Mutual connections are the strongest single signal.

Q: How do you avoid leakage in link prediction? **A:** Split edges by time, build the graph and all features from the past only, and evaluate on edges that formed later.

Q: How do you scale candidate generation for hub users? **A:** Cap and sample 2-hop expansion, precompute in batch, and add embedding-based ANN candidates instead of exhaustive traversal.

TL;DR

Generate candidates from 2-hop neighbours and embeddings, rank pairs with graph features, split by time, and guard against both leakage and harmful suggestions.

36. Trip ETA prediction

Level: Advanced · **Time:** 4–6 weeks · **Prerequisites:** Ch 12, 16

What it is

A service that predicts how long a taxi or delivery trip will take — the **ETA** (estimated time of arrival) — from origin, destination, time of day, and live traffic conditions. Following the pattern used by large ride-hailing companies, a simple routing estimate (distance-based or from a routing engine) gives a first guess, and an ML model predicts the **residual**: how much the real trip will differ from that guess. You build features, a gradient-boosting model and a neural model with embeddings for locations, serve it within strict latency, and monitor accuracy as conditions change.

The problem it solves

ETAs drive pricing, dispatch (which driver to send), customer expectations and courier batching. A few minutes of systematic error causes missed pickups, cold food and lost trust. Routing engines know the roads but not the messy reality (pickup delays, building entrances, local traffic patterns); ML learns those corrections from millions of completed trips.

Real-world examples

- Uber's DeepETA: How Uber Predicts Arrival Times Using Deep Learning describes predicting the residual on top of the routing engine's ETA, replacing gradient-boosted tree ensembles with a low-latency self-attention (linear transformer) network over discretised, embedded features.
- The NYC Taxi and Limousine Commission trip record data publishes pickup and drop-off times and zones for millions of trips per month; Kaggle's NYC Taxi Trip Duration offers a smaller version with coordinates.

Requirements

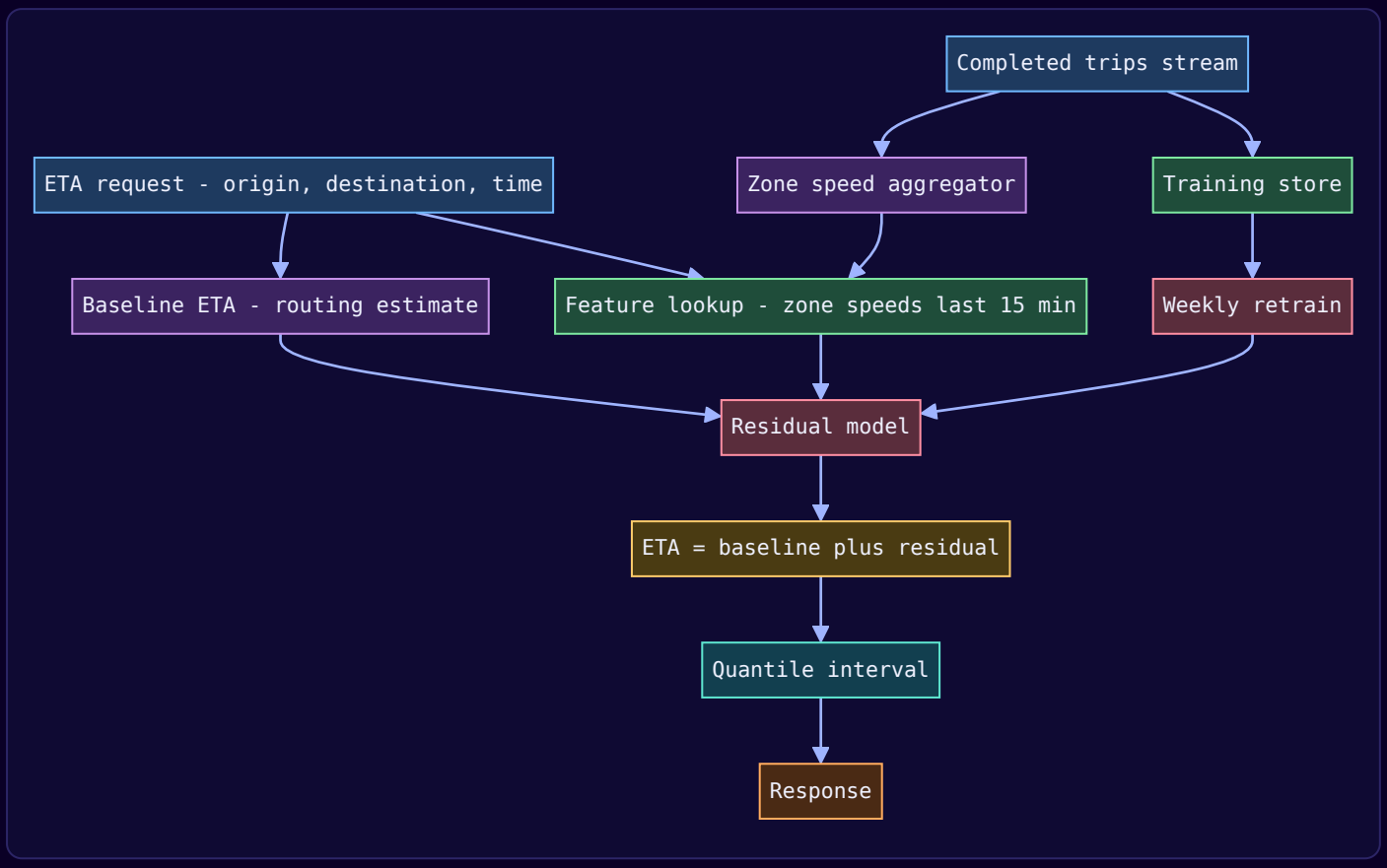
Functional

- Data: several months of NYC TLC trips (zone-level) or the Kaggle set (coordinates); clean impossible trips.
- Baseline ETA: haversine distance / zone-pair median speed by hour (stands in for a routing engine).
- Features: origin and destination zones, hour, weekday, holiday, weather, recent average speeds in the origin zone (streaming-style feature), baseline ETA.
- Models: LightGBM on residual; neural model with zone and time embeddings and an MLP (or small self-attention over feature embeddings).
- Loss: try MAE, Huber, and an **asymmetric** loss (being late is worse than early).
- Metrics: MAE, MAPE (mean absolute percentage error), error percentiles (p50, p90), and calibration of quantile predictions; slices by distance, borough, hour.
- API: `POST /eta` returns point ETA and p10–p90 interval.

Non-functional

- Inference under 10 ms p99 (ETA requests happen many times per trip).
- Fresh traffic features at most 5 minutes old.
- Model retrained weekly; daily accuracy dashboard.

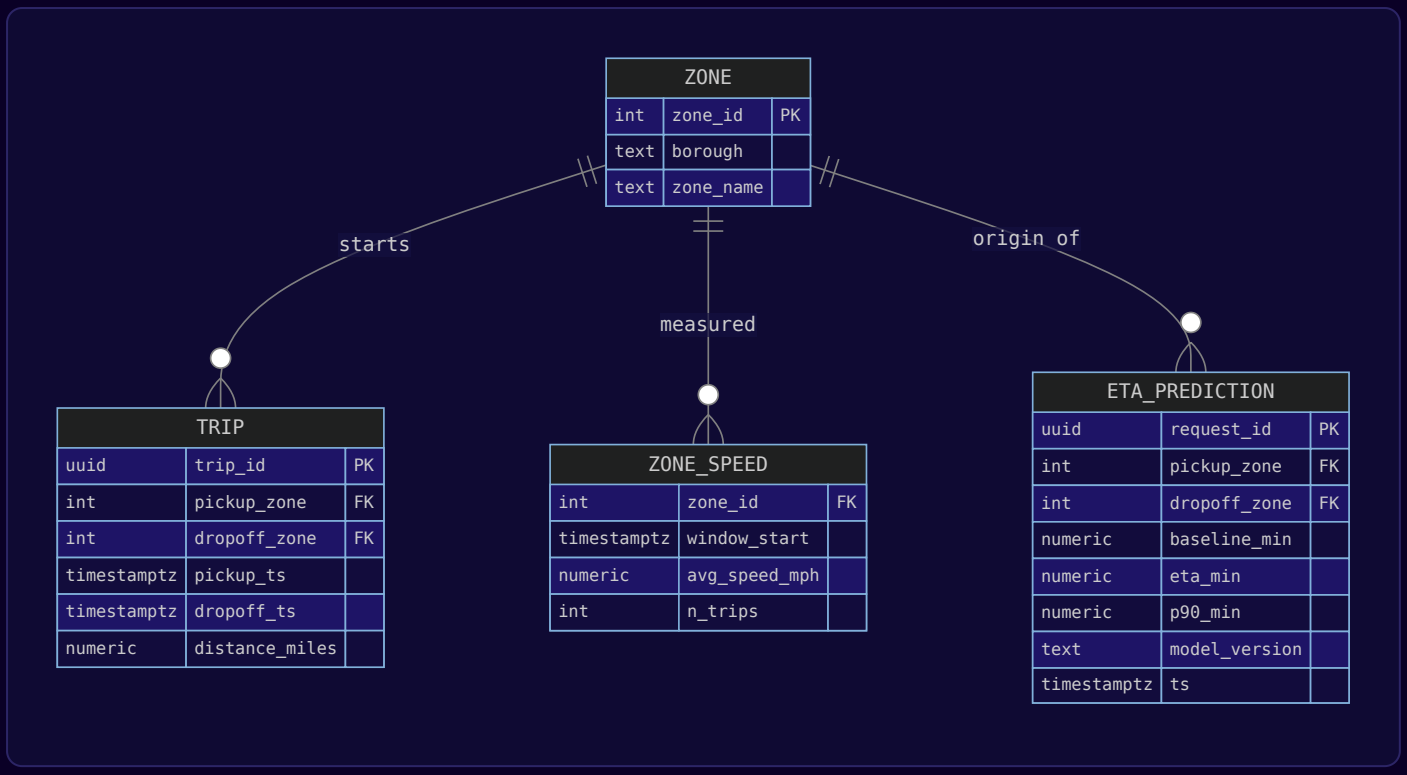
The architecture



- **Baseline ETA**: deterministic estimate; ML only learns corrections, which makes the problem easier and safer (fall back to baseline if the model fails).

- **Zone speed aggregator**: rolling average speeds of recently completed trips per zone and hour — a proxy for live traffic.
- **Residual model**: predicts actual minus baseline.
- **Quantile interval**: separate quantile heads or models give p10 and p90.

Data model



Tech stack

Component	Choice	Why
Data	polars + Parquet	Monthly files with millions of rows
Tree model	LightGBM	Strong residual baseline, quantile loss
Neural model	PyTorch with embeddings	Zone, time embeddings, asymmetric loss
Live features	Redis or PostgreSQL rolling table	Fresh zone speeds
Serving	FastAPI + ONNX Runtime	Low latency
Monitoring	daily SQL over predictions joined to actual trips	Accuracy tracking

Milestones

1. **Clean and baseline.** *Done when* outliers (zero distance, negative or multi-hour durations) are filtered with documented rules and the baseline ETA's MAE by hour is reported.
2. **Residual GBDT.** *Done when* LightGBM on residuals beats the baseline on MAE and p90 error with a time-based split.
3. **Live features.** *Done when* point-in-time zone speeds improve accuracy, especially at rush hours, without leakage.
4. **Neural model and losses.** *Done when* an embedding model is compared with GBDT and an asymmetric loss shifts errors toward early predictions as intended.
5. **Serve and monitor.** *Done when* the API meets 10 ms p99 with fallback to baseline, and a daily report compares predictions with actual durations.

Deep dive: the hard parts

- **Why predict the residual.** The baseline already captures most of the variance (distance). Learning the correction is an easier target, keeps predictions physically sensible, and degrades gracefully: if the model breaks, the baseline still works.
- **Leakage in live features.** "Average zone speed in the last 15 minutes" must use only trips *completed* before the request time, not trips that started before it but ended later. Get this wrong and offline accuracy is fantasy.
- **Asymmetric costs and quantiles.** Customers forgive early arrival more than late. Train with an asymmetric Huber or quantile loss, and give an interval rather than a falsely precise number. Check that 80% of trips fall inside the p10–p90 range.
- **Heavy tails.** A few trips take far longer (accidents, detours, waiting with the meter on). Squared loss chases them; MAE or Huber is more robust. Report p90 error so tail performance is visible.

Testing strategy

- Point-in-time test for zone-speed features on hand-made trip timelines.
- Baseline fallback test: model service down → API returns baseline with a flag.
- Interval coverage test on the latest week.
- Latency test with batch and single requests.
- Slice regression tests: no slice (airport trips, long trips) gets more than X% worse on a new model.

Trade-offs, scaling and failure modes

- Zone-level data is coarse; coordinate-level data and real routing engines (road segments) give better baselines but need map infrastructure.
- Events (marathons, storms) create rare conditions with little training data; fall back toward baseline plus live speeds.
- ETAs influence behaviour (customers choose based on ETA), creating feedback in the data — keep an eye on systematic bias by product.

Stretch goals

- Feature embeddings over discretised coordinates (hashed grid cells at several resolutions) and a self-attention layer over feature embeddings, in the spirit of DeepETA.
- Multi-task: predict pickup ETA and trip duration jointly.
- Conformal prediction for intervals with coverage guarantees.

Skills practised

- Ch 16 — ML in Production: real-time features, low-latency serving, monitoring, fallbacks.
- Ch 12 — Sequences, RNNs and Attention: embeddings and attention over features, time-series aggregation.
- Ch 7 — Decision Trees and Ensembles: gradient boosting with quantile and custom losses.
- Ch 6 — Evaluation and Tuning: MAE vs RMSE, percentiles, slices.

In the interview

Q: Why predict a residual on top of a routing ETA? **A:** The routing engine captures physics (distance, road speeds); ML learns systematic corrections. It is an easier target, more robust, and has a natural fallback.

Q: Which loss would you use? **A:** MAE or Huber for robustness to heavy tails, asymmetric if lateness costs more, and quantile losses for intervals.

Q: How do you keep latency under 10 ms? **A:** Pre-compute and cache features, use a compact model (trees or a small MLP exported to an optimised runtime), avoid network calls in the request path except one feature lookup, and batch where possible.

TL;DR

Predict the correction to a routing ETA with fresh, leak-free traffic features, a robust or asymmetric loss and honest intervals, served within milliseconds with a baseline fallback.

37. Session-based next-item recommender

Level: Advanced · **Time:** 4–5 weeks · **Prerequisites:** Ch 12, 13

What it is

A recommender that predicts the next item a user will view or buy from the sequence of items in their **current session** — useful when most visitors are anonymous and have no long history. You compare simple co-occurrence baselines, a GRU-based model (GRU4Rec style), and a **self-attentive** transformer model (SASRec), and serve "next item" recommendations that update after every click.

The problem it solves

On many e-commerce and media sites, most sessions are from logged-out or new users, so user-level collaborative filtering has nothing to work with. But the order of clicks within a session is strongly informative: someone who viewed a tent and then sleeping bags probably wants camping gear next. Sequence models capture that order.

Real-world examples

- Kang and McAuley's Self-Attentive Sequential Recommendation (SASRec, ICDM 2018) applied causal self-attention to item sequences and outperformed recurrent and Markov-chain baselines on several datasets.
- The RetailRocket e-commerce dataset contains view, add-to-cart and transaction events from a real online shop over about four and a half months.
- Moreira et al.'s News Session-Based Recommendations using Deep Neural Networks applies session-based recommendation to news, where items expire quickly.

Requirements

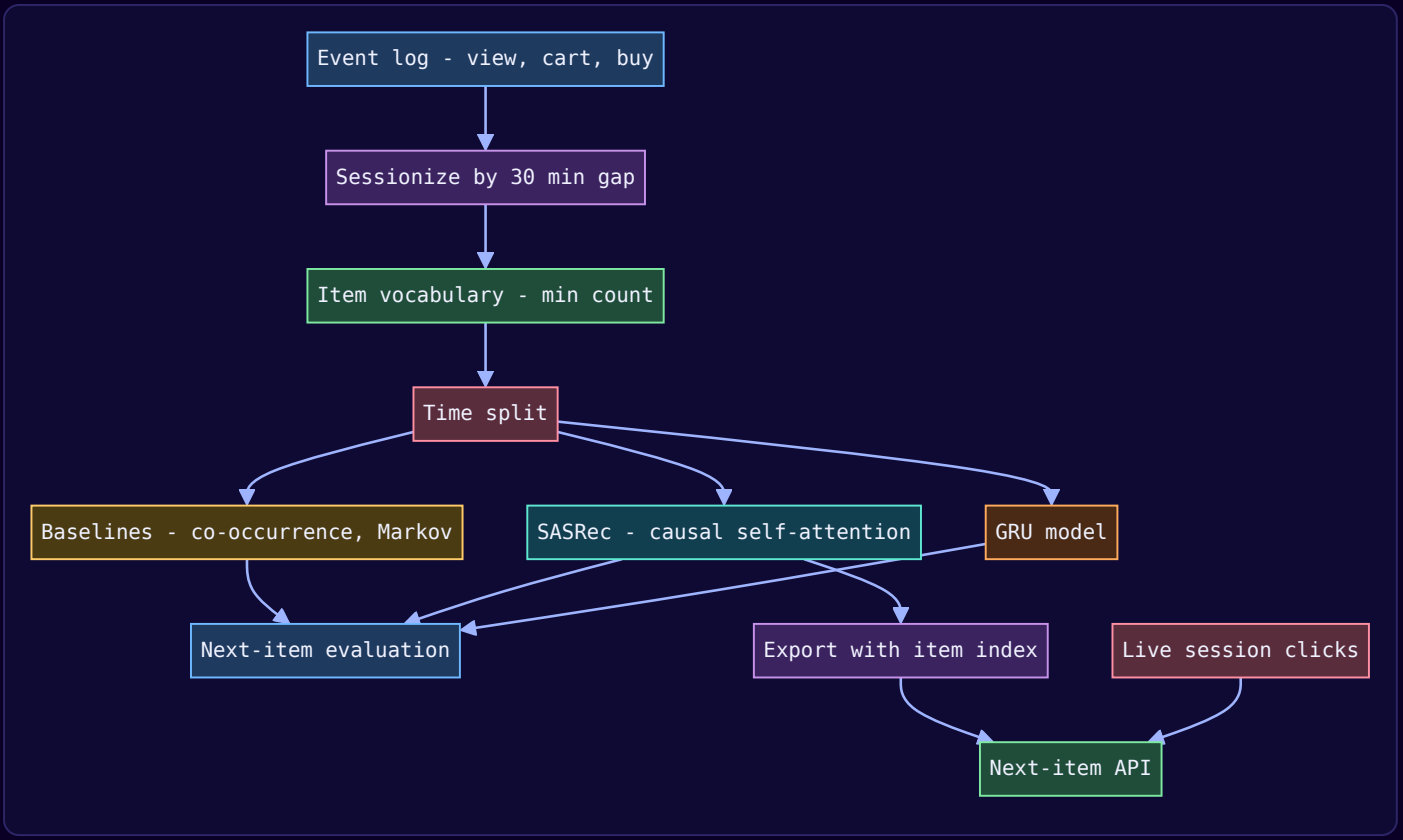
Functional

- Sessionise events (split a visitor's events where the gap exceeds 30 minutes); drop sessions of length 1 and very rare items.
- Time-based split: train on earlier sessions, test on the last weeks.
- Baselines: most popular, item-to-item co-occurrence ("viewed together"), first-order Markov (next-item transition counts).
- Models: GRU-based and SASRec in PyTorch, trained to predict the next item at every position.
- Metrics: Recall@20 and MRR@20 (mean reciprocal rank: 1 divided by the true next item's position, counted as 0 if it is outside the top 20) for the next item, evaluated at every step of test sessions.
- Serving: `POST /next` with the current session's item list returns top 20 items.

Non-functional

- Inference under 30 ms for sessions up to 50 items (CPU).
- Model refresh daily to include new items.
- Unknown items in the session are skipped gracefully.

The architecture



- **Sessionise**: turns a raw click stream into sequences.
- **SASRec**: item embeddings plus position embeddings, stacked causal self-attention blocks (each position may attend only to earlier positions), output scored against all item embeddings.
- **Evaluation**: for each prefix of a test session, predict the next item and check rank.
- **API**: stateless — the client (or a session cache) sends recent items; the model returns scores.

Data model

Item	Type	Notes
Event	visitor ID, item ID, event type, timestamp	Raw log
Session	ordered list of item IDs	Built by gap rule
Item index	dict item → int	Built from training data, min-count threshold
Training sample	input sequence, target sequence (shifted by 1)	Padded to max length
Model output	scores over all items	Top-k after masking current session items (optional)

Tech stack

Component	Choice	Why
Data	polars	Sessionisation with window functions
Models	PyTorch (`nn.TransformerEncoder` with causal mask, `nn.GRU`)	Sequence models
Evaluation	your own next-item harness	Exact protocol control
Serving	FastAPI + ONNX or TorchScript	Fast inference

Milestones

1. **Sessions and baselines.** *Done when* session length distribution is plotted and co-occurrence and Markov baselines have Recall@20 and MRR@20.

2. **GRU model.** *Done when* a GRU next-item model beats baselines on the same harness.
3. **SASRec.** *Done when* SASRec is trained with a causal mask and compared; you report the effect of max sequence length and number of blocks.
4. **Negative sampling vs full softmax.** *Done when* you compare sampled-softmax training with full softmax on speed and accuracy.
5. **Serve.** *Done when* the API returns updated recommendations after each simulated click within latency.

Deep dive: the hard parts

- **Causal masking.** During training, position t must not see positions after t. Forgetting the mask makes training loss look superb and test results collapse. Write a test that changes a future item and checks earlier outputs do not change.
- **Large item vocabularies.** Scoring every item at every position is expensive with hundreds of thousands of items. Sampled softmax or in-batch negatives make training feasible; at serving, compute full scores once per request (or use ANN).
- **Evaluation protocol choices.** Many papers evaluate against 100 sampled negatives, which inflates scores and can change model rankings. Evaluate against the full catalogue, and report what you did.
- **Repeat consumption.** In retail, users often re-view the same item. Decide whether to recommend items already in the session; measure both versions.

Testing strategy

- Causality test (as above).
- Sessionisation tests on hand-made event streams with known gaps.
- Metric harness test: a perfect oracle gets Recall@20 = 1.
- Time split test: no test-session timestamp precedes the training cutoff.

Trade-offs, scaling and failure modes

- Co-occurrence baselines are strong, cheap and easy to update in real time; neural models add accuracy but need retraining to learn new items.
- New items have no embeddings until retraining — add content features or blend with co-occurrence updated hourly.
- Sequence models can over-focus on the last click; check diversity of recommendations.

Stretch goals

- Combine session model with a long-term user embedding for logged-in users.
- Bidirectional training with masked item prediction (BERT4Rec style) and comparison with SASRec.
- Use add-to-cart and purchase events as weighted targets.

Skills practised

- Ch 12 — Sequences, RNNs and Attention: GRUs, self-attention, causal masks, transformers.
- Ch 13 — Recommender Systems and Embeddings: item embeddings, ranking metrics, cold start.
- Ch 6 — Evaluation and Tuning: Recall@k, MRR, honest evaluation protocols.

In the interview

Q: Why a sequence model instead of matrix factorisation? **A:** Most sessions are anonymous and short, and order matters: what you clicked last is highly predictive of what comes next. Sequence models use that order; MF needs user history.

Q: What does the causal mask do in SASRec? **A:** It prevents each position from attending to later positions, so the model learns to predict the next item only from the past, matching how it will be used.

Q: Why can sampled-negative evaluation be misleading? **A:** Ranking against 100 random items is much easier than against the full catalogue and can reorder which model looks best. Use full-catalogue metrics when possible.

TL;DR

Turn clicks into sessions, beat strong co-occurrence baselines with a causally masked self-attention model, and evaluate against the full catalogue.

38. Retrieval-augmented question answering with evaluation

Level: Advanced · **Time:** 4–6 weeks · **Prerequisites:** Ch 12, 13, 19

What it is

A **retrieval-augmented generation** (RAG) system: a user asks a question, the system retrieves relevant passages from a document collection, and a language model writes an answer grounded in those passages with citations. The emphasis of this capstone is the **evaluation harness** — measuring retrieval quality, answer faithfulness (does the answer only claim what the sources support?) and answer correctness — so that every change (chunk size, retriever, prompt, model) is judged by numbers, not by impressions.

The problem it solves

Language models alone can produce fluent but unsupported answers ("hallucinations") and do not know your private or recent documents. RAG grounds answers in retrieved text you control. But RAG systems have many moving parts, and without systematic evaluation teams cannot tell whether a change helped.

Real-world examples

- Lewis et al.'s Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (2020) introduced the RAG formulation combining a dense retriever with a generator.
- Ragas proposed automated metrics for RAG pipelines, such as faithfulness and context relevance; the Ragas docs describe them.
- BEIR, MS MARCO and Natural Questions provide labelled queries for retrieval and QA evaluation.

Requirements

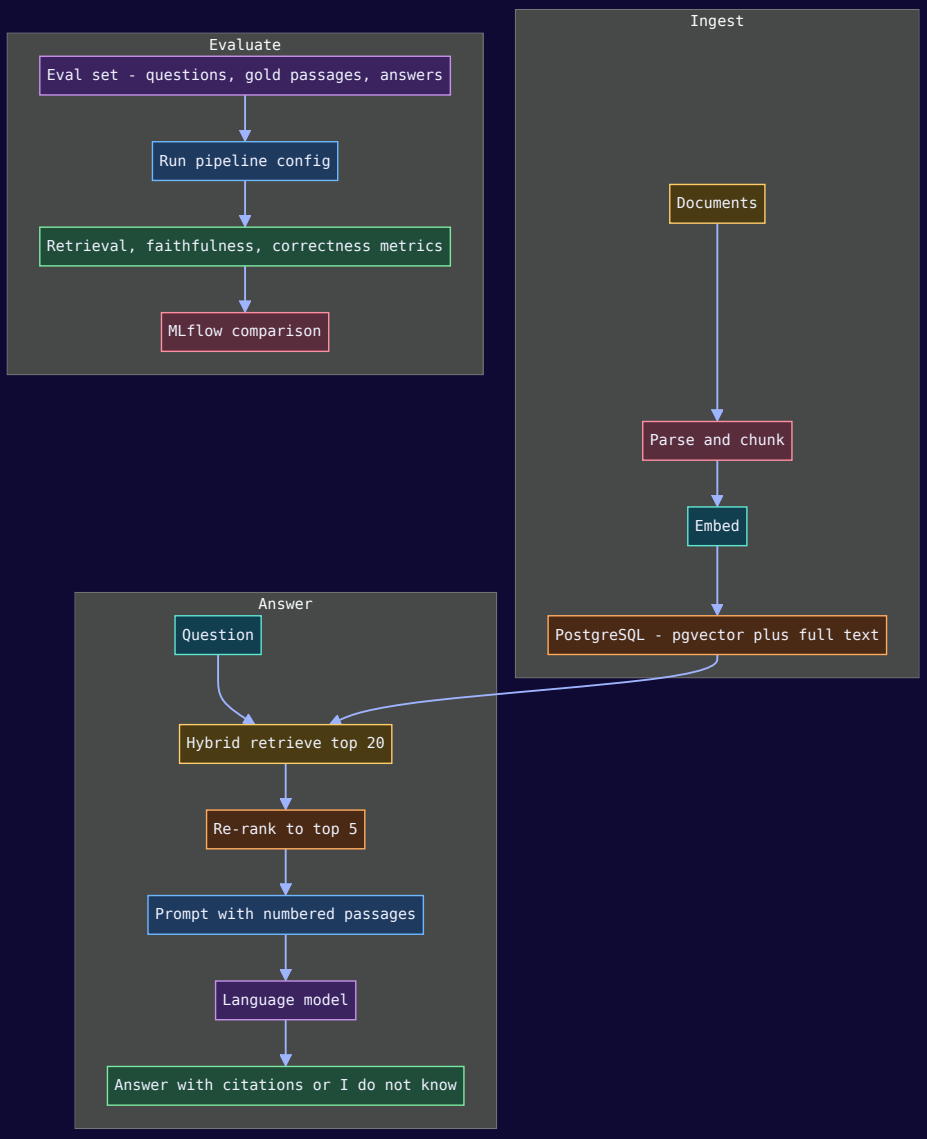
Functional

- Corpus: a documentation set you are allowed to use (for example the docs of an open-source project) or a BEIR/NQ subset with labelled answers.
- Ingestion: parse, chunk, embed, index in PostgreSQL + pgvector with full-text search (hybrid, as in Project 18).
- Answering: retrieve top k, optionally re-rank, build a prompt with numbered passages, generate an answer that cites passage numbers, or says "I don't know" when sources are insufficient.
- Evaluation set: at least 150 questions with reference answers and gold passages (from the dataset or written and reviewed by you).
- Metrics: retrieval Recall@k and MRR; answer correctness (exact match / F1 for short answers, or a rubric-based judge model for long answers, validated against your own labels); faithfulness (share of answer claims supported by retrieved passages); citation accuracy; "refuse when unanswerable" rate.
- Experiment runner comparing configurations, logged to MLflow.

Non-functional

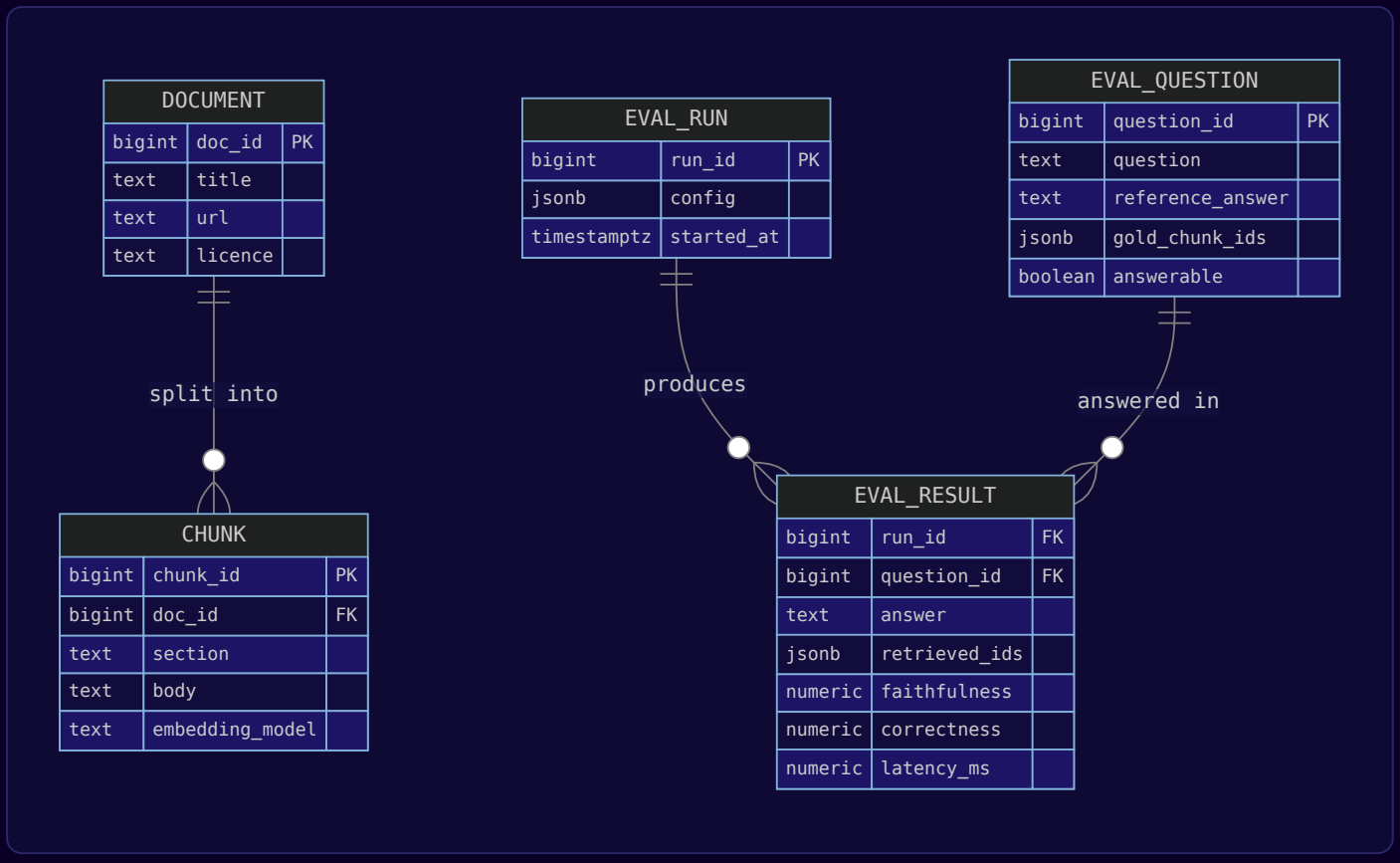
- End-to-end answer latency under 5 seconds p95 (model-dependent; report it).
- Cost per answer tracked.
- No document content sent to external services unless allowed by its licence and your settings (or use a locally run open model).
- Prompt-injection test cases included (documents containing instructions must not be followed).

The architecture



- **Ingest**: chunking and metadata (title, section, URL) matter for citation display.
- **Re-rank**: a cross-encoder improves which 5 passages reach the model, the main lever for faithfulness.
- **Prompt**: instructs the model to use only the passages, cite them, and say when it cannot answer.
- **Evaluate**: the same eval set and metrics for every configuration, so changes are comparable.

Data model



Tech stack

Component	Choice	Why
Embeddings and re-ranker	sentence-transformers models on PyTorch	Retrieval quality
Store	PostgreSQL + pgvector + full-text	One database, hybrid search
Generator	a hosted LLM API or a locally run open model	Swappable behind one interface
Orchestration	plain Python	Avoid framework lock-in while learning
Tracking	MLflow	Configs and metrics per run
API	FastAPI	Ask endpoint

Milestones

1. **Eval set first.** *Done when* 150+ questions with gold passages and reference answers exist, including 20 unanswerable ones and 10 prompt-injection cases.
2. **Retrieval.** *Done when* hybrid retrieval Recall@5 and MRR are measured on the eval set.
3. **Answering.** *Done when* answers with citations are produced and correctness and faithfulness are scored.
4. **Judge validation.** *Done when* any model-based judge is checked against your own labels on 50 answers and agreement is reported.
5. **Experiments.** *Done when* at least four configurations (chunk size, k, re-ranker on/off, prompt variant) are compared in MLflow with a clear winner per metric.
6. **Serve.** *Done when* the API answers within latency with citations, refusals and logging.

Deep dive: the hard parts

- **Retrieval is the ceiling.** If the right passage is not retrieved, the model cannot answer faithfully. Always report retrieval metrics separately, and fix retrieval (chunking, hybrid search, re-ranking) before tuning prompts.

- **Measuring faithfulness.** Split the answer into claims and check each against the retrieved passages — with a natural-language-inference model or an LLM judge. Judges have biases (favouring longer answers, their own style); validate on human labels and keep the judge fixed across experiments.
- **Knowing when not to answer.** A system that always answers will invent answers to unanswerable questions. Measure the refusal rate on unanswerable questions *and* the false-refusal rate on answerable ones.
- **Prompt injection.** Retrieved documents may contain text like "ignore previous instructions". Treat retrieved text as data: separate it clearly in the prompt, instruct the model not to follow it, and test with planted injections. Never give the generator tools that act on the world based on retrieved text without checks.

Testing strategy

- Deterministic unit tests for chunking, citation parsing and prompt building.
- Regression eval: a CI job runs a 30-question smoke subset on every change and fails on metric drops.
- Injection tests: planted instructions in documents never change behaviour.
- Privacy test: answers never include content from documents the user is not allowed to see (if you add access control).

Trade-offs, scaling and failure modes

- Larger k gives the model more evidence but more distraction and cost; re-ranking to a small k is usually better.
- Hosted models are strong and easy but send data outside; local models keep data private but need GPUs and are usually weaker.
- Stale index: documents change; schedule re-ingestion and show document dates in citations.

Stretch goals

- Query rewriting for multi-turn conversations.
- Fine-tune the embedding model on question–passage pairs from your eval data (with a held-out split).
- Access-controlled retrieval: filter chunks by user permissions before ranking.

Skills practised

- Ch 12 — Sequences, RNNs and Attention: transformers, embeddings, tokenisation.
- Ch 13 — Recommender Systems and Embeddings: embedding retrieval and ANN.
- Ch 6 — Evaluation and Tuning: ranking metrics, building evaluation sets.
- Ch 19 — Responsible ML: reliability, privacy and safety risks.

In the interview

Q: How do you evaluate a RAG system? **A:** Separately: retrieval (Recall@k, MRR against gold passages), generation faithfulness (claims supported by sources), correctness (against references), refusal behaviour, latency and cost — on a fixed eval set for every change.

Q: Answers are wrong. Where do you look first? **A:** Retrieval. Check whether the gold passage was retrieved; if not, fix chunking, hybrid search or re-ranking before touching prompts or models.

Q: What is prompt injection in RAG and how do you mitigate it? **A:** Retrieved text containing instructions the model might follow. Mitigate by clearly delimiting retrieved content, instructing the model to treat it as data, limiting tool access, and testing with planted injections.

TL;DR

Build the evaluation set first, measure retrieval and faithfulness separately, fix retrieval before prompts, and test refusals and prompt injection explicitly.

39. Federated next-word prediction

Level: Advanced · **Time:** 4–6 weeks · **Prerequisites:** Ch 12, 19

What it is

A next-word prediction model for a phone keyboard, trained with **federated learning**: instead of uploading users' typing to a server, each device trains on its own data and sends only model updates; the server averages them (**FedAvg**) into a new global model. You simulate hundreds of clients with non-identical data, add **secure aggregation** concepts and **differential privacy** (clipping each update and adding calibrated noise so no individual can be identified from the model), and measure the accuracy cost of privacy.

The problem it solves

Typing data is extremely personal — messages, passwords, health questions. Centralising it creates privacy and security risks. Federated learning lets the model learn from real usage while raw text stays on the device; differential privacy limits what even the updates reveal.

Real-world examples

- McMahan et al.'s Communication-Efficient Learning of Deep Networks from Decentralized Data (2016) introduced federated averaging.
- Hard et al.'s Federated Learning for Mobile Keyboard Prediction (Google, 2018) trained a recurrent next-word model for Gboard with federated learning and compared it to server-trained models.
- LEAF provides federated benchmarks including a Shakespeare next-character task partitioned by speaking role; Flower is an open-source federated learning framework.

Requirements

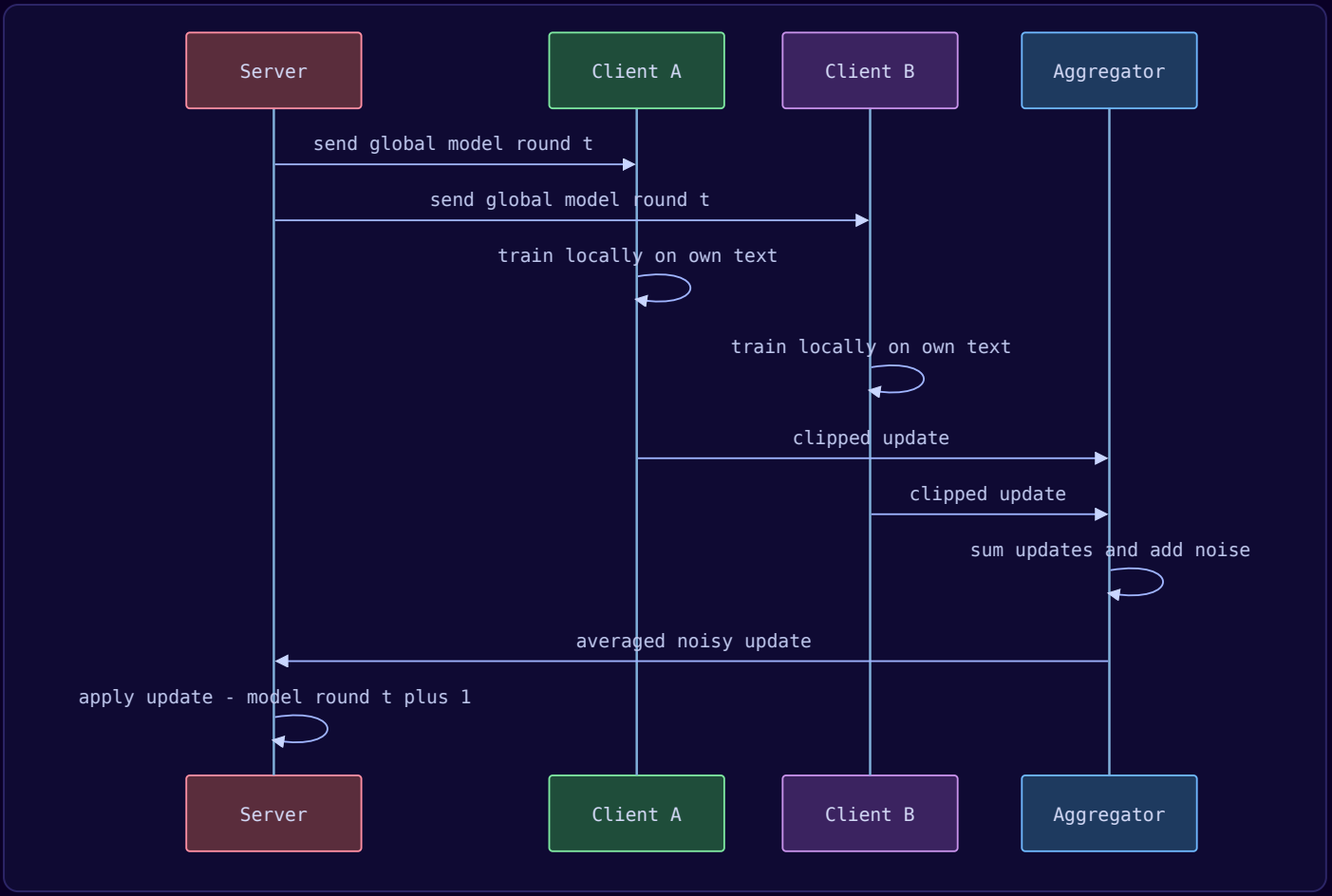
Functional

- Data: LEAF Shakespeare (clients = speaking roles) or a Reddit/StackOverflow-style corpus partitioned by user; keep natural non-IID partitions.
- Model: small LSTM or GRU language model (word- or character-level), with a vocabulary fixed in advance.
- Baseline: centralised training on pooled data (the upper bound).
- Federated: FedAvg with client sampling per round, local epochs, weighted averaging.
- Privacy: per-client update clipping and Gaussian noise (DP-FedAvg); report the privacy budget (ε, δ) from an accountant.
- Metrics: next-word top-1 and top-3 accuracy on held-out clients; rounds to reach a target; bytes communicated.
- Simulate stragglers and dropouts (clients that do not report back).

Non-functional

- Simulation of 500 clients × 200 rounds on one machine within a few hours.
- Model small enough for a phone (under 5 MB after quantisation).
- No raw client data ever leaves its client object in the simulation (enforced by code structure).

The architecture



- **Server**: picks a sample of available clients each round and broadcasts the model.
- **Clients**: train for a few local epochs on their own data, compute the weight difference, clip it to a maximum norm.
- **Aggregator**: in real systems, secure aggregation lets the server see only the sum, not individual updates; here simulate the interface. Gaussian noise added to the sum gives differential privacy.
- **Server update**: applies the averaged update (optionally with server-side momentum).

Data model

Item	Type	Notes
Client dataset	list of token sequences	Stays inside the client object
Global model	state dict	Broadcast each round
Client update	weight delta, clipped	Norm ≤ C
Round log	round, clients sampled, clients reported, metrics	For analysis
Privacy accountant state	noise multiplier, sampling rate, steps	Gives ε for δ

Tech stack

Component	Choice	Why
Model	PyTorch LSTM/GRU	Small language model
Federated simulation	your own loop first, then Flower (check Python 3.14 support)	Learn the mechanics
DP accounting	a DP library's RDP accountant (e.g. Opacus, if compatible)	Correct ε calculation
Tracking	MLflow	Rounds, accuracy, ε

Milestones

1. **Centralised baseline.** *Done when* the pooled-data model's next-word accuracy on held-out clients is recorded.
2. **FedAvg.** *Done when* federated training reaches a stated fraction of the centralised accuracy and you plot accuracy vs rounds.
3. **Non-IID and system effects.** *Done when* you compare IID vs natural partitions and show the effect of 30% client dropout and of local epochs.
4. **Differential privacy.** *Done when* clipping + noise is added, ε is reported, and an accuracy vs ε curve is plotted.
5. **Compression.** *Done when* update quantisation or sparsification reduces bytes per round with measured accuracy cost.

Deep dive: the hard parts

- **Non-IID data.** Each user's text is different (vocabulary, topics, language). Local training pulls models in different directions ("client drift"), slowing convergence. Fewer local epochs, client weighting and server momentum help; measure how far from centralised you end up.
- **Clipping and noise tuning.** Clip too tightly and you destroy signal; too loosely and noise must be large to protect outliers. Tune clip norm and noise multiplier together, and many clients per round reduce the relative noise.
- **What ε means.** A smaller ε is stronger privacy. Report ε with δ, the number of rounds and sampling rate; note that it bounds what an attacker can learn about any one client's participation.
- **Evaluation without central data.** In real deployments, evaluation also runs federated: clients compute metrics on their own held-out data and report aggregates.

Testing strategy

- FedAvg correctness: with IID data and one local step, FedAvg equals centralised mini-batch SGD (within numerical tolerance).
- Privacy plumbing: assert every update's norm ≤ clip value after clipping; noise standard deviation matches configuration.
- Data isolation: code review plus tests that the server object never holds client datasets.
- Dropout test: rounds with missing clients still produce valid averages.

Trade-offs, scaling and failure modes

- Federated training is slower and less accurate than centralised; the gain is privacy and access to on-device data that would never be uploaded.
- DP costs accuracy, especially for rare words (the very words that might identify someone).
- Real devices are available only when idle, charging and on Wi-Fi, biasing who participates — and so whose language the model learns.

Stretch goals

- Personalisation: fine-tune the global model locally per client and measure the gain.
- Membership inference attack against centralised vs DP models.
- FedProx or SCAFFOLD for non-IID robustness.

Skills practised

- Ch 19 — Responsible ML: privacy, federated learning, differential privacy.
- Ch 12 — Sequences, RNNs and Attention: RNN language models, tokenisation.
- Ch 10 — Neural Networks and Deep Learning: optimisation and gradient clipping.
- Ch 16 — ML in Production: on-device models and compression.

In the interview

Q: How does federated averaging work? **A:** The server sends the model to a sample of clients; each trains locally and returns its weight update; the server averages updates weighted by client data size and applies the result. Raw data never leaves devices.

Q: Does federated learning guarantee privacy? **A:** No. Updates can leak information. Secure aggregation hides individual updates from the server, and differential privacy (clipping + noise) limits what the final model reveals about any participant.

Q: What makes federated learning harder than centralised? **A:** Non-IID data, unreliable and slow clients, limited communication, privacy noise, and evaluation without central data.

TL;DR

Train on-device with FedAvg, protect updates with clipping, noise and secure aggregation, and report the accuracy cost against the privacy budget honestly.

40. Similar listings and real-time personalisation

Level: Advanced · **Time:** 4–6 weeks · **Prerequisites:** Ch 13, 17, 18

What it is

The capstone of capstones for a two-sided marketplace (homes, hotels or second-hand goods): learn **listing embeddings** from users' click sessions with a word2vec-style **skip-gram** model (listings clicked in the same session are treated like words in the same sentence), then use them twice — for a **similar listings** carousel on each listing page, and for **real-time personalisation** of search results, where features like "similarity of this result to listings the user clicked in the last 30 minutes" are added to a search ranker. You build the training pipeline, both product surfaces, and an evaluation plan.

The problem it solves

Marketplace listings are unique, short-lived and spread across many cities, so classic collaborative filtering struggles. Session-based embeddings capture what users treat as substitutes (similar price, style, neighbourhood), work across a huge catalogue and can be updated daily. Using them in real time lets search react to what a user is doing *right now*, not just their long-term profile.

Real-world examples

- Grbovic and Cheng's KDD 2018 paper Real-time Personalization using Embeddings for Search Ranking at Airbnb describes skip-gram listing embeddings trained on click sessions, with booked listings used as global context and same-market negatives.
- Airbnb's engineering post Listing Embeddings in Search Ranking explains how embeddings powered similar-listing recommendations and real-time search features.
- Inside Airbnb publishes listing and calendar data for many cities, usable for listing attributes (though not real click sessions — you will simulate those).

Requirements

Functional

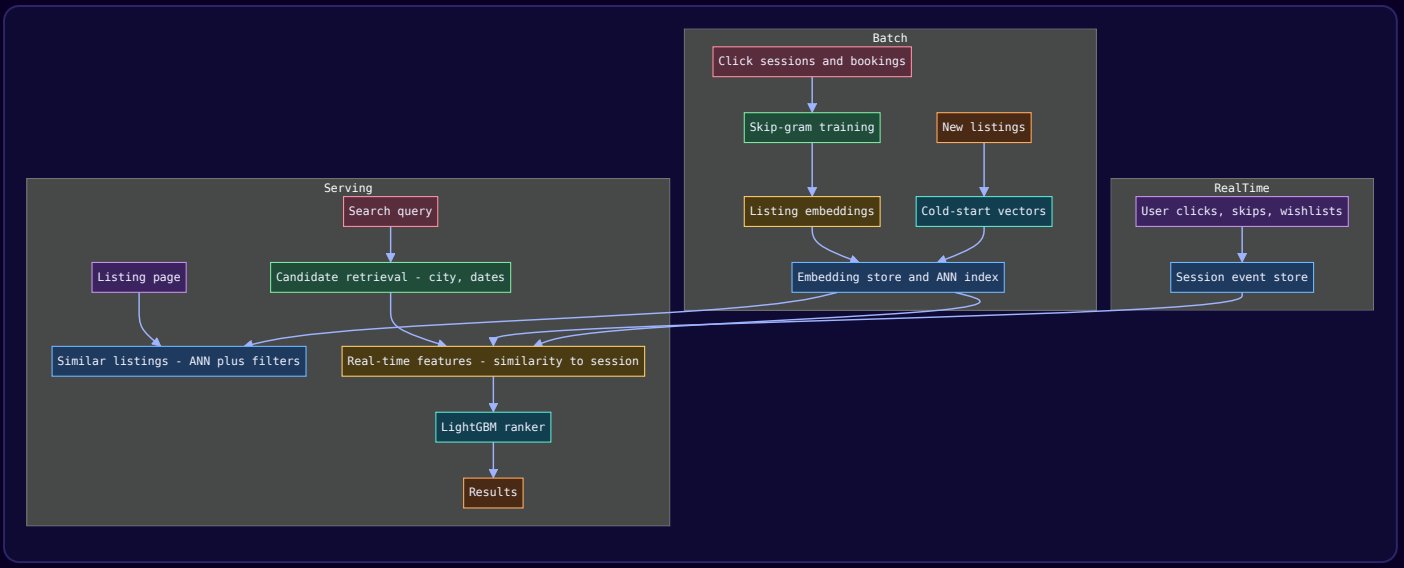
- Data: Inside Airbnb listings for two or three cities; a **session simulator** that generates browsing sessions from latent user preferences (price band, room type, neighbourhood, amenities) with occasional bookings.
- Embeddings: skip-gram with negative sampling on sessions; booked listing as a global context token; negatives sampled from the same city.
- Cold start: embeddings for new listings as the average of nearby, similar-type listings.
- Similar listings: k-nearest neighbours in embedding space, filtered by city, availability and guest capacity.
- Search personalisation: a LightGBM ranker with base features (price, reviews, distance) plus real-time embedding features (max/avg cosine to recently clicked, skipped and wish-listed listings).
- Metrics: offline — rank of the eventually booked listing within search results (NDCG/MRR), similar-listing click-through in simulation; embedding sanity checks (clusters by neighbourhood and price).
- APIs: `GET /similar/{listing_id}`, `POST /search` with session events.

Non-functional

- Similar listings under 30 ms; search re-rank of 200 listings under 80 ms.

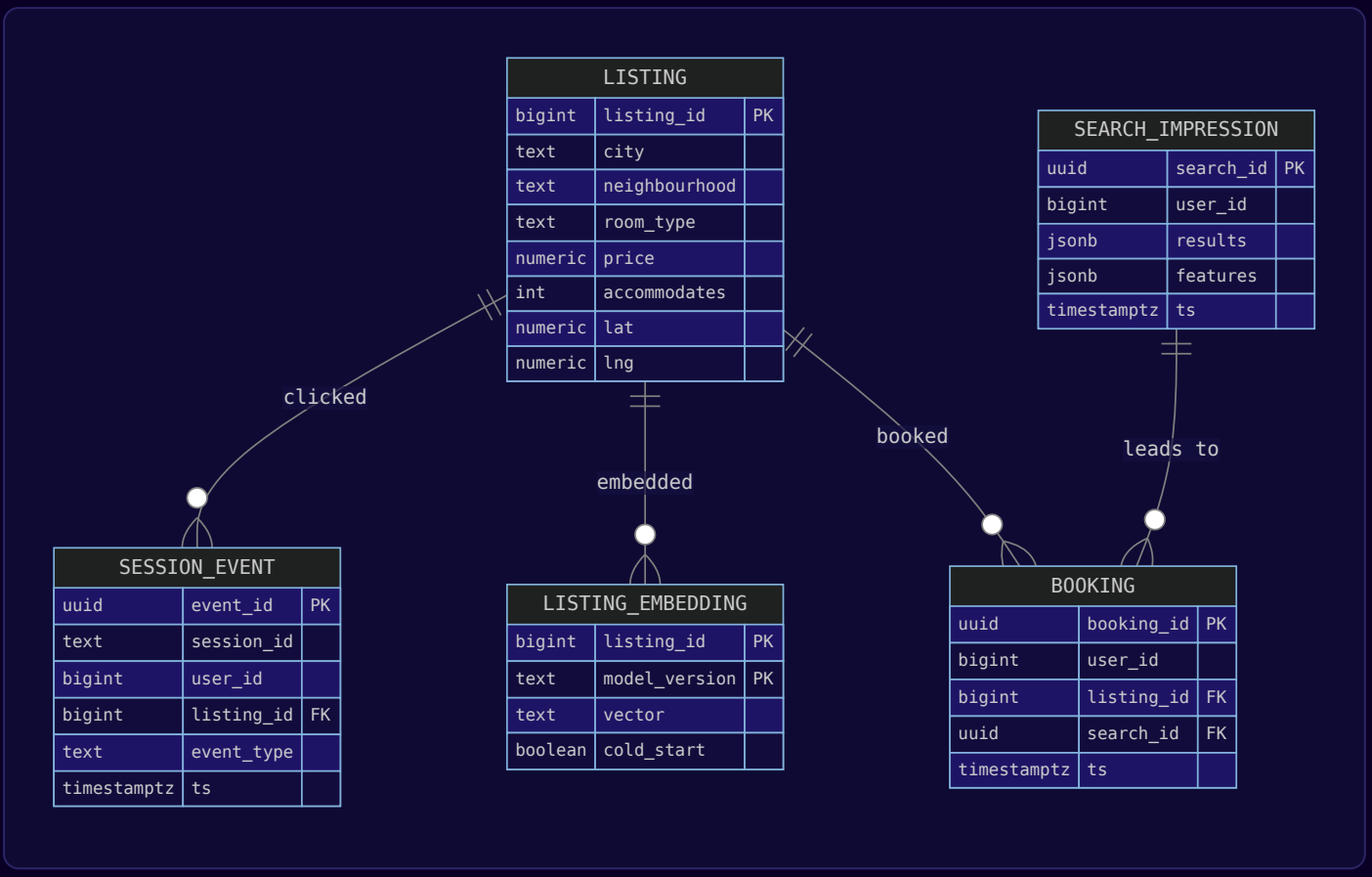
- Real-time features reflect a click within 2 seconds.
- Embeddings retrained daily; new listings get cold-start vectors within an hour.

The architecture



- **Skip-gram training**: for each clicked listing, predict nearby clicked listings in the same session (positives) against random and same-city listings (negatives); the booked listing is added as context for every listing in a booking session.
- **Cold-start vectors**: average the embeddings of the closest listings with matching type and price band.
- **Session event store**: recent clicks and skips per user, readable in milliseconds.
- **Real-time features**: cosine between each candidate and the user's recent positive and negative listings.
- **Ranker**: trained on simulated search logs where the label is booking (or long click).

Data model



Tech stack

Component	Choice	Why
Embeddings	PyTorch skip-gram (or a word2vec library on Python 3.14-compatible build)	Custom contexts and negatives
ANN	FAISS or pgvector HNSW	Similar-listing lookup
Session store	Redis	Millisecond reads and writes of recent events
Ranker	LightGBM (lambdarank)	Base + embedding features
Data	Inside Airbnb CSVs + polars	Real listing attributes
API	FastAPI	Similar and search endpoints

Milestones

1. **Simulator.** *Done when* simulated sessions show realistic patterns (users browse within a price band and area, with occasional switches) and bookings at a plausible rate.
2. **Embeddings.** *Done when* skip-gram embeddings are trained and sanity checks pass: nearest neighbours share neighbourhood, room type and price band far more than random pairs.
3. **Similar listings.** *Done when* the endpoint returns filtered neighbours and, in simulation, similar-listing clicks beat a "same neighbourhood, similar price" heuristic.
4. **Real-time features.** *Done when* session-similarity features are computed from Redis within latency and with no future leakage.
5. **Personalised ranking.** *Done when* the ranker with embedding features improves the booked listing's rank (NDCG/MRR) over the base ranker, on a time-based split.
6. **Experiment plan.** *Done when* you have written an A/B test design (Project 17) with primary metric, guardrails and duration.

Deep dive: the hard parts

- **Marketplace-specific negatives.** Random negatives mostly come from other cities, so the model learns geography and little else. Sampling extra negatives from the same city forces it to learn finer distinctions (style, price, amenities) — the trick Airbnb describes.
- **Bookings as global context.** Clicks are noisy; bookings are the real goal but rare. Adding the booked listing as context for every listing in the session pulls embeddings toward what led to bookings.
- **Leakage in real-time features.** When building training data for the ranker, session features must include only events *before* the search impression. Replaying the event store in time order (as in Project 28) avoids this.
- **Simulation honesty.** Results on simulated sessions reflect the simulator's assumptions. Use the simulator to test the pipeline and relative ideas, and say clearly that real gains require real logs and an online test.

Testing strategy

- Embedding sanity tests: same-neighbourhood neighbour rate above a threshold; vectors normalised; no NaNs.
- Cold-start test: a new listing gets a vector close to its attribute-matched neighbours.
- Point-in-time tests for session features using a scripted event sequence.
- API contract tests: similar listings respect city, capacity and availability filters.
- Latency tests for both endpoints.

Trade-offs, scaling and failure modes

- Session embeddings capture behavioural similarity but not explicit attributes; combine with content features for new and niche listings.
- Daily retraining changes the embedding space; never compare vectors across versions, and refresh all dependent features together.

- Two-sided fairness: personalisation can concentrate bookings on a few hosts. Track exposure distribution across hosts as a guardrail.
- Napkin math: 7 million listings × 32 floats × 4 bytes is under 1 GB — comfortably in memory on one server, so the ANN index is not the bottleneck.

Stretch goals

- User-type and listing-type embeddings for long-term preferences (as in the KDD paper) to handle users with few sessions.
- Multi-objective ranking balancing guest booking probability and host acceptance.
- Run the whole design through the seven-step framework (Ch 15) and present it as a mock interview.

Skills practised

- Ch 13 — Recommender Systems and Embeddings: skip-gram embeddings, ANN, cold start.
- Ch 18 — ML Case Studies II: similar listings case study.
- Ch 17 — ML Case Studies I: search ranking.
- Ch 16 — ML in Production: real-time features and point-in-time correctness.
- Ch 15 — ML System Design Framework: end-to-end design presentation.

In the interview

Q: How would you learn listing embeddings without content features? **A:** Treat click sessions as sentences and listings as words, train skip-gram with negative sampling, add booked listings as global context and same-market negatives.

Q: How do embeddings help search ranking in real time? **A:** For each candidate, compute similarity to listings the user clicked, skipped or saved in the current session, and feed those as features to the ranker, so results adapt within the session.

Q: How do you handle a brand-new listing? **A:** Initialise its embedding from similar nearby listings by attributes, then let real interactions update it at the next retrain.

TL;DR

Learn listing embeddings from sessions with marketplace-aware negatives, use them for similar listings and as real-time search features, and keep every session feature point-in-time correct.